

MODULE 1

1

Introduction to Statics



The Flemish mathematician and engineer Simon Stevinus (1548–1620) was the first to demonstrate resolution of forces, thereby establishing the foundation of modern statics.
© Bettmann/CORBIS

1.1 Introduction

a. *What is engineering mechanics?*

Statics and dynamics are among the first engineering topics encountered by most students. Therefore, it is appropriate that we begin with a brief exposition on the meaning of the term *engineering mechanics* and on the role that these courses play in engineering education. Before defining engineering mechanics, we must first consider the similarities and differences between physics and engineering.

In general terms, *physics* is the science that relates the properties of matter and energy, excluding biological and chemical effects. Physics includes the study

of mechanics,* thermodynamics, electricity and magnetism, and nuclear physics. On the other hand, *engineering* is the application of the mathematical and physical sciences (physics, chemistry, and biology) to the design and manufacture of items that benefit humanity. *Design* is the key concept that distinguishes engineers from scientists. According to the Accreditation Board for Engineering and Technology (ABET), engineering design is the process of devising a system, component, or process to meet desired needs.

Mechanics is the branch of physics that considers the action of forces on bodies or fluids that are *at rest* or *in motion*. Correspondingly, the primary topics of mechanics are statics and dynamics. The first topic that you studied in your initial physics course, in either high school or college, was undoubtedly mechanics. Thus, *engineering mechanics* is the branch of engineering that applies the principles of mechanics to mechanical design (i.e., any design that must take into account the effect of forces). The primary goal of engineering mechanics courses is to introduce the student to the engineering applications of mechanics. Statics and Dynamics are generally followed by one or more courses that introduce material properties and deformation, usually called Strength of Materials or Mechanics of Materials. This sequence of courses is then followed by formal training in mechanical design.

Of course, engineering mechanics is an integral component of the education of engineers whose disciplines are related to the mechanical sciences, such as aerospace engineering, architectural engineering, civil engineering, and mechanical engineering. However, a knowledge of engineering mechanics is also useful in most other engineering disciplines, because there, too, the mechanical behavior of a body or fluid must often be considered. Because mechanics was the first physical science to be applied to everyday life, it follows that engineering mechanics is the oldest branch of engineering. Given the interdisciplinary character of many engineering applications (e.g., robotics and manufacturing), a sound training in engineering mechanics continues to be one of the more important aspects of engineering education.

b. *Problem formulation and the accuracy of solutions*

Your mastery of the principles of engineering mechanics will be reflected in your ability to formulate and solve problems. Unfortunately, there is no simple method for teaching problem-solving skills. Nearly all individuals require a considerable amount of practice in solving problems before they begin to develop the analytical skills that are so necessary for success in engineering. For this reason, a relatively large number of sample problems and homework problems are placed at strategic points throughout this text.

To help you develop an “engineering approach” to problem analysis, you will find it instructive to divide your solution for each homework problem into the following parts:

1. **GIVEN:** After carefully reading the problem statement, list all the data provided. If a figure is required, sketch it neatly and approximately to scale.
2. **FIND:** State precisely the information that is to be determined.

*When discussing the topics included in physics, the term *mechanics* is used without a modifier. Quite naturally, this often leads to confusion between “mechanics” and “engineering mechanics.”

3. **SOLUTION:** Solve the problem, showing all the steps that you used in the analysis. Work neatly so that your work can be easily followed by others.
4. **VALIDATE:** Many times, an invalid solution can be uncovered by simply asking yourself, “Does the answer make sense?”

When reporting your answers, use only as many digits as the least accurate value in the given data. For example, suppose that you are required to convert 12523 m and 12 cm (assumed to be accurate to four significant digits) to kilometers. Using a calculator, you would divide 12523.12 by 100 m/km and report the answer as 12.52 (four significant digits), although the quotient displayed on the calculator would be 12.52312. Reporting the answer as 12.52312 implies that all seven digits are significant, which is, of course, untrue. It is your responsibility to round off the answer to the correct number of digits. *In this text*, you should assume that given data are accurate to three significant digits unless stated otherwise. For example, a length that is given as 3 m should be interpreted as 3.00 m.

When performing intermediate calculations, a good rule of thumb is to carry one more digit than will be reported in the final answer; for example, use four-digit intermediate values if the answer is to be significant to three digits. Furthermore, it is common practice to report four digits if the first digit in an answer is 1; for example, use 1.392 rather than 1.39.

1.2 Newtonian Mechanics

a. Scope of Newtonian mechanics

In 1687 Sir Isaac Newton (1642–1727) published his celebrated laws of motion in *Principia* (*Mathematical Principles of Natural Philosophy*). Without a doubt, this work ranks among the most influential scientific books ever published. We should not think, however, that its publication immediately established classical mechanics. Newton’s work on mechanics dealt primarily with celestial mechanics and was thus limited to particle motion. Another two hundred or so years elapsed before rigid-body dynamics, fluid mechanics, and the mechanics of deformable bodies were developed. Each of these areas required new axioms before it could assume a usable form.

Nevertheless, Newton’s work is the foundation of classical, or Newtonian, mechanics. His efforts have even influenced two other branches of mechanics, born at the beginning of the twentieth century: relativistic and quantum mechanics. *Relativistic mechanics* addresses phenomena that occur on a cosmic scale (velocities approaching the speed of light, strong gravitational fields, etc.). It removes two of the most objectionable postulates of Newtonian mechanics: the existence of a fixed or inertial reference frame and the assumption that time is an absolute variable, “running” at the same rate in all parts of the universe. (There is evidence that Newton himself was bothered by these two postulates.) *Quantum mechanics* is concerned with particles on the atomic or subatomic scale. It also removes two cherished concepts of classical mechanics: determinism and continuity. Quantum mechanics is essentially a probabilistic theory; instead of predicting an event, it determines the likelihood that an event will occur. Moreover, according to this theory, the events occur in discrete steps (called *quanta*) rather than in a continuous manner.

Relativistic and quantum mechanics, however, have by no means invalidated the principles of Newtonian mechanics. In the analysis of the motion of bodies encountered in our everyday experience, both theories converge on the equations of Newtonian mechanics. Thus the more esoteric theories actually reinforce the validity of Newton's laws of motion.

b. *Newton's laws for particle motion*

Using modern terminology, Newton's laws of particle motion may be stated as follows:

1. If a particle is at rest (or moving with constant velocity in a straight line), it will remain at rest (or continue to move with constant velocity in a straight line) unless acted upon by a force.
2. A particle acted upon by a force will accelerate in the direction of the force. The magnitude of the acceleration is proportional to the magnitude of the force and inversely proportional to the mass of the particle.
3. For every action, there is an equal and opposite reaction; that is, the forces of interaction between two particles are equal in magnitude and oppositely directed along the same line of action.

Although the first law is simply a special case of the second law, it is customary to state the first law separately because of its importance to the subject of statics.

c. *Inertial reference frames*

When applying Newton's second law, attention must be paid to the coordinate system in which the accelerations are measured. An *inertial reference frame* (also known as a Newtonian or Galilean reference frame) is defined to be any rigid coordinate system in which Newton's laws of particle motion relative to that frame are valid with an acceptable degree of accuracy. In most design applications used on the surface of the earth, an inertial frame can be approximated with sufficient accuracy by attaching the coordinate system to the earth. In the study of earth satellites, a coordinate system attached to the sun usually suffices. For interplanetary travel, it is necessary to use coordinate systems attached to the so-called fixed stars.

It can be shown that any frame that is translating with constant velocity relative to an inertial frame is itself an inertial frame. It is a common practice to omit the word *inertial* when referring to frames for which Newton's laws obviously apply.

d. *Units and dimensions*

The standards of measurement are called *units*. The term *dimension* refers to the type of measurement, regardless of the units used. For example, kilogram and m/s are units, whereas mass and length/time are dimensions. Throughout this text we use SI system (from *Système internationale d'unités*). The base dimensions in the SI system are mass $[M]$, length $[L]$, and time $[T]$, and the base units are kilogram (kg), meter (m), and second (s). All other dimensions or units are combinations

of the base quantities. For example, the dimension of velocity is $[L/T]$, the units being ft/s, m/s, and so on.

A system with the base dimensions $[FLT]$ is called a *gravitational system*. If the base dimensions are $[MLT]$ (as in the SI system), the system is known as an *absolute system*. In each system of measurement, the base units are defined by physically reproducible phenomena or physical objects. For example, the second is defined by the duration of a specified number of radiation cycles in a certain isotope, the kilogram is defined as the mass of a certain block of metal kept near Paris, France, and so on.

All equations representing physical phenomena must be *dimensionally homogeneous*; that is, each term of an equation must have the same dimension. Otherwise, the equation will not make physical sense (it would be meaningless, for example, to add a force to a length). Checking equations for dimensional homogeneity is a good habit to learn, as it can reveal mistakes made during algebraic manipulations.

e. Mass, force, and weight

If a force $\mathbf{F}$ acts on a particle of mass m , Newton's second law states that

$$\mathbf{F} = m\mathbf{a} \quad (1.1)$$

where $\mathbf{a}$ is the acceleration vector of the particle. For a gravitational $[FLT]$ system, dimensional homogeneity of Eq. (1.1) requires the dimension of mass to be

$$[M] = \left[\frac{FT^2}{L} \right] \quad (1.2a)$$

For an absolute $[MLT]$ system of units, dimensional homogeneity of Eq. (1.1) yields for the dimension of force

$$[F] = \left[\frac{ML}{T^2} \right] \quad (1.2b)$$

The derived unit of force in the SI system is a *newton* (N), defined as the force that accelerates a 1.0-kg mass at the rate of 1.0 m/s^2 . From Eq. (1.2b), we obtain

$$1.0 \text{ N} = 1.0 \text{ kg} \cdot \text{m/s}^2$$

Weight is the force of gravitation acting on a body. Denoting gravitational acceleration (free-fall acceleration of the body) by g , the weight W of a body of mass m is given by Newton's second law as

$$W = mg \quad (1.3)$$

Note that mass is a constant property of a body, whereas weight is a variable that depends on the local value of g . The gravitational acceleration on the surface of the earth is approximately 9.81 m/s^2 . Thus the mass of a body is 1.0 kg , its weight on earth is $(9.81 \text{ m/s}^2)(1.0 \text{ kg}) = 9.81 \text{ N}$.

f. Conversion of units

A convenient method for converting a measurement from one set of units to another is to multiply the measurement by appropriate conversion factors. For example, to convert 360 km/h into m/s , we proceed as follows:

$$360 \text{ km/h} = 360 \frac{\cancel{\text{km}}}{\cancel{\text{h}}} \times \frac{1.0\cancel{\text{h}}}{3600 \text{ s}} \times \frac{1000 \text{ m}}{1.0\cancel{\text{km}}} = 100 \text{ m/s}$$

where the multipliers $1.0 \text{ h}/3600 \text{ s}$ and $1000 \text{ m}/1.0 \text{ km}$ are conversion factors. Because $1.0 \text{ h} = 3600 \text{ s}$ and $1000 \text{ m} = 1.0 \text{ km}$, we see that each conversion factor is dimensionless and of magnitude 1. Therefore, a measurement is unchanged when it is multiplied by conversion factors—only its units are altered. Note that it is permissible to cancel units during the conversion as if they were algebraic quantities.

Conversion factors applicable to mechanics are listed inside the front cover of the book.

g. Law of gravitation

In addition to his many other accomplishments, Newton also proposed the law of universal gravitation. Consider two particles of mass m_A and m_B that are separated by a distance R , as shown in Fig. 1.1. The law of gravitation states that the two particles are attracted to each other by forces of magnitude F that act along the line connecting the particles, where

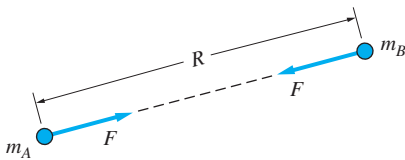


Fig. 1.1

$$F = G \frac{m_A m_B}{R^2} \quad (1.4)$$

The universal gravitational constant G is equal to $6.67 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$. Although this law is valid for particles, Newton showed that it is also applicable to spherical bodies, provided that their masses are distributed uniformly. (When attempting to derive this result, Newton was forced to develop calculus.)

If we let $m_A = M_e$ (the mass of the earth), $m_B = m$ (the mass of a body), and $R = R_e$ (the mean radius of the earth), then F in Eq. (1.4) will be the weight W of the body. Comparing $W = G M_e m / R_e^2$ with $W = mg$, we find that $g = G M_e / R_e^2$. Of course, adjustments may be necessary in the value of g for some applications in order to account for local variation of the gravitational attraction.

Sample Problem 1.1

Convert $2 \times 10^5 \text{ N/mm}^2$ to GPa ($1 \text{ Pa} = 1 \text{ N/m}^2$).

Solution

Using the conversion factor listed above, we obtain

$$\begin{aligned} 2 \times 10^5 \frac{\text{N}}{\text{mm}^2} &= \frac{2 \times 10^5 \left[\frac{\text{GN}}{10^9} \right]}{\left[\frac{\text{m}}{1000} \right]^2} \\ &= 200 \text{ GN/m}^2 \\ &= 200 \text{ GPa} \end{aligned}$$

Sample Problem 1.2

The acceleration a of a particle is related to its velocity v , its position coordinate x , and time t by the equation

$$a = Ax^3t + Bvt^2 \quad (\text{a})$$

where A and B are constants. The dimension of the acceleration is length per unit time squared; that is, $[a] = [L/T^2]$. The dimensions of the other variables are $[v] = [L/T]$, $[x] = [L]$, and $[t] = [T]$. Derive the dimensions of A and B if Eq. (a) is to be dimensionally homogeneous.

Solution

For Eq. (a) to be dimensionally homogeneous, the dimension of each term on the right-hand side of the equation must be $[L/T^2]$, the same as the dimension for a . Therefore, the dimension of the first term on the right-hand side of Eq. (a) becomes

$$[Ax^3t] = [A][x^3][t] = [A][L^3][T] = \left[\frac{L}{T^2} \right] \quad (\text{b})$$

Solving Eq.(b) for the dimension of A , we find

$$[A] = \frac{1}{[L^3][T]} \left[\frac{L}{T^2} \right] = \frac{1}{[L^2T^3]} \quad \text{Answer}$$

Performing a similar dimensional analysis on the second term on the right-hand side of Eq. (a) gives

$$[Bvt^2] = [B][v][t^2] = [B] \left[\frac{L}{T} \right] [T^2] = \left[\frac{L}{T^2} \right] \quad (c)$$

Solving Eq. (c) for the dimension of B , we find

$$[B] = \left[\frac{L}{T^2} \right] \left[\frac{T}{L} \right] \left[\frac{1}{T^2} \right] = \left[\frac{1}{T^3} \right] \quad \text{Answer}$$

Sample Problem 1.3

Find the gravitational force exerted by the earth on a 70-kg man whose elevation above the surface of the earth equals the radius of the earth. The mass and radius of the earth are $M_e = 5.9742 \times 10^{24}$ kg and $R_e = 6378$ km, respectively.

Solution

Consider a body of mass m located at the distance $2R_e$ from the center of the earth (of mass M_e). The law of universal gravitation, from Eq. (11.4), states that the body is attracted to the earth by the force F given by

$$F = G \frac{mM_e}{(2R_e)^2}$$

where $G = 6.67 \times 10^{-11} \text{ m}^3/(\text{kg} \cdot \text{s}^2)$ is the universal gravitational constant. Substituting the values for G and the given parameters, the earth's gravitational force acting on the 70-kg man is

$$F = (6.67 \times 10^{-11}) \frac{(70)(5.9742 \times 10^{24})}{[2(6378 \times 10^3)]^2} = 171.4 \text{ N} \quad \text{Answer}$$

1.3 *Fundamental Properties of Vectors*

A knowledge of vectors is a prerequisite for the study of statics. In this article, we describe the fundamental properties of vectors, with subsequent articles discussing some of the more important elements of vector algebra. (The calculus of vectors will be introduced as needed in *Dynamics*.) We assume that you are already familiar with vector algebra—our discussion is intended only to be a review of the basic concepts.

The differences between scalar and vector quantities must be understood:

A *scalar* is a quantity that has magnitude only. A *vector* is a quantity that possesses magnitude and direction and obeys the parallelogram law for addition.

Because scalars possess only magnitudes, they are real numbers that can be positive, negative, or zero. Physical quantities that are scalars include temperature, time, and speed. As shown later, force, velocity, and displacement are examples of physical quantities that are vectors. The magnitude of a vector is always taken to be a nonnegative number. When a vector represents a physical quantity, the units of the vector are taken to be the same as the units of its magnitude (pounds, meters per second, feet, etc.).

The algebraic notation used for a scalar quantity must, of course, be different from that used for a vector quantity. In this text, we adopt the following conventions: (1) scalars are written as italicized English or Greek letters—for example, t for time and θ for angle; (2) vectors are written as boldface letters—for example, $\mathbf{F}$ for force; and (3) the magnitude of a vector $\mathbf{A}$ is denoted as $|\mathbf{A}|$ or simply as A (italic).

There is no universal method for indicating vector quantities when writing by hand. The more common notations are $\vec{A}$, $\underline{A}$, $\overline{A}$, and $\underline{A}$. Unless instructed otherwise, you are free to use the convention that you find most comfortable. However, it is imperative that you take care to always distinguish between scalars and vectors when you write.

The following summarizes several important properties of vectors.

Vectors as Directed Line Segments Any vector $\mathbf{A}$ can be represented geometrically as a directed line segment (an arrow), as shown in Fig. 1.2(a). The magnitude of $\mathbf{A}$ is denoted by A , and the direction of $\mathbf{A}$ is specified by the sense of the arrow and the angle θ that it makes with a fixed reference line. When using graphical methods, the length of the arrow is drawn proportional to the magnitude of the vector. Observe that the representation shown in Fig. 1.2(a) is complete because both the magnitude and direction of the vector are indicated. In some instances, it is also convenient to use the representation shown in Fig. 1.2(b), where the vector character of $\mathbf{A}$ is given additional emphasis by using boldface. Both of these representations for vectors are used in this text.

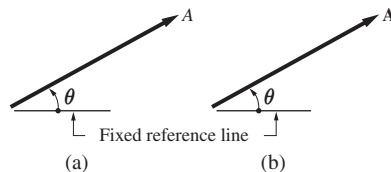


Fig. 1.2

We see that a vector does not possess a unique line of action, because moving a vector to a parallel line of action changes neither its magnitude nor its direction. In some engineering applications, the definition of a vector is more restrictive to include a line of action or even a point of application—see Art. 2.2.

Equality of Vectors Two vectors $\mathbf{A}$ and $\mathbf{B}$ are said to be equal, written as $\mathbf{A} = \mathbf{B}$, if (1) their magnitudes are equal—that is, $A = B$, and (2) they have the same direction.

Scalar-Vector Multiplication The multiplication of a scalar m and a vector $\mathbf{A}$, written as $m\mathbf{A}$ or as $\mathbf{A}m$, is defined as follows.

1. If m is positive, $m\mathbf{A}$ is the vector of magnitude mA that has the same direction as $\mathbf{A}$.
2. If m is negative, $m\mathbf{A}$ is the vector of magnitude $|m|A$ that is oppositely directed to $\mathbf{A}$.
3. If $m = 0$, $m\mathbf{A}$ (called the null or zero vector) is a vector of zero magnitude and arbitrary direction.

For $m = -1$, we see that $(-1)\mathbf{A}$ is the vector that has the same magnitude as $\mathbf{A}$ but is oppositely directed to $\mathbf{A}$. The vector $(-1)\mathbf{A}$, usually written as $-\mathbf{A}$, is called the *negative of $\mathbf{A}$* .

Unit Vectors A unit vector is a dimensionless vector with magnitude 1. Therefore, if λ represents a unit vector ($|\lambda| = 1$) with the same direction as $\mathbf{A}$, we can write

$$\mathbf{A} = A\lambda$$

This representation of a vector often is useful because it separates the magnitude A and the direction λ of the vector.

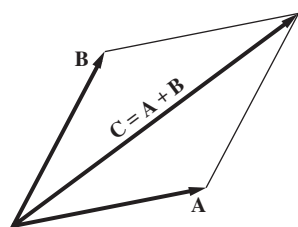
The Parallelogram Law for Addition and the Triangle Law The addition of two vectors $\mathbf{A}$ and $\mathbf{B}$ is defined to be the vector $\mathbf{C}$ that results from the geometric construction shown in Fig. 1.3(a). Observe that $\mathbf{C}$ is the diagonal of the parallelogram formed by $\mathbf{A}$ and $\mathbf{B}$. The operation depicted in Fig. 1.3(a), written as $\mathbf{A} + \mathbf{B} = \mathbf{C}$, is called the *parallelogram law for addition*. The vectors $\mathbf{A}$ and $\mathbf{B}$ are referred to as *components* of $\mathbf{C}$, and $\mathbf{C}$ is called the *resultant* of $\mathbf{A}$ and $\mathbf{B}$. The process of replacing a resultant with its components is called *resolution*. For example, $\mathbf{C}$ in Fig. 1.3(a) is resolved into its components $\mathbf{A}$ and $\mathbf{B}$.

An equivalent statement of the parallelogram law is the *triangle law*, which is shown in Fig. 1.3(b). Here the tail of $\mathbf{B}$ is placed at the tip of $\mathbf{A}$, and $\mathbf{C}$ is the vector that completes the triangle, drawn from the tail of $\mathbf{A}$ to the tip of $\mathbf{B}$. The result is identical if the tail of $\mathbf{A}$ is placed at the tip of $\mathbf{B}$ and $\mathbf{C}$ is drawn from the tail of $\mathbf{B}$ to the tip of $\mathbf{A}$.

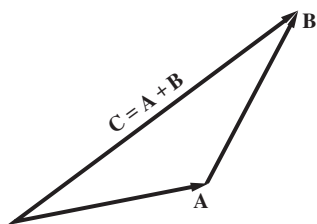
Letting $\mathbf{E}$, $\mathbf{F}$, and $\mathbf{G}$ represent any three vectors, we have the following two important properties (each follows directly from the parallelogram law):

- Addition is commutative: $\mathbf{E} + \mathbf{F} = \mathbf{F} + \mathbf{E}$
- Addition is associative: $\mathbf{E} + (\mathbf{F} + \mathbf{G}) = (\mathbf{E} + \mathbf{F}) + \mathbf{G}$

It is often convenient to find the sum $\mathbf{E} + \mathbf{F} + \mathbf{G}$ (no parentheses are needed) by adding the vectors from tip to tail, as shown in Fig. 1.4. The sum of the three vectors is seen to be the vector drawn from the tail of the first vector ($\mathbf{E}$) to the tip of the last vector ($\mathbf{G}$). This method, called the *polygon rule for addition*, can easily be extended to any number of vectors.



(a) Parallelogram law



(b) Triangle law

Fig. 1.3

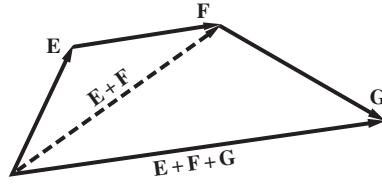


Fig. 1.4

The subtraction of two vectors $\mathbf{A}$ and $\mathbf{B}$, written as $\mathbf{A} - \mathbf{B}$, is defined as $\mathbf{A} - \mathbf{B} = \mathbf{A} + (-\mathbf{B})$, as shown in Fig. 1.5.

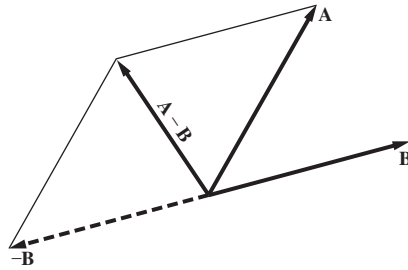
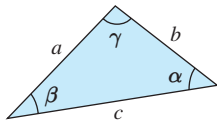


Fig. 1.5

Because of the geometric nature of the parallelogram law and the triangle law, vector addition can be accomplished graphically. A second technique is to determine the relationships between the various magnitudes and angles analytically by applying the *laws of sines and cosines* to a sketch of the parallelogram (or the triangle)—see Table 1.1. Both the graphical and the analytical methods are illustrated in Sample Problem 1.4.



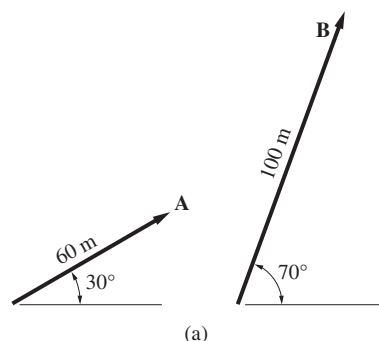
Law of sines	$\frac{a}{\sin \alpha} = \frac{b}{\sin \beta} = \frac{c}{\sin \gamma}$
Law of cosines	$a^2 = b^2 + c^2 - 2bc \cos \alpha$ $b^2 = c^2 + a^2 - 2ca \cos \beta$ $c^2 = a^2 + b^2 - 2ab \cos \gamma$

Table 1.1

Some words of caution: It is unfortunate that the symbols $+$, $-$, and $=$ are commonly used in both scalar algebra and vector algebra, because they have completely different meanings in the two systems. For example, note the different meanings for $+$ and $=$ in the following two equations: $\mathbf{A} + \mathbf{B} = \mathbf{C}$ and $1 + 2 = 3$. In computer programming, this is known as *operator overloading*, where the rules of the operation depend on the operands involved in the process. Unless you are extremely careful, this double meaning for symbols can easily lead to invalid expressions—for example, $\mathbf{A} + 5$ (a vector cannot be added to a scalar!) and $\mathbf{A} = 1$ (a vector cannot equal a scalar!).

Sample Problem 1.4

Figure (a) shows two position vectors of magnitudes $A = 60$ m and $B = 100$ m. (A position vector is a vector drawn between two points in space.) Determine the resultant $\mathbf{R} = \mathbf{A} + \mathbf{B}$ using the following methods: (1) analytically, using the triangle law; and (2) graphically, using the triangle law.



Solution

Part 1

The first step in the analytical solution is to draw a sketch (approximately to scale) of the triangle law. The magnitude and direction of the resultant are then found by applying the laws of sines and cosines to the triangle.

In this problem, the triangle law for the vector addition of $\mathbf{A}$ and $\mathbf{B}$ is shown in Fig. (b). The magnitude R of the resultant and the angle α are the unknowns to be determined. Applying the law of cosines, we obtain

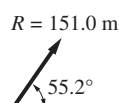
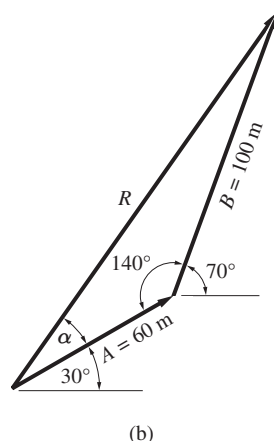
$$R^2 = 60^2 + 100^2 - 2(60)(100) \cos 140^\circ$$

which yields $R = 151.0$ m.

The angle α can now be found from the law of sines:

$$\frac{100}{\sin \alpha} = \frac{R}{\sin 140^\circ}$$

Substituting $R = 151.0$ m and solving for α , we get $\alpha = 25.2^\circ$. Referring to Fig. (b), we see that the angle that $\mathbf{R}$ makes with the horizontal is $30^\circ + \alpha = 30^\circ + 25.2^\circ = 55.2^\circ$. Therefore, the resultant of $\mathbf{A}$ and $\mathbf{B}$ is



Answer

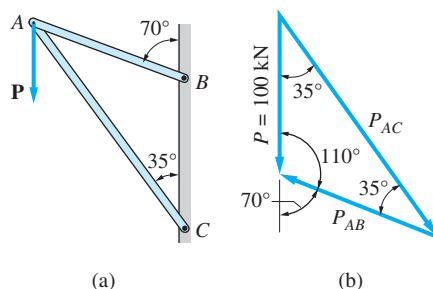
Part 2

In the graphical solution, Fig. (b) is drawn to scale with the aid of a ruler and a protractor. We first draw the vector **A** at 30° to the horizontal and then append vector **B** at 70° to the horizontal. The resultant **R** is then obtained by drawing a line from the tail of **A** to the head of **B**. The magnitude of **R** and the angle it makes with the horizontal can now be measured directly from the figure.

Of course, the results would not be as accurate as those obtained in the analytical solution. If care is taken in making the drawing, two-digit accuracy is the best we can hope for. In this problem we should get $R \approx 150$ m, inclined at 55° to the horizontal.

Sample Problem 1.5

The vertical force **P** of magnitude 100 kN is applied to the frame shown in Fig. (a). Resolve **P** into components that are parallel to the members *AB* and *AC* of the truss.



Solution

The force triangle in Fig. (b) represents the vector addition $\mathbf{P} = \mathbf{P}_{AC} + \mathbf{P}_{AB}$. The angles in the figure were derived from the inclinations of *AC* and *AB* with the vertical: $\mathbf{P}_{AC}$ is inclined at 35° (parallel to *AC*), and $\mathbf{P}_{AB}$ is inclined at 70° (parallel to *AB*). Applying the law of sines to the triangle, we obtain

$$\frac{100}{\sin 35^\circ} = \frac{P_{AB}}{\sin 35^\circ} = \frac{P_{AC}}{\sin 110^\circ}$$

which yields for the magnitudes of the components

$$P_{AB} = 100.0 \text{ kN} \quad P_{BC} = 163.8 \text{ kN}$$

Answer

1.4 Representation of Vectors Using Rectangular Components

The fundamental properties of vectors discussed in the preceding article are independent of coordinate systems. However, in engineering applications, it is customary to describe vectors using their rectangular components and then to perform vector operations, such as addition, in terms of these components.

a. Rectangular components and direction cosines

The reference frame we use throughout this book is shown in Fig. 1.6(a). It is a right-handed, rectangular Cartesian coordinate system. To test for right-handedness, curl the fingers of your right hand and extend your thumb, as shown in the figure. The coordinate axes now should line up with your hand as

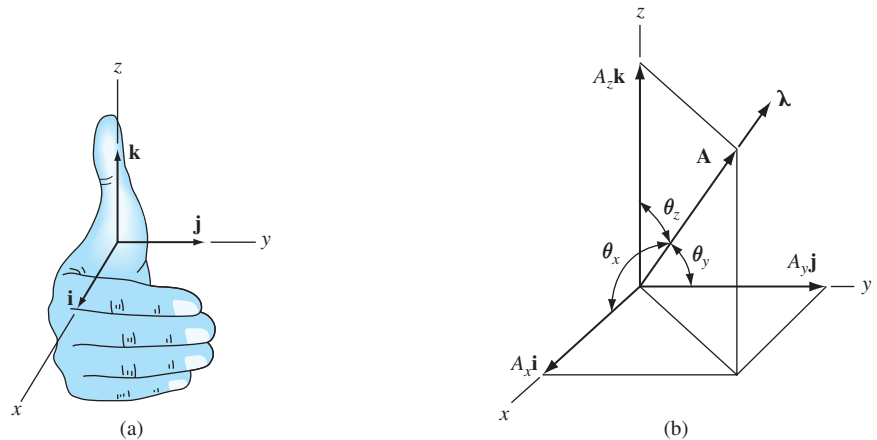


Fig. 1.6

follows: the palm is parallel to the x -direction, the fingers point in the y -direction, and the thumb lies in the z -direction. Figure 1.6(a) also shows the *base vectors* $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ of the coordinate system. They are dimensionless vectors of unit magnitude directed in the positive coordinate directions.

If a vector $\mathbf{A}$ is resolved into its rectangular components, as illustrated in Fig. 1.6(b), it can be written as

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k} \quad (1.5)$$

where $A_x \mathbf{i}$, $A_y \mathbf{j}$, and $A_z \mathbf{k}$ are the *vector components* of $\mathbf{A}$. The *scalar components* of $\mathbf{A}$ are

$$A_x = A \cos \theta_x \quad A_y = A \cos \theta_y \quad A_z = A \cos \theta_z \quad (1.6)$$

where θ_x , θ_y , and θ_z are the angles between $\mathbf{A}$ and the positive coordinate axes. The scalar components can be positive or negative, depending upon whether the corresponding vector component points in the positive or negative coordinate direction. The magnitude of $\mathbf{A}$ is related to its scalar components by

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad (1.7)$$

The direction of $\mathbf{A}$ customarily is specified by its *direction cosines* defined as

$$\lambda_x = \cos \theta_x \quad \lambda_y = \cos \theta_y \quad \lambda_z = \cos \theta_z \quad (1.8)$$

The scalar components of $\mathbf{A}$ in Eq. (1.6) now become $A_x = A\lambda_x$, $A_y = A\lambda_y$, and $A_z = A\lambda_z$, so Eq. (1.5) takes the form

$$\mathbf{A} = A(\lambda_x \mathbf{i} + \lambda_y \mathbf{j} + \lambda_z \mathbf{k}) = A\boldsymbol{\lambda} \quad (1.9)$$

where

$$\boldsymbol{\lambda} = \lambda_x \mathbf{i} + \lambda_y \mathbf{j} + \lambda_z \mathbf{k} \quad (1.10)$$

is a unit vector in the direction of $\mathbf{A}$, as shown in Fig. 1.6(b). Because the magnitude of $\boldsymbol{\lambda}$ is one, its components satisfy the identity

$$\lambda_x^2 + \lambda_y^2 + \lambda_z^2 = 1 \quad (1.11)$$

b. Vector addition using rectangular components

Consider the two vectors $\mathbf{A} = A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}$ and $\mathbf{B} = B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k}$. Letting $\mathbf{C}$ be the sum of $\mathbf{A}$ and $\mathbf{B}$, we have

$$\mathbf{C} = \mathbf{A} + \mathbf{B} = (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) + (B_x\mathbf{i} + B_y\mathbf{j} + B_z\mathbf{k})$$

which can be written as

$$\begin{aligned}\mathbf{C} &= C_x\mathbf{i} + C_y\mathbf{j} + C_z\mathbf{k} \\ &= (A_x + B_x)\mathbf{i} + (A_y + B_y)\mathbf{j} + (A_z + B_z)\mathbf{k}\end{aligned}\quad (1.12)$$

Equating like components, we find that the rectangular components of $\mathbf{C}$ are

$$C_x = A_x + B_x \quad C_y = A_y + B_y \quad C_z = A_z + B_z \quad (1.13)$$

Equations (1.13) show that each component of the sum equals the sum of the components. This result is depicted in Fig. 1.7, where, for simplicity's sake, the xy -plane has been chosen as a plane that contains the vectors $\mathbf{A}$ and $\mathbf{B}$. Equations (1.12) and (1.13) can, of course, be extended to include the sum of any number of vectors.

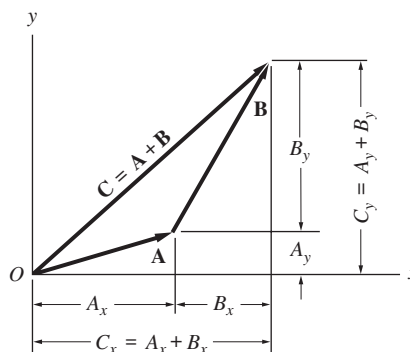


Fig. 1.7

c. Relative position vectors

The vector drawn from the origin O of a coordinate system to point B , denoted by $\vec{OB}$, is called the *position vector of B*. The vector $\vec{AB}$, drawn from point A to point B , is called the *position vector of B relative to A*. (Note that the position vector of B relative to A is the negative of the position vector of A relative to B ; that is, $\vec{AB} = -\vec{BA}$.)

Figure 1.8 shows the relative position vector $\vec{AB}$: the vector drawn from $A(x_A, y_A, z_A)$ to $B(x_B, y_B, z_B)$. The rectangular representation of this vector is

$$\vec{AB} = (x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k} \quad (1.14)$$

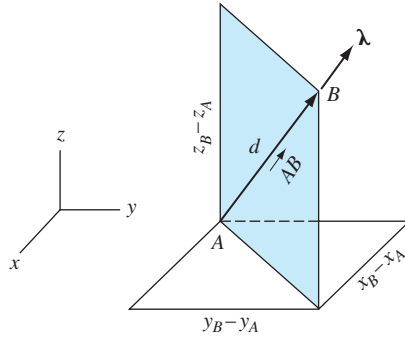


Fig. 1.8

The magnitude of $\vec{AB}$ (the distance d in Fig. 1.8) is

$$|\vec{AB}| = d = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2} \quad (1.15)$$

The unit vector λ in the direction of $\vec{AB}$ can be found by dividing the vector $\vec{AB}$ by its magnitude:

$$\lambda = \frac{\vec{AB}}{|\vec{AB}|} = \frac{(x_B - x_A)\mathbf{i} + (y_B - y_A)\mathbf{j} + (z_B - z_A)\mathbf{k}}{d} \quad (1.16)$$

Therefore, the components of λ are

$$\lambda_x = \frac{x_B - x_A}{d} \quad \lambda_y = \frac{y_B - y_A}{d} \quad \lambda_z = \frac{z_B - z_A}{d} \quad (1.17)$$

d. How to write a vector in rectangular form

In statics, we frequently encounter the following problem: Given the magnitude of a vector and two points on its line of action, determine the rectangular representation of the vector. Figure 1.9 shows a vector $\mathbf{F}$ that acts along the line AB . Suppose that the magnitude of $\mathbf{F}$ and the coordinates of A and B are known, and we want to write the vector $\mathbf{F}$ in the rectangular form $\mathbf{F} = F_x\mathbf{i} + F_y\mathbf{j} + F_z\mathbf{k}$. The recommended procedure is

1. Write the relative position vector $\vec{AB}$. The rectangular components of $\vec{AB}$ can be obtained by inspection of a sketch similar to Fig. 1.9 or by substituting the coordinates of points A and B into Eq. (1.14).
2. Evaluate the unit vector $\lambda = \vec{AB}/|\vec{AB}|$.
3. Write $\mathbf{F}$ in the rectangular form

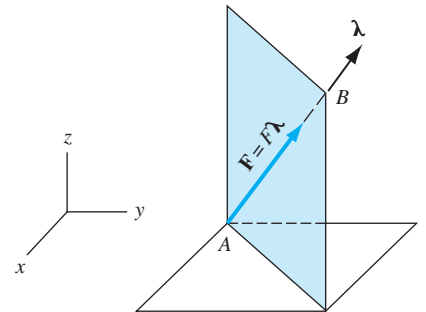
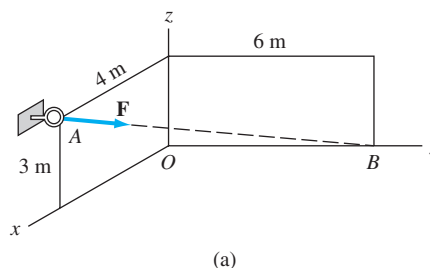


Fig. 1.9

$$\mathbf{F} = F\lambda = F(\lambda_x\mathbf{i} + \lambda_y\mathbf{j} + \lambda_z\mathbf{k}) \quad (1.18)$$

Sample Problem 1.6

The cable attached to the eyebolt in Fig. (a) is pulled with the force $\mathbf{F}$ of magnitude 500 N. Determine the rectangular representation of this force.

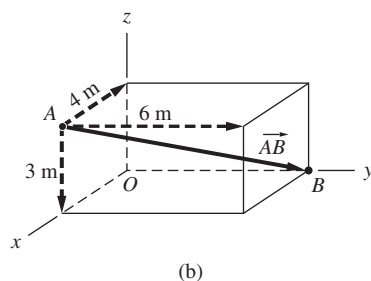


Solution

Because the coordinates of points A and B on the line of action of $\mathbf{F}$ are known, the following is a convenient method for obtaining the rectangular representation of $\mathbf{F}$.

1. Write $\vec{AB}$, the vector from A to B , in rectangular form.

The vector $\vec{AB}$ and its rectangular components are shown in Fig. (b). Two common errors made by students at this point are choosing the wrong signs and mixing up the scalar components. You can avoid both of these difficulties by taking the time to show the vector on a carefully drawn sketch of the appropriate parallelepiped. From Fig. (b) we see that



$$\vec{AB} = -4\mathbf{i} + 6\mathbf{j} - 3\mathbf{k} \text{ m}$$

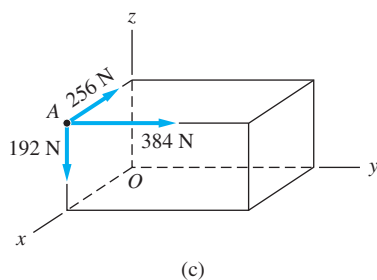
2. Evaluate λ , the unit vector from A toward B :

$$\begin{aligned}\lambda &= \frac{\vec{AB}}{|\vec{AB}|} = \frac{-4\mathbf{i} + 6\mathbf{j} - 3\mathbf{k}}{\sqrt{(-4)^2 + 6^2 + (-3)^2}} \\ &= -0.5122\mathbf{i} + 0.7682\mathbf{j} - 0.3841\mathbf{k}\end{aligned}$$

3. Write $\mathbf{F} = F\lambda$:

$$\begin{aligned}\mathbf{F} &= 500(-0.5122\mathbf{i} + 0.7682\mathbf{j} - 0.3841\mathbf{k}) \\ &= -256\mathbf{i} + 384\mathbf{j} - 192\mathbf{k} \text{ N}\end{aligned}$$

Answer



The rectangular components of $\mathbf{F}$ are shown in Fig. (c).

Sample Problem 1.7

Referring to Fig. (a), determine (1) the rectangular representation of the position vector $\mathbf{A}$; and (2) the angles between $\mathbf{A}$ and each of the positive coordinate axes.

Solution

Part 1

We first resolve $\mathbf{A}$ into two components as shown in Fig. (b): A_z along the z -axis and A_{xy} in the xy -plane. (Once again we see that a carefully drawn sketch is an essential aid in performing vector resolution.) Because $\mathbf{A}$, A_z , and A_{xy} lie in the same plane (a diagonal plane of the parallelepiped), we obtain by trigonometry

$$A_z = A \cos 30^\circ = 12 \cos 30^\circ = 10.392 \text{ m}$$

$$A_{xy} = A \sin 30^\circ = 12 \sin 30^\circ = 6 \text{ m}$$

The next step, illustrated in Fig. (c), is to resolve A_{xy} into the components along the coordinate axes:

$$A_x = A_{xy} \cos 40^\circ = 6 \cos 40^\circ = 4.596 \text{ m}$$

$$A_y = A_{xy} \sin 40^\circ = 6 \sin 40^\circ = 3.857 \text{ m}$$

Therefore, the rectangular representation of $\mathbf{A}$ is

$$\mathbf{A} = A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k} = 4.60 \mathbf{i} + 3.86 \mathbf{j} + 10.39 \mathbf{k} \text{ m}$$

Answer

Part 2

The angles between $\mathbf{A}$ and the coordinate axes can be computed from Eqs. (1.6):

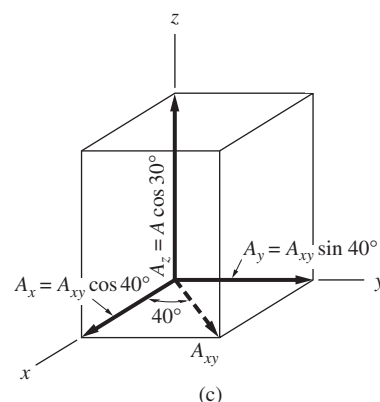
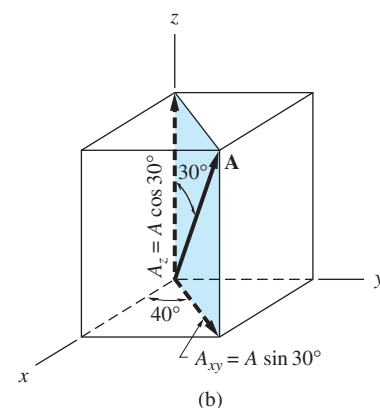
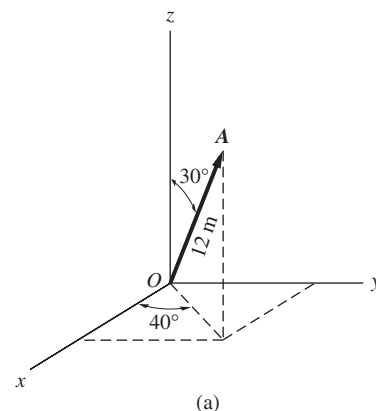
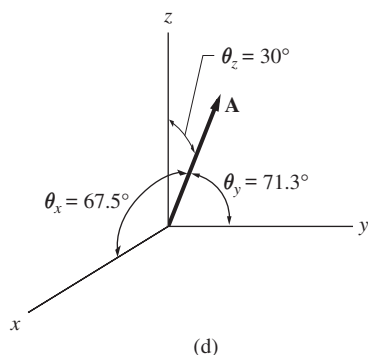
$$\theta_x = \cos^{-1} \frac{A_x}{A} = \cos^{-1} \frac{4.596}{12} = 67.5^\circ$$

$$\theta_y = \cos^{-1} \frac{A_y}{A} = \cos^{-1} \frac{3.857}{12} = 71.3^\circ$$

$$\theta_z = \cos^{-1} \frac{A_z}{A} = \cos^{-1} \frac{10.392}{12} = 30.0^\circ$$

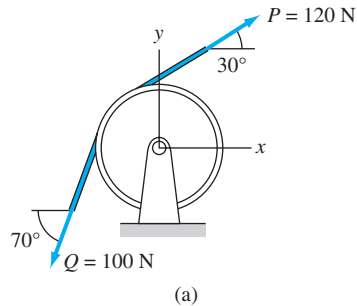
Answer

These angles are shown in Fig. (d). Note that it was not necessary to compute θ_z , because it was already given in Fig (a).



Sample Problem 1.8

The pulley shown in Fig. (a) is subjected to the belt forces **P** and **Q**. Using rectangular components, determine the magnitude and direction of the resultant force.

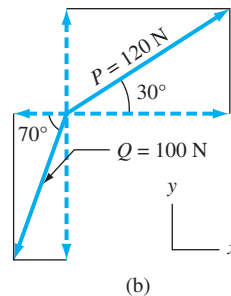


Solution

Referring to Fig. (b), the rectangular representations of **P** and **Q** are

$$\mathbf{P} = 120 \cos 30^\circ \mathbf{i} + 120 \sin 30^\circ \mathbf{j} = 103.9\mathbf{i} + 60.0\mathbf{j} \text{ N}$$

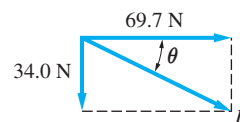
$$\mathbf{Q} = -100 \cos 70^\circ \mathbf{i} - 100 \sin 70^\circ \mathbf{j} = -34.2\mathbf{i} - 94.0\mathbf{j} \text{ N}$$



The resultant of **P** and **Q** is found by adding their components:

$$\begin{aligned} \mathbf{R} &= \mathbf{P} + \mathbf{Q} = (103.9 - 34.2)\mathbf{i} + (60.0 - 94.0)\mathbf{j} \\ &= 69.7\mathbf{i} - 34.0\mathbf{j} \text{ N} \end{aligned}$$

Calculating the magnitude and direction of **R**, we obtain



Answer

$$R = \sqrt{34.0^2 + 69.7^2} = 77.6 \text{ N} \quad \theta = \tan^{-1} \frac{34.0}{69.7} = 26.0^\circ$$

1.5 Vector Multiplication*

a. Dot (scalar) product

Figure 1.10 shows two vectors $\mathbf{A}$ and $\mathbf{B}$, with θ being the angle between their positive directions. The *dot product* of $\mathbf{A}$ and $\mathbf{B}$ is defined as

$$\mathbf{A} \cdot \mathbf{B} = AB \cos \theta \quad (0 \leq \theta \leq 180^\circ) \quad (1.19)$$

Because the dot product is a scalar, it is also called the *scalar product*. Note that the dot product is positive if $\theta < 90^\circ$, negative if $\theta > 90^\circ$, and zero if $\theta = 90^\circ$.

The following two properties of the dot product follow from its definition in Eq. (1.19).

- The dot product is commutative: $\mathbf{A} \cdot \mathbf{B} = \mathbf{B} \cdot \mathbf{A}$
- The dot product is distributive: $\mathbf{A} \cdot (\mathbf{B} + \mathbf{C}) = \mathbf{A} \cdot \mathbf{B} + \mathbf{A} \cdot \mathbf{C}$

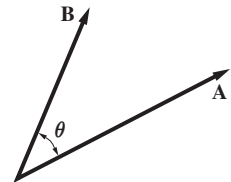


Fig. 1.10

*Note that division by a vector, such as $1/\mathbf{A}$ or $\mathbf{B}/\mathbf{A}$, is not defined.

From the definition of the dot product, we also note that the base vectors of a rectangular coordinate system satisfy the following identities:

$$\begin{aligned}\mathbf{i} \cdot \mathbf{i} &= \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1 \\ \mathbf{i} \cdot \mathbf{j} &= \mathbf{j} \cdot \mathbf{k} = \mathbf{k} \cdot \mathbf{i} = 0\end{aligned}\quad (1.20)$$

When $\mathbf{A}$ and $\mathbf{B}$ are expressed in rectangular form, their dot product becomes

$$\mathbf{A} \cdot \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \cdot (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k})$$

which, using the distributive property of the dot product and Eqs. (1.20), reduces to

$$\mathbf{A} \cdot \mathbf{B} = A_x B_x + A_y B_y + A_z B_z \quad (1.21)$$

Equation (1.21) is a powerful and relatively simple method for computing the dot product of two vectors that are given in rectangular form.

The following are two of the more important applications of the dot product.

Finding the Angle Between Two Vectors The angle θ between the two vectors $\mathbf{A}$ and $\mathbf{B}$ in Fig. 1.11 can be found from the definition of the dot product in Eq. (1.19), which can be rewritten as

$$\cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{AB} = \frac{\mathbf{A}}{A} \cdot \frac{\mathbf{B}}{B}$$

Letting $\lambda_A = \mathbf{A}/A$ and $\lambda_B = \mathbf{B}/B$ be the unit vectors that have the same directions as $\mathbf{A}$ and $\mathbf{B}$, as shown in Fig. 1.11, the last equation becomes

$$\cos \theta = \lambda_A \cdot \lambda_B \quad (1.22)$$

If the unit vectors are written in rectangular form, this dot product is easily evaluated using Eq. (1.21).

Determining the Orthogonal Component of a Vector in a Given Direction If we project $\mathbf{B}$ onto $\mathbf{A}$ as in Fig. 1.12, the projected length $B \cos \theta$ is called the *orthogonal component of $\mathbf{B}$ in the direction of $\mathbf{A}$* . Because θ is the angle between $\mathbf{A}$ and $\mathbf{B}$, the definition of the dot product, $\mathbf{A} \cdot \mathbf{B} = AB \cos \theta$, yields

$$B \cos \theta = \frac{\mathbf{A} \cdot \mathbf{B}}{A} = \mathbf{B} \cdot \frac{\mathbf{A}}{A}$$

Because $\mathbf{A}/A = \lambda_A$ (the unit vector in the direction of $\mathbf{A}$), as shown in Fig. 1.12, the last equation becomes

$$B \cos \theta = \mathbf{B} \cdot \lambda_A \quad (1.23)$$

Therefore,

$$\text{The orthogonal component of } \mathbf{B} \text{ in the direction of } \mathbf{A} \text{ equals } \mathbf{B} \cdot \lambda_A. \quad (1.24)$$

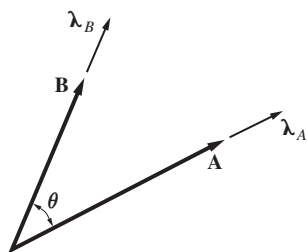


Fig. 1.11

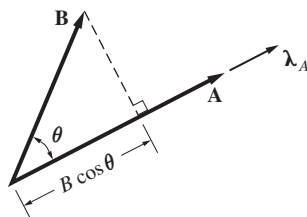


Fig. 1.12

b. Cross (vector) product

The *cross product* $\mathbf{C}$ of two vectors $\mathbf{A}$ and $\mathbf{B}$, denoted by

$$\mathbf{C} = \mathbf{A} \times \mathbf{B}$$

has the following characteristics (see Fig. 1.13):

- The magnitude of $\mathbf{C}$ is

$$C = AB \sin \theta \quad (1.25)$$

where θ ($0 \leq \theta \leq 180^\circ$) is the angle between the positive directions of $\mathbf{A}$ and $\mathbf{B}$. (Note that C is always a positive number.)

- $\mathbf{C}$ is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$.
- The sense of $\mathbf{C}$ is determined by the right-hand rule, which states that when the fingers of your right hand are curled in the direction of the angle θ (directed from $\mathbf{A}$ toward $\mathbf{B}$), your thumb points in the direction of $\mathbf{C}$.*

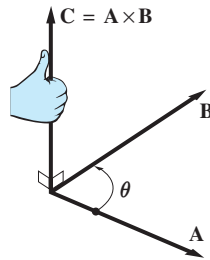


Fig. 1.13

The cross product of two vectors is also called their *vector product*.

It can be shown that the cross product is distributive; that is,

$$\mathbf{A} \times (\mathbf{B} + \mathbf{C}) = (\mathbf{A} \times \mathbf{B}) + (\mathbf{A} \times \mathbf{C})$$

However, the cross product is neither associative nor commutative. In other words,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) \neq (\mathbf{A} \times \mathbf{B}) \times \mathbf{C}$$

$$\mathbf{A} \times \mathbf{B} \neq \mathbf{B} \times \mathbf{A}$$

In fact, it can be deduced from the right-hand rule that $\mathbf{A} \times \mathbf{B} = -\mathbf{B} \times \mathbf{A}$.

From the definition of the cross product $\mathbf{C} = \mathbf{A} \times \mathbf{B}$, we see that (1) if $\mathbf{A}$ and $\mathbf{B}$ are perpendicular ($\theta = 90^\circ$), then $C = AB$; and (2) if $\mathbf{A}$ and $\mathbf{B}$ are parallel ($\theta = 0^\circ$ or 180°), then $C = 0$.

*An alternative statement of the right-hand rule is this: The direction of $\mathbf{C}$ is the direction in which a right-hand screw would advance when turned in the direction of θ (directed from $\mathbf{A}$ toward $\mathbf{B}$).

From the properties of the cross product, we deduce that the base vectors of a rectangular coordinate system satisfy the following identities:

$$\begin{aligned} \mathbf{i} \times \mathbf{i} &= \mathbf{0} & \mathbf{j} \times \mathbf{j} &= \mathbf{0} & \mathbf{k} \times \mathbf{k} &= \mathbf{0} \\ \mathbf{i} \times \mathbf{j} &= \mathbf{k} & \mathbf{j} \times \mathbf{k} &= \mathbf{i} & \mathbf{k} \times \mathbf{i} &= \mathbf{j} \end{aligned} \quad (1.26)$$

where the equations in the bottom row are valid in a *right-handed* coordinate system. If the coordinate axes are labeled such that $\mathbf{i} \times \mathbf{j} = -\mathbf{k}$, $\mathbf{j} \times \mathbf{k} = -\mathbf{i}$, and $\mathbf{k} \times \mathbf{i} = -\mathbf{j}$, the system is said to be *left-handed*. Examples of both right- and left-handed coordinate systems are shown in Fig. 1.14.*

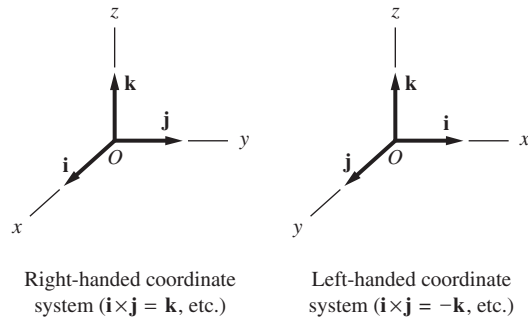


Fig. 1.14

When **A** and **B** are expressed in rectangular form, their cross product becomes

$$\mathbf{A} \times \mathbf{B} = (A_x \mathbf{i} + A_y \mathbf{j} + A_z \mathbf{k}) \times (B_x \mathbf{i} + B_y \mathbf{j} + B_z \mathbf{k})$$

Using the distributive property of the cross product and Eqs. (1.26), this equation becomes

$$\begin{aligned} \mathbf{A} \times \mathbf{B} &= (A_y B_z - A_z B_y) \mathbf{i} \\ &\quad - (A_x B_z - A_z B_x) \mathbf{j} \\ &\quad + (A_x B_y - A_y B_x) \mathbf{k} \end{aligned} \quad (1.27)$$

The identical expression is obtained when the rules for expanding a 3×3 determinant are applied to the following array of nine terms (because the terms are not all scalars, the array is not a true determinant):

$$\mathbf{A} \times \mathbf{B} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} \quad (1.28)$$

*In this text, we assume that all rectangular coordinate systems are right-handed.

You may use any method for determinant expansion, but you will find that the following technique, called *expansion by minors using the first row*, is very convenient.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = a \begin{vmatrix} e & f \\ h & i \end{vmatrix} - b \begin{vmatrix} d & f \\ g & i \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

$$= a(ei - fh) - b(di - fg) + c(dh - eg)$$

Expanding Eq. (1.28) by this method, we find that the 2×2 determinants equal the **i**, **j**, and **k** components of the cross product.

c. Scalar triple product

Of the vector products that involve three or more vectors, the one that is most useful in statics is the scalar triple product. The *scalar triple product* arises when the cross product of two vectors is dotted with a third vector—for example, $\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$. When writing this product, it is not necessary to show the parentheses, because $\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$ can be interpreted only in one way—the cross product must be done first; otherwise the expression is meaningless.

Assuming that **A**, **B**, and **C** are expressed in rectangular form and recalling Eq. (1.27), the scalar triple product becomes

$$\begin{aligned} \mathbf{A} \times \mathbf{B} \cdot \mathbf{C} = & [(A_y B_z - A_z B_y) \mathbf{i} - (A_x B_z - A_z B_x) \mathbf{j} \\ & + (A_x B_y - A_y B_x) \mathbf{k}] \cdot (C_x \mathbf{i} + C_y \mathbf{j} + C_z \mathbf{k}) \end{aligned}$$

Using Eq. (1.21) and the properties of the dot products of the rectangular base vectors, this expression simplifies to

$$\begin{aligned} \mathbf{A} \times \mathbf{B} \cdot \mathbf{C} = & (A_y B_z - A_z B_y) C_x - (A_x B_z - A_z B_x) C_y \\ & + (A_x B_y - A_y B_x) C_z \end{aligned} \quad (1.29)$$

Therefore, the scalar triple product can be written in the following determinant form, which is easy to remember:

$$\mathbf{A} \times \mathbf{B} \cdot \mathbf{C} = \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} \quad (1.30)$$

The following identities relating to the scalar triple product are useful:

$$\mathbf{A} \times \mathbf{B} \cdot \mathbf{C} = \mathbf{A} \cdot \mathbf{B} \times \mathbf{C} = \mathbf{B} \cdot \mathbf{C} \times \mathbf{A} = \mathbf{C} \cdot \mathbf{A} \times \mathbf{B} \quad (1.31)$$

Observe that the value of the scalar triple product is not altered if the locations of the dot and cross are interchanged or if the positions of **A**, **B**, and **C** are changed—provided that the cyclic order $A-B-C$ is maintained.

Sample Problem 1.9

Given the vectors

$$\mathbf{A} = 8\mathbf{i} + 4\mathbf{j} - 2\mathbf{k} \text{ N}$$

$$\mathbf{B} = 2\mathbf{j} + 6\mathbf{k} \text{ m}$$

$$\mathbf{C} = 3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k} \text{ m}$$

calculate the following: (1) $\mathbf{A} \cdot \mathbf{B}$; (2) the orthogonal component of $\mathbf{B}$ in the direction of $\mathbf{C}$; (3) the angle between $\mathbf{A}$ and $\mathbf{C}$; (4) $\mathbf{A} \times \mathbf{B}$; (5) a unit vector $\boldsymbol{\lambda}$ that is perpendicular to both $\mathbf{A}$ and $\mathbf{B}$; and (6) $\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$.

Solution

Part 1

From Eq. (1.21), the dot product of $\mathbf{A}$ and $\mathbf{B}$ is

$$\begin{aligned}\mathbf{A} \cdot \mathbf{B} &= A_x B_x + A_y B_y + A_z B_z = 8(0) + 4(2) + (-2)(6) \\ &= -4 \text{ N} \cdot \text{m} \quad \text{Answer}\end{aligned}$$

The negative sign indicates that the angle between $\mathbf{A}$ and $\mathbf{B}$ is greater than 90° .

Part 2

Letting θ be the angle between $\mathbf{B}$ and $\mathbf{C}$, we obtain from Eq. (1.23)

$$\begin{aligned}B \cos \theta &= \mathbf{B} \cdot \boldsymbol{\lambda}_C = \mathbf{B} \cdot \frac{\mathbf{C}}{C} = (2\mathbf{j} + 6\mathbf{k}) \cdot \frac{3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}}{\sqrt{3^2 + (-2)^2 + 4^2}} \\ &= \frac{(0)(3) + (2)(-2) + (6)(4)}{\sqrt{29}} = 3.71 \text{ m} \quad \text{Answer}\end{aligned}$$

Part 3

Letting α be the angle between $\mathbf{A}$ and $\mathbf{C}$, we find from Eq. (1.22)

$$\begin{aligned}\cos \alpha &= \boldsymbol{\lambda}_A \cdot \boldsymbol{\lambda}_C = \frac{\mathbf{A}}{A} \cdot \frac{\mathbf{C}}{C} \\ &= \frac{8\mathbf{i} + 4\mathbf{j} - 2\mathbf{k}}{\sqrt{8^2 + 4^2 + (-2)^2}} \cdot \frac{3\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}}{\sqrt{3^2 + (-2)^2 + 4^2}} \\ &= \frac{(8)(3) + (4)(-2) + (-2)(4)}{\sqrt{84}\sqrt{29}} = 0.16209\end{aligned}$$

which yields

$$\alpha = 80.7^\circ$$

Answer

Part 4

Referring to Eq. (1.28), the cross product of **A** and **B** is

$$\begin{aligned}\mathbf{A} \times \mathbf{B} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ A_x & A_y & A_z \\ B_x & B_y & B_z \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 8 & 4 & -2 \\ 0 & 2 & 6 \end{vmatrix} \\ &= \mathbf{i} \begin{vmatrix} 4 & -2 \\ 2 & 6 \end{vmatrix} - \mathbf{j} \begin{vmatrix} 8 & -2 \\ 0 & 6 \end{vmatrix} + \mathbf{k} \begin{vmatrix} 8 & 4 \\ 0 & 2 \end{vmatrix} \\ &= 28\mathbf{i} - 48\mathbf{j} + 16\mathbf{k} \text{ N} \cdot \text{m}\end{aligned}$$

Answer

Part 5

The cross product $\mathbf{A} \times \mathbf{B}$ is perpendicular to both **A** and **B**. Therefore, a unit vector in that direction is obtained by dividing $\mathbf{A} \times \mathbf{B}$, which was evaluated above, by its magnitude

$$\begin{aligned}\frac{\mathbf{A} \times \mathbf{B}}{|\mathbf{A} \times \mathbf{B}|} &= \frac{28\mathbf{i} - 48\mathbf{j} + 16\mathbf{k}}{\sqrt{28^2 + (-48)^2 + 16^2}} \\ &= 0.484\mathbf{i} - 0.830\mathbf{j} + 0.277\mathbf{k}\end{aligned}$$

Because the negative of this vector is also a unit vector that is perpendicular to both **A** and **B**, we obtain

$$\boldsymbol{\lambda} = \pm (0.484\mathbf{i} - 0.830\mathbf{j} + 0.277\mathbf{k})$$

Answer

Part 6

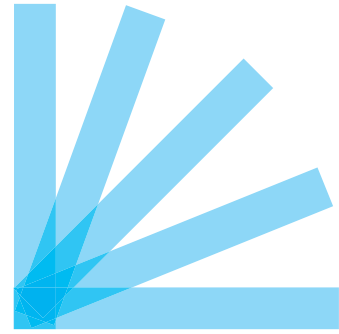
The scalar triple product $\mathbf{A} \times \mathbf{B} \cdot \mathbf{C}$ is evaluated using Eq. (1.30).

$$\begin{aligned}\mathbf{A} \times \mathbf{B} \cdot \mathbf{C} &= \begin{vmatrix} A_x & A_y & A_z \\ B_x & B_y & B_z \\ C_x & C_y & C_z \end{vmatrix} = \begin{vmatrix} 8 & 4 & -2 \\ 0 & 2 & 6 \\ 3 & -2 & 4 \end{vmatrix} \\ &= 8 \begin{vmatrix} 2 & 6 \\ -2 & 4 \end{vmatrix} - 4 \begin{vmatrix} 0 & 6 \\ 3 & 4 \end{vmatrix} + (-2) \begin{vmatrix} 0 & 2 \\ 3 & -2 \end{vmatrix} \\ &= 160 + 72 + 12 = 244 \text{ N} \cdot \text{m}^2\end{aligned}$$

Answer

2

Basic Operations with Force Systems



2.1 Introduction

The usefulness of vector algebra in real-world problems stems from the fact that several commonly encountered physical quantities possess the properties of vectors. One such quantity is force, which was shown to obey the parallelogram law of addition by Stevinus (1548–1620).

In this chapter we begin to study the effects of forces on particles and rigid bodies. In particular, we learn how to use vector algebra to reduce a system of forces to a simpler, equivalent system. If the forces are *concurrent* (all forces intersect at the same point), we show that the equivalent system is a single force. The reduction of a nonconcurrent force system requires two additional vector concepts: the moment of a force and the couple. Both of these concepts are introduced in this chapter.

A fundamental concept of statics is the equivalence of forces. For example, a single force can produce the same effect on the floating drilling platform as the two forces applied by the tugboats. Equivalence of forces is one of the topics discussed in this chapter. Don Farrall/Photodisc/Getty Images

2.2 Equivalence of Vectors

We recall that vectors are quantities that have magnitude and direction, and combine according to the parallelogram law for addition. Two vectors that have the same magnitude and direction are said to be *equal*.

In mechanics, the term *equivalence* implies interchangeability; two vectors are considered to be equivalent if they can be interchanged without changing the outcome of the problem. Equality does not always result in equivalence. For example, a force applied to a certain point in a body does not necessarily produce the same effect on the body as an equal force acting at a different point.

From the viewpoint of equivalence, vectors representing physical quantities are classified into the following three types:

- *Fixed vectors*: Equivalent vectors have the same magnitude, direction, and point of application.
- *Sliding vectors*: Equivalent vectors have the same magnitude, direction, and line of action.
- *Free vectors*: Equivalent vectors have the same magnitude and direction.

It is possible for a physical quantity to be one type of vector—say, fixed—in one application and another type of vector, such as sliding, in another application. In vector algebra, reviewed in Chapter 1, all vectors were treated as free vectors.

2.3 Force

Force is the term assigned to mechanical interaction between bodies. A force can affect both the motion and the deformation of the body on which it acts. Forces may arise from direct contact between bodies, or they may be applied at a distance (such as gravitational attraction). Contact forces are distributed over a surface area of the body, whereas forces acting at a distance are distributed over the volume of the body.

Sometimes the area over which a contact force is applied is so small that it may be approximated by a point, in which case the force is said to be *concentrated* at the point of contact. The contact point is also called the *point of application* of the force. The *line of action* of a concentrated force is the line that passes through the point of application and is parallel to the force. In this chapter we consider only concentrated forces; the discussion of distributed forces begins in the next chapter.

Force is a *fixed vector*, because one of its characteristics (in addition to its magnitude and direction) is its point of application. As an informal proof, consider the three identical bars in Fig. 2.1, each loaded by two equal but opposite forces of magnitude P . If the forces are applied as shown Fig. 2.1(a), the bar is under tension, and its deformation is an elongation. By interchanging the forces, as seen in Fig. 2.1(b), the bar is placed in compression, resulting in its shortening. The loading in Fig. 2.1(c), where both forces are acting at point A, produces no deformation. Note that the forces in all three cases have the same line of action and the same zero resultant; only the points of application are different. Therefore, we conclude that the point of application is a characteristic of a force, as far as deformation is concerned.

If the bar is rigid, however (meaning that the deformation is negligible), there will be no observable differences in the behavior of the three bars in Fig. 2.1.

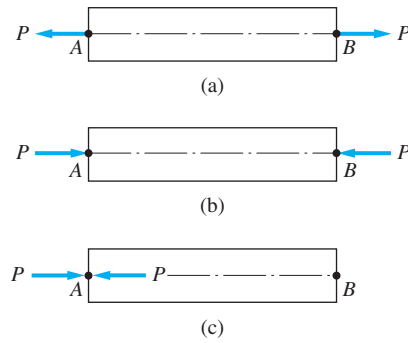


Fig. 2.1

In other words, the *external effects** of the three loadings are identical. It follows that if we are interested only in the external effects, a force can be treated as a *sliding vector*. The above conclusion is summarized by the *principle of transmissibility*:

A force may be moved anywhere along its line of action without changing its external effects on a rigid body.

Two force systems that produce the same external effects on a rigid body are said to be *equivalent*. (Sometimes the term *rigid-body equivalent* is used.)

In summary, a force is a fixed vector tied to a point of application, but if one is interested only in its external effect on a rigid body, a force may be treated as a sliding vector.

As a further illustration of the principle of transmissibility, consider the rigid block shown in Fig. 2.2. The block is subjected to three forces **P**, **Q**, and **S**, each with magnitude 20 N. The three forces are equal in the mathematical sense: $\mathbf{P} = \mathbf{Q} = \mathbf{S}$. However, only **P** and **Q** would produce identical external effects because they have the same line of action. Because **S** has a different line of action, its external effect would be different.

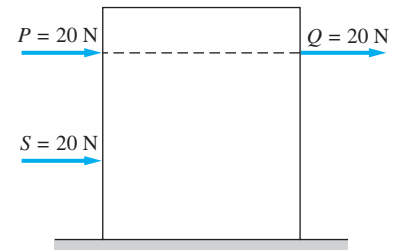


Fig. 2.2

2.4 Reduction of Concurrent Force Systems

In this article, we discuss the method for replacing a system of concurrent forces with a single equivalent force.

Consider the forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$ acting on the rigid body in Fig. 2.3(a) (for convenience, only three of the forces are shown). All the forces are concurrent at point *O*. (Their lines of action intersect at *O*.) These forces can be reduced to a single, equivalent force by the following two steps.

1. Move the forces along their lines of action to the point of concurrency *O*, as indicated in Fig. 2.3(b). According to the principle of transmissibility, this

*The external effects that concern us most are the motion (or state of rest) of the body, and the support reactions.

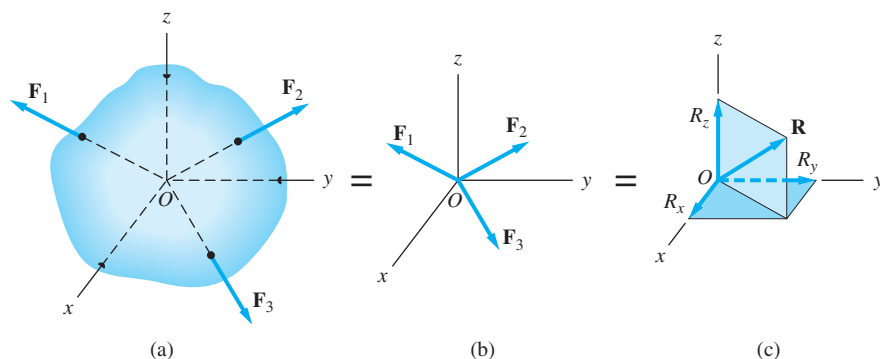


Fig. 2.3

operation does not change the external effects on the body. Therefore, the force systems in Figs. 2.3(a) and (b) are equivalent, which is indicated by the equal sign between the figures.

2. With the forces now at the common point O , compute their resultant $\mathbf{R}$ from the vector sum

$$\mathbf{R} = \Sigma \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \cdots \quad (2.1)$$

This resultant, which is also equivalent to the original force system, is shown in Fig. 2.3(c) together with its rectangular components. Note that Eq. (2.1) determines only the magnitude and direction of the resultant. The line of action of $\mathbf{R}$ must pass through the point of concurrency O in order for the equivalence to be valid.

When evaluating Eq. (2.1), any of the graphical or analytical methods for vector addition discussed in Chapter 1 may be used. If rectangular components are chosen, the equivalent scalar equations for determining the resultant force $\mathbf{R}$ are

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad R_z = \Sigma F_z \quad (2.2)$$

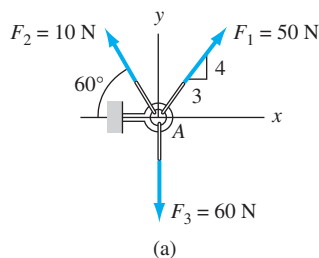
Thus we see that three scalar equations are required to determine the resultant force for a concurrent system of forces. If the original forces lie in a common plane—say, the xy -plane—the equation $R_z = \Sigma F_z$ yields no independent information and only the following two equations are necessary to determine the resultant force.

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad (2.3)$$

We emphasize that the method described here for determining the resultant force is valid only for forces that are concurrent. Because a force is tied to its line of action, the reduction of nonconcurrent force systems will require additional concepts, which are discussed later.

Sample Problem 2.1

Determine the resultant of the three concurrent forces shown in Fig. (a).



Solution

Because the three forces are concurrent at point A, they may be added immediately to obtain the resultant force **R**.

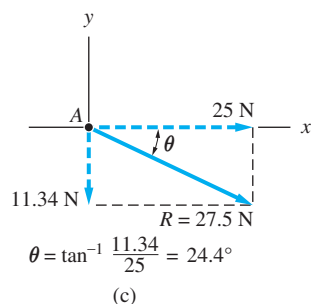
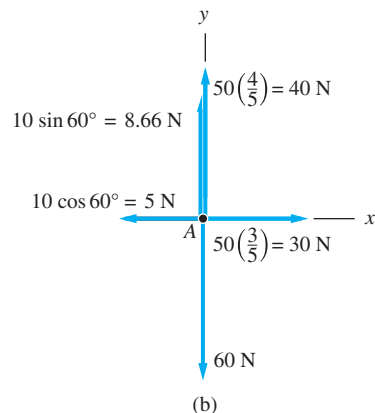
The rectangular components of each of the three forces are shown in Fig. (b). Using Eqs. (2.3) to determine the components of the resultant, we have

$$R_x = \Sigma F_x \quad \rightarrow \quad R_x = 30 - 5 = 25 \text{ N}$$

and

$$R_y = \Sigma F_y \quad \uparrow \quad R_y = 40 + 8.66 - 60 = -11.34 \text{ N}$$

The signs in these equations indicate that R_x acts to the right and R_y acts downward. The resultant force **R** is shown in Fig. (c). Note that the magnitude of the resultant is 27.5 N and that it acts through point A (the original point of concurrency) at the 24.4° angle shown.



The foregoing solution could also have been accomplished using vector notation. The forces would first be written in vector form as follows,

$$\mathbf{F}_1 = 30\mathbf{i} + 40\mathbf{j} \text{ N}$$

$$\mathbf{F}_2 = -5\mathbf{i} + 8.66\mathbf{j} \text{ N}$$

$$\mathbf{F}_3 = -60\mathbf{j} \text{ N}$$

and the resultant force $\mathbf{R}$ would then be determined from the vector equation

$$\mathbf{R} = \Sigma \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$$

$$\mathbf{R} = (30\mathbf{i} + 40\mathbf{j}) + (-5\mathbf{i} + 8.66\mathbf{j}) + (-60\mathbf{j})$$

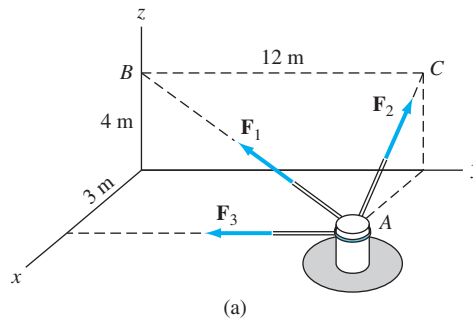
$$\mathbf{R} = 25\mathbf{i} - 11.34\mathbf{j} \text{ N}$$

Answer

Whether you use scalar or vector notation is a matter of personal preference.

Sample Problem 2.2

Three ropes are attached to the post at A in Fig. (a). The forces in the ropes are $F_1 = 260 \text{ N}$, $F_2 = 75 \text{ N}$, and $F_3 = 60 \text{ N}$. Determine (1) the magnitude of the force $\mathbf{R}$ that is equivalent to the three forces shown, and (2) the coordinates of the point where the line of action of $\mathbf{R}$ intersects the yz -plane.



Solution

Part 1

The forces are concurrent at point A and thus may be added immediately. Because the forces do not lie in a coordinate plane, it is convenient to use vector notation.

One method for expressing each of the forces in vector notation is to use the form $\mathbf{F} = F\boldsymbol{\lambda}$, where $\boldsymbol{\lambda}$ is the unit vector in the direction of the force $\mathbf{F}$. Thus

$$\begin{aligned} \mathbf{F}_1 &= 260\boldsymbol{\lambda}_{AB} = 260 \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = 260 \left(\frac{-3\mathbf{i} - 12\mathbf{j} + 4\mathbf{k}}{13} \right) \\ &= -60\mathbf{i} - 240\mathbf{j} + 80\mathbf{k} \text{ N} \end{aligned}$$

$$\begin{aligned} \mathbf{F}_2 &= 75\boldsymbol{\lambda}_{AC} = 75 \frac{\overrightarrow{AC}}{|\overrightarrow{AC}|} = 75 \left(\frac{-3\mathbf{i} + 4\mathbf{k}}{5} \right) \\ &= -45\mathbf{i} + 60\mathbf{k} \text{ N} \end{aligned}$$

$$\mathbf{F}_3 = -60\mathbf{j} \text{ N}$$

The resultant force is given by

$$\begin{aligned}\mathbf{R} &= \Sigma \mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 \\ &= (-60\mathbf{i} - 240\mathbf{j} + 80\mathbf{k}) + (-45\mathbf{i} + 60\mathbf{k}) + (-60\mathbf{j}) \\ &= -105\mathbf{i} - 300\mathbf{j} + 140\mathbf{k} \text{ N}\end{aligned}$$

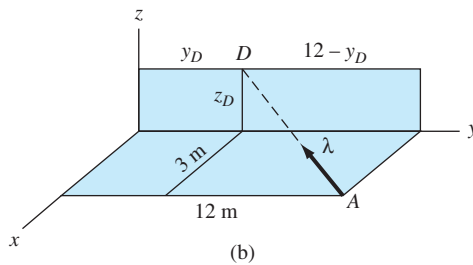
The magnitude of $\mathbf{R}$ is

$$R = \sqrt{(-105)^2 + (-300)^2 + (140)^2} = 347.3 \text{ N} \quad \text{Answer}$$

Part 2

The unit vector $\boldsymbol{\lambda}$ in the direction of $\mathbf{R}$ is

$$\begin{aligned}\boldsymbol{\lambda} &= \frac{\mathbf{R}}{R} = \frac{-105\mathbf{i} - 300\mathbf{j} + 140\mathbf{k}}{347.3} \\ &= -0.3023\mathbf{i} - 0.8638\mathbf{j} + 0.4031\mathbf{k}\end{aligned}$$



Let D be the point where $\boldsymbol{\lambda}$ intersects the yz -plane, as shown in Fig. (b). The coordinates of D can be determined by proportions:

$$\frac{|\lambda_x|}{3} = \frac{|\lambda_y|}{12 - y_D} = \frac{|\lambda_z|}{z_D}$$

Substituting the components of $\boldsymbol{\lambda}$, this becomes

$$\frac{0.3023}{3} = \frac{0.8638}{12 - y_D} = \frac{0.4031}{z_D}$$

yielding

$$y_D = 3.43 \text{ m} \quad z_D = 4.0 \text{ m} \quad \text{Answer}$$

2.5 Moment of a Force about a Point

In general, a force acting on a rigid body tends to rotate, as well as translate, the body. The force itself is the translational effect—the body tends to move in the direction of the force, and the magnitude of the force is proportional to its ability to translate the body. (The formal statement of this relationship is Newton's second law: Force equals mass times acceleration.) Here we introduce the tendency of a force to rotate a body, called the *moment of a force about a point*. This rotational effect depends on the magnitude of the force and the distance between the point and the line of action of the force. The tendency of a force to rotate a body about an axis, called the *moment of a force about an axis*, is discussed in the next article.

a. Definition

Let $\mathbf{F}$ be a force and O a point that is not on the line of action of $\mathbf{F}$, as shown in Fig. 2.4. Note that the force $\mathbf{F}$ and the point O determine a unique plane. We let A be any point on the line of action of $\mathbf{F}$ and define $\mathbf{r}$ to be the vector from point O to point A .

The moment of the force $\mathbf{F}$ about point O , called the *moment center*, is defined as

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} \quad (2.4)$$

Note that moment about a point has the dimension $[ML^2T^{-2}]$. In SI units, moment is measured in newton-meters ($\text{N} \cdot \text{m}$).

The moment of $\mathbf{F}$ about point O is a vector by definition. From the properties of the cross product of two vectors, $\mathbf{M}_O$ is perpendicular to both $\mathbf{r}$ and $\mathbf{F}$, with its sense determined by the right-hand rule, as shown in Fig. 2.4.*

b. Geometric interpretation

The moment of a force about a point can always be computed using the cross product in Eq. (2.4). However, a scalar computation of the magnitude of the moment can be obtained from the geometric interpretation of Eq. (2.4).

Observe that the magnitude of $\mathbf{M}_O$ is given by

$$M_O = |\mathbf{M}_O| = |\mathbf{r} \times \mathbf{F}| = rF \sin \theta \quad (2.5)$$

in which θ is the angle between $\mathbf{r}$ and $\mathbf{F}$. Returning to Fig. 2.4, we see that

$$r \sin \theta = d \quad (2.6)$$

where d is the perpendicular distance from the moment center to the line of action of the force $\mathbf{F}$ called the *moment arm* of the force. Therefore, the magnitude of $\mathbf{M}_O$ is

$$M_O = Fd \quad (2.7)$$

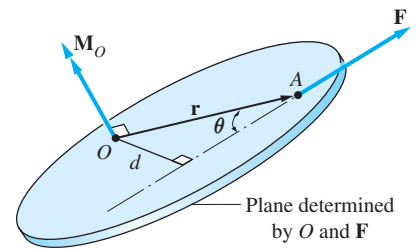


Fig. 2.4

*Moment vectors are drawn as double-headed arrows throughout this text.

Because the magnitude of $\mathbf{M}_O$ depends only on the magnitude of the force and the perpendicular distance d , a force may be moved anywhere along its line of action without changing its moment about a point. Therefore, in this application, a force may be treated as a sliding vector. This explains why any point A on the line of action of the force may be chosen when determining the vector $\mathbf{r}$ in Eq. (2.4).

Equation (2.7) is convenient only when the moment arm can be easily determined. Furthermore, when using Eq. (2.7), the direction of $\mathbf{M}_O$ must be found by inspection. For example, the magnitude of the moment of the 100-N force about the point O in Fig. 2.5(a) is $(100)(2) = 200 \text{ N} \cdot \text{m}$, and its direction is counter-clockwise, as viewed from the positive z -axis. Using the right-hand rule, the vector representation of this moment is $\mathbf{M}_O = 200\mathbf{k} \text{ N} \cdot \text{m}$, as shown in Fig. 2.5(b). The magnitude of the moment about point O for the 100-N force in Fig. 2.5(c) is also $200 \text{ N} \cdot \text{m}$, but in this case its direction is clockwise, as viewed from the positive z -axis. For this force, $\mathbf{M}_O = -200\mathbf{k} \text{ N} \cdot \text{m}$, as shown in Fig. 2.5(d). Although the vector description for both forces is $-100\mathbf{i} \text{ N}$, their moments about point O are oppositely directed.

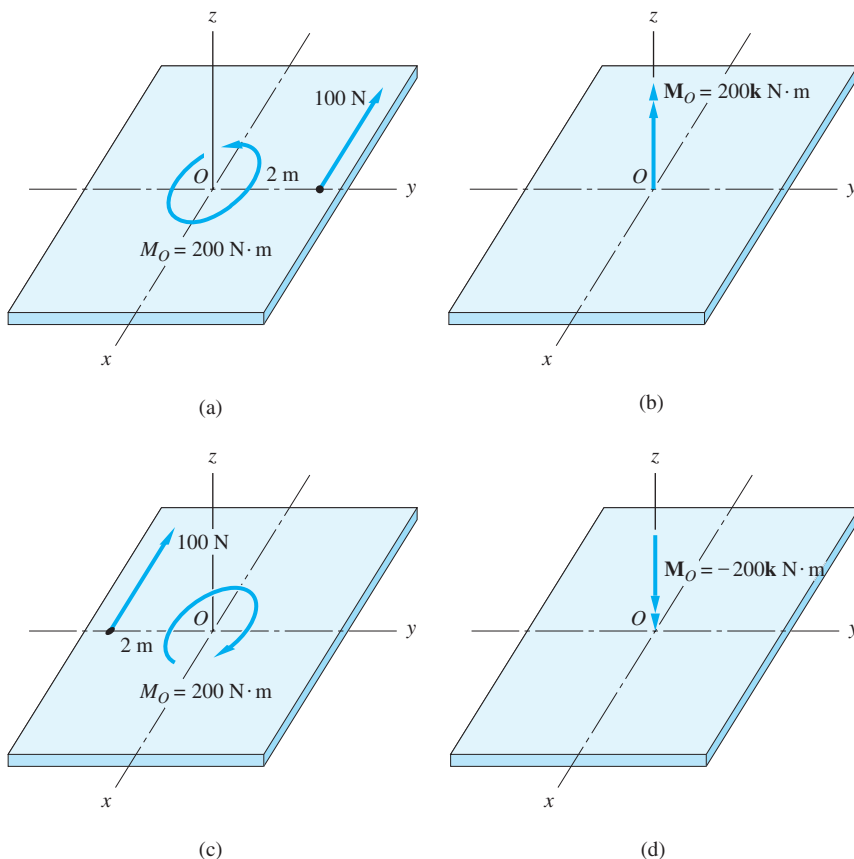


Fig. 2.5

c. Principle of moments

When determining the moment of a force about a point, it is often convenient to use the *principle of moments*, also known as *Varignon's theorem*:

The moment of a force about a point is equal to the sum of the moments of its components about that point.

Proof

To prove Varignon's theorem, consider the three forces $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ concurrent at point A , as shown in Fig. 2.6, where $\mathbf{r}$ is the vector from point O to point A . The sum of the moments about point O for the three forces is

$$\mathbf{M}_O = \Sigma(\mathbf{r} \times \mathbf{F}) = (\mathbf{r} \times \mathbf{F}_1) + (\mathbf{r} \times \mathbf{F}_2) + (\mathbf{r} \times \mathbf{F}_3) \quad (\text{a})$$

Using the properties of the cross product, Eq. (a) may be written as

$$\mathbf{M}_O = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3) = \mathbf{r} \times \mathbf{R} \quad (\text{b})$$

where $\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ is the resultant force for the three original forces. Equation (b) proves the principle of moments: The moment of $\mathbf{R}$ equals the moments of the components of $\mathbf{R}$. (Although the preceding proof has used only three components, it may obviously be extended to any number of components.)

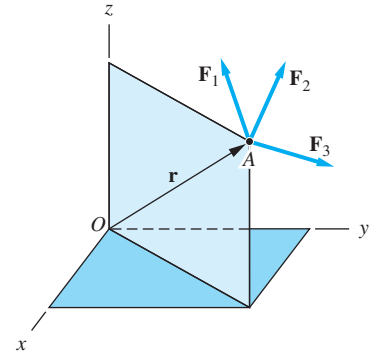


Fig. 2.6

d. Vector and scalar methods

From the preceding discussion, we observe that the following are equivalent methods for computing the moment of a force $\mathbf{F}$ about a point O .

Vector Method The vector method uses $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ is a vector from point O to any point on the line of action of $\mathbf{F}$. The most efficient technique for using the vector method (with rectangular components) is the following: (1) Write $\mathbf{F}$ in vector form; (2) choose an $\mathbf{r}$, and write it in vector form; and (3) use the determinant form of $\mathbf{r} \times \mathbf{F}$ to evaluate $\mathbf{M}_O$:

$$\mathbf{M}_O = \mathbf{r} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix} \quad (2.8)$$

where the second and third lines in the determinant are the rectangular components of $\mathbf{r}$ and $\mathbf{F}$, respectively. These components are shown in Fig. 2.7. Expansion of the determinant in Eq. (2.8) yields

$$\mathbf{M}_O = (yF_z - zF_y)\mathbf{i} + (zF_x - xF_z)\mathbf{j} + (xF_y - yF_x)\mathbf{k} \quad (2.9)$$

Scalar Method In the scalar method, the magnitude of the moment of the force $\mathbf{F}$ about the point O is found from $M_O = Fd$, where d is the moment arm of the force. In this method, the sense of the moment must be determined by inspection. As mentioned previously, the scalar method is convenient only when the moment arm d can be easily determined.

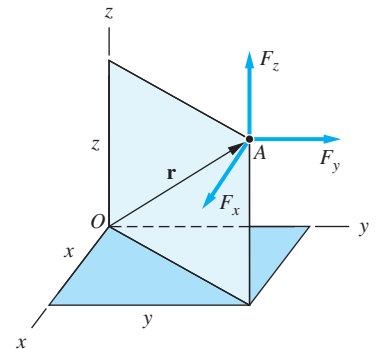
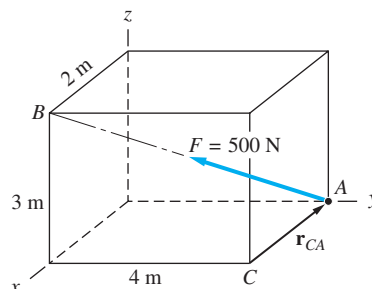


Fig. 2.7

Sample Problem 2.3

Determine (1) the moment of the force $\mathbf{F}$ about point C ; and (2) the perpendicular distance between C and the line of action of $\mathbf{F}$.



Solution

Part 1

The moment of a force about point C can be computed by either the scalar method ($M_C = Fd$), or the vector method ($\mathbf{M}_C = \mathbf{r} \times \mathbf{F}$). In this problem the scalar method would be inconvenient, because we have no easy means of determining d (the perpendicular distance between C and the line AB). Therefore, we use the vector method, which consists of the following three steps: (1) write $\mathbf{F}$ in vector form; (2) choose an $\mathbf{r}$, and write it in vector form; and (3) compute $\mathbf{M}_C = \mathbf{r} \times \mathbf{F}$.

Step 1: Write $\mathbf{F}$ in vector form.

Referring to the figure, we obtain

$$\mathbf{F} = 500\lambda_{AB} = 500 \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = 500 \left(\frac{2\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}}{5.385} \right)$$

which yields

$$\mathbf{F} = 185.7\mathbf{i} - 371.4\mathbf{j} + 278.6\mathbf{k} \text{ N}$$

Step 2: Choose an $\mathbf{r}$, and write it in vector form.

The vector $\mathbf{r}$ is a vector from point C to any point on the line of action of $\mathbf{F}$. From the figure we see that there are two convenient choices for $\mathbf{r}$ —the vector from point C to either point A or point B . As shown in the figure, let us choose $\mathbf{r}$ to be $\mathbf{r}_{CA}$. (As an exercise, you may wish to solve this problem by choosing $\mathbf{r}$ to be the vector from point C to point B .) Now we have

$$\mathbf{r} = \mathbf{r}_{CA} = -2\mathbf{i} \text{ m}$$

Step 3: Calculate $\mathbf{M}_C = \mathbf{r} \times \mathbf{F}$.

The easiest method for evaluating the cross product is to use the determinant expansion:

$$\mathbf{M}_C = \mathbf{r} \times \mathbf{F} = \mathbf{r}_{CA} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 0 & 0 \\ 185.7 & -371.4 & 278.6 \end{vmatrix}$$

Expanding this determinant gives

$$\mathbf{M}_C = 557.2\mathbf{j} + 742.8\mathbf{k} \text{ N} \cdot \text{m} \quad \text{Answer}$$

Part 2

The magnitude of $\mathbf{M}_C$ is

$$M_C = \sqrt{(557.2)^2 + (742.8)^2} = 928.6 \text{ N} \cdot \text{m}$$

The perpendicular distance d from point C to the line of action of $\mathbf{F}$ may be determined by

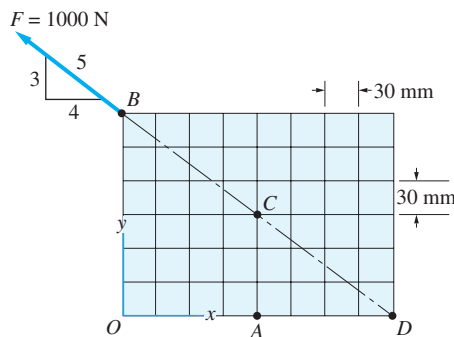
$$d = \frac{M_C}{F} = \frac{928.6}{500} = 1.857 \text{ m} \quad \text{Answer}$$

Observe that, instead of using the perpendicular distance to determine the moment, we have used the moment to determine the perpendicular distance.

Caution A common mistake is choosing the wrong sense for $\mathbf{r}$ in Eq. (2.4). Note that $\mathbf{r}$ is directed from the moment center to the line of action of $\mathbf{F}$. If the sense of $\mathbf{r}$ is reversed, $\mathbf{r} \times \mathbf{F}$ will yield the correct magnitude of the moment, but the wrong sense. To avoid this pitfall, it is strongly recommended that you draw $\mathbf{r}$ on your sketch before attempting to write it in vector form.

Sample Problem 2.4

Determine the moment of the force $\mathbf{F}$ in Fig. (a) about point A .



(a)

Solution

The force $\mathbf{F}$ and point A lie in the xy -plane. Problems of this type may be solved using either the vector method ($\mathbf{r} \times \mathbf{F}$) or the scalar method (Fd). For illustrative purposes, we use both methods.

Vector Solution

Recall that the three steps in the vector method are to write $\mathbf{F}$ in vector form, choose $\mathbf{r}$ and write it in vector form, and then evaluate the cross product $\mathbf{r} \times \mathbf{F}$.

Writing $\mathbf{F}$ in vector form, we get

$$\begin{aligned}\mathbf{F} &= -\left(\frac{4}{5}\right) 1000\mathbf{i} + \left(\frac{3}{5}\right) 1000\mathbf{j} \\ &= -800\mathbf{i} + 600\mathbf{j} \text{ N}\end{aligned}$$

There are several good choices for $\mathbf{r}$ in this problem, three of which are $\mathbf{r}_{AB}$, $\mathbf{r}_{AC}$, and $\mathbf{r}_{AD}$. Choosing

$$\mathbf{r} = \mathbf{r}_{AB} = -120\mathbf{i} + 180\mathbf{j}$$

the moment about point A is

$$\mathbf{M}_A = \mathbf{r} \times \mathbf{F} = \mathbf{r}_{AB} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -120 & 180 & 0 \\ -800 & 600 & 0 \end{vmatrix}$$

Expanding this determinant, we obtain

$$\mathbf{M}_A = \mathbf{k}[(800)(-120) + (800)(180)] = 72000\mathbf{k} \text{ N} \cdot \text{mm} \quad \text{Answer}$$

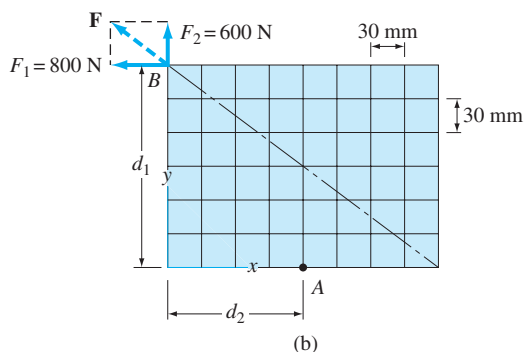
The magnitude of $\mathbf{M}_A$ is $72000 \text{ N} \cdot \text{mm}$. Note that the direction of $\mathbf{M}_A$ is the positive z direction, which by the right-hand rule means that the moment about point A is counterclockwise.

Scalar Solution

In Fig. (b), we have resolved the force into the rectangular components $\mathbf{F}_1$ and $\mathbf{F}_2$ at point B . The moment arm of each component about point A (the perpendicular distance between A and the line of action of the force) can be determined by inspection. The moment arms are $d_1 = 180 \text{ mm}$ for $\mathbf{F}_1$ and $d_2 = 120 \text{ mm}$ for $\mathbf{F}_2$, as shown in Fig. (b).

The moment of $\mathbf{F}$ about A now can be obtained by the principle of moments. Nothing that the moment of $\mathbf{F}_1$ is counterclockwise, whereas the moment of $\mathbf{F}_2$ is clockwise, we obtain

$$\begin{aligned}\curvearrowright M_A &= F_1 d_1 - F_2 d_2 \\ &= 800(180) - 600(120) = 72000 \text{ N} \cdot \text{mm}\end{aligned}$$



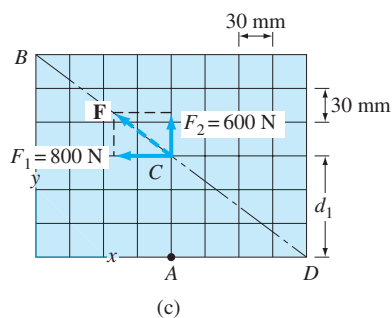
Note that the sense of M_A is counterclockwise. Applying the right-hand rule, the vector representation of the moment is

$$\mathbf{M}_A = 72000\mathbf{k} \text{ N}\cdot\text{mm} \quad \text{Answer}$$

Recall that a force, being a sliding vector, can be moved to any point on its line of action without changing its moment. In Fig. (c) we have moved $\mathbf{F}$ to point C . Now the moment arm of $\mathbf{F}_1$ about A is $d_1 = 90 \text{ mm}$, and the moment arm of $\mathbf{F}_2$ is zero. Hence, the moment of $\mathbf{F}$ about A is

$$\curvearrowleft M_A = F_1 d_1 = 800(90) = 72000 \text{ N}\cdot\text{mm}$$

counterclockwise, as before.



Another convenient location for $\mathbf{F}$ would be point D in Fig. (c). Here the moment arm of $\mathbf{F}_1$ about A is zero, whereas the moment arm of $\mathbf{F}_2$ is 120 mm , which again yields $M_A = 72000 \text{ N}\cdot\text{mm}$ counterclockwise.

2.6 *Moment of a Force about an Axis*

Whereas the preceding article defined the moment of a force about a point, this article discusses the moment of a force about an axis. Because moment about an axis is a measure of the tendency of a force to rotate a body about the axis, it is fundamental to the study of engineering mechanics. We begin with a formal definition of the moment about an axis, and we then examine its geometric interpretation.

a. Definition

The moment of a force about an axis, called the *moment axis*, is most easily defined in terms of the moment of the force about a point on the axis. Figure 2.8 shows the force $\mathbf{F}$ and its moment $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$ about point O , where O is any point on the axis AB . We define the moment about an axis as follows:

The moment of $\mathbf{F}$ about the axis AB is the orthogonal component of $\mathbf{M}_O$ along the axis AB , where O is any point on AB .

Letting $\boldsymbol{\lambda}$ be a unit vector directed from A toward B , this definition gives for the moment of $\mathbf{F}$ about the axis AB

$$M_{AB} = M_O \cos \alpha \quad (2.10)$$

where α is the angle between $\mathbf{M}_O$ and $\boldsymbol{\lambda}$, as shown in Fig. 2.8.

Because $M_O \cos \alpha = \mathbf{M}_O \cdot \boldsymbol{\lambda}$ (from the definition of the dot product), Eq. (2.10) can also be expressed in the form

$$M_{AB} = \mathbf{M}_O \cdot \boldsymbol{\lambda} = \mathbf{r} \times \mathbf{F} \cdot \boldsymbol{\lambda} \quad (2.11)$$

Let us review each of the terms appearing in this equation:

- M_{AB} is the moment (actually, the magnitude of the moment) of the force $\mathbf{F}$ about the axis AB .
- $\mathbf{M}_O$ represents the moment of $\mathbf{F}$ about the point O , where O is any point on the axis AB .*
- $\boldsymbol{\lambda}$ is the unit vector directed from A toward B .
- $\mathbf{r}$ is the position vector drawn from O to any point on the line of action of $\mathbf{F}$.

Note that the direction of $\boldsymbol{\lambda}$ determines the positive sense of M_{AB} by the right-hand rule, as illustrated in Fig. 2.9. Paying heed to this sign convention will enable you to interpret correctly the sign of M_{AB} in Eqs. (2.10) and (2.11).

Sometimes we wish to express the moment of $\mathbf{F}$ about the axis AB as a vector. We can do this by multiplying M_{AB} by the unit vector $\boldsymbol{\lambda}$ that specifies the direction of the moment axis, yielding

$$\mathbf{M}_{AB} = M_{AB} \boldsymbol{\lambda} = (\mathbf{r} \times \mathbf{F} \cdot \boldsymbol{\lambda}) \boldsymbol{\lambda} \quad (2.12)$$

Rectangular components of $\mathbf{M}_O$ Let $\mathbf{M}_O$ be the moment of a force $\mathbf{F}$ about O , where O is the origin of the xyz -coordinate system shown in Fig. 2.10. The

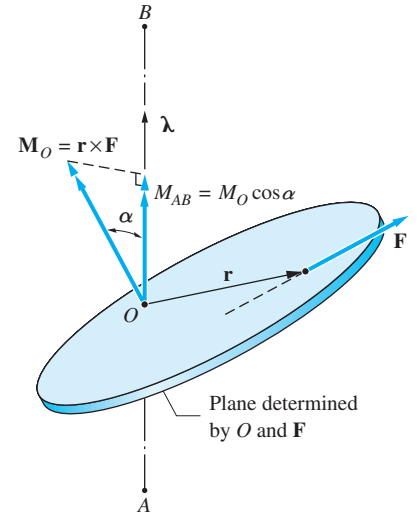


Fig. 2.8

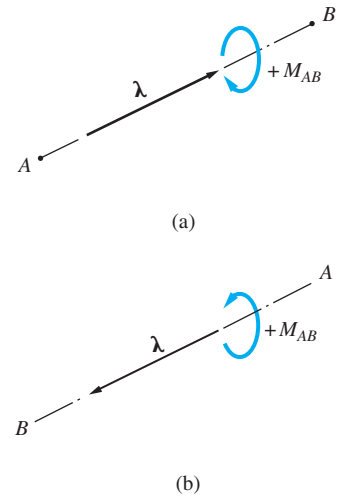


Fig. 2.9

*If we let O and C be two points on the moment axis AB , then $\mathbf{M}_O$ and $\mathbf{M}_C$ will, in general, be different. However, it can be shown that $\mathbf{M}_O \cdot \boldsymbol{\lambda} = \mathbf{M}_C \cdot \boldsymbol{\lambda}$, where $\boldsymbol{\lambda}$ is a unit vector parallel to AB . For this reason, O in Eq. (2.10) can be any point on AB .

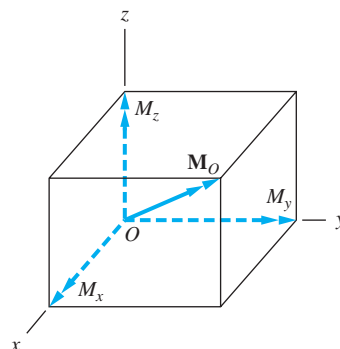


Fig. 2.10

moments of $\mathbf{F}$ about the three coordinate axes can be obtained from Eq. (2.11) by substituting $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ in turn for $\boldsymbol{\lambda}$. The results are

$$M_x = \mathbf{M}_O \cdot \mathbf{i} \quad M_y = \mathbf{M}_O \cdot \mathbf{j} \quad M_z = \mathbf{M}_O \cdot \mathbf{k}$$

from which we draw the following conclusion:

The rectangular components of the moment of a force about the origin O are equal to the moments of the force about the coordinate axes.

In other words,

$$\mathbf{M}_O = M_x \mathbf{i} + M_y \mathbf{j} + M_z \mathbf{k} \quad (2.13)$$

where M_x , M_y , and M_z , shown in Fig. 2.10, are equal to the moments of the force about the coordinate axes.

Special Case: Moment Axis Perpendicular to $\mathbf{F}$ Consider the case where the moment axis is perpendicular to the plane containing the force $\mathbf{F}$ and the point O , as shown in Fig. 2.11(a). Because the directions of $\mathbf{M}_O$ and $\mathbf{M}_{AB}$ now coincide, $\boldsymbol{\lambda}$ in Eq. (2.11) is in the direction of $\mathbf{M}_O$. Consequently, Eq. (2.11) yields

$$M_O = M_{AB} \quad (2.14)$$

That is, the moment of $\mathbf{F}$ about point O equals the moment of $\mathbf{F}$ about the axis AB .

A two-dimensional representation of Fig. 2.11(a), viewed along the moment axis AB , is shown in Fig. 2.11(b). We will frequently use a similar figure in the solution of two-dimensional problems (problems where all forces lie in the same plane). In problems of this type, it is customary to use the term *moment about a point* (M_O), rather than *moment about an axis* (M_{AB}).

b. Geometric interpretation

It is instructive to examine the geometric interpretation of the equation $M_{AB} = \mathbf{r} \times \mathbf{F} \cdot \boldsymbol{\lambda}$.

Suppose we are given an arbitrary force $\mathbf{F}$ and an arbitrary axis AB , as shown in Fig. 2.12. We construct a plane $\mathcal{P}$ that is perpendicular to the AB axis and let O and C be the points where the axis and the line of action of the force intersect $\mathcal{P}$,

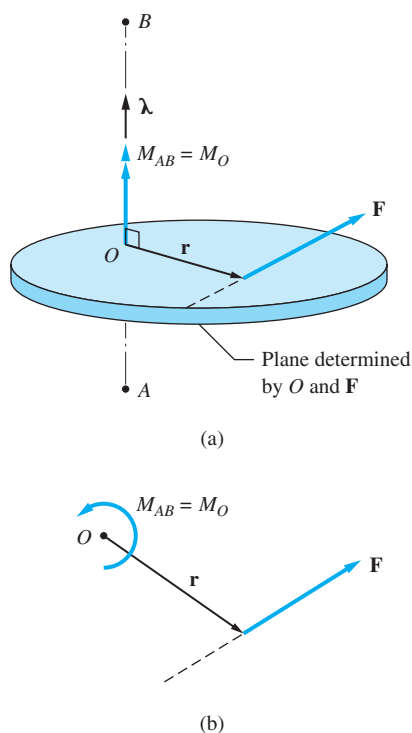


Fig. 2.11

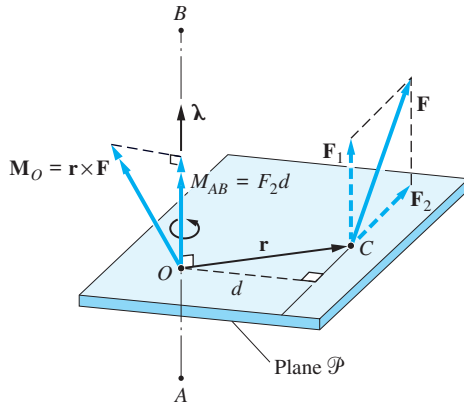


Fig. 2.12

respectively. The vector from O to C is denoted by $\mathbf{r}$, and $\boldsymbol{\lambda}$ is the unit vector along the axis AB . We then resolve $\mathbf{F}$ into two components: $\mathbf{F}_1$ and $\mathbf{F}_2$, which are parallel and perpendicular to the axis AB , respectively (observe that $\mathbf{F}_2$ lies in plane $\mathcal{P}$). In terms of these components, the moment of $\mathbf{F}$ about the axis AB is

$$\begin{aligned} M_{AB} &= \mathbf{r} \times \mathbf{F} \cdot \boldsymbol{\lambda} = \mathbf{r} \times (\mathbf{F}_1 + \mathbf{F}_2) \cdot \boldsymbol{\lambda} \\ &= \mathbf{r} \times \mathbf{F}_1 \cdot \boldsymbol{\lambda} + \mathbf{r} \times \mathbf{F}_2 \cdot \boldsymbol{\lambda} \end{aligned}$$

Because $\mathbf{r} \times \mathbf{F}_1$ is perpendicular to $\boldsymbol{\lambda}$, $\mathbf{r} \times \mathbf{F}_1 \cdot \boldsymbol{\lambda} = 0$, and we get

$$M_{AB} = \mathbf{r} \times \mathbf{F}_2 \cdot \boldsymbol{\lambda}$$

Substitution of $\mathbf{r} \times \mathbf{F}_2 \cdot \boldsymbol{\lambda} = F_2 d$, where d is the perpendicular distance from O to the line of action of $\mathbf{F}_2$, yields

$$M_{AB} = F_2 d \quad (2.15)$$

We see that the moment of $\mathbf{F}$ about the axis AB equals the product of the component of $\mathbf{F}$ that is perpendicular to AB and the perpendicular distance of this component from AB . Observe that Eq. (2.15) gives only the magnitude of the moment about the axis; its sense must be determined by inspection.

Consideration of Eq. (2.15) reveals that the moment of a force about an axis, as defined in Eq. (2.10), possesses the following physical characteristics:

- A force that is parallel to the moment axis (such as F_1) has no moment about that axis.
- If the line of action of a force intersects the moment axis ($d = 0$), the force has no moment about that axis.
- The moment of a force is proportional to its component that is perpendicular to the moment axis (such as F_2), and the moment arm (d) of that component.
- The sense of the moment is consistent with the direction in which the force would tend to rotate a body.

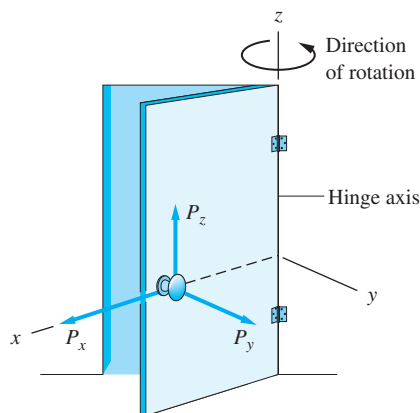


Fig. 2.13

To illustrate the above characteristics, consider opening the door in Fig. 2.13 by applying a force $\mathbf{P}$ to the handle. In the figure, $\mathbf{P}$ is resolved into the following rectangular components: P_x intersects the hinge axis, P_y is perpendicular to the door, and P_z is parallel to the hinge axis.

To open the door, we must apply a moment about the z -axis (the hinge axis). Experience tells us that P_y is the only component of the force that would accomplish this task. The components P_x and P_z are ineffective, because their moments about the z -axis are zero. We also know that it is easier to open the door if we increase the distance between the handle and the hinge axis (the moment arm) or if the magnitude of P_y is increased. Finally, observe that P_y causes the door to rotate in the direction shown in the figure, which is also the sense of the moment about the z -axis.

c. Vector and scalar methods

From the preceding discussion we see that the moment of the force $\mathbf{F}$ about an axis AB can be computed by two methods.

Vector Method The moment of $\mathbf{F}$ about AB is obtained from the triple scalar product $M_{AB} = \mathbf{r} \times \mathbf{F} \cdot \boldsymbol{\lambda}$, where $\mathbf{r}$ is a vector drawn from any point on the moment axis AB to any point on the line of action of $\mathbf{F}$ and $\boldsymbol{\lambda}$ represents a unit directed from A toward B . A convenient means of evaluating the scalar triple product is its determinant form

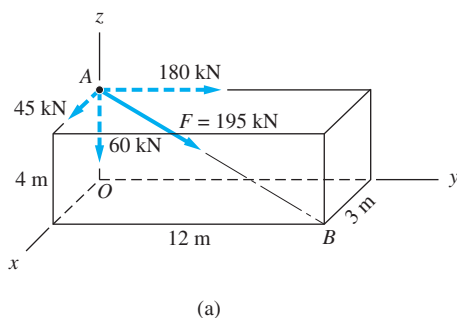
$$M_{AB} = \begin{vmatrix} x & y & z \\ F_x & F_y & F_z \\ \lambda_x & \lambda_y & \lambda_z \end{vmatrix} \quad (2.16)$$

where x , y , and z are the rectangular components of $\mathbf{r}$.

Scalar Method The moment of $\mathbf{F}$ about AB is obtained from the scalar expression $M_{AB} = F_2 d$. The sense of the moment must be determined by inspection. This method is convenient if AB is parallel to one of the coordinate axes (which is always the case in two-dimensional problems).

Sample Problem 2.5

The force $\mathbf{F}$ of magnitude 195 kN acts along the line AB . (1) Determine the moments M_x , M_y , and M_z of $\mathbf{F}$ about the coordinate axes by the scalar method; and (2) find the moment of $\mathbf{F}$ about point O by the vector method and verify that $\mathbf{M}_O = M_x\mathbf{i} + M_y\mathbf{j} + M_z\mathbf{k}$.



Solution

We start by computing the rectangular components of $\mathbf{F}$:

$$\begin{aligned}\mathbf{F} &= F\lambda_{AB} = F \frac{\overrightarrow{AB}}{|\overrightarrow{AB}|} = 195 \left(\frac{3\mathbf{i} + 12\mathbf{j} - 4\mathbf{k}}{\sqrt{3^2 + 12^2 + (-4)^2}} \right) \\ &= 45\mathbf{i} + 180\mathbf{j} - 60\mathbf{k} \text{ kN}\end{aligned}$$

When calculating the moment of a force, the force may be placed at any point on its line of action. As shown in Fig. (a), we chose to have the force acting at point A .

Part 1

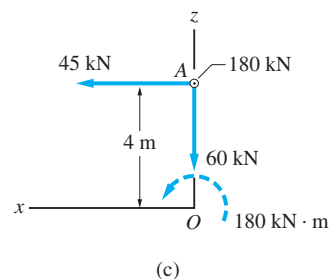
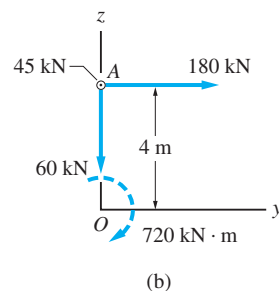
The moment of $\mathbf{F}$ about a coordinate axis can be computed by summing the moments of the components of $\mathbf{F}$ about that axis (the principle of moments).

Moment about the x -Axis Figure (b) represents a two-dimensional version of Fig. (a), showing the yz -plane. We see that the 45-kN and the 60-kN components of the force contribute nothing to the moment about the x -axis (the former is parallel to the axis, and the latter intersects the axis). The perpendicular distance (moment arm) between the 180-kN component and the x -axis is 4 m. Therefore, the moment of this component about the x -axis (which is also the moment of $\mathbf{F}$) is $180(4) = 720 \text{ kN} \cdot \text{m}$, clockwise. According to the right-hand rule, the positive sense of M_x is counterclockwise, which means that M_x is negative; that is,

$$M_x = -720 \text{ kN} \cdot \text{m}$$

Answer

Moment about the y -Axis To compute the moment about the y -axis, we refer to Fig. (c), which represents the xz -plane. We note that only the 45-kN force component has a moment about the y -axis, because the 180-kN component is parallel to the y -axis and the 60-kN component intersects the y -axis. Because the



moment arm of the 45-kN component is 4 m, the moment of $\mathbf{F}$ about the y-axis is $45(4) = 180 \text{ kN} \cdot \text{m}$, counterclockwise. Therefore, we have

$$M_y = 45(4) = 180 \text{ kN} \cdot \text{m} \quad \text{Answer}$$

The sign of the moment is positive, because the right-hand rule determines positive M_y to be counterclockwise.

Moment about the z-Axis The moment of $\mathbf{F}$ about the z-axis is zero, because $\mathbf{F}$ intersects that axis. Hence

$$M_z = 0 \quad \text{Answer}$$

Part 2

Recognizing that the vector from O to A in Fig. (a) is $\mathbf{r}_{OA} = 4\mathbf{k}$ m, the moment of $\mathbf{F}$ about point O can be computed as follows.

$$\begin{aligned} \mathbf{M}_O = \mathbf{r}_{OA} \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & 4 \\ 45 & 180 & -60 \end{vmatrix} = -\mathbf{i}(4)(180) + \mathbf{j}(4)(45) \\ &= -720\mathbf{i} + 180\mathbf{j} \text{ kN} \cdot \text{m} \quad \text{Answer} \end{aligned}$$

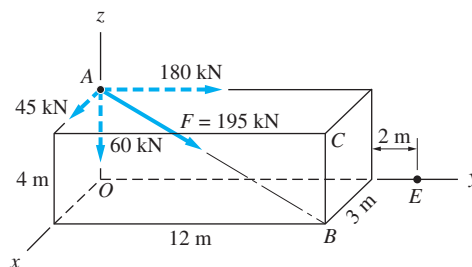
Comparing with $\mathbf{M}_O = M_x\mathbf{i} + M_y\mathbf{j} + M_z\mathbf{k}$, we see that

$$M_x = -720 \text{ kN} \cdot \text{m} \quad M_y = 180 \text{ kN} \cdot \text{m} \quad M_z = 0$$

which agree with the results obtained in Part 1.

Sample Problem 2.6

The force $\mathbf{F}$ of Sample Problem 2.5 is shown again in Fig. (a). (1) Determine the moment of $\mathbf{F}$ about the axis CE ; and (2) express the moment found in Part 1 in vector form.



(a)

Solution

Part 1

Referring to Fig. (a), we see that it is not practical to calculate the moment about the axis CE by the scalar method. Because the axis CE is not parallel to a coordinate axis, the task of determining the perpendicular distance between $\mathbf{F}$ and CE would be tedious. However, if the vector method is used, the calculation of the moment is straightforward.

To employ the vector method we first express the force $\mathbf{F}$ in vector form. This was already done in the solution to Sample Problem 2.5:

$$\mathbf{F} = 45\mathbf{i} + 180\mathbf{j} - 60\mathbf{k} \text{ kN}$$

Next we calculate the moment of $\mathbf{F}$ about any convenient point on the axis CE . Inspection of Fig. (a) reveals that there are only two convenient points from which to choose—points C and E . Let us choose point C . Because we will use the cross product $\mathbf{r} \times \mathbf{F}$ to compute the moment about C , our next step is to choose the vector $\mathbf{r}$ and to write it in vector form (remember that $\mathbf{r}$ must be a vector from point C to any point on the line of action of $\mathbf{F}$). From Fig. (a) we see that there are two convenient choices for $\mathbf{r}$: either the vector from C to A or the vector from C to B . Choosing the latter, we have

$$\mathbf{r} = \mathbf{r}_{CB} = -4\mathbf{k} \text{ m}$$

The moment of $\mathbf{F}$ about point C then becomes

$$\begin{aligned} \mathbf{M}_C &= \mathbf{r}_{CB} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 0 & -4 \\ 45 & 180 & -60 \end{vmatrix} \\ &= 720\mathbf{i} - 180\mathbf{j} \text{ kN} \cdot \text{m} \end{aligned}$$

Note that the z -component of $\mathbf{M}_C$ is zero. To understand this result, recall that the z -component of $\mathbf{M}_C$ equals the moment of $\mathbf{F}$ about the axis BC (the line parallel to the z -axis passing through C). Because $\mathbf{F}$ intersects BC , its moment about BC is expected to be zero.

Next, we calculate the unit vector λ_{CE} directed from point C toward point E :

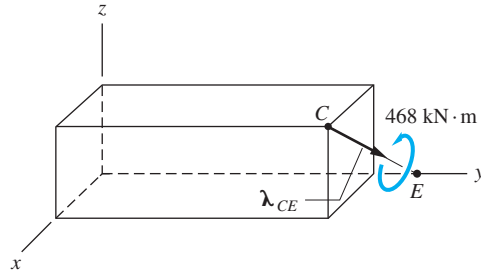
$$\lambda_{CE} = \frac{\overrightarrow{CE}}{|\overrightarrow{CE}|} = \frac{-3\mathbf{i} + 2\mathbf{j} - 4\mathbf{k}}{\sqrt{(-3)^2 + 2^2 + (-4)^2}} = -0.5571\mathbf{i} + 0.3714\mathbf{j} - 0.7428\mathbf{k}$$

The moment of $\mathbf{M}_C$ about the axis CE can now be obtained from Eq. (2.11):

$$\begin{aligned} M_{CE} &= \mathbf{M}_C \cdot \lambda_{CE} \\ &= (720\mathbf{i} - 180\mathbf{j}) \cdot (-0.5571\mathbf{i} + 0.3714\mathbf{j} - 0.7428\mathbf{k}) \\ &= -468 \text{ kN} \cdot \text{m} \end{aligned}$$

Answer

The negative sign indicates that the sense of the moment is as shown in Fig. (b)—that is, opposite to the sense associated with λ_{CE} .



(b)

We could also compute M_{CE} without first determining $\mathbf{M}_C$ by using the scalar triple product:

$$\begin{aligned} M_{CE} &= \mathbf{r}_{BC} \times \mathbf{F} \cdot \lambda_{CE} = \begin{vmatrix} 0 & 0 & -4 \\ 45 & 180 & -60 \\ -0.5571 & 0.3714 & -0.7428 \end{vmatrix} \\ &= -468 \text{ kN} \cdot \text{m} \end{aligned}$$

This agrees, of course, with the result determined previously.

Part 2

To express the moment of $\mathbf{F}$ about the axis CE in vector form, we multiply M_{CE} by the unit vector λ_{CE} , which gives

$$\begin{aligned} \mathbf{M}_{CE} &= M_{CE} \lambda_{CE} = -468(-0.5571\mathbf{i} + 0.3714\mathbf{j} - 0.7428\mathbf{k}) \\ &= 261\mathbf{i} - 174\mathbf{j} + 348\mathbf{k} \text{ kN} \cdot \text{m} \end{aligned} \quad \text{Answer}$$

There is no doubt that using the vector method is convenient when one wishes to calculate the moment about an axis such as CE , which is skewed relative to the coordinate system. However, there is a drawback to vector formalism: You can easily lose appreciation for the physical nature of the problem.

2.7 Couples

As pointed out before, a force has two effects on a rigid body: translation due to the force itself and rotation due to the moment of the force. A couple, on the other hand, is a purely rotational effect—it has a moment but no resultant force. Couples play an important role in the analysis of force systems.

a. Definition

Two parallel, noncollinear forces that are equal in magnitude and opposite in direction are known as a *couple*.

A typical couple is shown in Fig. 2.14. The two forces of equal magnitude F are oppositely directed along lines of action that are separated by the perpendicular distance d . (In a vector description of the forces, one of the forces would be labeled $\mathbf{F}$ and the other $-\mathbf{F}$.) The lines of action of the two forces determine a plane that we call the *plane of the couple*. The two forces that form a couple have some interesting properties, which will become apparent when we calculate their combined moment about a point.

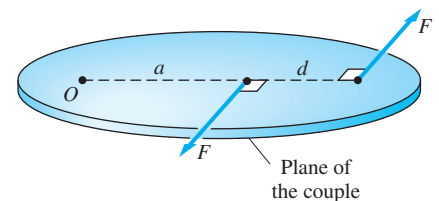


Fig. 2.14

b. Moment of a couple about a point

The moment of a couple about a point is the sum of the moments of the two forces that form the couple. When calculating the moment of a couple about a point, either the scalar method (force times perpendicular distance) or the vector method ($\mathbf{r} \times \mathbf{F}$) may be used. For illustrative purposes, we will calculate the moment of a couple using both methods. Using two methods of analysis to determine the same quantity may appear redundant, but it is instructive because each method emphasizes different characteristics of a couple.

Scalar Calculation Let us calculate the moment of the couple shown in Fig. 2.14 about the point O . Note that O is an arbitrary point in the plane of

the couple and that it is located a distance a from the force on the left. The sum of the moments about point O for the two forces is

$$\curvearrowright M_O = F(a + d) - F(a) = Fd \quad (2.17)$$

Observe that the moment of the couple about point O is *independent* of the location of O , because the result is independent of the distance a .

From the foregoing discussion, we see that a couple possesses two important characteristics: (1) A couple has no resultant force ($\Sigma \mathbf{F} = \mathbf{0}$), and (2) the moment of a couple is the same about any point in the plane of the couple.

Vector Calculation When the two forces that form the couple are expressed as vectors, they can be denoted by $\mathbf{F}$ and $-\mathbf{F}$, as shown in Fig. 2.15. The points labeled in the figure are A , any point on the line of action of $\mathbf{F}$; B , any point on the line of action of $-\mathbf{F}$; and O , an arbitrary point in space (not necessarily lying in the plane of the couple). The vectors $\mathbf{r}_{OA}$ and $\mathbf{r}_{OB}$ are drawn from point O to points A and B , respectively. The vector $\mathbf{r}_{BA}$ connects points B and A . Using the cross product to evaluate the moment of the couple about point O , we get

$$\mathbf{M}_O = [\mathbf{r}_{OA} \times \mathbf{F}] + [\mathbf{r}_{OB} \times (-\mathbf{F})] = (\mathbf{r}_{OA} - \mathbf{r}_{OB}) \times \mathbf{F}$$

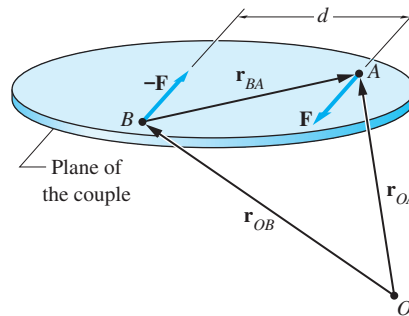


Fig. 2.15

Since $\mathbf{r}_{OA} - \mathbf{r}_{OB} = \mathbf{r}_{BA}$, the moment of the couple about point O reduces to

$$\mathbf{M}_O = \mathbf{r}_{BA} \times \mathbf{F} \quad (2.18)$$

which confirms that the moment of the couple about point O is independent of the location of O . Although the choice of point O determines $\mathbf{r}_{OA}$ and $\mathbf{r}_{OB}$, neither of these vectors appear in Eq. (2.18). Thus we conclude the following:

The moment of a couple is the same about every point.

In other words, the moment of a couple is a *free vector*. (Recall that, in the scalar calculation, point O was restricted to points in the plane of the couple. We see now that this restriction is unnecessary; that is, O can be any point in space.) In contrast, the moment of a force about a point (the moment center) is a fixed vector, because the moment depends on the location of the moment center.

c. Equivalent couples

Because a couple has no resultant force, its only effect on a rigid body is its moment. For this reason, two couples that have the same moment are said to be *equivalent* (have the same effect on a rigid body). Figure 2.16 illustrates the four operations that may be performed on a couple without changing its moment; all couples shown in the figure are equivalent. The operations are

1. Changing the magnitude F of each force and the perpendicular distance d while keeping the product Fd constant
2. Rotating the couple in its plane
3. Moving the couple to a parallel position in its plane
4. Moving the couple to a parallel plane

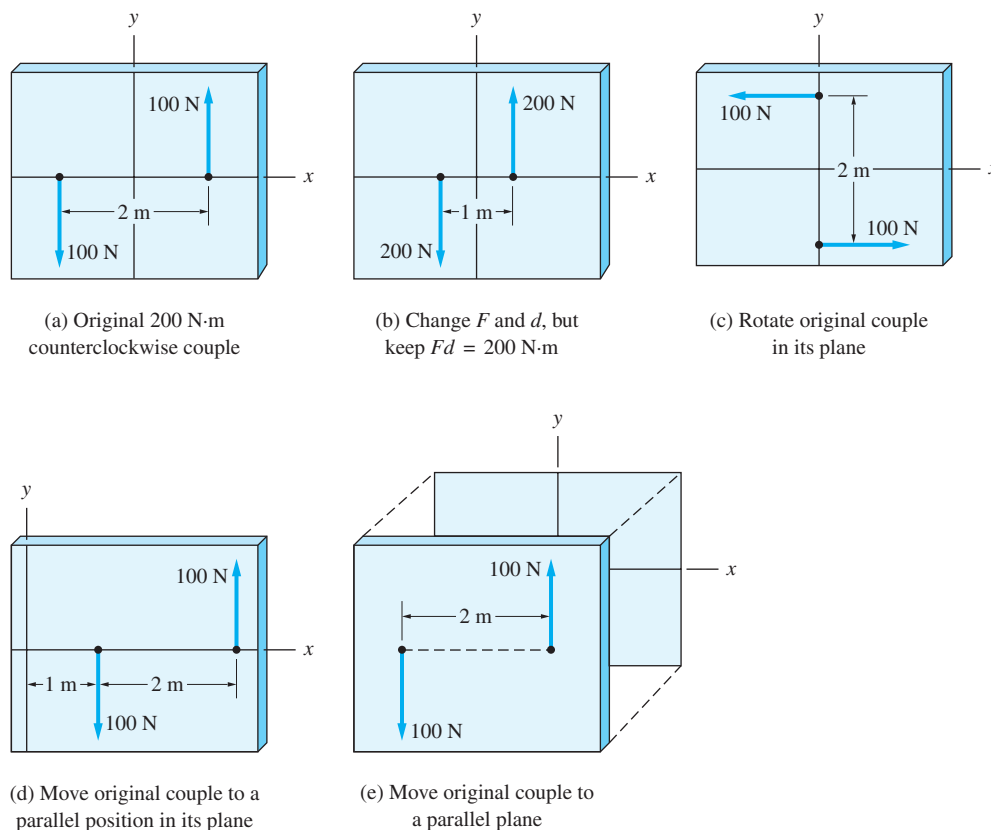


Fig. 2.16

d. Notation and terminology

Consider the couple shown in Fig. 2.17(a). The moment of this couple, shown in Fig. 2.17(b), has a magnitude of $C = 1800 \text{ N}\cdot\text{m}$ and is directed counterclockwise in the xy -plane. Because the only rigid-body effect of a couple is its moment, the representations in Figs. 2.17(a) and (b) are equivalent. That is, we can replace

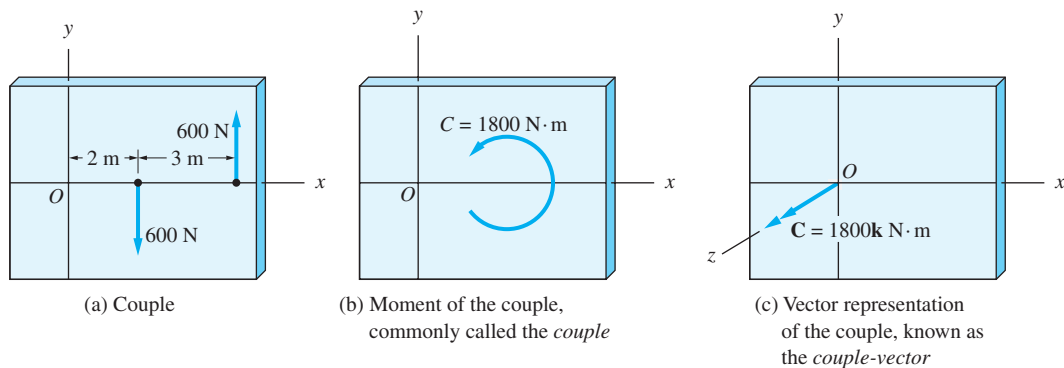


Fig. 2.17

a couple that acts on a rigid body by its moment without changing the external effect on the body. This equivalence also applies to the terminology—rather than referring to C as the *moment of the couple*, it usually is called simply the *couple*.

Figure 2.17(c) shows the same couple (strictly speaking, the moment of the couple) as a vector, which we call the *couple-vector*. The couple-vector is perpendicular to the plane of the couple, and its direction is determined by the right-hand rule. The choice of point O for the location of the couple-vector was arbitrary. Being a free vector, the couple-vector could be placed anywhere in the figure.

We will use the scalar and vector representations in Figs. 2.17(b) and (c) interchangeably, choosing the one that is more convenient for the problem at hand.

e. The addition and resolution of couples

Because couples are vectors, they may be added by the usual rules of vector addition. Being free vectors, the requirement that the couples to be added must have a common point of application does not apply. This is in contrast to the addition of forces, which can be added only if they are concurrent. Concurrency is also required for the addition of moments of forces about points, because these are fixed to a moment center. It follows that we must be careful when representing moments of forces and couples as vectors—it is easy to confuse these two concepts. To minimize the possibility of confusion, we will use $\mathbf{M}$ to denote moments of forces and reserve $\mathbf{C}$ for couples.

The resolution of couples is no different than the resolution of moments of forces. For example, the moment of a couple $\mathbf{C}$ about an axis AB can be computed from Eq. (2.11) by replacing $\mathbf{M}_O$ with $\mathbf{C}$:

$$M_{AB} = \mathbf{C} \cdot \boldsymbol{\lambda} \quad (2.19)$$

where $\boldsymbol{\lambda}$ is the unit vector in the direction of the axis. Note that the subscript O , which indicated that the moment must be taken about point O lying on the axis AB , is no longer present in Eq. (2.19). The reason is, of course, that the moment of $\mathbf{C}$ is the same about every point. As in the case of moments of forces, M_{AB} is equal to the rectangular component of $\mathbf{C}$ in the direction of AB , and is a measure of the tendency of $\mathbf{C}$ to rotate a body about the axis AB .

Sample Problem 2.7

For the couple shown in Fig. (a), determine (1) the corresponding couple-vector; and (2) the moment of the couple about the axis GH .

Solution

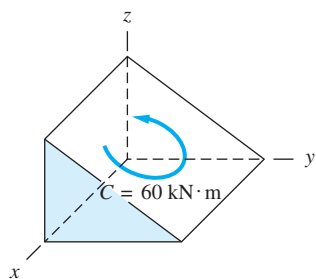
Part 1

One method for determining the couple-vector is to multiply the magnitude of the couple by the unit vector in its direction. The magnitude of the couple is

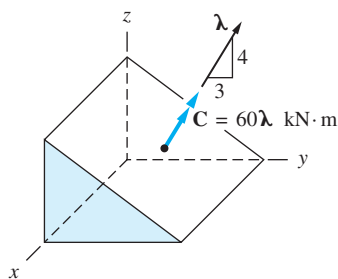
$$Fd = 100(0.6) = 60 \text{ kN} \cdot \text{m}$$

The sense of the couple is shown in Fig. (b)—counterclockwise looking down on the plane of the couple. Letting λ be the unit vector perpendicular to the plane of the couple, as shown in Fig. (c), the couple-vector $\mathbf{C}$ may be written as $\mathbf{C} = 60\lambda \text{ kN} \cdot \text{m}$. Because λ is perpendicular to the line AB , it can be seen that $\lambda = (3\mathbf{j} + 4\mathbf{k})/5$ (recalling that perpendicular lines have negative reciprocal slopes). Therefore, the couple-vector is

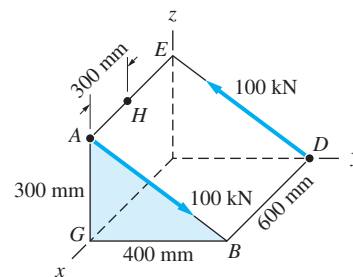
$$\mathbf{C} = 60\lambda = 60 \left(\frac{3\mathbf{j} + 4\mathbf{k}}{5} \right) = 36\mathbf{j} + 48\mathbf{k} \text{ kN} \cdot \text{m} \quad \text{Answer}$$



(b)



(c)



(a)

Alternative Solution Because the couple-vector is equal to the moment of the couple about any point, it can also be determined by adding the moments of the two forces forming the couple about any convenient point, such as point B . Letting $\mathbf{F}$ be the 100-kN force that acts along the line DE , we have

$$\begin{aligned} \mathbf{F} &= 100\lambda_{DE} = 100 \frac{\overrightarrow{DE}}{|\overrightarrow{DE}|} = 100 \left(\frac{-0.4\mathbf{j} + 0.3\mathbf{k}}{0.5} \right) \\ &= -80\mathbf{j} + 60\mathbf{k} \text{ kN} \end{aligned}$$

Equating $\mathbf{C}$ to the moment of $\mathbf{F}$ about point B (the other force of the couple passes through B), we obtain

$$\begin{aligned}\mathbf{C} = \mathbf{r}_{BD} \times \mathbf{F} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -0.6 & 0 & 0 \\ 0 & -80 & 60 \end{vmatrix} \\ &= 36\mathbf{j} + 48\mathbf{k} \text{ kN} \cdot \text{m}\end{aligned}$$

which agrees with the answer determined previously.

In this solution, the choice of point B as the moment center was arbitrary. Because the moment of a couple is the same about every point, the same result would have been obtained no matter which point had been chosen as the moment center.

Part 2

The most direct method for determining the moment of the couple about the axis GH is $M_{GH} = \mathbf{C} \cdot \boldsymbol{\lambda}_{GH}$. Because $\mathbf{C}$ has already been computed, all we need to do is compute the unit vector $\boldsymbol{\lambda}_{GH}$ and evaluate the dot product. Referring to Fig. (a), we have

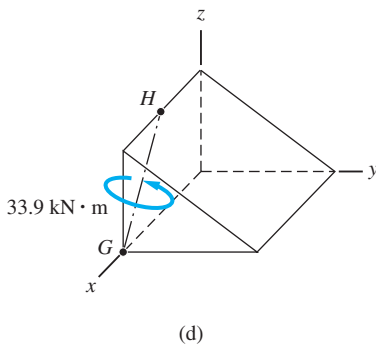
$$\boldsymbol{\lambda}_{GH} = \frac{\overrightarrow{GH}}{|\overrightarrow{GH}|} = \frac{-0.3\mathbf{i} + 0.3\mathbf{k}}{0.3\sqrt{2}} = -0.7071\mathbf{i} + 0.7071\mathbf{k}$$

Hence the moment of the couple about axis GH is

$$\begin{aligned}M_{GH} &= \mathbf{C} \cdot \boldsymbol{\lambda}_{GH} = (36\mathbf{j} + 48\mathbf{k}) \cdot (-0.7071\mathbf{i} + 0.7071\mathbf{k}) \\ &= +33.9 \text{ kN} \cdot \text{m}\end{aligned}$$

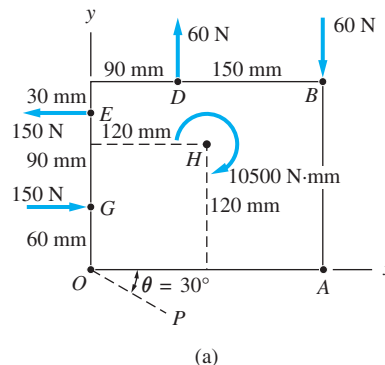
Answer

The result is illustrated in Fig. (d). If you need help in interpreting the positive sign in the answer, you should refer back to Fig. 2.9.



Sample Problem 2.8

The flat plate shown in Fig. (a) is acted on by the three couples. Replace the three couples with (1) a couple-vector; (2) two forces, one acting along the line OP



and the other acting at point A ; and (3) the smallest pair of forces, with one force acting at point O and the other at point A .

Solution

Part 1

The magnitudes (Fd) and senses of the couples, all of which lie in the xy -plane, are listed below.

- Couple at H : $10500 \text{ N} \cdot \text{mm}$ clockwise.
- Couple acting on GE : $(150)(90) = 13500 \text{ N} \cdot \text{mm}$ counterclockwise.
- Couple acting on DB : $(60)(150) = 9000 \text{ N} \cdot \text{mm}$ clockwise.

Because all three couples lie in the same plane, they can be added algebraically, their sum being the resultant couple C^R . Choosing the counterclockwise sense as positive, we get

$$\curvearrowleft C^R = -10500 + 13500 - 9000 = -6000 \text{ N} \cdot \text{mm}$$

The negative sign shows that the sense of C^R is clockwise. Therefore, the corresponding couple-vector $\mathbf{C}^R$ is, according to the right-hand rule, in the negative z -direction. It follows that

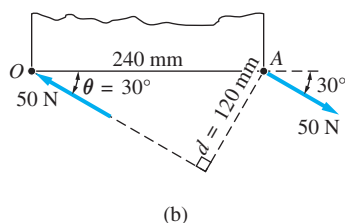
$$\mathbf{C}^R = -6000\mathbf{k} \text{ N} \cdot \text{mm} \quad \text{Answer}$$

Note that more dimensions are given in Fig. (a) than are needed for the solution. The only relevant dimensions are the distances between the 60-N forces (150 mm) and the 150-N forces (90 mm).

Part 2

Two forces that are equivalent to the three couples shown in Fig. (a) must, of course, form a couple. The problem states that one of the forces acts along the line OP and the other acts at point A .

Because the two forces that form a couple must have parallel lines of action, the line of action of the force at point A must also be parallel to OP . From Fig. (b), we see that the perpendicular distance d between the lines of action of the two forces is $d = 240 \sin 30^\circ = 120 \text{ mm}$. Having already determined that the magnitude of the resultant couple is $6000 \text{ N} \cdot \text{mm}$, the magnitudes of the forces that form the couple are given by $C^R/d = 6000/120 = 50 \text{ N}$. The sense of each force must be consistent with the clockwise sense of C^R . The final result is shown in Fig. (b).

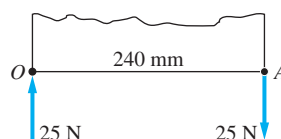


Answer

Part 3

Here we are to determine the smallest two forces acting at points O and A that are equivalent to the three couples shown in Fig. (a). Therefore, the two forces to be determined must form a couple that is equivalent to the resultant couple ($6000 \text{ N} \cdot \text{mm}$, clockwise).

The magnitude of a couple (Fd) equals the product of the magnitude of the forces that form the couple (F) and the perpendicular distance (d) between the forces. For a couple of given magnitude, the smallest forces will be obtained when the perpendicular distance d is as large as possible. From Fig. (b) it can be seen that for forces acting at points O and A , the largest d will correspond to $\theta = 90^\circ$, giving $d = 240 \text{ mm}$. Therefore, the magnitudes of the smallest forces are given by $C^R/d = 6000/240 = 25 \text{ N}$. These results are shown in Fig. (c), where again note should be taken of the senses of the forces.



(c)

Answer

Sample Problem 2.9

A section of a piping system is acted on by the three couples shown in Fig. (a). Determine the magnitude of the resultant couple-vector $\mathbf{C}^R$ and its direction cosines, given that the magnitudes of the applied couples are $C_1 = 50 \text{ N} \cdot \text{m}$, $C_2 = 90 \text{ N} \cdot \text{m}$, and $C_3 = 140 \text{ N} \cdot \text{m}$.

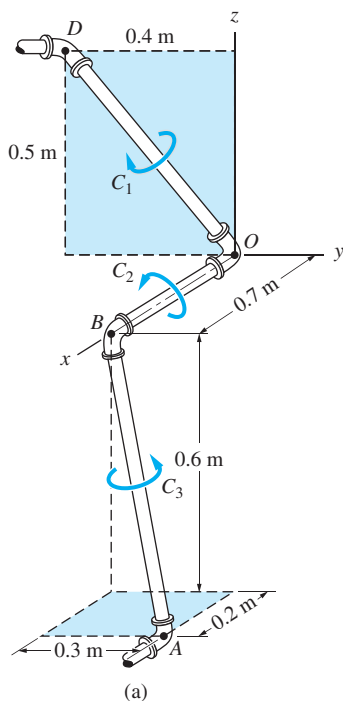
Solution

Applying the right-hand rule to each of the three couples in Fig. (a), we see that the corresponding couple-vectors will be directed as follows: $\mathbf{C}_1$, from point D toward point O ; $\mathbf{C}_2$, from point O toward point B ; and $\mathbf{C}_3$, from point A toward point B . Because these couple-vectors do not have the same directions, the most practical method of determining their resultant is to use the vector equation

$$\mathbf{C}^R = \mathbf{C}_1 + \mathbf{C}_2 + \mathbf{C}_3$$

Using the three unit vectors shown in Fig. (b), the couple-vectors $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{C}_3$ can be written as

$$\begin{aligned} \mathbf{C}_1 &= C_1 \lambda_{DO} = 50 \frac{\overrightarrow{DO}}{|\overrightarrow{DO}|} = 50 \left(\frac{0.4\mathbf{j} - 0.5\mathbf{k}}{0.6403} \right) \\ &= 31.24\mathbf{j} - 39.04\mathbf{k} \text{ N} \cdot \text{m} \end{aligned}$$



$$\mathbf{C}_2 = C_2 \lambda_{OB} = 90\mathbf{i} \text{ N} \cdot \text{m}$$

$$\begin{aligned} \mathbf{C}_3 &= C_3 \lambda_{AB} = 140 \frac{\vec{AB}}{|\vec{AB}|} = 140 \left(\frac{-0.2\mathbf{i} - 0.3\mathbf{j} + 0.6\mathbf{k}}{0.7000} \right) \\ &= -40\mathbf{i} - 60\mathbf{j} + 120\mathbf{k} \text{ N} \cdot \text{m} \end{aligned}$$

Adding these three couple-vectors gives

$$\mathbf{C}^R = 50\mathbf{i} - 28.76\mathbf{j} + 80.96\mathbf{k} \text{ N} \cdot \text{m}$$

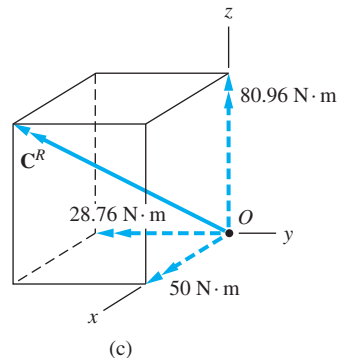
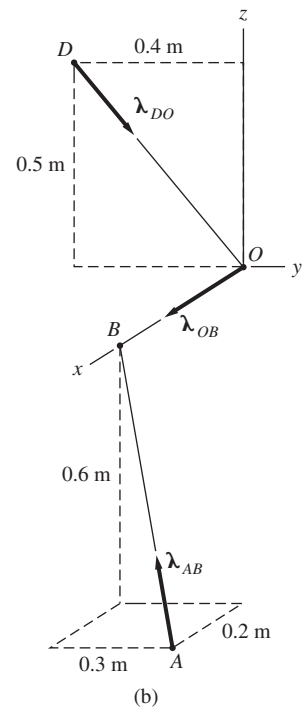
The magnitude of $\mathbf{C}^R$ is

$$C^R = \sqrt{(50)^2 + (-28.76)^2 + (80.96)^2} = 99.41 \text{ N} \cdot \text{m} \quad \text{Answer}$$

and the direction cosines of $\mathbf{C}^R$ are the components of the unit vector λ directed along $\mathbf{C}^R$:

$$\lambda_x = \frac{50}{99.41} = 0.503 \quad \lambda_y = -\frac{28.76}{99.41} = -0.289 \quad \lambda_z = \frac{80.96}{99.41} = 0.814 \quad \text{Answer}$$

The resultant couple-vector is shown in Fig. (c). Although $\mathbf{C}^R$ is shown at point O , it must be remembered that couples are free vectors, so that $\mathbf{C}^R$ could be shown acting anywhere.



The couple-vector $\mathbf{C}^R$ can be represented as two equal and opposite parallel forces. However, because the two forces will lie in a plane perpendicular to the couple-vector, in this case a skewed plane, this representation is inconvenient here.

In general, given two forces that form a couple, the corresponding couple-vector is easily determined (e.g., by summing the moments of the two forces about any point). However, given a couple-vector, it is not always convenient (or even desirable) to determine two equivalent forces.

2.8 *Changing the Line of Action of a Force*

In this article we show how to change the line of action of a force without affecting its external effect on a rigid body. This topic lays the foundation for the next chapter, in which we discuss the resultants of force systems.

Referring to Fig. 2.18(a), consider the problem of moving the force of magnitude F from point B to point A . We cannot simply move the force to A , because this would change its line of action, thereby altering the rotational effect (the

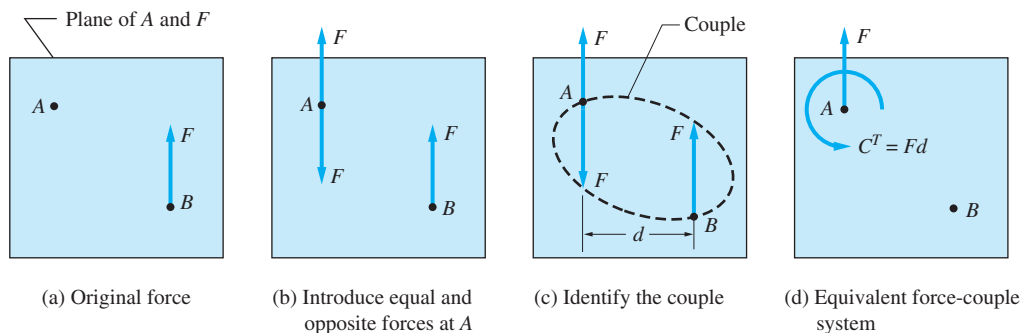


Fig. P2.18

moment) of the force. We can, however, counteract this change by introducing a couple that restores the rotational effect to its original state. The construction for determining this couple is illustrated in Fig. 2.18. It consists of the following two steps:

- Introduce two equal and opposite forces of magnitude F at point A , as shown in Fig. 2.18(b). These forces are parallel to the original force at B . Because the forces at A have no net external effect on a rigid body, the force systems in Figs. 2.18(a) and (b) are equivalent.
- Identify the two forces that form a couple, as has been done in Fig. 2.18(c). The magnitude of this couple is $C^T = Fd$, where d is the distance between the lines of action of the forces at A and B . The third force and C^T thus constitute the *force-couple system* shown in Fig. 2.18(d), which is equivalent to the original force in Fig. 2.18(a).

We refer to the couple C^T as the *couple of transfer*, because it is the couple that must be introduced when a force is transferred from one line of action to another. From the construction in Fig. 2.18 we note the following:

The couple of transfer is equal to the moment of the original force (acting at B) about the transfer point A .

In vector terminology, the line of action of a force $\mathbf{F}$ can be changed to a parallel line, provided that we introduce the couple of transfer

$$\mathbf{C}^T = \mathbf{r} \times \mathbf{F} \quad (2.20)$$

where $\mathbf{r}$ is the vector drawn from the transfer point A to the point of application B of the original force, as illustrated in Fig. 2.19. It is conventional to show $\mathbf{C}^T$ acting at the transfer point, as in Fig. 2.19(b), but you must not forget that a couple is a free vector that could be placed anywhere.

According to the properties of the cross product in Eq. (2.20), the couple-vector $\mathbf{C}^T$ is perpendicular to $\mathbf{F}$. Thus a force at a given point can always be replaced by a force at a different point and a couple-vector that is perpendicular to the force. The converse is also true: A force and a couple-vector that are mutually perpendicular can always be reduced to a single, equivalent force by reversing the construction outlined in Figs. 2.18 and 2.19.

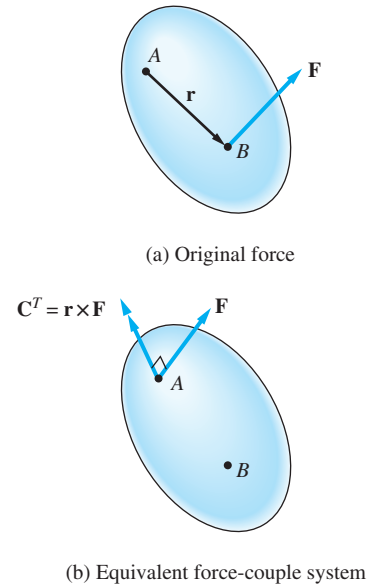
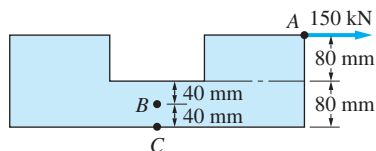


Fig. 2.19

Sample Problem 2.10

For the machine part shown in Fig. (a), replace the applied load of 150 kN acting at point *A* by (1) an equivalent force-couple system with the force acting at point *B*; and (2) two horizontal forces, one acting at point *B* and the other acting at point *C*.



(a)

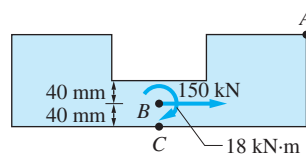
Solution

Part 1

First we move the 150-kN force to point *B*, and then we introduce the couple of transfer equal to the moment of the 150-kN force in Fig. (a) about point *B*, given by

$$\curvearrowright C^T = M_B = -150(0.080 + 0.040) = -18 \text{ kN} \cdot \text{m}$$

The negative sign indicates that the sense of the couple is clockwise. The equivalent force-couple system is shown in Fig. (b).

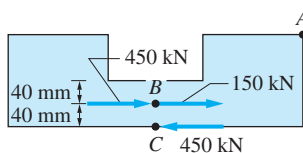


(b)

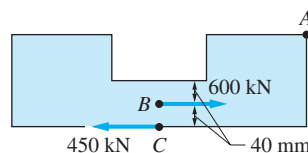
Answer

Part 2

The 18 kN·m clockwise couple in Fig. (b) can be replaced by two 450-kN forces, one acting at point *B* and the other at point *C*, as shown in Fig. (c). (The couple represented by these two forces is $450(0.040) = 18 \text{ kN} \cdot \text{m}$ in the clockwise direction.) The two forces acting at point *B* can be added to get the system shown in Fig. (d). This is the answer, because we have replaced the original force with two horizontal forces, one at point *B* and the other at point *C*, as required.



(c)



(d)

Answer

Sample Problem 2.11

Replace the force-couple system shown in Fig. (a) with an equivalent force-couple system, with the force acting at point A, given that $F = 100 \text{ N}$ and $C = 3600 \text{ N} \cdot \text{mm}$.

Solution

Moving the given force $\mathbf{F}$ from point B to point A requires the introduction of a couple of transfer $\mathbf{C}^T$. This couple is then added to the given couple-vector $\mathbf{C}$, thereby obtaining the resultant couple-vector, which we label $\mathbf{C}^R$. The couple-vector $\mathbf{C}^R$ and the force $\mathbf{F}$ located at point A will then be the required force-couple system.

Owing to the three-dimensional nature of this problem, it is convenient to use vector methods in the solution. Writing $\mathbf{F}$ in vector form, we obtain

$$\begin{aligned}\mathbf{F} &= 100\lambda_{BE} = 100 \frac{\overrightarrow{BE}}{|\overrightarrow{BE}|} = 100 \left(\frac{-120\mathbf{i} + 60\mathbf{k}}{134.164} \right) \\ &= -89.44\mathbf{i} + 44.72\mathbf{k} \text{ N}\end{aligned}$$

The position vector from A to B is $\mathbf{r}_{AB} = 120\mathbf{j} - 60\mathbf{k} \text{ m}$. The couple of transfer is equal to the moment of the given force $\mathbf{F}$ about point A, so we have

$$\begin{aligned}\mathbf{C}^T = \mathbf{M}_A &= \mathbf{r}_{AB} \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & 120 & -60 \\ -89.44 & 0 & 44.72 \end{vmatrix} \\ &= 5366.4\mathbf{i} + 5366.4\mathbf{j} + 10732.8\mathbf{k} \text{ N} \cdot \text{mm}\end{aligned}$$

Expressing the given couple-vector $\mathbf{C}$ shown in Fig. (a) in vector form,

$$\begin{aligned}\mathbf{C} &= 3600\lambda_{DB} = 3600 \frac{\overrightarrow{DB}}{|\overrightarrow{DB}|} = 3600 \left(\frac{120\mathbf{i} + 120\mathbf{j} - 60\mathbf{k}}{180} \right) \\ &= 2400\mathbf{i} + 2400\mathbf{j} - 1200\mathbf{k} \text{ N} \cdot \text{mm}\end{aligned}$$

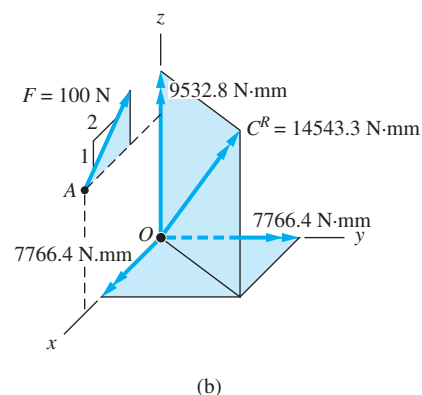
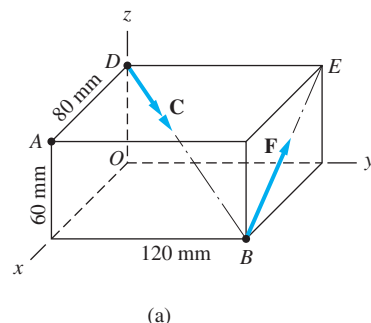
Adding $\mathbf{C}^T$ and $\mathbf{C}$ (remember that couple-vectors are free vectors), the resultant couple-vector is

$$\mathbf{C}^R = \mathbf{C}^T + \mathbf{C} = 7766.4\mathbf{i} + 7766.4\mathbf{j} + 9532.8\mathbf{k} \text{ N} \cdot \text{mm}$$

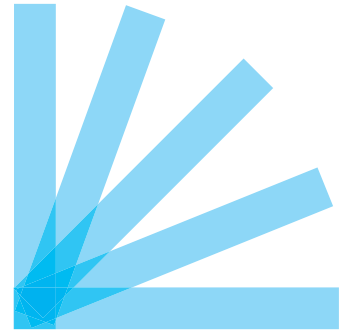
The magnitude of $\mathbf{C}^R$ is given by

$$C^R = \sqrt{(7766.4)^2 + (7766.4)^2 + (9532.8)^2} = 14543.3 \text{ N} \cdot \text{mm}$$

The equivalent force-couple system is shown in Fig. (b). Note that the force acts at point A. For convenience of representation, $\mathbf{C}^R$ is shown at point O, but being a free vector, it could be placed anywhere.



3



Resultants of Force Systems



Strong winds result in significant loads on high-rise buildings. If the pressure distribution is known, the resultant force of the wind and the location of the corresponding pressure center can be computed using the principles of statics—see Problem P3.52. Visions of America/Joe Sohm/Digital Vision/Getty Images

3.1 Introduction

In order to investigate the effects of a system of forces on a body, it is often convenient to reduce the force system to its simplest equivalent representation. Some of these simplifications have been discussed in the preceding chapter. For example, you have learned that a system of concurrent forces can be replaced by a single force and that a system of couples can be replaced by a single couple.

The next article explains how an arbitrary force system can be reduced to a force and a couple. Subsequent articles discuss applications of the force-couple system to the determination of the resultants of force systems.

3.2 Reduction of a Force System to a Force and a Couple

Here we show how a system of forces can be reduced to an equivalent system consisting of a force acting at an arbitrary point, plus a couple.

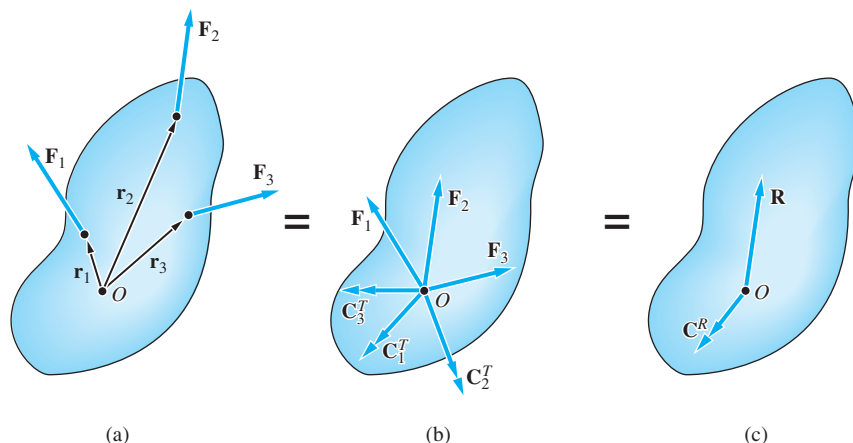


Fig. 3.1

Consider the force system shown in Fig. 3.1(a), consisting of the forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$. The position vectors $\mathbf{r}_1, \mathbf{r}_2, \mathbf{r}_3, \dots$ of the points where the forces act are measured from an arbitrarily chosen base point O . We can reduce this force system to an equivalent force-couple system, with the force acting at O , by the following procedure:

- Move each force to point O . As explained in Art. 2.8, the force $\mathbf{F}_1$ can be moved to O if we introduce the couple of transfer $\mathbf{C}_1^T = \mathbf{r}_1 \times \mathbf{F}_1$ (the moment of $\mathbf{F}_1$ about O). The forces $\mathbf{F}_2, \mathbf{F}_3, \dots$ can be moved in the same manner, their couples of transfer being $\mathbf{C}_2^T = \mathbf{r}_2 \times \mathbf{F}_2, \mathbf{C}_3^T = \mathbf{r}_3 \times \mathbf{F}_3, \dots$. After all the forces have been moved, we end up with the force system in Fig. 3.1(b), which is equivalent to the original system. (The equal signs between the figures signify equivalence.)
- Because the forces are now concurrent at point O , they can be added to yield the resultant force $\mathbf{R}$:

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3 + \dots = \Sigma \mathbf{F} \quad (3.1)$$

The couples of transfer can also be added, their sum being the resultant couple-vector $\mathbf{C}^R$:

$$\mathbf{C}^R = \mathbf{r}_1 \times \mathbf{F}_1 + \mathbf{r}_2 \times \mathbf{F}_2 + \mathbf{r}_3 \times \mathbf{F}_3 + \dots = \Sigma \mathbf{M}_O \quad (3.2)$$

The resultant force-couple system is displayed in Fig. 3.1(c), with both $\mathbf{R}$ and $\mathbf{C}^R$ shown acting at point O . It should be noted, however, that $\mathbf{R}$ is a sliding vector (its line of action must pass through O), whereas $\mathbf{C}^R$ is a free vector. Although $\mathbf{C}_1^T$ is perpendicular to $\mathbf{F}_1$, and so on, as pointed out in Art. 2.8, $\mathbf{C}^R$ is generally not perpendicular to $\mathbf{R}$.

Using Eqs. (3.1) and (3.2), any force system can be reduced to an equivalent force-couple system, with the force acting at a reference point of your choosing. The resultant force $\mathbf{R}$ is simply the vector sum of all the forces—therefore, it is not affected by the location of the reference point. However, the resultant couple-vector $\mathbf{C}^R$, being the sum of the moments of all the forces about the reference point,^{*} does depend on the choice of the reference point.

If we choose the reference point O be the origin of a rectangular coordinate system, the two vector equations in Eqs. (3.1) and (3.2) are equivalent to the following six scalar equations:

$$\begin{aligned} R_x &= \Sigma F_x & R_y &= \Sigma F_y & R_z &= \Sigma F_z \\ C_x^R &= \Sigma M_x & C_y^R &= \Sigma M_y & C_z^R &= \Sigma M_z \end{aligned} \quad (3.3)$$

If the forces of the original system lie in a plane—say, the xy -plane—the following three scalar equations are necessary to determine the force-couple system.

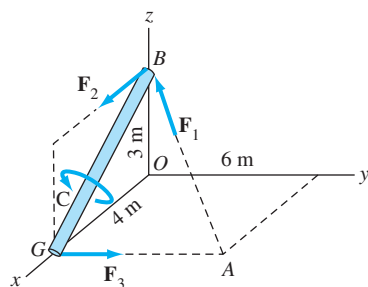
$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad C^R = \Sigma M_O \quad (3.4)$$

The couple-vector $\mathbf{C}^R$ will always be in the z -direction, because the plane of the couple is the xy -plane. Because the resultant force $\mathbf{R}$ lies in the xy -plane, $\mathbf{R}$ and $\mathbf{C}^R$ will be mutually perpendicular. The last observation is significant—it implies that a coplanar force system can be further reduced to a single force or a single couple. This topic is discussed in more detail in the next article.

^{*}If the original force system contains couples, their moments must be included in the sum.

Sample Problem 3.1

The force system acting on a structural member consists of the couple C and the forces $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$. Determine the equivalent force-couple system with the force acting at point G . Use $C = 200 \text{ N} \cdot \text{m}$, $F_1 = 100 \text{ N}$, $F_2 = 90 \text{ N}$, and $F_3 = 120 \text{ N}$.



Solution

Because of the three-dimensional nature of this problem, we will use vector algebra to solve it. The first step is to express the three forces and the couple in vector form:

$$\mathbf{F}_1 = 100\lambda_{AB} = 100 \frac{\vec{AB}}{|\vec{AB}|} = 100 \left(\frac{-4\mathbf{i} - 6\mathbf{j} + 3\mathbf{k}}{\sqrt{(-4)^2 + (-6)^2 + 3^2}} \right)$$

$$= -51.22\mathbf{i} - 76.82\mathbf{j} + 38.41\mathbf{k} \text{ N}$$

$$\mathbf{F}_2 = 90\mathbf{i} \text{ N}$$

$$\mathbf{F}_3 = 120\mathbf{j} \text{ N}$$

$$\mathbf{C} = 200\lambda_{BG} = 200 \frac{\vec{BG}}{|\vec{BG}|} = 200 \left(\frac{4\mathbf{i} - 3\mathbf{k}}{\sqrt{4^2 + (-3)^2}} \right)$$

$$= 160\mathbf{i} - 120\mathbf{k} \text{ N} \cdot \text{m}$$

When we move the forces $\mathbf{F}_1$ and $\mathbf{F}_2$ to point G , they are concurrent with $\mathbf{F}_3$. Adding the three forces, we get for the resultant force

$$\mathbf{R} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$$

$$= (-51.22\mathbf{i} - 76.82\mathbf{j} + 38.41\mathbf{k}) + 90\mathbf{i} + 120\mathbf{j}$$

$$= 38.78\mathbf{i} + 43.18\mathbf{j} + 38.41\mathbf{k} \text{ N} \quad \text{Answer}$$

The couple of transfer that arises from moving $\mathbf{F}_1$ and $\mathbf{F}_2$ is the moment about G of

$$\mathbf{F}_1 + \mathbf{F}_2 = (-51.22\mathbf{i} - 76.82\mathbf{j} + 38.41\mathbf{k}) + 90\mathbf{i}$$

$$= 38.78\mathbf{i} - 76.82\mathbf{j} + 38.41\mathbf{k} \text{ N}$$

The resultant couple is the sum of the couple of transfer and the couple $\mathbf{C}$, resulting in

$$\mathbf{C}^R = \mathbf{r}_{GB} \times (\mathbf{F}_1 + \mathbf{F}_2) + \mathbf{C}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -4 & 3 & 0 \\ 38.78 & -76.82 & 38.41 \end{vmatrix} + 160\mathbf{i} - 120\mathbf{k}$$

$$= 275\mathbf{i} + 153.6\mathbf{j} + 70.9\mathbf{k} \text{ N} \cdot \text{m} \quad \text{Answer}$$

Sample Problem 3.2

The coplanar force system in Fig. (a) consists of three forces and one couple. Determine the equivalent force-couple system with the force acting at point O .

Solution

We will solve this problem with scalar algebra; however, the use of vector algebra would be almost as convenient.

The components of the resultant force $\mathbf{R}$ are

$$R_x = \Sigma F_x \quad \xrightarrow{+} \quad R_x = 50 \cos 40^\circ + 90 - \frac{3}{5}(100) = 68.30 \text{ N}$$

$$R_y = \Sigma F_y \quad \xrightarrow{+} \quad R_y = 50 \sin 40^\circ - \frac{4}{5}(100) = -47.86 \text{ N}$$

Thus the resultant force is

$$\mathbf{R} = 68.3\mathbf{i} - 47.9\mathbf{j} \text{ N}$$

Answer

The magnitude of $\mathbf{R}$ is

$$R = \sqrt{(68.30)^2 + (-47.86)^2} = 83.4 \text{ N}$$

and the angle that $\mathbf{R}$ makes with the x -axis is

$$\theta = \tan^{-1} \frac{47.86}{68.30} = 35.0^\circ$$

The force $\mathbf{R}$ acting at point O is shown in Fig. (b).

The magnitude of the resultant couple equals the magnitude of the total moment about point O of the original force system. Referring to Fig. (a), we have

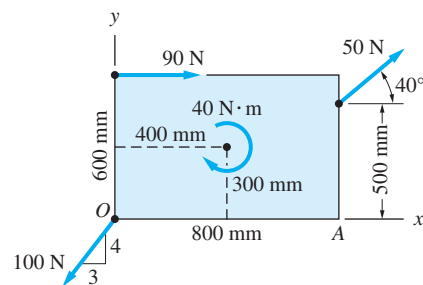
$$\begin{aligned} C^R = \Sigma M_O \quad \curvearrowright \quad C^R &= 50 \sin 40^\circ (0.800) - 50 \cos 40^\circ (0.500) \\ &\quad - 90(0.600) - 40 \\ &= -87.44 \text{ N} \cdot \text{m} \end{aligned}$$

Therefore,

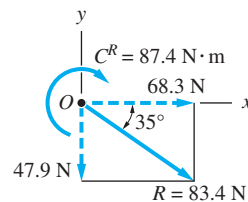
$$C^R = 87.4 \text{ N} \cdot \text{m} \quad \text{clockwise}$$

Answer

The resultant force-couple system is shown in Fig. (b).



(a)



(b)

3.3 *Definition of Resultant*

The *resultant of a force system* is defined to be the simplest system that can replace the original system without changing its external effect on a rigid body.

The word *simplest* is used in the sense that one force is simpler than two forces, one couple is simpler than two couples, a force is simpler than a force and a couple, and so on.

The resultant of a force system is generally a force-couple system $\mathbf{R}$ and $\mathbf{C}^R$. As explained in Art. 2.8, if $\mathbf{R}$ and $\mathbf{C}^R$ are mutually perpendicular, they can be further reduced to a single force. It follows that the resultant of a force system must be one of the following:*

- A resultant force $\mathbf{R}$ (if $\mathbf{C}^R = \mathbf{0}$ or if $\mathbf{R}$ and $\mathbf{C}^R$ are perpendicular)
- A resultant couple-vector $\mathbf{C}^R$ (if $\mathbf{R} = \mathbf{0}$)
- A resultant force-couple system (if $\mathbf{R}$ and $\mathbf{C}^R$ are not mutually perpendicular)

Force systems that have the same resultant are called *statically equivalent*.

The remainder of this chapter discusses the procedures for determining the resultants of two- and three-dimensional force systems.

3.4 Resultants of Coplanar Force Systems

This article investigates the resultants of force systems in which all the forces lie in a single plane, chosen as the xy -coordinate plane. We begin with a discussion of the resultants of general coplanar force systems and then consider two special cases: concurrent force systems and parallel force systems.

a. General coplanar force system

A general coplanar force system is shown in Fig. 3.2(a), with all the forces lying in the xy -plane. The origin O is located at any convenient point in the plane. The reduction of this force system to its resultant (simplest equivalent force system) is accomplished by the following procedure.

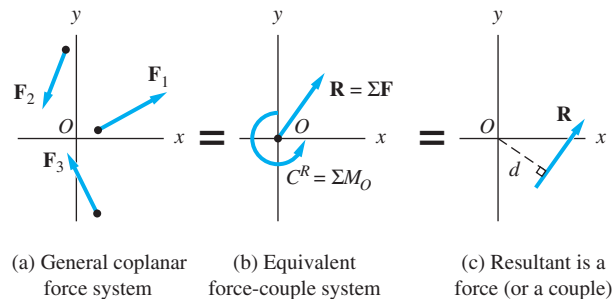


Fig. 3.2

*It is important that you pay particular attention to the use of the terms *resultant*, *resultant force* $\mathbf{R}$, and *resultant couple-vector* $\mathbf{C}^R$.

Replace the original force system with the equivalent system consisting of the resultant force $\mathbf{R} = \Sigma \mathbf{F}$ (or $R_x = \Sigma F_x$ and $R_y = \Sigma F_y$) acting at O and the resultant couple $C^R = \Sigma M_O$, as shown in Fig. 3.2(b). This procedure has three possible outcomes:

- $\mathbf{R} = \mathbf{0}$. The resultant is the couple C^R .
- $C^R = 0$. The resultant is the force $\mathbf{R}$ acting through O .
- $\mathbf{R} \neq \mathbf{0}$ and $C^R \neq 0$. Because $\mathbf{R}$ and C^R are perpendicular to each other, the system can be reduced to a single force $\mathbf{R}$ acting at a point different from O , as illustrated in Fig. 3.2(c). The perpendicular distance d between O and the line of action of $\mathbf{R}$ is determined by the requirement that moments about O of the force systems in Figs. 3.2(b) and (c) must be the same; that is, $\Sigma M_O = Rd$.

In summary, the resultant of the general coplanar force system shown in Fig. 3.2(a) is either a force or a couple. If $\Sigma \mathbf{F} \neq \mathbf{0}$, then the resultant is a force $\mathbf{R}$ determined by

$$R_x = \Sigma F_x \quad R_y = \Sigma F_y \quad \Sigma M_O = Rd \quad (3.5)$$

Note that the moment equation locates the line of action of $\mathbf{R}$.

If $\Sigma \mathbf{F} = \mathbf{0}$ and $\Sigma M_O \neq 0$, then the resultant is the couple

$$C^R = \Sigma M_O \quad (3.6)$$

b. Concurrent, coplanar force system

The resultant of a concurrent, coplanar force system is the force $\mathbf{R} = \Sigma \mathbf{F}$ ($R_x = \Sigma F_x$, $R_y = \Sigma F_y$) acting through the point of concurrency O , as indicated in Fig. 3.3. This conclusion follows from Eq. (3.5): Because $\Sigma M_O = 0$ for a force system that is concurrent at O , the moment equation $\Sigma M_O = Rd$ yields $d = 0$.

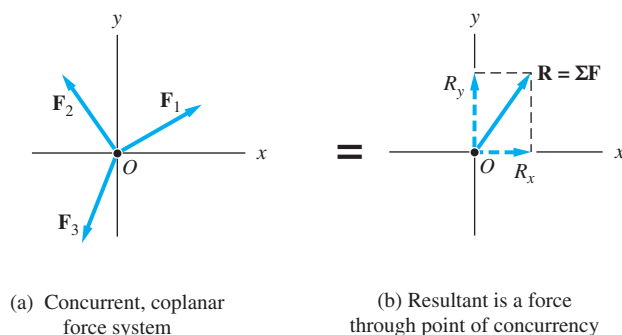


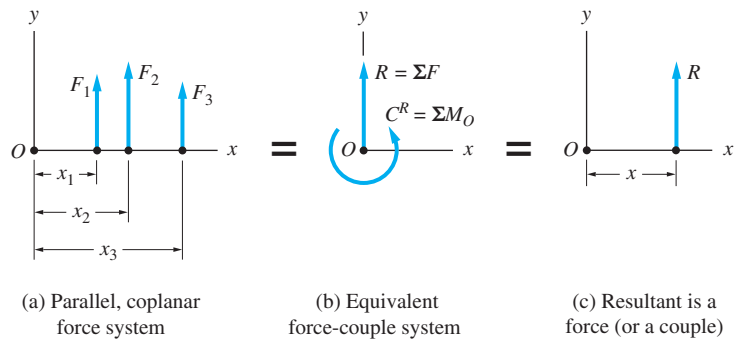
Fig. 3.3

c. Parallel, coplanar force system

Figure 3.4(a) shows a coplanar force system, where the forces $F_1, F_2, F_3, \dots$ are parallel to the y -axis. The equivalent force-couple system at point O is shown in Fig. 3.4(b), where

$$R = F_1 + F_2 + F_3 + \dots = \Sigma F$$

$$C^R = F_1x_1 + F_2x_2 + F_3x_3 + \dots = \Sigma M_O$$

**Fig. 3.4**

If $\Sigma F \neq 0$, the resultant is a force R located at the distance x from O , as indicated in Fig. 3.4(c). The value of x is obtained by equating the moments about O in Figs. 3.4(b) and (c):

$$\Sigma M_O = Rx \quad (3.7)$$

If, on the other hand, $\Sigma F = 0$ and $\Sigma M_O \neq 0$, then the resultant is the couple $C^R = \Sigma M_O$.

Sample Problem 3.3

The values of $R_x = \Sigma F_x$, $R_y = \Sigma F_y$, and ΣM_O for five force systems lying in the xy -plane are listed in the following table. Point O is the origin of the coordinate system, and positive moments are counterclockwise. Determine the resultant for each force system, and show it on a sketch of the coordinate system.

Part	R_x	R_y	ΣM_O
1	0	200 N	400 N · m
2	0	200 N	−400 N · m
3	300 N	400 N	600 N · m
4	400 N	−600 N	−900 N · m
5	0	0	−200 N · m

Solution

Part 1

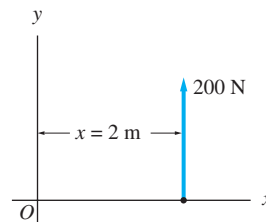
$$R_x = 0 \quad R_y = 200 \text{ N} \quad \Sigma M_O = 400 \text{ N} \cdot \text{m}$$

The resultant is a 200-N force that is parallel to the y -axis, as shown in Fig. (a). Letting x be the distance from point O to the line of action of the resultant, as shown in Fig. (a), and using Eq. (3.7), we have

$$\Sigma M_O = Rx \quad \curvearrowleft \quad 400 = 200x$$

which gives

$$x = 2 \text{ m}$$



(a)

Part 2

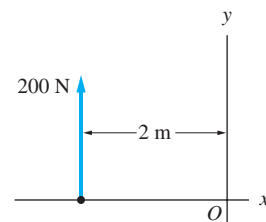
$$R_x = 0 \quad R_y = 200 \text{ N} \quad \Sigma M_O = -400 \text{ N} \cdot \text{m}$$

The resultant is the same 200-N force as in Part 1, but here the moment equation gives

$$\Sigma M_O = Rx \quad \curvearrowleft \quad -400 = 200x$$

or

$$x = -2 \text{ m}$$



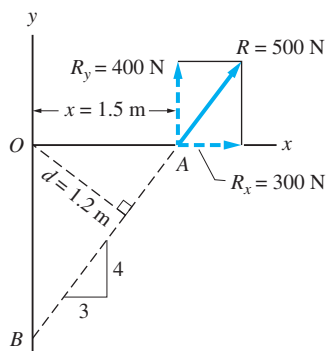
(b)

The negative sign indicates that x lies to the left of point O , as shown in Fig. (b).

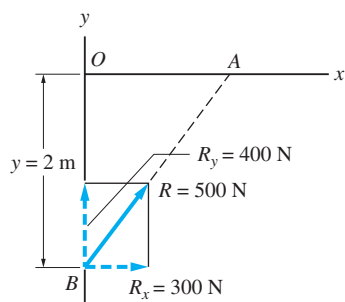
Part 3

$$R_x = 300 \text{ N} \quad R_y = 400 \text{ N} \quad \Sigma M_O = 600 \text{ N} \cdot \text{m}$$

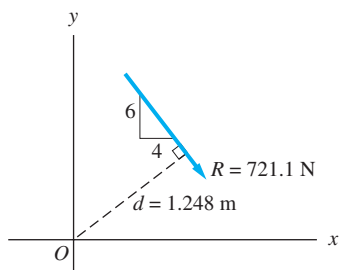
The resultant is the force $\mathbf{R} = 300\mathbf{i} + 400\mathbf{j}$ N. Its magnitude is $R = \sqrt{(300)^2 + (400)^2} = 500$ N. The moment equation of Eqs. (3.5) must be used to



(c)



(d)



(e)

determine the line of action of **R**. Letting d be the perpendicular distance from point O to the line of action of **R**, as shown in Fig. (c), we have

$$\Sigma M_O = R d \quad \curvearrowright \quad 600 = 500d$$

which yields

$$d = 1.2 \text{ m}$$

The points where **R** intersects the coordinate axes can then be determined by trigonometry or by using the principle of moments, as follows.

With **R** placed at A , as in Fig. (c): With **R** placed at B , as in Fig. (d):

$$\Sigma M_O = R_y x$$

$$\curvearrowright \quad 600 = 400x$$

$$x = 1.5 \text{ m}$$

$$\Sigma M_O = R_x y$$

$$\curvearrowright \quad 600 = 300y$$

$$y = 2 \text{ m}$$

Part 4

$$R_x = 400 \text{ N} \quad R_y = -600 \text{ N} \quad \Sigma M_O = -900 \text{ N} \cdot \text{m}$$

The resultant is the force **R** = $400\mathbf{i} - 600\mathbf{j}$ N; its magnitude is $R = \sqrt{(400)^2 + (600)^2} = 721.1 \text{ N}$. Letting d be the perpendicular distance from point O to the line of action of **R**, as shown in Fig. (e), we have

$$\Sigma M_O = R d \quad \curvearrowright \quad -900 = -721.1d$$

which gives

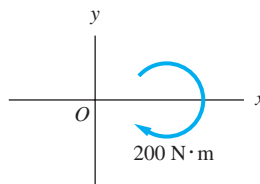
$$d = 1.248 \text{ m}$$

Note that the line of action of **R** must be placed to the right of the origin, so that its moment about point O has the same sense as ΣM_O —that is, clockwise.

Part 5

$$R_x = 0 \quad R_y = 0 \quad \Sigma M_O = -200 \text{ N} \cdot \text{m}$$

Because the sum of the forces is zero, the resultant of this force system is a $200\text{-N} \cdot \text{m}$ clockwise couple, as shown in Fig. (f).



(f)

Sample Problem 3.4

The force **R** is the resultant of the other three concurrent forces shown. Determine **P** and **R**.

Solution

The three applied forces represent a concurrent, coplanar force system. Therefore, the components of the resultant force are determined by two scalar equations: $R_x = \Sigma F_x$ and $R_y = \Sigma F_y$. Because the directions of all the forces are known, there are two unknowns in this problem—the magnitudes P and R . The most direct method for determining these two unknowns is to solve the following two scalar equations (comparing the number of unknowns with the number of available equations is often a valuable aid in the solution of problems):

$$R_x = \Sigma F_x \quad \xrightarrow{+} \quad R \cos 30^\circ = P \sin 20^\circ - 80 \quad (1)$$

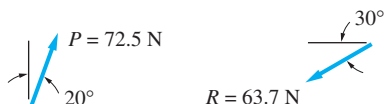
$$R_y = \Sigma F_y \quad \xrightarrow{+} \quad R \sin 30^\circ = P \cos 20^\circ - 100 \quad (2)$$

Solving Eqs. (1) and (2) simultaneously gives

$$P = 72.5 \text{ N} \quad \text{and} \quad R = -63.7 \text{ N}$$

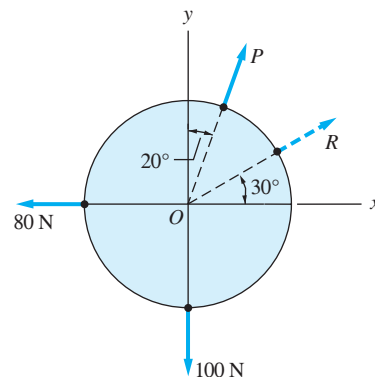
The positive value of P indicates that **P** is directed as shown in the figure. The negative sign associated with R means that **R** acts in the direction opposite to that shown in the figure.

Therefore, the forces **P** and **R** are



Answer

Of course, the lines of action of **P** and **R** pass through O , the point of concurrency.



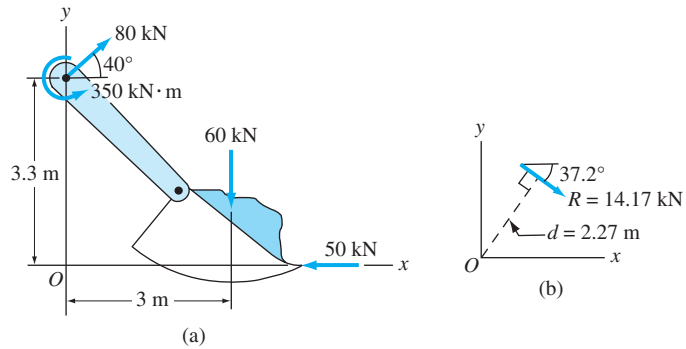
Sample Problem 3.5

Determine the resultant of the coplanar force system shown in Fig. (a) that acts on the arm of an excavator. Show your answer on a sketch of the coordinate system.

Solution

For a coplanar force system, the resultant is either a force or a couple. If the resultant is a force, then Eqs. (3.5) provide three scalar equations: $R_x = \Sigma F_x$, $R_y = \Sigma F_y$, and $\Sigma M_O = Rd$.

We see that there are no unknown quantities in the original force system. Therefore, if the resultant is a force, the three unknowns in this problem will



be R_x , R_y , and d , which could be determined from the three scalar equations. Referring to Fig. (a), the three equations become

$$R_x = \Sigma F_x \quad \rightarrow \quad R_x = 80 \cos 40^\circ - 50 = 11.284 \text{ kN}$$

This equation is sufficient to tell us that the resultant is a force, not a couple; if the resultant were a couple, ΣF_x would be zero.

$$R_y = \Sigma F_y \quad \uparrow \quad R_y = 80 \sin 40^\circ - 60 = -8.577 \text{ kN}$$

$$\Sigma M_O = Rd \quad \curvearrowright \quad \Sigma M_O = 350 - (80 \cos 40^\circ)(3.3) - 60(3) = -32.24 \text{ kN} \cdot \text{m}$$

Therefore, the resultant **R** is

$$\theta = \tan^{-1} \frac{8.577}{11.284} = 37.2^\circ$$

Because ΣM_O is negative (i.e., clockwise), the resultant **R** must also provide a clockwise moment about O , as shown in Fig. (b). Therefore we obtain

$$\Sigma M_O = Rd \quad \curvearrowright \quad 32.24 = 14.174d$$

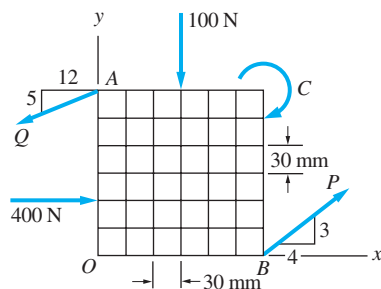
which gives

$$d = 2.27 \text{ m}$$

The final result is shown in Fig. (b).

Sample Problem 3.6

The force system shown consists of the couple C and four forces. If the resultant of this system is a $75000\text{-N} \cdot \text{mm}$ counterclockwise couple, determine P , Q , and C .



Solution

This problem contains three unknowns: P , Q , and C . Because the force system is the general coplanar case, three equations are available to determine the resultant. Since the resultant is a couple, the first two of Eqs. (3.5) become

$$R_x = \Sigma F_x = 0 \quad \rightarrow \quad -\frac{12}{13}Q + \frac{4}{5}P + 400 = 0 \quad (a)$$

$$R_y = \Sigma F_y = 0 \quad \uparrow \quad -\frac{5}{13}Q + \frac{3}{5}P - 100 = 0 \quad (b)$$

Solving Eqs. (a) and (b) simultaneously gives

$$P = 1000 \text{ N} \quad \text{and} \quad Q = 1300 \text{ N} \quad \text{Answer}$$

The third equation is Eq. (3.6), $C^R = \Sigma M_O$. Because a couple is a free vector, the moment center can be any point. Given that $C^R = 75000 \text{ N} \cdot \text{mm}$, counterclockwise, and choosing point A as the moment center, we have

$$C^R = \Sigma M_A \quad \curvearrowright \quad 75000 = -100(90) - C + 400(120) + \frac{3}{5}P(180) + \frac{4}{5}P(180)$$

Substituting $P = 1000 \text{ N}$ and solving yields

$$C = 216000 \text{ N} \cdot \text{mm} \quad \text{Answer}$$

Because the values for P , Q , and C are positive, each force acts in the direction shown in the figure.

3.5 *Resultants of Three-Dimensional Systems*

In general, a three-dimensional force system cannot be simplified beyond a force-couple system. Exceptions are systems in which the forces are either concurrent or parallel. In this article, we discuss these two special cases, together with a special form of the force-couple system called the *wrench*.

a. Concurrent, three-dimensional force system

A concurrent, three-dimensional force system is shown in Fig. 3.5(a). As in the case of concurrent, coplanar forces, this system can be reduced to the resultant force $\mathbf{R} = \Sigma \mathbf{F}$ ($R_x = \Sigma F_x$, $R_y = \Sigma F_y$, $R_z = \Sigma F_z$) acting through the point of concurrency O , as indicated in Fig. 3.5(b).

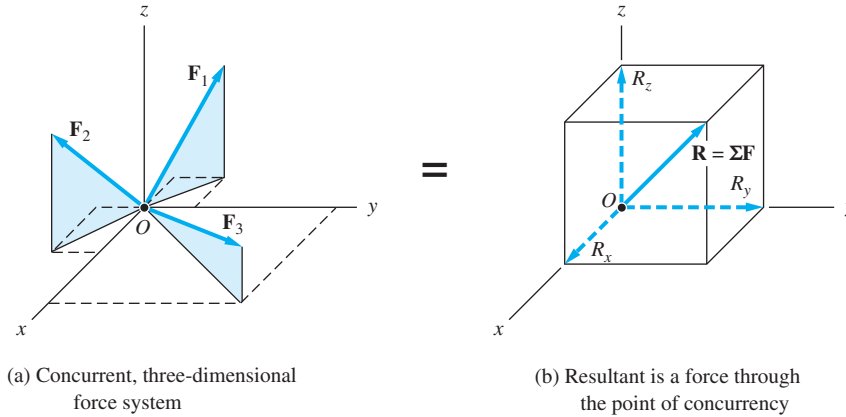


Fig. 3.5

b. Parallel, three-dimensional force system

Consider the force system in Fig. 3.6(a), where the forces $F_1, F_2, F_3, \dots$ are parallel to the z -axis. To find the resultant, we begin by replacing the forces with an equivalent force-couple system, with the force acting at the origin O , as shown in Fig. 3.6(b). The magnitude of the resultant force $\mathbf{R}$, which is also parallel to the z -axis, and the resultant couple-vector $\mathbf{C}^R$ are given by

$$R = \Sigma F \quad (3.8)$$

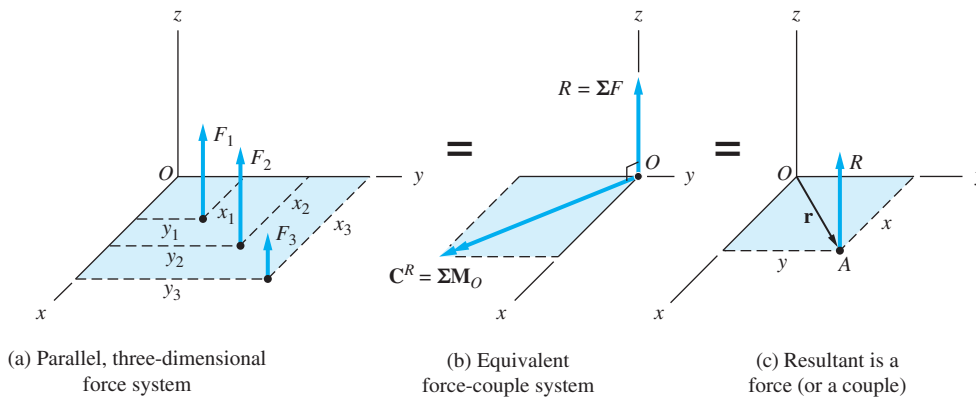


Fig. 3.6

and

$$\mathbf{C}^R = \Sigma \mathbf{M}_O \quad (3.9)$$

where $\Sigma \mathbf{M}_O$ is the sum of the moments of $F_1, F_2, F_3, \dots$ about O . The resultant couple-vector $\mathbf{C}^R$ lies in the xy -plane ($\mathbf{C}^R$ has no z -component because forces parallel to an axis have no moment about that axis).

Since $\mathbf{R}$ and $\mathbf{C}^R$ are mutually perpendicular, the force system in Fig. 3.6(b) can be further simplified. If $\Sigma F = 0$, then the resultant is the couple $\mathbf{C}^R = \Sigma \mathbf{M}_O$. If $\Sigma F \neq 0$, the resultant is the force $\mathbf{R}$ acting through the unique point A in the xy -plane, as shown in Fig. 3.6(c). The vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ that locates this point is obtained by equating the moments about point O of the force-couple system in Fig. 3.6(b) and the force R in Fig. 3.6(c):

$$\Sigma \mathbf{M}_O = \mathbf{r} \times \mathbf{R} \quad (3.10)$$

The scalar components of this vector equation are

$$\Sigma M_x = F_1 y_1 + F_2 y_2 + F_3 y_3 + \dots = Ry$$

$$\Sigma M_y = -F_1 x_1 - F_2 x_2 - F_3 x_3 - \dots = -Rx$$

where ΣM_x and ΣM_y are the moments of the original forces about the x - and y -axes, respectively. Therefore, the coordinates x and y become

$$x = -\frac{\Sigma M_y}{R} \quad y = \frac{\Sigma M_x}{R} \quad (3.11)$$

c. General three-dimensional force system: The wrench

It was shown in Art. 3.2 that a given force system can always be reduced to a force-couple system consisting of a resultant force $\mathbf{R} = \Sigma \mathbf{F}$, acting at an arbitrary point O , and a resultant couple-vector $\mathbf{C}^R = \Sigma \mathbf{M}_O$, as shown in Fig. 3.7(a). If $\mathbf{R}$ and $\mathbf{C}^R$ are mutually perpendicular, they can be reduced to a single force $\mathbf{R}$, acting

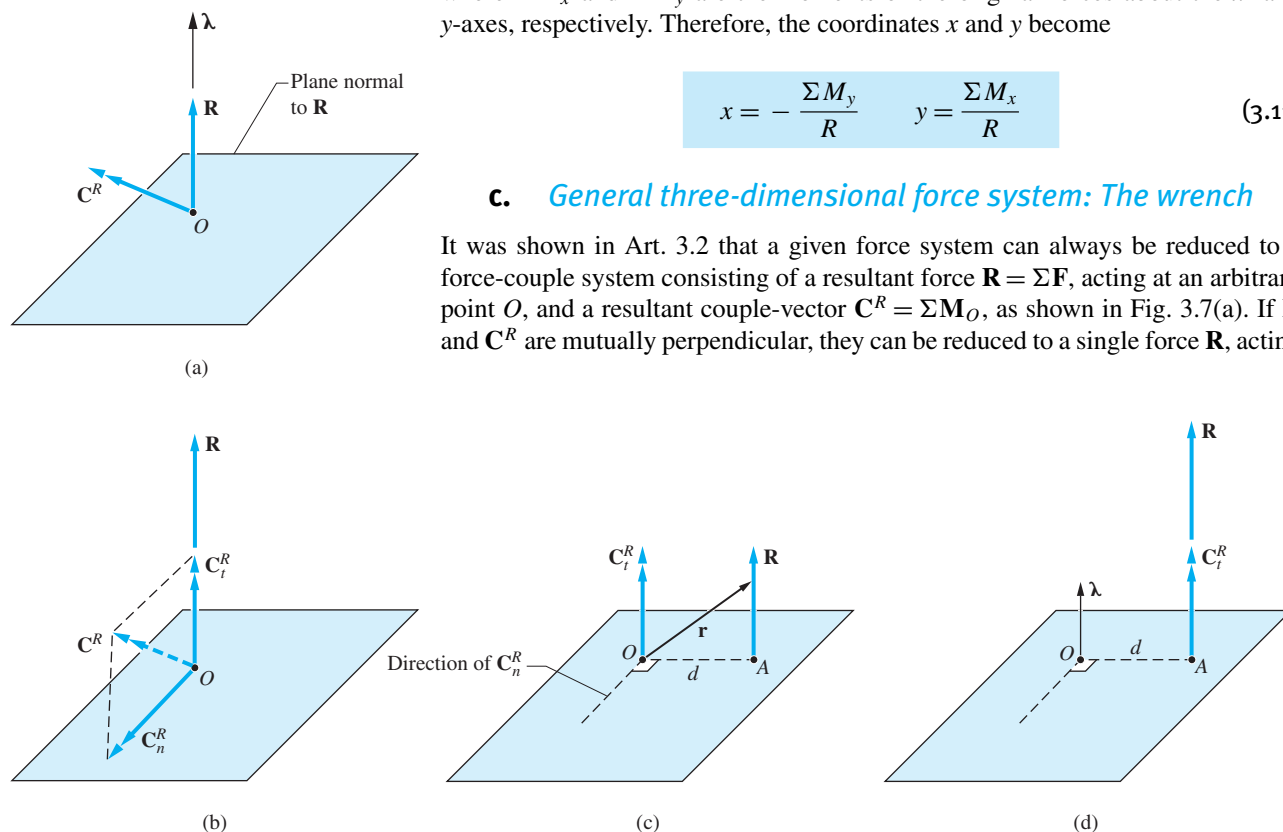


Fig. 3.7

through a unique point (this property was used in the special cases of coplanar and parallel force systems). In the general case, $\mathbf{R}$ and $\mathbf{C}^R$ will not be perpendicular to each other, and thus they will not be reducible to a single force. However, a general force system can always be represented by a force and a *parallel* couple-vector by the procedure described next.

- Resolve $\mathbf{C}^R$ into the orthogonal components $\mathbf{C}_t^R$ and $\mathbf{C}_n^R$, which are parallel and perpendicular to $\mathbf{R}$, respectively. The result is shown in Fig. 3.7(b). The parallel component can be computed from Eq. (2.11):

$$\mathbf{C}_t^R = (\mathbf{C}^R \cdot \boldsymbol{\lambda})\boldsymbol{\lambda} \quad (3.12)$$

where $\boldsymbol{\lambda}$ is the unit vector in the direction of $\mathbf{R}$. The normal component is then found from

$$\mathbf{C}_n^R = \mathbf{C}^R - \mathbf{C}_t^R \quad (3.13)$$

- Because $\mathbf{C}_n^R$ and $\mathbf{R}$ are mutually perpendicular, they can be replaced by the single force $\mathbf{R}$ acting at point A , as illustrated in Fig. 3.7(c). The line of action of this force is determined by the requirement that its moment about O must be equal to $\mathbf{C}_n^R$. In other words,

$$\mathbf{r} \times \mathbf{R} = \mathbf{C}_n^R \quad (3.14)$$

where $\mathbf{r}$ is the vector drawn from O to any point on the new line of action of $\mathbf{R}$. The scalar form of Eq. (3.14) is $Rd = C_n^R$, where d is the distance between O and A , as indicated in Fig. 3.7(c). This equation yields

$$d = \frac{C_n^R}{R} \quad (3.15)$$

Note that the line OA is perpendicular to $\mathbf{C}_n^R$.

- Move $\mathbf{C}_t^R$ to point A , as shown in Fig. 3.7(d) (we can do this because a couple is a free vector). The result is a collinear force-couple system, called the *wrench*. The direction of the wrench, also known as the *axis of the wrench*, is specified by the vector $\boldsymbol{\lambda}$.

A physical example of a wrench is the operation of a screwdriver. You exert a force along the axis of the screwdriver to hold its tip against the screw, while applying a couple about the same axis to turn the screw. Because the force and the couple-vector are parallel, they constitute a wrench.

Sample Problem 3.7

The values of ΣF_z , ΣM_x , and ΣM_y for three force systems that are parallel to the z -axis are as follows:

Part	ΣF_z	ΣM_x	ΣM_y
1	50 kN	60 kN · m	−125 kN · m
2	−600 N	0	−1200 N · m
3	0	600 N · mm	−800 N · cm

Determine the resultant of each force system and show it on a sketch of the coordinate system.

Solution

Part 1

$$\Sigma F_z = 50 \text{ kN} \quad \Sigma M_x = 60 \text{ kN} \cdot \text{m} \quad \Sigma M_y = -125 \text{ kN} \cdot \text{m}$$

The resultant is the force $\mathbf{R} = 50\mathbf{k}$ kN. With $\Sigma \mathbf{M}_O = \Sigma M_x \mathbf{i} + \Sigma M_y \mathbf{j}$ and $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$, Eq. (3.10) can be used to determine the line of action of $\mathbf{R}$:

$$\Sigma \mathbf{M}_O = \mathbf{r} \times \mathbf{R}$$

$$60\mathbf{i} - 125\mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & 0 \\ 0 & 0 & 50 \end{vmatrix} = 50y\mathbf{i} - 50x\mathbf{j}$$

Equating like components gives the following scalar equations:

$$60 = 50y \quad \text{and} \quad -125 = -50x$$

which gives $x = 2.5$ m and $y = 1.2$ m. The sketch of the resultant is shown in Fig. (a).

Identical results for x and y are obtained if one uses Eqs. (3.11):

$$x = -\frac{\Sigma M_y}{R} = -\frac{-125}{50} = 2.5 \text{ m}$$

$$y = \frac{\Sigma M_x}{R} = \frac{60}{50} = 1.2 \text{ m}$$

Part 2

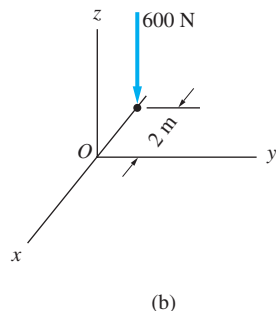
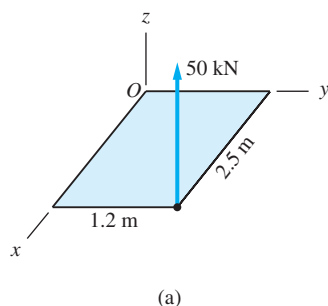
$$\Sigma F_z = -600 \text{ N} \quad \Sigma M_x = 0 \quad \Sigma M_y = -1200 \text{ N} \cdot \text{m}$$

The resultant is the force $\mathbf{R} = -600\mathbf{k}$ N. In this case, Eq. (3.10) yields

$$\Sigma \mathbf{M}_O = \mathbf{r} \times \mathbf{R}$$

$$-1200\mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & 0 \\ 0 & 0 & -600 \end{vmatrix} = -600y\mathbf{i} + 600x\mathbf{j}$$

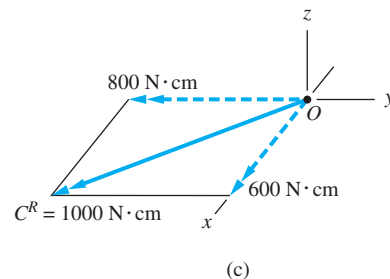
Equating like components gives $x = -2$ m and $y = 0$. The resultant is shown in Fig. (b).



Part 3

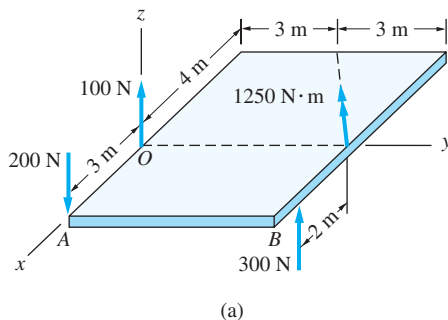
$$\Sigma F_z = 0 \quad \Sigma M_x = 600 \text{ N} \cdot \text{cm} \quad \Sigma M_y = -800 \text{ N} \cdot \text{cm}$$

Because the sum of the forces is zero and the sum of the moments is not zero, the resultant is the couple-vector $\mathbf{C}^R = \Sigma M_x \mathbf{i} + \Sigma M_y \mathbf{j} = 600\mathbf{i} - 800\mathbf{j} \text{ N} \cdot \text{cm}$, shown in Fig. (c). The magnitude of this couple-vector is $1000 \text{ N} \cdot \text{cm}$.



Sample Problem 3.8

The parallel force system in Fig. (a) consists of the three forces shown and the $1250\text{-N}\cdot\text{m}$ couple. (1) Determine the resultant, and show it on a sketch of the coordinate system. (2) Determine the resultant if the direction of the 100-N force is reversed.



Solution

Part 1

The resultant of the force system shown in Fig. (a) will be either a force or a couple. We begin by summing the forces.

$$+\uparrow R = \Sigma F_z = 100 - 200 + 300 = 200 \text{ N}$$

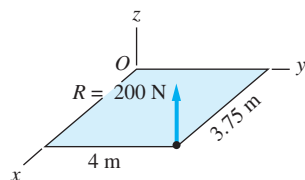
Therefore, the resultant is the force $\mathbf{R} = 200\mathbf{k} \text{ N}$.

We must use a moment equation to find the line of action of $\mathbf{R}$. Using the origin O as the moment center and assuming that $\mathbf{R}$ intersects the xy -plane at the point $(x, y, 0)$, Eq. (3.10) becomes

$$\Sigma \mathbf{M}_O = \mathbf{r} \times \mathbf{R}$$

$$3\mathbf{i} \times (-200\mathbf{k}) + [(2\mathbf{i} + 6\mathbf{j}) \times 300\mathbf{k}]$$

$$- \left(\frac{4}{5}\right) 1250\mathbf{i} - \left(\frac{3}{5}\right) 1250\mathbf{j} = (x\mathbf{i} + y\mathbf{j}) \times 200\mathbf{k}$$



(b)

Expanding the cross products and simplifying, we obtain

$$800\mathbf{i} - 750\mathbf{j} = 200y\mathbf{i} - 200x\mathbf{j}$$

Equating like components yields $x = 3.75$ m and $y = 4$ m. The resultant is shown in Fig. (b).

Part 2

If the direction of the 100-N force is reversed, the sum of the forces will be zero, which means that the resultant is not a force. To determine the resultant couple, we must compute the moment about any point. We choose the origin O as the moment center. Because reversing the direction of the 100-N force has no effect on the moment about O , we conclude that the moment is the same as that found in the solution to Part 1. Therefore, the resultant is the couple-vector $\mathbf{C}^R = \Sigma \mathbf{M}_O = 800\mathbf{i} - 750\mathbf{j} \text{ N} \cdot \text{m}$.

Sample Problem 3.9

The plate is acted on by four parallel forces, three of which are shown in Fig. (a). The fourth force $\mathbf{P}$ and its line of action are unknown. The resultant of this force system is the couple-vector $\mathbf{C}^R = -1100\mathbf{i} + 1500\mathbf{j} \text{ N} \cdot \text{m}$. Determine $\mathbf{P}$ and its line of action.

Solution

Because the resultant is a couple, the sum of the forces must be zero:

$$+\uparrow R = \Sigma F_z = P + 300 + 400 - 200 = 0$$

from which $P = -500 \text{ N}$. Therefore, the force $\mathbf{P}$ is

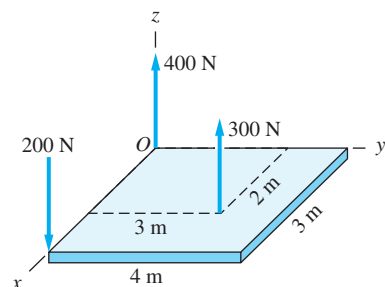
$$\mathbf{P} = -500\mathbf{k} \text{ N}$$

Answer

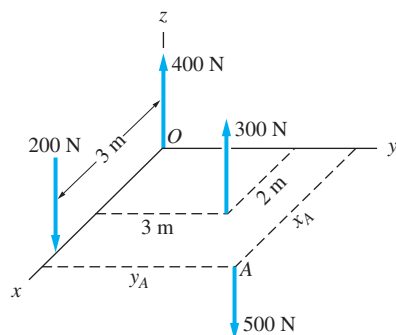
As shown in Fig. (b), we let A be the point where $\mathbf{P}$ intersects the xy -plane. To determine the location of A , we equate the sum of the moments of the original forces about any point to the moment of the resultant about that point (in this case, the moment of the resultant about every point is simply $\mathbf{C}^R$). Choosing point O as the moment center and noting that $x_A\mathbf{i} + y_A\mathbf{j}$ is the vector from point O to point A , the moment equation becomes

$$\mathbf{C}^R = \Sigma \mathbf{M}_O = \Sigma \mathbf{r} \times \mathbf{F}$$

$$-1100\mathbf{i} + 1500\mathbf{j} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 0 & 0 \\ 0 & 0 & -200 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 2 & 3 & 0 \\ 0 & 0 & 300 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x_A & y_A & 0 \\ 0 & 0 & -500 \end{vmatrix}$$



(a)



(b)

Expanding the above determinants and equating like components gives

$$\begin{aligned}-1100 &= 900 - 500y_A \\ 1500 &= 600 - 600 + 500x_A\end{aligned}$$

from which

$$x_A = 3 \text{ m} \quad \text{and} \quad y_A = 4 \text{ m} \quad \text{Answer}$$

Sample Problem 3.10

Determine the wrench that is equivalent to the force system described in Sample Problem 3.1. Find the coordinates of the point where the axis of the wrench crosses the xy -plane.

Solution

As explained in the solution to Sample Problem 3.1, the original force system can be reduced to the force-couple system shown in Fig. (a): the force $\mathbf{R}$, acting at the origin O , and the couple $\mathbf{C}^R$, where

$$\begin{aligned}\mathbf{R} &= 38.8\mathbf{i} + 43.2\mathbf{j} + 38.4\mathbf{k} \text{ N} \\ \mathbf{C}^R &= 390\mathbf{i} + 116\mathbf{j} + 360\mathbf{k} \text{ N} \cdot \text{m}\end{aligned} \quad (a)$$

The magnitude of $\mathbf{R}$ is

$$R = \sqrt{(38.8)^2 + (43.2)^2 + (38.4)^2} = 69.6 \text{ N}$$

We begin by determining the axis of the wrench, defined by the unit vector $\boldsymbol{\lambda}$ in the direction of $\mathbf{R}$:

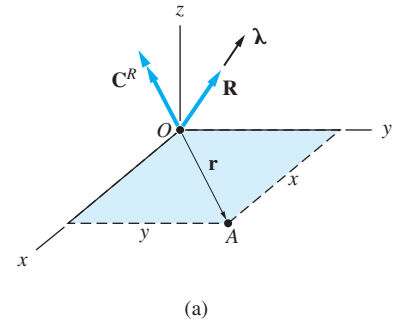
$$\begin{aligned}\boldsymbol{\lambda} &= \frac{\mathbf{R}}{R} = \frac{38.8\mathbf{i} + 43.2\mathbf{j} + 38.4\mathbf{k}}{69.6} \\ &= 0.557\mathbf{i} + 0.621\mathbf{j} + 0.552\mathbf{k}\end{aligned}$$

The component of $\mathbf{C}^R$ in the direction of $\boldsymbol{\lambda}$ can now be obtained from Eq. (3.12): $\mathbf{C}_t^R = (\mathbf{C}^R \cdot \boldsymbol{\lambda})\boldsymbol{\lambda}$. The magnitude of this vector is

$$\begin{aligned}C_t^R &= \mathbf{C}^R \cdot \boldsymbol{\lambda} \\ &= (390\mathbf{i} + 116\mathbf{j} + 360\mathbf{k}) \cdot (0.557\mathbf{i} + 0.621\mathbf{j} + 0.552\mathbf{k}) \\ &= 488 \text{ N} \cdot \text{m}\end{aligned}$$

which gives

$$\begin{aligned}\mathbf{C}_t^R &= C_t^R \boldsymbol{\lambda} = 488(0.557\mathbf{i} + 0.621\mathbf{j} + 0.552\mathbf{k}) \\ &= 272\mathbf{i} + 303\mathbf{j} + 269\mathbf{k} \text{ N} \cdot \text{m}\end{aligned}$$



Therefore, the wrench consists of the force-couple system

$$\mathbf{R} = 38.8\mathbf{i} + 43.2\mathbf{j} + 38.4\mathbf{k} \text{ N} \quad \text{Answer}$$

$$\mathbf{C}_t^R = 272\mathbf{i} + 303\mathbf{j} + 269\mathbf{k} \text{ N} \cdot \text{m} \quad \text{Answer}$$

To find the coordinates of the point where the axis of the wrench intersects the xy -plane, we must find $\mathbf{C}_n^R$, the component of $\mathbf{C}^R$ that is normal to λ . From Eq. (3.13), we obtain

$$\begin{aligned} \mathbf{C}_n^R &= \mathbf{C}^R - \mathbf{C}_t^R = (390\mathbf{i} + 116\mathbf{j} + 360\mathbf{k}) - (272\mathbf{i} + 303\mathbf{j} + 269\mathbf{k}) \\ &= 118\mathbf{i} - 187\mathbf{j} + 91\mathbf{k} \text{ N} \cdot \text{m} \end{aligned}$$

Referring to Fig. (a), we let $\mathbf{r} = x\mathbf{i} + y\mathbf{j}$ be the vector from the origin O to A , the point where the wrench intersects the xy -plane. Using Eq. (3.14), we have

$$\begin{aligned} \mathbf{r} \times \mathbf{R} &= \mathbf{C}_n^R \\ \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ x & y & 0 \\ 38.8 & 43.2 & 38.4 \end{vmatrix} &= 118\mathbf{i} - 187\mathbf{j} + 91\mathbf{k} \end{aligned}$$

After expanding the determinant, we get

$$38.4y\mathbf{i} - 38.4x\mathbf{j} + (43.2x - 38.8y)\mathbf{k} = 118\mathbf{i} - 187\mathbf{j} + 91\mathbf{k}$$

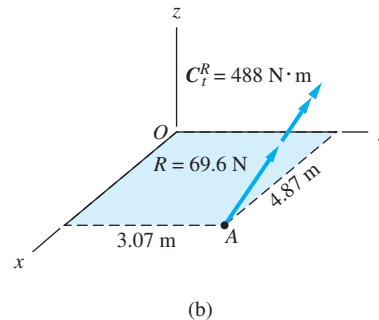
Equating the coefficients of $\mathbf{i}$ and $\mathbf{j}$ yields

$$38.4y = 118 \quad y = 3.07 \text{ m} \quad \text{Answer}$$

$$-38.4x = -187 \quad x = 4.87 \text{ m} \quad \text{Answer}$$

The third equation, obtained by equating the coefficients of $\mathbf{k}$, is not independent of the preceding two equations, as can be easily verified.

The resultant wrench is depicted in Fig. (b), which shows the magnitudes of the force and the couple-vector.



4



Coplanar Equilibrium Analysis



Principles of statics often enable us to determine quantities that cannot be easily measured directly. For example, measuring the axle loads of a dump truck with the tray in the up and down positions enables us to compute the weight of the chassis and the weight of the tray. This is illustrated in Problem P4.59. Lester Lefkowitz/The Image Bank/Getty Images

4.1 Introduction

The first three chapters of this text were devoted to mastering the elements of vector algebra, with emphasis on forces and couples. Proficiency in vector algebra is a prerequisite to the study of statics and most other areas of engineering mechanics.

With this chapter, we begin the application of vector methods to the equilibrium analysis of engineering problems. We introduce the free-body diagram, which is perhaps the most important physical concept found in this text. We show how the free-body diagram is used to obtain the equations that relate the forces acting on a body in equilibrium.

For the present, we restrict our attention to the analysis of bodies that are held in equilibrium by coplanar force systems. The subject is divided into three parts: analysis of single bodies, analysis of composite bodies (called *frames* and *machines* in some texts), and analysis of plane trusses.

4.2 Definition of Equilibrium

A body is said to be in equilibrium if the resultant of the force system that acts on the body vanishes. Equilibrium means that both the resultant force and the resultant couple are zero.

When a force system acts on a body that is initially at rest, the absence of a resultant means that the body has no tendency to move. The analysis of problems of this type is the focus of statics; dynamics is concerned with the response of bodies to force systems that are not in equilibrium.

We showed in Chapter 3 that a coplanar force system always can be represented as a resultant force $\mathbf{R}$ passing through an arbitrary point O and a couple C^R that lies in the plane of the forces. Assuming that the forces lie in the xy -plane, $\mathbf{R}$ and C^R can be determined from $R_x = \Sigma F_x$, $R_y = \Sigma F_y$, and $C^R = \Sigma M_O$. Therefore, the equations of equilibrium are

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_O = 0 \quad (4.1)$$

The summations in Eqs. (4.1) must, of course, include *all* the forces that act on the body—both the applied forces and the reactions (the forces provided by supports).

PART A: Analysis of Single Bodies

4.3 Free-Body Diagram of a Body

The first step in equilibrium analysis is to identify all the forces that act on the body. This is accomplished by means of a *free-body diagram*.

The *free-body diagram* (FBD) of a body is a sketch of the body showing all forces that act on it. The term *free* implies that all supports have been removed and replaced by the forces (reactions) that they exert on the body.

The importance of mastering the FBD technique cannot be overemphasized. Free-body diagrams are fundamental to all engineering disciplines that are concerned with the effects that forces have on bodies. The construction of an FBD is the key step that translates a physical problem into a form that can be analyzed mathematically.

Forces that act on a body can be divided into two general categories—*reactive forces* (or, simply, *reactions*) and *applied forces*. Reactions are those forces that are exerted on a body by the supports to which it is attached. Forces acting on a body that are not provided by the supports are called *applied forces*. Of course, *all* forces, both reactive and applied, must be shown on free-body diagrams.

The following is the general procedure for constructing a free-body diagram.

1. A sketch of the body is drawn assuming that all supports (surfaces of contact, supporting cables, etc.) have been removed.
2. All applied forces are drawn and labeled on the sketch. The weight of the body is considered to be an applied force acting at the center of gravity. As shown in Chapter 8, the *center of gravity* of a homogeneous body coincides with the *centroid* of its volume.
3. The support reactions are drawn and labeled on the sketch. If the sense of a reaction is unknown, it should be assumed. The solution will determine the correct sense: A positive result indicates that the assumed sense is correct, whereas a negative result means that the correct sense is opposite to the assumed sense.
4. All relevant angles and dimensions are shown on the sketch.

When you have completed this procedure, you will have a drawing (i.e., a free-body diagram) that contains all of the information necessary for writing the equilibrium equations for the body.

The most difficult step to master in the construction of FBDs is the determination of the support reactions. Table 4.1 shows the reactions exerted by various coplanar supports; it also lists the number of unknowns that are introduced on an FBD by the removal of each support. To be successful at drawing FBDs, you must be completely familiar with the contents of Table 4.1. It is also helpful to understand the physical reasoning that determines the reactions at each support, which are described below.

- (a) *Flexible Cable (Negligible Weight).* A flexible cable exerts a pull, or tensile force, in the direction of the cable. With the weight of the cable neglected, the cable forms a straight line. If its direction is known, removal of the cable introduces one unknown in a free-body diagram—the magnitude of the force exerted by the cable.
- (b) *Frictionless Surface: Single Point of Contact.* When a body is in contact with a frictionless surface at only one point, the reaction is a force that is perpendicular to the surface, acting at the point of contact. This reaction is often referred to simply as the *normal force*. (Walking on an icy sidewalk is treacherous because it is difficult to generate a force in any direction except perpendicular to the sidewalk.) Therefore, removing such a surface introduces one unknown in a free-body diagram—the magnitude of the normal force. If contact between the body and the surface occurs across a finite area, rather than at one point, the line of action of the resultant normal force will also be unknown.
- (c) *Roller Support.* A roller support is equivalent to a frictionless surface: It can only exert a force that is perpendicular to the supporting surface. The magnitude of the force is thus the only unknown introduced in a free-body diagram when the support is removed.
- (d) *Surface with Friction: Single Point of Contact.* A friction surface can exert a force that acts at an angle to the surface. The unknowns may be taken to be

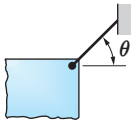
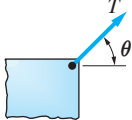
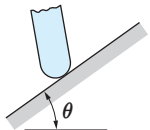
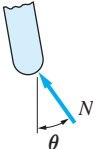
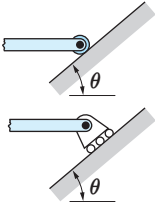
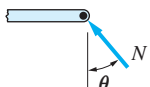
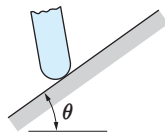
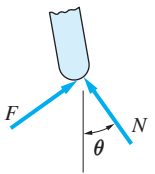
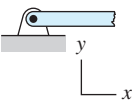
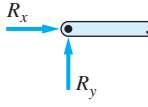
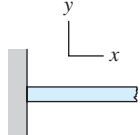
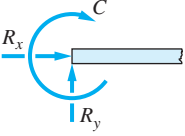
Support	Reaction(s)	Description of reaction(s)	Number of unknowns
(a)  Flexible cable of negligible weight		Tension of unknown magnitude T in the direction of the cable	One
(b)  Frictionless surface (single point of contact)		Force of unknown magnitude N directed normal to the surface	One
(c)  Roller support		Force of unknown magnitude N normal to the surface supporting the roller	One
(d)  Surface with friction (single point of contact)		Force of unknown magnitude N normal to the surface and a friction force of unknown magnitude F parallel to the surface	Two
(e)  Pin support		Unknown force $\mathbf{R}$	Two
(f)  Built-in (cantilever) support		Unknown force $\mathbf{R}$ and a couple of unknown magnitude C	Three

Table 4.1 Reactions of Coplanar Supports

the magnitude and direction of the force. However, it is usually advantageous to represent the unknowns as N and F , the components that are perpendicular and parallel to the surface, respectively. The component N is called the *normal force*, and F is known as the *friction force*. If there is an area of contact, the line of action of N will also be unknown.

- (e) **Pin Support.** A pin is a cylinder that is slightly smaller than the hole into which it is inserted, as shown in Fig. 4.1(a). Neglecting friction, the pin can only exert a force that is normal to the contact surface, shown as $\mathbf{R}$ in Fig. 4.1(b). A pin support thus introduces two unknowns: the magnitude of $\mathbf{R}$ and the angle α that specifies the direction of $\mathbf{R}$ (α is unknown because the point where the pin contacts the surface of the hole is not known). More commonly, the two unknowns are chosen to be perpendicular components of $\mathbf{R}$, such as R_x and R_y shown in Fig. 4.1(c).
- (f) **Built-in (Cantilever) Support.** A built-in support, also known as a *cantilever support*, prevents all motion of the body at the support. Translation (horizontal or vertical movement) is prevented by a force, and a couple prohibits rotation. Therefore, a built-in support introduces three unknowns in a free-body diagram: the magnitude and direction of the reactive force $\mathbf{R}$ (these unknowns are commonly chosen to be two components of $\mathbf{R}$, such as R_x and R_y) and the magnitude C of the reactive couple.

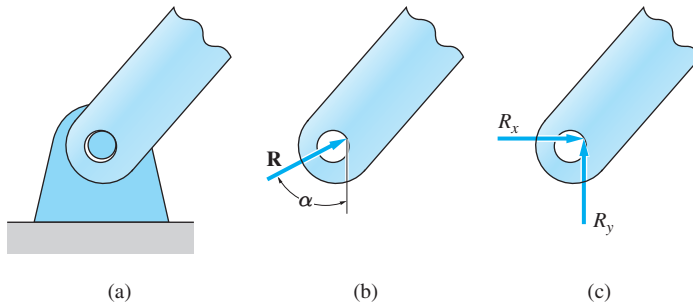


Fig. 4.1

You should keep the following points in mind when you are drawing free-body diagrams.

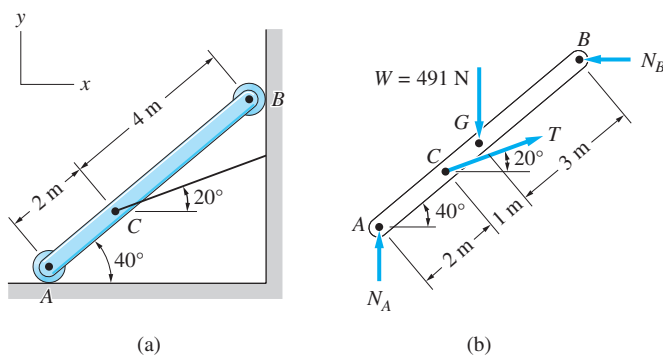
1. Be neat. Because the equilibrium equations will be derived directly from the free-body diagram, it is essential that the diagram be readable.
2. Clearly label all forces, angles, and distances with values (if known) or symbols (if the values are not known).
3. The support reactions must be consistent with the information presented in Table 4.1.
4. Show only forces that are external to the body (this includes support reactions and the weight). Internal forces occur in equal and opposite pairs and thus will not appear on free-body diagrams.

Sample Problem 4.1

The homogeneous 6-m bar AB in Fig. (a) is supported in the vertical plane by rollers at A and B and by a cable at C . The mass of the bar is 50 kg. Draw the FBD of bar AB . Determine the number of unknowns on the FBD.

Solution

The FBD of bar AB is shown in Fig. (b). The first step in the construction of this diagram is to sketch the bar, assuming the supports have been removed. Then the following forces are added to the sketch.



W : The Weight of the Bar

The weight W is shown as a vertical force acting at G , the center of gravity of the bar. Because the bar is homogeneous, G is located at the center of the bar. The magnitude of the weight is $W = mg = (50)(9.81) = 491$ N.

N_A : The Normal Reaction at A

Removal of the roller support at A dictates that we show the force that this support can exert on the bar. From Table 4.1, we note that a roller support can exert a single force that is normal to the supporting surface. Therefore, on the FBD we show the reaction at A as a vertical force and label its magnitude as N_A .

N_B : The Normal Reaction at B

Following an argument similar to that for N_A , we conclude that the removal of the roller support at B means that we must show a horizontal force at that point. On the FBD, we label this reaction as N_B .

T : The Tension in the Cable at C

From Table 4.1, the force exerted by a cable is a tensile force acting in the direction of the cable. Therefore, the force exerted on the bar by the cable is shown as a force of magnitude T , acting at 20° to the horizontal.

We note that there are three unknowns on the FBD: the magnitudes of the three reactions (N_A , N_B , and T).

Sample Problem 4.2

The homogeneous, 250-kg triangular plate in Fig. (a) is supported by a pin at A and a roller at C . Draw the FBD of the plate and determine the number of unknowns.

Solution

The FBD of the plate is shown in Fig. (b). The pin and roller supports have been removed and replaced by the reactive forces. The forces acting on the plate are described below.

W : The Weight of the Plate

The weight of the plate is $W = mg = (250)(9.81) = 2453 \text{ N}$. It acts at the centroid G of the triangle ABC , the location of which was determined from Table 3.1. Only the horizontal location of G is shown in the figure, because it is sufficient to determine the line of action of W .

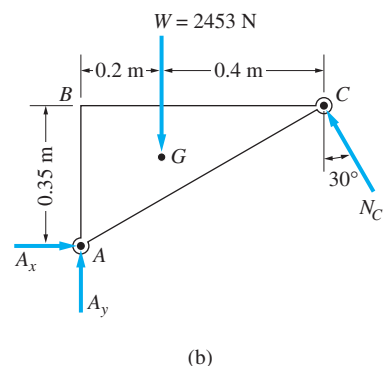
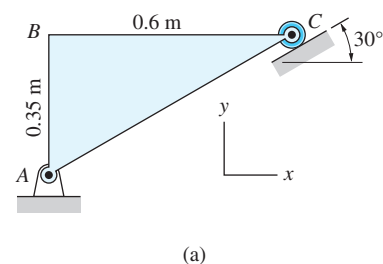
A_x and A_y : The Components of the Pin Reaction at A

From Table 4.1, we see that a pin reaction can be shown as two components A_x and A_y , which are equivalent to an unknown force acting at an unknown angle. We have shown A_x acting to the right and A_y acting upward. These directions were chosen arbitrarily; the solution of the equilibrium equations will determine the correct sense for each force. Therefore, the free-body diagram would be correct even if A_x or A_y had been chosen to act in directions opposite to those shown in Fig. (b).

N_C : The Normal Reaction at C

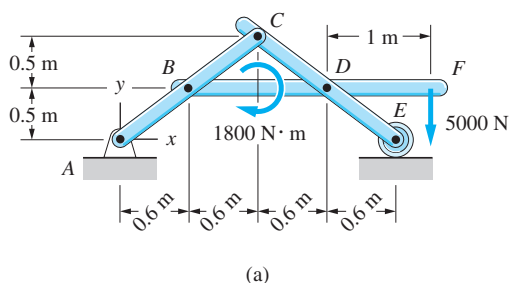
From Table 4.1, the force exerted by a roller support is normal to the inclined surface. Therefore, on the FBD we show the force N_C at C , inclined at 30° to the vertical.

The FBD contains three unknowns: A_x , A_y , and N_C .



Sample Problem 4.3

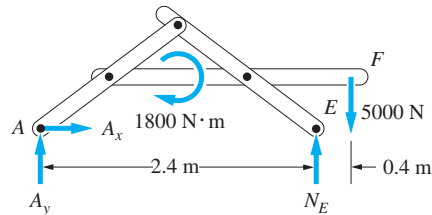
A rigid frame is fabricated by joining the three bars with pins at B , C , and D , as shown in Fig. (a). The frame is loaded by the 5000-N force and the 1800-N·m couple. The supports consist of a pin at A and a roller support at E . Draw the FBD



of the frame, neglecting the weights of the members. How many unknowns are on the FBD?

Solution

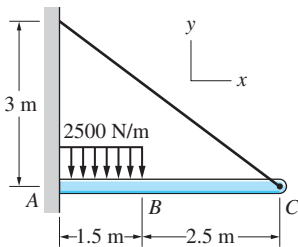
The FBD of the entire frame is shown in Fig. (b). In addition to the applied force and couple, the diagram shows the pin reaction at A (A_x and A_y) and the normal force at roller E (N_E).



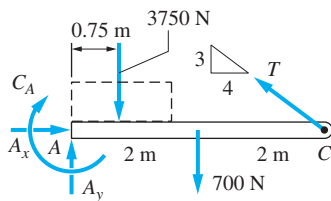
(b)

It is important to realize that the forces at pins B , C , and D do not appear on the FBD of the frame. These pin forces, as well as the forces inside the bars themselves, are internal to the frame (recall that only external forces are shown on FBDs).

We note that there are three unknowns on the FBD: A_x , A_y , and N_E .



(a)



(b)

Sample Problem 4.4

The beam ABC , built into the wall at A and supported by a cable at C , carries a distributed load over part of its length, as shown in Fig. (a). The weight of the beam is 175 N/m . Draw the FBD of the beam.

Solution

The FBD of the beam is shown in Fig. (b). Because a built-in, or cantilever, support can exert a force and a couple, the reactions at the wall are shown as the force components A_x and A_y and the couple C_A . The tension in the cable is labeled T . Also shown on the FBD are the weight of the beam ($175 \text{ N/m} \times 4 \text{ m} = 700 \text{ N}$) and the resultant of the distributed load (3750 N , acting at the centroid of the loading diagram).

Observe that the FBD contains four unknowns whereas the number of equilibrium equations in Eqs. (4.1) is three. Therefore, it would not be possible to calculate all of the unknowns using only equilibrium analysis. The reason for the indeterminacy is that the beam is oversupported; it would be in equilibrium even if the cable at C were removed or if the built-in support were replaced by a pin connection.

4.4 Coplanar Equilibrium Equations

a. General case

As stated in Art. 4.2, a body is in equilibrium under a coplanar force system if both the resultant force $\mathbf{R}$ and the resultant couple C^R of the force system are zero. It follows that the following three conditions are necessary for equilibrium:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_O = 0 \quad (4.1 \text{ repeated})$$

where the moment center O and the orientation of the xy -coordinate system can be chosen arbitrarily. The two force equations are equivalent to $\mathbf{R} = \mathbf{0}$, and the moment equation assures us that $C^R = 0$.

It often is convenient to use a set of three independent equations different from those in Eqs. (4.1). The alternative equations are described next.

- Two force equations and one moment equation** The x - and y -directions in Eqs. (4.1) do not have to be mutually perpendicular—as long as they are *not parallel*. Hence, the equilibrium equations can be restated as

$$\Sigma F_{x'} = 0 \quad \Sigma F_{y'} = 0 \quad \Sigma M_O = 0 \quad (4.2)$$

where x' and y' are any two non-parallel directions and O is an arbitrary point.

- Two moment equations and one force equation** It is possible to replace one of the force equations in Eqs. (4.2) by a moment equation, obtaining

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad \Sigma F_{x'} = 0 \quad (4.3)$$

Here, A and B are any two distinct points, and x' is any direction that is *not perpendicular* to the line AB . Note that if $\Sigma M_A = 0$ and $\Sigma M_B = 0$ are satisfied, the resultant only can be a force $\mathbf{R}$ that lies along the line AB , as shown in Fig. 4.2. The equation $\Sigma F_{x'} = 0$ (x' not perpendicular to AB) then can be satisfied only if $\mathbf{R} = \mathbf{0}$.

- Three moment equations** We also can replace both force equations in Eqs. (4.2) by two moment equations. The result is

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad \Sigma M_C = 0 \quad (4.4)$$

where A , B , and C are any three distinct, *non-collinear* points, as indicated in Fig. 4.2. Again the equations $\Sigma M_A = 0$ and $\Sigma M_B = 0$ are satisfied only if the resultant is a force $\mathbf{R}$ that lies along the line AB . The third equation $\Sigma M_C = 0$ (C not on the line AB) then guarantees that $\mathbf{R} = \mathbf{0}$.

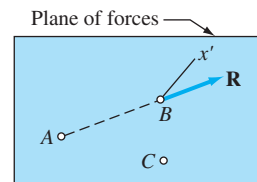


Fig. 4.2

b. Concurrent force system

Recall that the resultant of a concurrent force system is a force $\mathbf{R}$ that passes through the point of concurrency, which we label as point O . The moment equation $\Sigma M_O = 0$ now is satisfied trivially, so that the number of independent equilibrium equations is reduced from three to two. Using the arguments in Part a, it is straightforward to verify that the following are valid choices for independent equilibrium equations, each set consisting of two equations.

1. Two force equations.

$$\Sigma F_{x'} = 0 \quad \Sigma F_{y'} = 0 \quad (4.5)$$

where x' and y' are any two non-parallel directions in the xy -plane.

2. Two moment equations.

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad (4.6)$$

where A and B are any two points in the xy -plane (except point O) provided that A , B , and O do not lie on a straight line.

3. One force equation and one moment equation.

$$\Sigma F_{x'} = 0 \quad \Sigma M_A = 0 \quad (4.7)$$

where A is any point in the xy -plane (except point O) and x' is any direction that is not perpendicular to the line OA .

c. Parallel force system

Assume that all the forces lying in the xy -plane are parallel to the y -axis. The equation $\Sigma F_x = 0$ is automatically satisfied, and the number of independent equilibrium equations is again reduced from three to two. Using the reasoning in Part a, it can be shown that there are two choices for independent equilibrium equations, each containing two equations:

1. One force equation and one moment equation.

$$\Sigma F_{y'} = 0 \quad \Sigma M_A = 0 \quad (4.8)$$

where y' is any direction in the xy -plane except the x -direction, and A is any point in the xy -plane.

2. Two moment equations.

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad (4.9)$$

where A and B are any two points in the xy -plane, provided that the line AB is not parallel to the y -axis.

4.5 Writing and Solving Equilibrium Equations

The three steps in the equilibrium analysis of a body are:

- Step 1:** Draw a free-body diagram (FBD) of the body that shows all of the forces and couples that act on the body.
- Step 2:** Write the equilibrium equations in terms of the forces and couples that appear on the free-body diagram.
- Step 3:** Solve the equilibrium equations for the unknowns.

In this article, we assume that the correct free-body diagram has already been drawn, so that we can concentrate on Steps 2 and 3—writing and solving the equilibrium equations.

The force system that holds a body in equilibrium is said to be *statically determinate* if the number of independent equilibrium equations equals the number of unknowns that appear on its free-body diagram. Statically determinate problems can therefore be solved by equilibrium analysis alone. If the number of unknowns exceeds the number of independent equilibrium equations, the problem is called *statically indeterminate*. The solution of statically indeterminate problems requires the use of additional principles that are beyond the scope of this text.

When analyzing a force system that holds a body in equilibrium, you should first determine the number of independent equilibrium equations and count the number of unknowns. If the force system is statically determinate, these two numbers will be equal. It is then best to outline a *method of analysis*, or plan of attack, which specifies the sequence in which the equations are to be written and lists the unknowns that will appear in each equation. After you have determined a viable method of analysis, you can then proceed to the *mathematical details* of the solution.

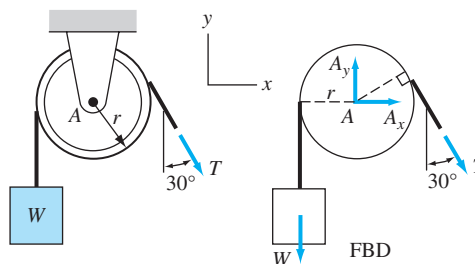
One word of caution—the set of equilibrium equations used in the analysis *must* be independent. An attempt to solve a dependent set of equations will, at some stage, yield a useless identity, such as $0 = 0$.

By now you should realize that, although the solution of a statically determinate problem is unique, the set of equations used to determine that solution is not unique. For example, there is an infinite number of choices for point O in the equilibrium equation $\Sigma M_O = 0$.

With an infinite number of equilibrium equations from which to choose, how are you to decide which equations to use for a given problem? The answer is to base your choice on mathematical convenience. If you intend to solve the equations by hand, try to select equations that involve as few unknowns as possible, thus simplifying the algebraic manipulations required. However, if you have access to a computer or a programmable calculator with equation-solving capability, the solution of simultaneous equations is not burdensome and the choice of equations is therefore not critical. It cannot be overemphasized that the set of chosen equations must be independent.

Sample Problem 4.5

The weight W is attached to one end of a rope that passes over a pulley that is free to rotate about the pin at A . The weight is held at rest by the force T applied to the other end of the rope. Using the given FBD, show that $T = W$ and compute the pin reactions at A .



Solution

Method of Analysis

The forces shown on the FBD are the weight W , the pull T acting at the end of the rope, and the reactive forces applied to the pulley by the pin at A . Because the force system is coplanar, there are three independent equilibrium equations. The number of unknowns is also three (T , A_x , and A_y), which means that the problem is statically determinate.

The equilibrium equation $\Sigma M_A = 0$ is a convenient starting point. Since the pin reactions have zero moments about A , the only unknown in this equation is T . Having computed T , we can then determine A_x from $\Sigma F_x = 0$ and A_y from $\Sigma F_y = 0$.

Mathematical Details

$$\Sigma M_A = 0 \quad \textcircled{+} \quad Tr - Wr = 0$$

$$T = W \quad \text{Q.E.D.}$$

This result is significant because it shows that the *tension in a rope does not change when the rope passes over a pulley that is supported by a frictionless pin*.

$$\Sigma F_x = 0 \quad \text{---}\rightarrow \quad A_x + T \sin 30^\circ = 0$$

$$\Sigma F_y = 0 \quad \uparrow \quad A_y - W - T \cos 30^\circ = 0$$

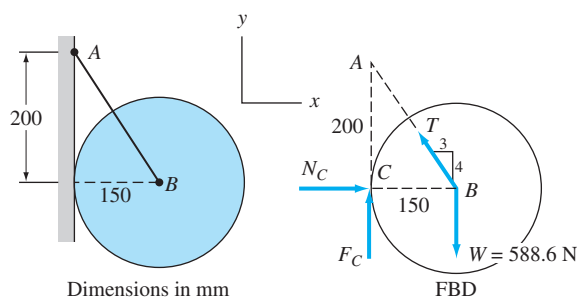
With $T = W$, the last two equations yield

$$A_x = -0.5W \quad A_y = 1.866W \quad \text{Answer}$$

The minus sign indicates that A_x acts to the left; that is, in the direction opposite to what is shown on the FED.

Sample Problem 4.6

The homogeneous 60-kg disk supported by the rope AB rests against a rough vertical wall. Using the given FBD, determine the force in the rope and the reaction at the wall.



Solution

Method of Analysis

The FBD contains the weight $W = 60(9.81) = 588.6 \text{ N}$, acting at the center of the disk. The other forces shown on the FBD are the force T applied by the rope and the reactive forces N_C (the normal reaction) and F_C (the friction force) applied by the wall. The result is a general, coplanar force system for which there are three independent equilibrium equations. Since the number of unknown forces is also three, the problem is statically determinate.

Because N_C and T intersect at point B , the moment equation $\Sigma M_B = 0$ will yield F_C . Then the two force equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$ can be used to calculate N_C and T .

Mathematical Details

$$\Sigma M_B = 0 \quad \text{gives} \quad F_C = 0 \quad \text{Answer}$$

$$\Sigma F_y = 0 \quad \uparrow \quad \frac{4}{5}T - 588.6 = 0$$

$$T = 735.8 \text{ N} \quad \text{Answer}$$

$$\Sigma F_x = 0 \quad \rightarrow \quad N_C - \frac{3}{5}T = 0 \quad N_C - \frac{3}{5}(735.8) = 0$$

$$N_C = 441 \text{ N} \quad \text{Answer}$$

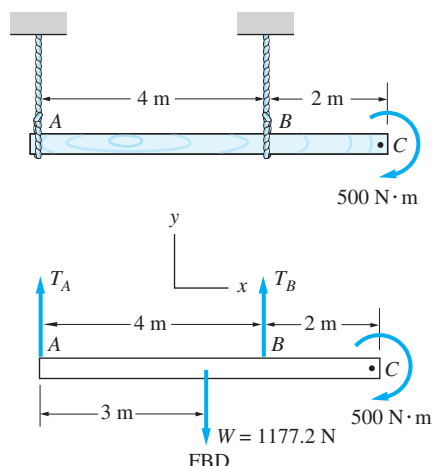
Another Method of Analysis

$$\Sigma M_A = 0 \quad \curvearrowright \quad 200N_C - 150(588.6) = 0$$

$$N_C = 441 \text{ N}$$

Sample Problem 4.7

The homogeneous, 120-kg wooden beam is suspended from ropes at A and B . A power wrench applies the 500-N · m clockwise couple to tighten a bolt at C . Use the given FBD to determine the tensions in the ropes.



Solution

Method of Analysis

The FBD of the beam contains the weight $W = mg = 120(9.81) = 1177.2$ N acting at the center of the beam, the 500-N · m couple applied by the wrench, and the unknown tensions T_A and T_B in the ropes. Because all the forces lie in the xy -plane and are parallel to the y -axis, there are two independent equilibrium equations. There are also two unknowns: T_A and T_B . Therefore, the problem is statically determinate.

It is convenient to start the analysis with the equilibrium equation $\Sigma M_A = 0$. Because this equation does not contain T_A , we can immediately solve it for T_B . We can then use the equation $\Sigma F_y = 0$ to calculate T_A .

Mathematical Details

$$\Sigma M_A = 0 \quad \curvearrowright + \quad 4T_B - 1177.2(3) - 500 = 0$$

$$T_B = 1007.9 \text{ N} \quad \text{Answer}$$

$$\Sigma F_y = 0 \quad +\uparrow \quad T_A + T_B - 1177.2 = 0$$

Substituting $T_B = 1007.9$ N and solving for T_A , we get

$$T_A = 169.3 \text{ N} \quad \text{Answer}$$

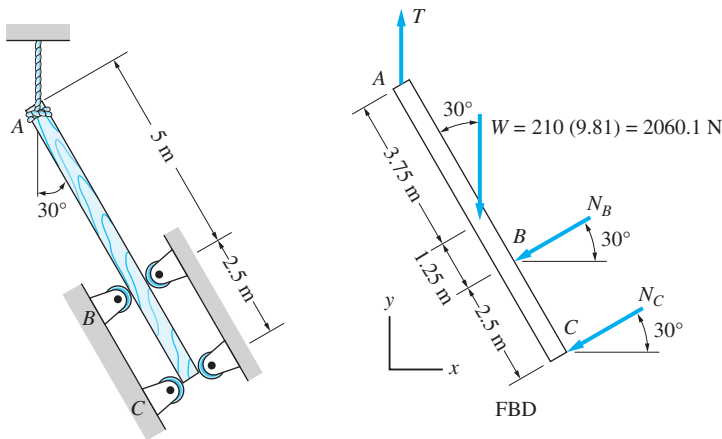
Other Methods of Analysis

Another, equally convenient option is to compute T_A using $\Sigma M_B = 0$ as

$$\begin{aligned}\Sigma M_B = 0 \quad \curvearrowright \quad & -4T_A + 1177.2(1.0) - 500 = 0 \\ & T_A = 169.3 \text{ N}\end{aligned}$$

Sample Problem 4.8

The 210-kg homogeneous log is supported by a rope at A and loose-fitting rollers at B and C as it is being fed into a sawmill. Calculate the tension in the rope and the reactions at the rollers, using the given FBD. Which rollers are in contact with the log?



Solution

Method of Analysis

The FBD contains the weight W of the log and three unknown forces: the tension T in rope and the roller reactions N_B and N_C perpendicular to the log. The sense of each roller reaction indicates that we have assumed the upper rollers to be in contact with the lumber.

The force system in the FBD is the general coplanar case, for which three independent equilibrium equations are available. Because there are also three unknowns, the problem is statically determinate.

In general, equilibrium analysis would require the solution of three simultaneous equations, with all three unknowns appearing in each equation. With planning, it usually is possible to reduce the number of unknowns that must be solved simultaneously. Referring to the FBD, we could start with $\Sigma F_x = 0$, which would contain only two unknowns: N_B and N_C . Then we would look for another equation that contains only these two unknowns. Inspection of the FBD reveals that the equation $\Sigma M_A = 0$ would not contain T , because this force passes through A .

The equations ΣF_x and ΣM_A could thus be solved simultaneously for N_A and N_B . Finally, ΣF_y would be used to compute T .

Mathematical Details

$$\Sigma F_x = 0 \quad \rightarrow \quad -N_B \cos 30^\circ - N_C \cos 30^\circ = 0 \quad (a)$$

$$\Sigma M_A = 0 \quad \curvearrowright \quad 2060.1 (3.75 \sin 30^\circ) + 5N_B + 7.5 N_C = 0 \quad (b)$$

The solution of Eqs. (a) and (b) is

$$N_B = 1545 \text{ N} \quad \text{and} \quad N_C = -1545 \text{ N} \quad \text{Answer}$$

The signs indicate that the sense of N_B is as shown on the FBD, whereas the sense of N_C is opposite to that shown. Therefore, the upper roller at B and the lower roller at C are in contact with the log.

$$\Sigma F_y = 0 \quad \uparrow \quad T - 2060.1 - N_B \sin 30^\circ - N_C \sin 30^\circ = 0$$

Because $N_B = -N_C$, this equation yields

$$T = 2060.1 \text{ N} \quad \text{Answer}$$

Other Methods of Analysis

The above solution used the equations $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M_A = 0$. There are other sets of independent equilibrium equations that would serve equally well. For example, we could find T from just a single equation—summation of forces parallel to the log equals zero. Because N_B and N_C are perpendicular to the log, T would be the only unknown in this equation. The reaction N_C could also be computed independently from the other unknowns by setting the sum of the moments about the point where T and N_B intersect to zero. Similarly, we could find N_B from a single equation—summation of moments about the point where T and N_C intersect equals zero.

It is important to realize that the equilibrium equations must be independent. Referring to the FBD, you might be tempted to use the three moment equations $\Sigma M_A = 0$, $\Sigma M_B = 0$, and $\Sigma M_C = 0$. Although each is a valid equation, they are not independent of each other. Why not? What would happen if you tried to solve these equations for N_B , N_C , and T ?

4.6 *Equilibrium Analysis for Single-Body Problems*

We learned that the three steps in the equilibrium analysis of a body are:

1. Draw the free-body diagram (FBD).
2. Write the equilibrium equations.
3. Solve the equations for the unknowns.

The individual steps were introduced separately in the preceding articles. The purpose of this article is to give you experience in the entire process of equilibrium analysis.

Always begin by drawing the FBD; *there are no exceptions*. The FBD is the very key to equilibrium analysis, so it should be drawn with great care. We recommend that you use a straightedge and circle template. After the FBD has been drawn, the remainder of the solution, consisting of writing and solving equilibrium equations, will be straight forward.

It must be reiterated that if the number of unknowns on the FBD equals the number of independent equations (statically determinate problem), you will be able to calculate all of the unknowns. Conversely, if the number of unknowns exceeds the number of independent equations (statically indeterminate problem), *all* of the unknowns cannot be determined by using equilibrium analysis alone.

Although there are many statically indeterminate problems of practical importance, you will find that nearly all problems in this text are statically determinate. To solve a statically indeterminate problem, one must consider deformations of the body, as well as equations of equilibrium. The solution of statically indeterminate problems is discussed in texts with such titles as *Strength of Materials* or *Mechanics of Materials*, the understanding of which requires a prior knowledge of statics.

Sample Problem 4.9

The telephone cable spool in Fig. (a) weighs 1500 N and is held at rest on a 40° incline by the horizontal cable. The cable is wound around the inner hub of the spool and attached to the support at *B*. Assume that *G*, the center of gravity of the spool, is located at the center of the spool. Find all forces acting on the spool.

Solution

Method of Analysis

The first step is, of course, to draw the FBD of the spool, which is shown in Fig. (b). In addition to its weight, the spool is acted on by the normal contact force *N* and friction force *F* (both acting at the point of contact *A*) and by the cable tension *T*. Note that the magnitudes *T*, *N*, and *F* are the only unknowns and that there are three independent equilibrium equations (general coplanar force system). Therefore, the problem is statically determinate. We illustrate one method of solution in detail and then discuss several other methods that could be used.

We start with the equation

$$\Sigma M_A = 0$$

The tension *T* can be calculated using this equation because it will be the only unknown (*N* and *F* do not have moments about point *A*). The next equation is

$$\Sigma M_G = 0$$

The unknowns in this equation will be *T* and *F*, because *N* has no moment about *G*. Because *T* has already been found, this equation can be solved for *F*. Finally, we use the equation

$$\Sigma F_{y'} = 0$$

The unknowns in this equation will be *T* and *N* (*F* is perpendicular to the *y'*-direction). Again, with *T* already computed, *N* can be found.

Mathematical Details

To help you follow the details of the preceding analysis, the FBD of the spool has been redrawn in Fig. (c). Note that the 1500 N weight of the spool has been replaced by its *x'*- and *y'*-components and that the vertical distance between *A* and *G* ($1.5 \cos 40^\circ$ m) has been added. The analysis now proceeds as follows:

$$\Sigma M_A = 0 \quad \curvearrowleft \quad 1500 \sin 40^\circ (1.5) - T(0.8 + 1.5 \cos 40^\circ) = 0$$

$$T = 742 \text{ N}$$

Answer

$$\Sigma M_G = 0 \quad \curvearrowleft \quad F(1.5) - T(0.8) = 0$$

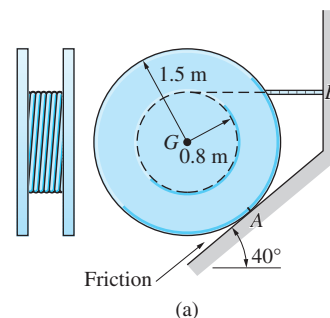
$$F = \frac{742(0.8)}{1.5} = 395.7 \text{ N}$$

Answer

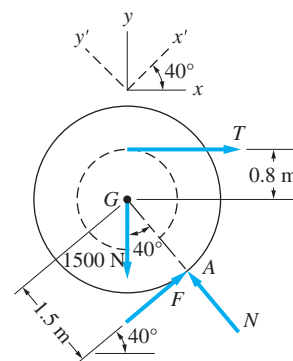
$$\Sigma F_{y'} = 0 \quad \nearrow \quad N - 1500 \cos 40^\circ - T \sin 40^\circ = 0$$

$$N = 1500 \cos 40^\circ + 742 \sin 40^\circ = 1626 \text{ N}$$

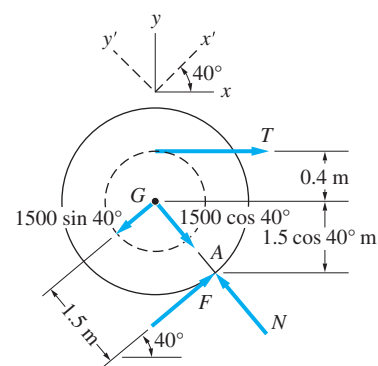
Answer



(a)



(b)



(c)

The positive signs determined for T , F , and N indicate that the correct sense for each force was assumed on the FBD.

As a check on this solution, we can verify that the above answers satisfy a fourth equilibrium equation. For example,

$$\begin{aligned}\Sigma F_x = 0 \quad \rightarrow \quad & F \cos 40^\circ - N \sin 40^\circ + T \\ & = 395.7 \cos 40^\circ - 1626 \sin 40^\circ + 742 \approx 0 \quad \text{Check}\end{aligned}$$

Other Methods of Analysis

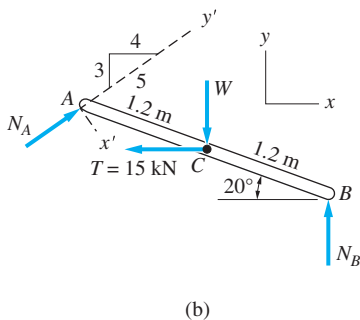
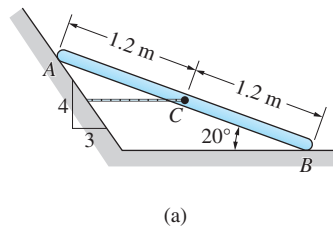
Two additional methods of analysis are outlined in the table below, with the mathematical details omitted.

Equation	Unknowns	Solution
$\Sigma M_G = 0$	T and F	Solve simultaneously for T and F
$\Sigma F_{x'} = 0$	T and F	
$\Sigma F_{y'} = 0$	T and N	
$\Sigma M_A = 0$	T	Solve for T
$\Sigma F_{x'} = 0$	T and F	Knowing T , solve for F
$\Sigma F_y = 0$	T , N , and F	Knowing T and F , solve for N

In this sample problem, we have illustrated only three of the many sets of equations that can be used to analyze this problem. You may find it beneficial to outline one or more additional analyses. Outlining the solution will permit you to concentrate on the method of analysis without becoming too involved with the mathematical details of the solution.

Sample Problem 4.10

Determine the mass of the heaviest uniform bar that can be supported in the position shown in Fig. (a) if the breaking strength of the horizontal cable attached at C is 15 kN. Neglect friction.



Solution

Method of Analysis

We begin by drawing the FBD of the bar as shown in Fig. (b). The weight W of the heaviest bar that can be supported will be obtained when the tension T is set

equal to 15 kN. A heavier bar would result in a cable tension greater than 15 kN, and the cable would break.

There are three unknowns in the FBD: the normal contact forces N_A and N_B , and W . Note that the directions of all these forces are known. The unknowns can, therefore, be found using the three independent equilibrium equations that are available for a general coplanar force system. However, because we are seeking W only, it may not be necessary to use all three equations.

In our analysis, we will use the following two equations.

$$\Sigma F_x = 0$$

The force N_A can be found from this equation (W and N_B will not appear because they are perpendicular to the x -direction).

$$\Sigma M_B = 0$$

This equation will contain the two unknowns W and N_A . Because N_A already has been determined, W now can be found.

Mathematical Details

Referring to the FBD in Fig. (b), the mathematical details of the preceding analysis are as follows:

$$\begin{aligned} \Sigma F_x = 0 \quad \rightarrow \quad \frac{4}{5}N_A - 15 &= 0 \\ N_A &= 18.75 \text{ kN} \end{aligned} \quad (a)$$

$$\begin{aligned} \Sigma M_B = 0 \quad \curvearrowright \quad W(1.2 \cos 20^\circ) + 15(1.2 \sin 20^\circ) - \frac{3}{5}N_A(2.4 \cos 20^\circ) \\ - \frac{4}{5}N_A(2.4 \sin 20^\circ) &= 0 \end{aligned} \quad (b)$$

Substituting $N_A = 18.75 \text{ kN}$ from Eq. (a) into Eq. (b) gives $W = 28.0 \text{ kN}$. Therefore, the mass of the heaviest bar that can be supported without breaking the cable is

$$m = \frac{W}{g} = \frac{28.0 \times 10^3}{9.81} = 2850 \text{ kg} \quad \text{Answer}$$

Other Methods of Analysis

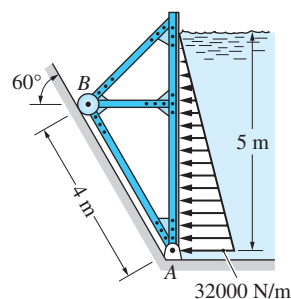
Another method that could be used to calculate W is outlined in the following table.

Equation	Unknowns	Solution
$\Sigma M_A = 0$	N_B and W	} Solve simultaneously for N_B and W
$\Sigma F_{x'} = 0$	N_B and W	

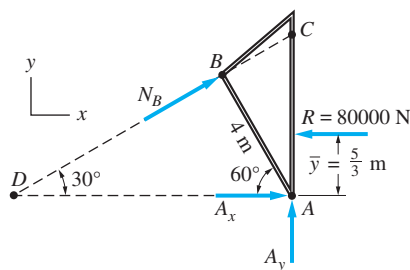
There are, of course, many other sets of equations that could be used to compute W . It is even possible to determine W using only one equilibrium equation—a moment equation taken about the point where N_A and N_B intersect.

Sample Problem 4.11

Figure (a) shows the distributed loading due to water pressure that is acting on the upstream side of the flood barrier. Determine the support reactions acting on the barrier at A and B. Neglect the weight of the barrier.



(a)



(b)

Solution

Method of Analysis

The FBD of the barrier is shown in Fig. (b), where N_B is the reaction at B, acting perpendicular to the inclined surface, and A_x and A_y are the components of the pin reaction at A. Any three independent equilibrium equations can be used to determine these three unknowns. As explained in Art. 3.6, the resultant of a distributed load is equal to the area under the loading diagram, acting at the centroid of that area. Therefore, we obtain

$$R = \frac{1}{2}(5)(32000) = 80000 \text{ N}$$

and from Table 3.1, we find

$$\bar{y} = \frac{5}{3} \text{ m}$$

Because the unknown forces A_x and A_y intersect at A, a convenient starting point is

$$\Sigma M_A = 0$$

This equation will determine N_B . We then use

$$\Sigma F_x = 0$$

Having previously determined N_B , this equation will give A_x . The final equation is

$$\Sigma F_y = 0$$

With N_B previously computed, A_y can be found from this equation.

Mathematical Details

$$\Sigma M_A = 0 \quad \curvearrowleft \quad 80000 \left(\frac{5}{3} \right) - N_B(4) = 0$$

$$N_B = 33333.3 \text{ N}$$

Answer

$$\Sigma F_x = 0 \quad \rightarrow \quad N_B \cos 30^\circ + A_x - 80000 = 0$$

$$A_x = 80000 - (33333.3) \cos 30^\circ = 51132.5 \text{ N}$$

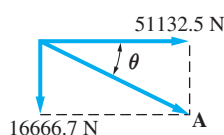
Answer

$$\Sigma F_y = 0 \quad \uparrow \quad A_y + N_B \sin 30^\circ = 0$$

$$A_y = - (33333.3) \sin 30^\circ = - 16666.7 \text{ N}$$

Answer

The signs indicate that N_B and A_x are directed as shown on the FBD, whereas the correct direction of A_y is opposite the direction shown on the FBD. Therefore, the force that acts on the barrier at A is



$$|A| = \sqrt{(51132.5)^2 + (16666.7)^2} = 53780 \text{ N}$$

$$\theta = \tan^{-1} \left(\frac{16666.7}{51132.5} \right) = 18.1^\circ$$

Answer

and the force at B is



Answer

Other Methods of Analysis

There are, of course, many other independent equations that could be used to solve this problem. Referring to the FBD in Fig. (b), the following set of equations has the advantage of determining each unknown independently of the other two.

Equation	Unknowns	Solution
$\Sigma M_A = 0$	N_B	Solve for N_B
$\Sigma M_C = 0$	A_x	Solve for A_x
$\Sigma M_D = 0$	A_y	Solve for A_y

PART C: Analysis of Plane Trusses

4.10 Description of a Truss

A *truss* is a structure that is made of straight, slender bars that are joined together to form a pattern of triangles. Trusses are usually designed to transmit forces over relatively long spans; common examples are bridge trusses and roof trusses. A typical bridge truss is shown in Fig. 4.15(a).

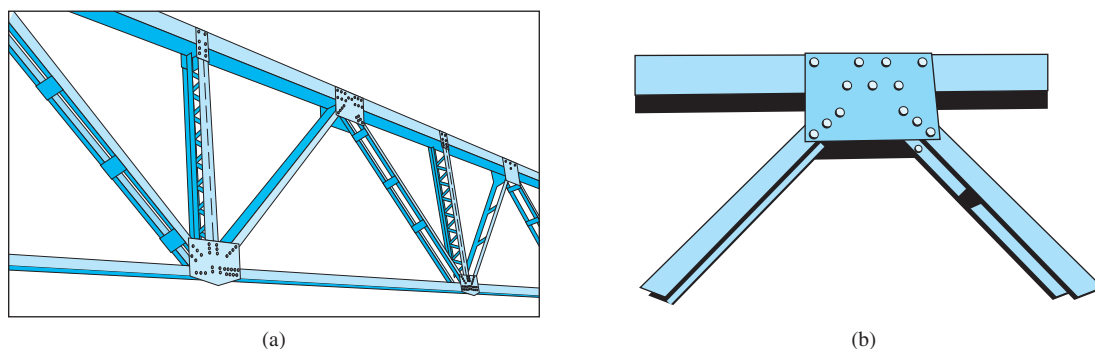


Fig. 4.15

The analysis of trusses is based on the following three assumptions:

1. **The weights of the members are negligible.** A truss can be classified as a lightweight structure, meaning that the weights of its members are generally much smaller than the loads that it is designed to carry.
2. **All joints are pins.** In practice, the members at each joint are usually riveted or welded to a plate, called a *gusset plate*, as shown in Fig. 4.15(b).

However, if the members at a joint are aligned so that their centroidal axes (axes that pass through the centroids of the cross-sectional areas of the members) intersect at a common point, advanced methods of analysis indicate that the assumption of pins is justified.

3. **The applied forces act at the joints.** Because the members of a truss are slender, they may fail in bending when subjected to loads applied at locations other than the joints. Therefore, trusses are designed so that the major applied loads act at the joints.

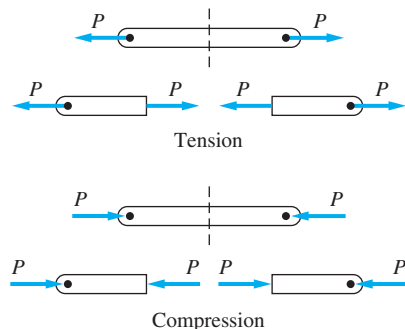


Fig. 4.16

Although these assumptions may appear to oversimplify the real situation, they lead to results that are adequate in most applications.

Using the assumptions, the free-body diagram for any member of a truss will contain only two forces—the forces exerted on the member by the pin at each end. Therefore, *each member of a truss is a two-force body*.

When dealing with the internal force in a two-force body, engineers commonly distinguish between *tension* and *compression*. Figure 4.16 shows the external and internal forces in tension and compression. Tensile forces elongate (stretch) the member, whereas compressive forces compress (shorten) it. Because

the forces act along the longitudinal axis of the member, they are often called *axial forces*. Note that internal forces always occur as equal and opposite pairs on the two faces of an internal cross section.

The two common techniques for computing the internal forces in a truss are the method of joints and the method of sections, each of which is discussed in the following articles.

4.11 Method of Joints

When using the *method of joints* to calculate the forces in the members of a truss, the equilibrium equations are applied to individual joints (or pins) of the truss. Because the members are two-force bodies, the forces in the FBD of a joint are concurrent. Consequently, two independent equilibrium equations are available for each joint.

To illustrate this method of analysis, consider the truss shown in Fig. 4.17(a). The supports consist of a pin at A and a roller at E (one of the supports is usually designed to be equivalent to a roller, in order to permit the elongation and contraction of the truss with temperature changes).

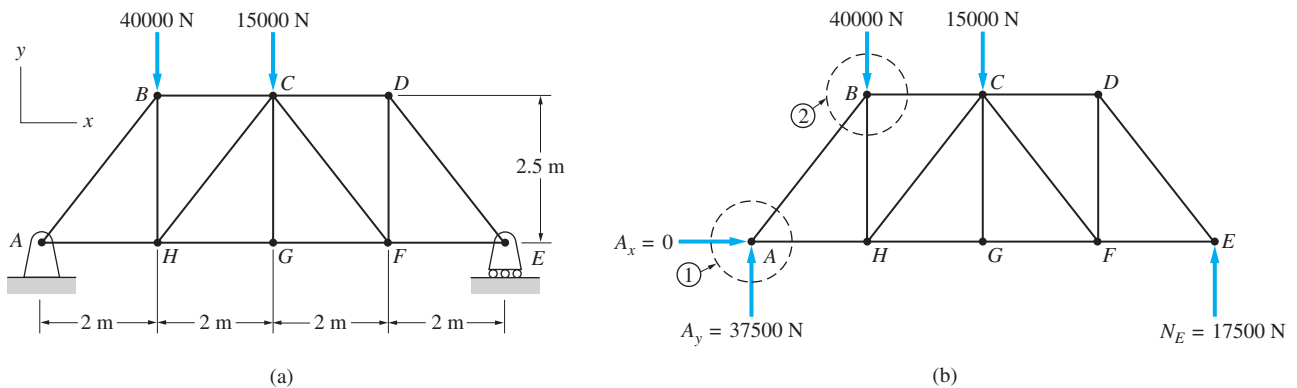


Fig. 4.17

a. Support reactions

It is usually a good idea to start the analysis by determining the reactions at the supports using the FBD of the entire truss. The FBD of the truss in Fig. 4.17(b) contains three unknown reactions: A_x , A_y , and N_E , which can be found from the three available equilibrium equations. The results of the computation are shown in Fig. 4.17(b).

Note that A_x , the horizontal reaction at A , is zero. This result indicates that the truss would be in equilibrium under the given loading even if the pin at A were replaced by a roller. However, we would then have an *improper constraint*, because an incidental horizontal force would cause the truss to move horizontally. Therefore, a pin support at A (or B) is necessary to properly constrain the truss.

Sometimes the number of unknown reactions on the FBD of the entire truss is greater than three. In this case, all the reactions cannot be found at the outset.

b. Equilibrium analysis of joints

Let us now determine the forces in the individual members of the truss in Fig. 4.17. Because the force in a member is internal to the truss, it will appear on a FBD only if the FBD “cuts” the member, thereby separating it from the rest of the truss. For example, to determine the force in members AB and AH , we can draw the FBD of joint A —that is, the portion of the truss encircled by the dashed line 1 in Fig. 4.17(b). This FBD, shown in Fig. 4.18(a), contains the external reactions A_x and A_y and the member forces P_{AB} and P_{AH} (the subscripts identify the member). Note that we have assumed the forces in the members to be tensile. If the solution yields a negative value for a force, the force is compressive. By assuming the members to be in tension, we are using an established convention for which positive results indicate tension and negative results indicate compression.

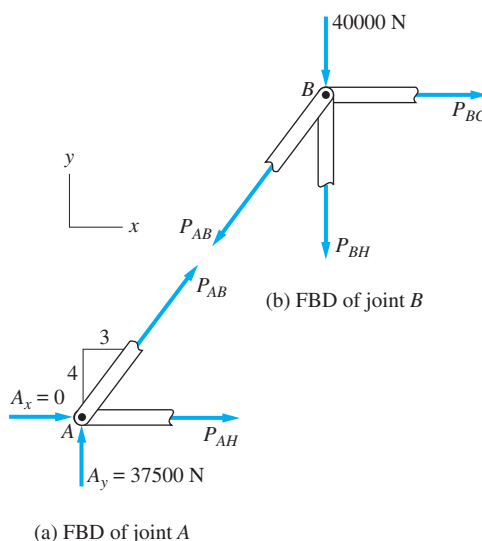


Fig. 4.18

Having previously computed A_x and A_y , the forces P_{AB} and P_{AH} are the only unknowns in the FBD for joint A . Therefore, they can be determined from the two independent equilibrium equations for the joint, as follows.

$$\begin{aligned}
 \Sigma F_y = 0 \quad +\uparrow \quad & 37500 + \frac{4}{5}P_{AB} = 0 \\
 & P_{AB} = -\frac{5}{4}(37500) = -46875 \text{ N} \\
 \Sigma F_x = 0 \quad +\rightarrow \quad & \frac{3}{5}P_{AB} + P_{AH} + A_x = 0 \\
 & \frac{3}{5}(-46875) + P_{AH} + 0 = 0 \\
 & P_{AH} = -\frac{3}{5}(-46875) = 28125 \text{ N}
 \end{aligned}$$

The negative value for P_{AB} indicates that the force in member AB is compressive; the positive value for P_{AH} means that the force in member AH is tensile.

To compute the forces in members BC and BH , we draw the FBD of joint B —the portion of the truss encircled by the dashed line 2 in Fig. 4.17(b). This FBD is shown in Fig. 4.18(b). Note that the force P_{AB} is equal and opposite to the corresponding force in Fig. 4.18(a), and that we again assumed P_{BC} and P_{BH} to be tensile. Knowing that $P_{AB} = -46875$ N, P_{BC} and P_{BH} are the only unknowns in this FBD. The equilibrium equations of the joint yield

$$\begin{aligned}\Sigma F_x = 0 \quad & \rightarrow P_{BC} - \frac{3}{5}P_{AB} = 0 \\ P_{BC} &= \frac{3}{5}P_{AB} = \frac{3}{5}(-46875) = -28125 \text{ N} \\ \Sigma F_y = 0 \quad & \uparrow -\frac{4}{5}P_{AB} - P_{BH} - 40000 = 0 \\ P_{BH} &= -40000 - \frac{4}{5}P_{AB} \\ &= -40000 - \frac{4}{5}(-46875) = -2500 \text{ N}\end{aligned}$$

The negative values indicate that both P_{BC} and P_{BH} are compressive.

We could continue the procedure, moving from joint to joint, until the forces in all the members are determined. In order to show that this is feasible, we count the number of unknowns and the number of independent equilibrium equations:

$$\begin{aligned}13 \text{ member forces} + 3 \text{ support reactions} &= 16 \text{ unknowns} \\ 8 \text{ joints, each yielding 2 equilibrium equations} &= 16 \text{ equations}\end{aligned}$$

Because the number of equations equals the number of unknowns, the truss is statically determinate. The three equilibrium equations of the entire truss were not counted, because they are not independent of the joint equilibrium equations (recall that a structure is in equilibrium if each of its components is in equilibrium).

c. Equilibrium analysis of pins

In the above example, the FBD of a joint contained a finite portion of the truss surrounding the joint. This required “cutting” the members attached to the joint, so that the internal forces in the members would appear on the FBD. An alternative approach, preferred by many engineers, is to draw the FBDs of the “pins,” as illustrated in Fig. 4.19. In this case, the internal forces in the members appear as forces acting on the pin. For all practical purposes, the FBDs in Figs. 4.18 and 4.19 are identical. The FBD of a pin is easier to draw, but the FBD of a joint is somewhat more meaningful, particularly when it comes to determining whether the member forces are tensile or compressive.

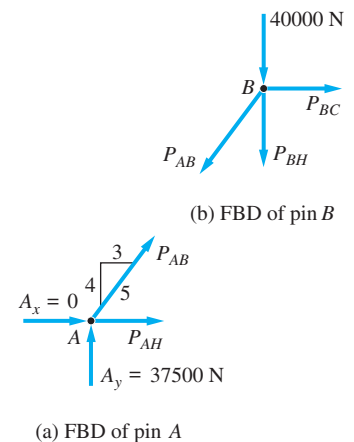
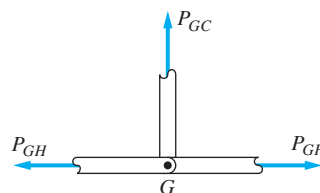
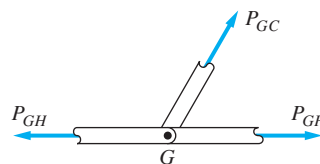


Fig. 4.19

d. Zero-force members

There is a special case that occurs frequently enough to warrant special attention. Figure 4.20(a) shows the FBD for joint G of the truss in Fig. 4.17. Because no external loads are applied at G , the joint equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$ yield $P_{GH} = P_{GF}$ and $P_{GC} = 0$. Because member GC does not carry a force, it is called a *zero-force member*. It is easily verified that the results remain unchanged if member GC is inclined to GH and GF , as shown in Fig. 4.20(b). When analyzing a truss, it is often advantageous to begin by identifying zero-force members, thereby simplifying the solution.

(a) FBD of joint G 

(b)

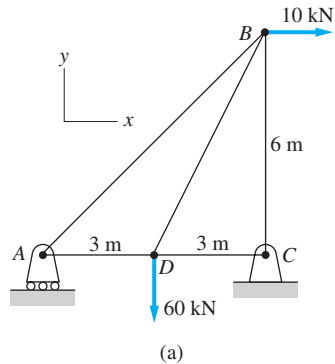
Fig. 4.20

You may wonder why a member, such as GC , is included in the truss if it carries no force. The explanation is the same as the one given for providing a pin support—rather than a roller—at A for the truss in Fig. 4.17(a): It is necessary to ensure the proper constraint of joint G . If member GC were removed, the truss would theoretically remain in equilibrium for the loading shown.* However, the slightest vertical load applied to the joint at G would cause the truss to deform excessively, or even collapse. Moreover, it is unlikely that the loads shown in Fig. 4.17(a) will be the only forces acting on the truss during its lifetime. Should a vertical load be suspended from joint G at some future time, member GC would be essential for equilibrium.

*The word *theoretically* is to be interpreted as “in accordance with the assumptions.” Our mathematical model for a truss assumes that the weights of the members are negligible. In practice, the force in a so-called zero-force member is not exactly zero but is determined by the weights of the members.

Sample Problem 4.18

Using the method of joints, determine the force in each member of the truss shown in Fig. (a). Indicate whether the members are in tension or compression.



Solution

The FBD of the entire truss is shown in Fig. (b). The three unknowns (N_A , C_x , and C_y) can be computed from the three equilibrium equations

$$\Sigma M_C = 0 \quad \curvearrowright \quad -N_A(6) + 60(3) - 10(6) = 0$$

$$N_A = 20 \text{ kN}$$

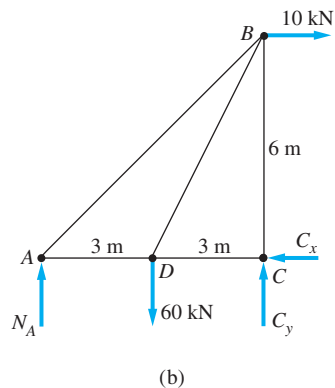
$$\Sigma F_y = 0 \quad \uparrow \quad N_A - 60 + C_y = 0$$

$$C_y = 60 - N_A = 60 - 20 = 40 \text{ kN}$$

$$\Sigma F_x = 0 \quad \rightarrow \quad 10 - C_x = 0$$

$$C_x = 10 \text{ kN}$$

We now proceed to the computation of the internal forces by analyzing the FBDs of various pins.



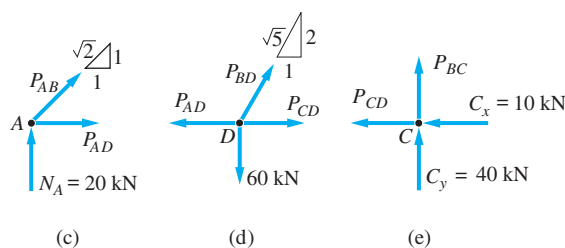
Method of Analysis

In the following discussion, the external reactions are treated as knowns, because they have already been calculated. It is convenient to assume the force in each member to be tensile. Therefore, positive values of the forces indicate tension, and negative values denote compression.

The FBD of pin A, shown in Fig. (c), contains two unknowns: P_{AB} and P_{AD} . We can compute these two forces immediately, because two independent equilibrium equations are available from this FBD.

The FBD of pin D, in Fig. (d), contains the forces P_{AD} , P_{BD} , and P_{CD} . Because P_{AD} has already been found, we have once again two equations that can be solved for the two unknowns.

Figure (e) shows the FBD of pin C. With P_{CD} previously found, the only remaining unknown is P_{BC} , which can be easily computed.



Mathematical Details

From the FBD of pin A, Fig. (c),

$$\begin{aligned}\Sigma F_y = 0 \quad +\uparrow \quad & \frac{1}{\sqrt{2}} P_{AB} + N_A = 0 \\ & P_{AB} = -\sqrt{2}(20) = -28.3 \text{ kN} \\ & P_{AB} = 28.3 \text{ kN} \quad (\text{compression}) \quad \text{Answer} \\ \Sigma F_x = 0 \quad +\rightarrow \quad & \frac{1}{\sqrt{2}} P_{AB} + P_{AD} = 0 \\ & P_{AD} = -\frac{1}{\sqrt{2}} P_{AB} = -\frac{1}{\sqrt{2}}(-28.3) \\ & = 20.0 \text{ kN} \quad (\text{tension}) \quad \text{Answer}\end{aligned}$$

From the FBD of pin D, Fig. (d),

$$\begin{aligned}\Sigma F_y = 0 \quad +\uparrow \quad & \frac{2}{\sqrt{5}} P_{BD} - 60 = 0 \\ & P_{BD} = \frac{\sqrt{5}}{2}(60) = 67.1 \text{ kN} \quad (\text{tension}) \quad \text{Answer}\end{aligned}$$

$$\Sigma F_x = 0 \quad \rightarrow \quad -P_{AD} + \frac{1}{\sqrt{5}}P_{BD} + P_{CD} = 0$$

$$P_{CD} = P_{AD} - \frac{1}{\sqrt{5}}P_{BD}$$

$$P_{CD} = 20.0 - \frac{1}{\sqrt{5}}(67.1) = -10.0 \text{ kN}$$

$$P_{CD} = 10.0 \text{ kN} \quad (\text{compression}) \quad \textbf{Answer}$$

From the FBD of pin *C*, Fig. (e),

$$\Sigma F_y = 0 \quad \uparrow \quad C_y + P_{BC} = 0$$

$$P_{BC} = -C_y = -40 \text{ kN}$$

$$P_{BC} = 40 \text{ kN} \quad (\text{compression}) \quad \textbf{Answer}$$

Note that the equation $\Sigma F_x = 0$ yields $P_{CD} = -10.0 \text{ kN}$, a value that has been found before. Therefore, this equation is not independent of the equations used previously. The reason for the dependence is that the external reactions were determined by analyzing the FBD of the entire truss. However, the equations for the pins and those for the entire truss are not independent of each other.

Other Methods of Analysis

In the preceding analysis, the pins were considered in the following order: *A*, *D*, and *C* (the FBD of pin *B* was not used). Another sequence that could be used is

1. FBD of pin *A*: Calculate P_{AB} and P_{BD} (as before).
2. FBD of pin *B*: With P_{AB} already found, calculate P_{BD} and P_{BC} .
3. FBD of pin *C*: Calculate P_{CD} .

In this analysis, the FBD of pin *D* would not be used.

Yet another approach would be to compute the three external reactions and the forces in the five members (a total of eight unknowns) by using the equilibrium equations for all the pins (a total of eight equations, two for each pin).

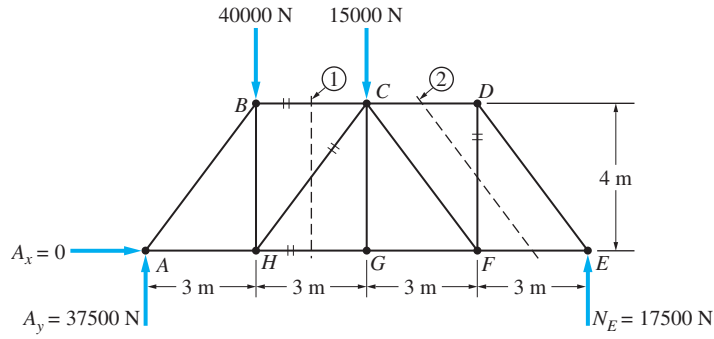
4.12 *Method of Sections*

Truss analysis by the method of joints is based on the FBDs of individual joints. Analyzing the free-body diagram of a part of a truss that contains two or more joints is called the *method of sections*. The FBD for a single joint results in a concurrent, coplanar force system (two independent equilibrium equations). When applying the method of sections, the force system will generally be nonconcurrent, coplanar (three independent equilibrium equations).

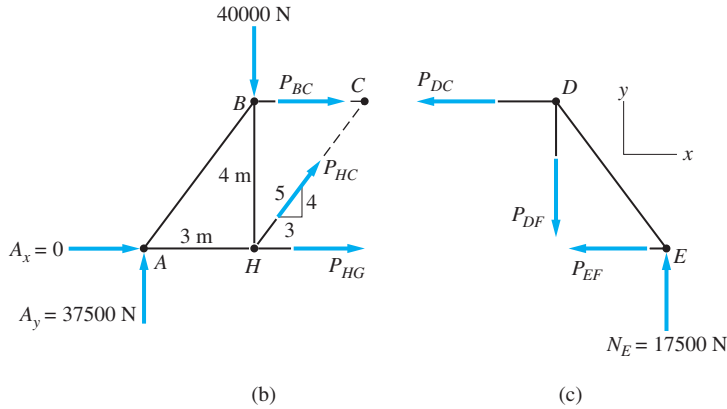
In the method of sections, a part of the truss is isolated on an FBD so that it exposes the forces to be computed. If the FBD for the isolated portion contains three unknowns, all of them can usually be computed from the three available equilibrium equations. If the number of unknowns exceeds three, one or more of the unknowns must be found by analyzing a different part of the truss. If you are skillful in writing and solving equilibrium equations, the only challenge in using the method of sections is selecting a convenient part of the truss for the FBD.

Consider once again the truss discussed in the preceding article [its FBD is repeated in Fig. 4.21(a)]. We now use the method of sections to determine the forces in members BC , HC , HG , and DF —each of these members is identified by two short parallel lines in Fig. 4.21(a).

Assuming that the external reactions have been previously computed, the first and most important step is the selection of the part of the truss to be analyzed. We note that the section labeled 1 in Fig. 4.21(a) passes through members BC , HC , and HG . The forces in these three members are the only unknowns if the FBD is drawn for a part of the truss that is isolated by this section. Note that after the section has been chosen, the portion of the truss on either side of the cut may be used for the FBD. The forces inside the members occur in equal and opposite pairs, so the same results will be obtained regardless of which part is analyzed. Of course, given a choice, one would naturally select the less complicated part.



(a)



(b)

(c)

Fig. 4.21

For our truss, using either the portion to the left or to the right of section 1 is equally convenient. For no particular reason, we choose to analyze the left portion, with its FBD shown in Fig. 4.21(b) (we have again assumed the members to be in tension). Note that the force system is nonconcurrent and coplanar, so that any set of three independent equations can be used to compute the three unknown forces.

The following is an efficient method of solution.

- $\Sigma F_y = 0$ —determines P_{HC} , because P_{BC} and P_{HG} have no y-components
- $\Sigma M_H = 0$ —determines P_{BC} , because P_{HC} and P_{HG} have no moment about point H
- $\Sigma M_C = 0$ —determines P_{HG} , because P_{BC} and P_{HC} have no moment about point C

The details of this analysis are as follows:

$$\Sigma F_y = 0 \quad \uparrow \quad \frac{4}{5} P_{HC} - 40000 + 37500 = 0$$

$$P_{HC} = \frac{5}{4} (40000 - 37500)$$

$$= 3125 \text{ N (tension)}$$

Answer

$$\Sigma M_H = 0 \quad \curvearrowright + \quad - 37500(3) - P_{BC}(4) = 0$$

$$P_{BC} = - 28125 \text{ N}$$

$$= 28125 \text{ N} \quad (\text{compression}) \quad \textbf{Answer}$$

$$\Sigma M_C = 0 \quad \curvearrowright + \quad 40000(3) - 37500(6) + P_{HG}(4) = 0$$

$$P_{HG} = \frac{1}{4.0} [37500(6) - 40000(3)]$$

$$= 26250 \text{ N} \quad (\text{tension}) \quad \textbf{Answer}$$

To determine the force in member DF by the method of sections, consider once again the FBD of the entire truss in Fig. 4.21(a). Our intention is to isolate a part of the truss by cutting only three members, one of which is the member DF . It can be seen that section 2 accomplishes this task. We choose to analyze the part to the right of this section because it contains fewer forces. The FBD is shown in Fig. 4.21(c), with the unknown forces again assumed to be tensile. Note that P_{DF} can be computed from the equation

$$\Sigma F_y = 0 \quad \uparrow + \quad 17500 - P_{DF} = 0$$

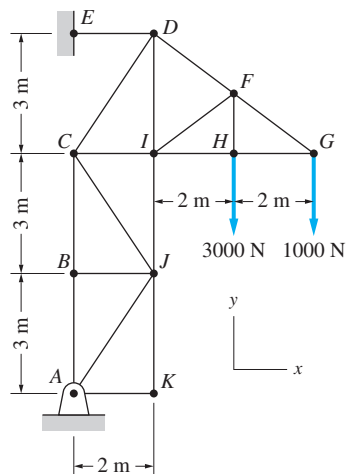
$$P_{DF} = 17500 \text{ N} \quad (\text{tension}) \quad \textbf{Answer}$$

If desired, P_{EF} could now be calculated using $\Sigma M_D = 0$, and $\Sigma M_E = 0$ would give P_{DC} .

As you see, the forces in the members of a truss can be found by either the method of joints or the method of sections. Selecting the method that results in the most straightforward analysis is usually not difficult. For example, for the truss shown in Fig. 4.21(a), the FBD of joint A is convenient for computing P_{AB} , whereas the method of sections is more advantageous for calculating P_{BC} .

Sample Problem 4.19

Using the method of sections, determine the forces in the following members of the truss in Fig. (a): FI and JC . Indicate tension or compression.



(a)

Solution

The FBD of the entire truss is shown in Fig. (b). The three external reactions (A_x , A_y , and P_{DE}) can be calculated using the following equations (mathematical details have been omitted):

$$\Sigma M_A = 0 \quad \text{gives} \quad P_{DE} = 2000 \text{ N}$$

$$\Sigma F_x = 0 \quad \text{gives} \quad A_x = 2000 \text{ N}$$

$$\Sigma F_y = 0 \quad \text{gives} \quad A_y = 4000 \text{ N}$$

These forces are shown in Fig. (b), and from now on we treat them as known quantities.

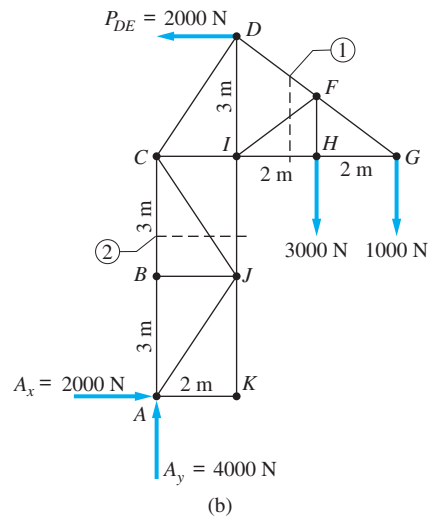
Member FI

Method of Analysis

Referring to the FBD in Fig. (b), it can be seen that section 1 cuts through members FD , FI , and HI . Because there are three equilibrium equations available for a portion of the truss separated by this section, we could find the forces in all three members.

Mathematical Details

Having chosen section 1, we must now decide which portion of the truss to analyze. We select the portion lying to the right of the section, because it is somewhat



(b)

less complicated than the portion on the left. Of course, identical results will be obtained regardless of which part of the truss is analyzed.

The FBD for the part of the truss lying to the right of section 1 is shown in Fig. (c), with the three unknowns being P_{FD} , P_{FI} , and P_{HI} . Any set of three independent equations can be used to solve for these unknowns. A convenient solution that yields P_{FI} directly is the following:

$$\Sigma M_G = 0 \quad \text{gives } P_{FI}, \text{ because } P_{FD} \text{ and } P_{HI} \text{ pass through point } G.$$

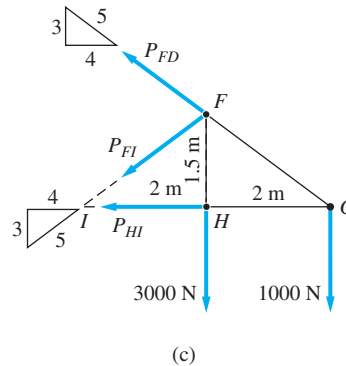
$$\curvearrowleft \quad 3000(2) + \frac{3}{5}P_{FI}(4) = 0$$

$$P_{FI} = -2500 \text{ N}$$

$$= 2500 \text{ N} \quad (\text{compression})$$

Answer

Note: The moment of P_{FI} about point G was computed by replacing P_{FI} with its x - and y -components acting at point I .



Member JC

Method of Analysis

In Fig. (b), section 2 passes through only three members, one of which is member JC . Therefore, we can find the force in that member (and the forces in the other two members if desired) from one FBD.

Mathematical Details

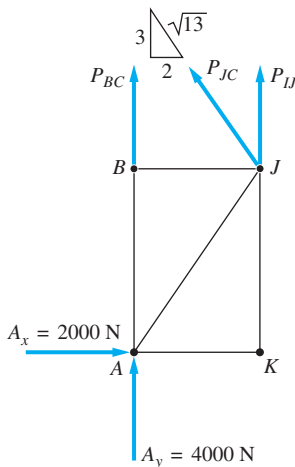
The FBD for the portion of the truss lying below section 2 is shown in Fig. (d). The three unknowns are P_{BC} , P_{JC} , and P_{IJ} . The force P_{JC} can be obtained from the following equation:

$$\Sigma F_x = 0 \quad \rightarrow \quad A_x - \frac{2}{\sqrt{13}}P_{JC} = 0$$

$$P_{JC} = \frac{\sqrt{13}}{2}A_x = \frac{\sqrt{13}}{2}(2000)$$

$$= 3610 \text{ N} \quad (\text{tension})$$

Answer



(d)

SIMPLE STRESS AND STRAIN

1.1. Load

In any engineering structure or mechanism the individual components will be subjected to external forces arising from the service conditions or environment in which the component works. If the component or member is in equilibrium, the resultant of the external forces will be zero but, nevertheless, they together place a load on the member which tends to deform that member and which must be reacted by internal forces which are set up within the material.

If a cylindrical bar is subjected to a direct pull or push along its axis as shown in Fig. 1.1, then it is said to be subjected to *tension* or *compression*. Typical examples of tension are the forces present in towing ropes or lifting hoists, whilst compression occurs in the legs of your chair as you sit on it or in the support pillars of buildings.

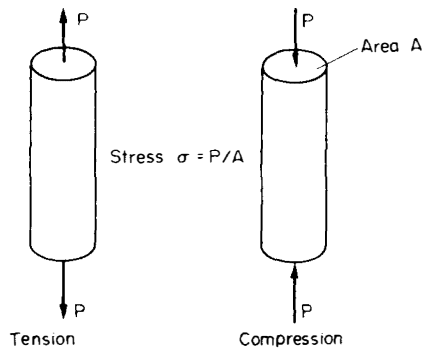


Fig. 1.1. Types of direct stress.

In the SI system of units load is measured in *newtons*, although a single newton, in engineering terms, is a very small load. In most engineering applications, therefore, loads appear in SI multiples, i.e. kilonewtons (kN) or meganewtons (MN).

There are a number of different ways in which load can be applied to a member. Typical loading types are:

- (a) *Static* or *dead* loads, i.e. non-fluctuating loads, generally caused by gravity effects.
- (b) *Live* loads, as produced by, for example, lorries crossing a bridge.
- (c) *Impact* or *shock* loads caused by sudden blows.
- (d) *Fatigue*, *fluctuating* or *alternating* loads, the magnitude and sign of the load changing with time.

1.2. Direct or normal stress (σ)

It has been noted above that external force applied to a body in equilibrium is reacted by internal forces set up within the material. If, therefore, a bar is subjected to a uniform tension or compression, i.e. a direct force, which is uniformly or equally applied across the cross-section, then the internal forces set up are also distributed uniformly and the bar is said to be subjected to a uniform *direct or normal stress*, the stress being defined as

$$\text{stress } (\sigma) = \frac{\text{load}}{\text{area}} = \frac{P}{A}$$

Stress σ may thus be compressive or tensile depending on the nature of the load and will be measured in units of newtons per square metre (N/m^2) or multiples of this.

In some cases the loading situation is such that the stress will vary across any given section, and in such cases the stress at any point is given by the limiting value of $\delta P / \delta A$ as δA tends to zero.

1.3. Direct strain (ϵ)

If a bar is subjected to a direct load, and hence a stress, the bar will change in length. If the bar has an original length L and changes in length by an amount δL , the *strain* produced is defined as follows:

$$\text{strain } (\epsilon) = \frac{\text{change in length}}{\text{original length}} = \frac{\delta L}{L}$$

Strain is thus a measure of the deformation of the material and is non-dimensional, i.e. it has no units; it is simply a ratio of two quantities with the same unit (Fig. 1.2).

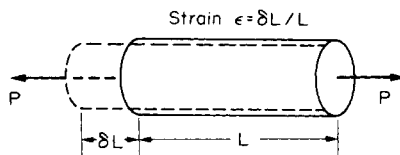


Fig. 1.2.

Since, in practice, the extensions of materials under load are very small, it is often convenient to measure the strains in the form of $\text{strain} \times 10^{-6}$, i.e. *microstrain*, when the symbol used becomes $\mu\epsilon$.

Alternatively, strain can be expressed as a *percentage strain*

i.e.
$$\text{strain } (\epsilon) = \frac{\delta L}{L} \times 100\%$$

1.4. Sign convention for direct stress and strain

Tensile stresses and strains are considered **POSITIVE** in sense producing an *increase* in length. Compressive stresses and strains are considered **NEGATIVE** in sense producing a *decrease* in length.

1.5. Elastic materials – Hooke's law

A material is said to be *elastic* if it returns to its original, unloaded dimensions when load is removed. A particular form of elasticity which applies to a large range of engineering materials, at least over part of their load range, produces deformations which are proportional to the loads producing them. Since loads are proportional to the stresses they produce and deformations are proportional to the strains, this also implies that, whilst materials are elastic, stress is proportional to strain. *Hooke's law*, in its simplest form*, therefore states that

$$\text{stress } (\sigma) \propto \text{strain } (\varepsilon)$$

i.e.
$$\frac{\text{stress}}{\text{strain}} = \text{constant}^*$$

It will be seen in later sections that this law is obeyed within certain limits by most ferrous alloys and it can even be assumed to apply to other engineering materials such as concrete, timber and non-ferrous alloys with reasonable accuracy. Whilst a material is elastic the deformation produced by any load will be *completely* recovered when the load is removed; there is no permanent deformation.

Other classifications of materials with which the reader should be acquainted are as follows:

A material which has a uniform structure throughout without any flaws or discontinuities is termed a *homogeneous* material. *Non-homogeneous* or *inhomogeneous* materials such as concrete and poor-quality cast iron will thus have a structure which varies from point to point depending on its constituents and the presence of casting flaws or impurities.

If a material exhibits uniform properties throughout in all directions it is said to be *isotropic*; conversely one which does not exhibit this uniform behaviour is said to be *non-isotropic* or *anisotropic*.

An *orthotropic* material is one which has different properties in different planes. A typical example of such a material is wood, although some composites which contain systematically orientated "inhomogeneities" may also be considered to fall into this category.

1.6. Modulus of elasticity – Young's modulus

Within the elastic limits of materials, i.e. within the limits in which Hooke's law applies, it has been shown that

$$\frac{\text{stress}}{\text{strain}} = \text{constant}$$

This constant is given the symbol E and termed the *modulus of elasticity* or *Young's modulus*.

Thus
$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\varepsilon} \quad (1.1)$$

$$= \frac{P}{A} \div \frac{\delta L}{L} = \frac{PL}{A \delta L} \quad (1.2)$$

* Readers should be warned that in more complex stress cases this simple form of Hooke's law will not apply and mis-application could prove dangerous; see §14.1, page 361.

Young's modulus E is generally assumed to be the same in tension or compression and for most engineering materials has a high numerical value. Typically, $E = 200 \times 10^9 \text{ N/m}^2$ for steel, so that it will be observed from (1.1) that strains are normally very small since

$$\epsilon = \frac{\sigma}{E} \quad (1.3)$$

In most common engineering applications strains do not often exceed 0.003 or 0.3% so that the assumption used later in the text that deformations are small in relation to original dimensions is generally well founded.

The actual value of Young's modulus for any material is normally determined by carrying out a standard tensile test on a specimen of the material as described below.

1.7. Tensile test

In order to compare the strengths of various materials it is necessary to carry out some standard form of test to establish their relative properties. One such test is the standard tensile test in which a circular bar of uniform cross-section is subjected to a gradually increasing tensile load until failure occurs. Measurements of the change in length of a selected *gauge length* of the bar are recorded throughout the loading operation by means of extensometers and a graph of load against extension or stress against strain is produced as shown in Fig. 1.3; this shows a typical result for a test on a mild (low carbon) steel bar; other materials will exhibit different graphs but of a similar general form see Figs 1.5 to 1.7.

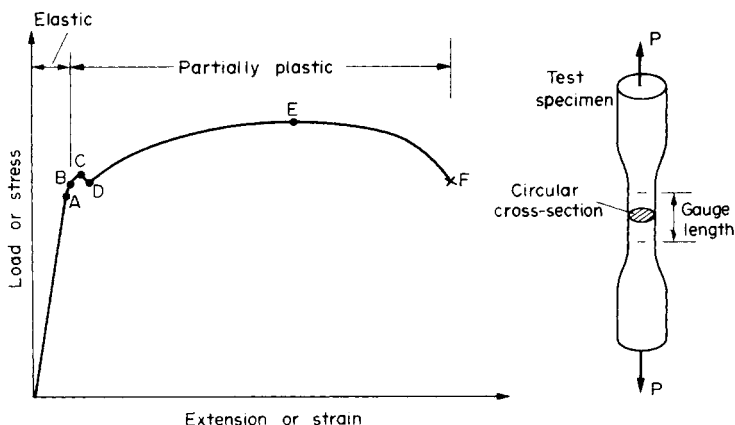


Fig. 1.3. Typical tensile test curve for mild steel.

For the first part of the test it will be observed that Hooke's law is obeyed, i.e. the material behaves elastically and stress is proportional to strain, giving the straight-line graph indicated. Some point A is eventually reached, however, when the linear nature of the graph ceases and this point is termed the *limit of proportionality*.

For a short period beyond this point the material may still be elastic in the sense that deformations are completely recovered when load is removed (i.e. strain returns to zero) but

Hooke's law does not apply. The limiting point *B* for this condition is termed the *elastic limit*. For most practical purposes it can often be assumed that points *A* and *B* are coincident.

Beyond the elastic limit *plastic deformation* occurs and strains are not totally recoverable. There will thus be some permanent deformation or *permanent set* when load is removed. After the points *C*, termed the *upper yield point*, and *D*, the *lower yield point*, relatively rapid increases in strain occur without correspondingly high increases in load or stress. The graph thus becomes much more shallow and covers a much greater portion of the strain axis than does the elastic range of the material. The capacity of a material to allow these large plastic deformations is a measure of the so-called *ductility* of the material, and this will be discussed in greater detail below.

For certain materials, for example, high carbon steels and non-ferrous metals, it is not possible to detect any difference between the upper and lower yield points and in some cases no yield point exists at all. In such cases a *proof stress* is used to indicate the onset of plastic strain or as a comparison of the relative properties with another similar material. This involves a measure of the permanent deformation produced by a loading cycle; the 0.1 % proof stress, for example, is that stress which, when removed, produces a permanent strain or "set" of 0.1 % of the original gauge length – see Fig. 1.4(a).

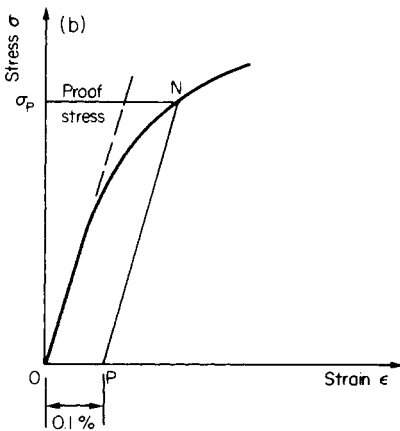


Fig. 1.4. (a) Determination of 0.1 % proof stress.

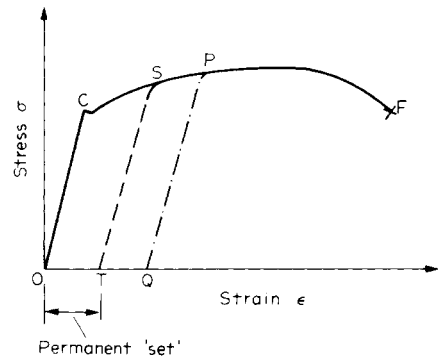


Fig. 1.4. (b) Permanent deformation or "set" after straining beyond the yield point.

The 0.1 % proof stress value may be determined from the tensile test curve for the material in question as follows:

Mark the point *P* on the strain axis which is equivalent to 0.1 % strain. From *P* draw a line parallel with the initial straight line portion of the tensile test curve to cut the curve in *N*. The stress corresponding to *N* is then the 0.1 % *proof stress*. A material is considered to satisfy its specification if the permanent set is no more than 0.1 % after the proof stress has been applied for 15 seconds and removed.

Beyond the yield point some increase in load is required to take the strain to point *E* on the graph. Between *D* and *E* the material is said to be in the *elastic-plastic* state, some of the section remaining elastic and hence contributing to recovery of the original dimensions if load is removed, the remainder being plastic. Beyond *E* the cross-sectional area of the bar

begins to reduce rapidly over a relatively small length of the bar and the bar is said to *neck*. This necking takes place whilst the load reduces, and fracture of the bar finally occurs at point *F*.

The nominal stress at failure, termed the *maximum* or *ultimate tensile stress*, is given by the load at *E* divided by the original cross-sectional area of the bar. (This is also known as the *tensile strength* of the material of the bar.) Owing to the large reduction in area produced by the necking process the actual stress at fracture is often greater than the above value. Since, however, designers are interested in maximum loads which can be carried by the complete cross-section, the stress at fracture is seldom of any practical value.

If load is removed from the test specimen after the yield point *C* has been passed, e.g. to some position *S*, Fig. 1.4(b), the unloading line *ST* can, for most practical purposes, be taken to be linear. Thus, despite the fact that loading to *S* comprises both elastic (*OC*) and partially plastic (*CS*) portions, the unloading procedure is totally elastic. A second load cycle, commencing with the permanent elongation associated with the strain *OT*, would then follow the line *TS* and continue along the previous curve to failure at *F*. It will be observed, however, that the repeated load cycle has the effect of increasing the elastic range of the material, i.e. raising the effective yield point from *C* to *S*, while the tensile strength is unaltered. The procedure could be repeated along the line *PQ*, etc., and the material is said to have been *work hardened*.

In fact, careful observation shows that the material will no longer exhibit true elasticity since the unloading and reloading lines will form a small *hysteresis loop*, neither being precisely linear. Repeated loading and unloading will produce a yield point approaching the ultimate stress value but the elongation or strain to failure will be much reduced.

Typical stress-strain curves resulting from tensile tests on other engineering materials are shown in Figs 1.5 to 1.7.

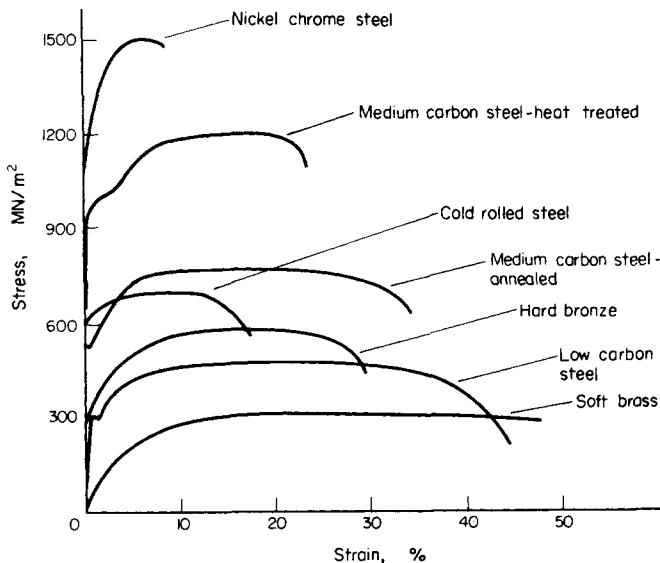


Fig. 1.5. Tensile test curves for various metals.

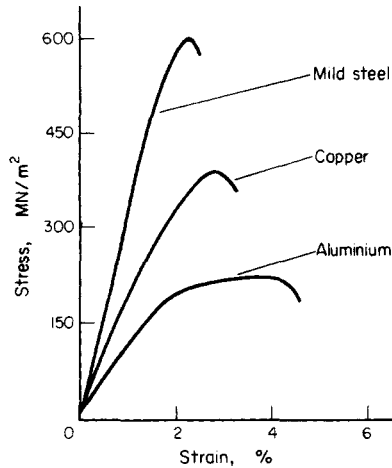


Fig. 1.6. Typical stress-strain curves for hard drawn wire materials—note large reduction in strain values from those of Fig. 1.5.

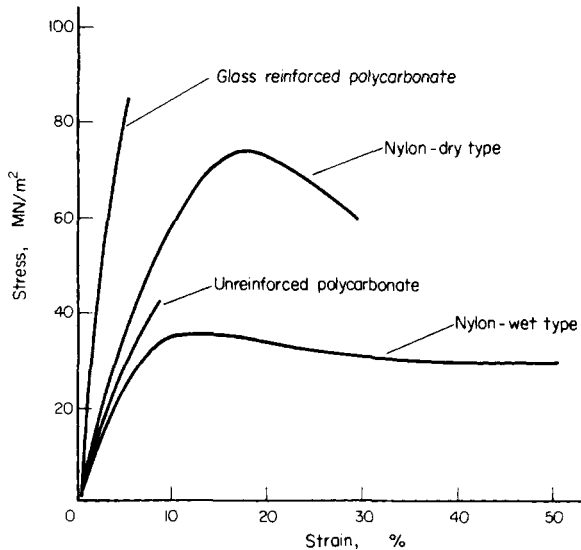


Fig. 1.7. Typical tension test results for various types of nylon and polycarbonate.

After completing the standard tensile test it is usually necessary to refer to some “British Standard Specification” or “Code of Practice” to ensure that the material tested satisfies the requirements, for example:

BS 4360	British Standard Specification for Weldable Structural Steels.
BS 970	British Standard Specification for Wrought Steels.
BS 153	British Standard Specification for Steel Girder Bridges.
BS 449	British Standard Specification for the use of Structural Steel in Building, etc.

1.8. Ductile materials

It has been observed above that the partially plastic range of the graph of Fig. 1.3 covers a much wider part of the strain axis than does the elastic range. Thus the extension of the material over this range is considerably in excess of that associated with elastic loading. The capacity of a material to allow these large extensions, i.e. the ability to be drawn out plastically, is termed its *ductility*. Materials with high ductility are termed *ductile* materials, members with low ductility are termed *brittle* materials. A quantitative value of the ductility is obtained by measurements of the *percentage elongation* or *percentage reduction in area*, both being defined below.

$$\text{Percentage elongation} = \frac{\text{increase in gauge length to fracture}}{\text{original gauge length}} \times 100$$

$$\text{Percentage reduction in area} = \frac{\text{reduction in cross-sectional area of necked portion}}{\text{original area}} \times 100$$

The latter value, being independent of any selected gauge length, is generally taken to be the more useful measure of ductility for reference purposes.

A property closely related to ductility is *malleability*, which defines a material's ability to be hammered out into thin sheets. A typical example of a malleable material is lead. This is used extensively in the plumbing trade where it is hammered or beaten into corners or joints to provide a weatherproof seal. Malleability thus represents the ability of a material to allow permanent extensions in all lateral directions under compressive loadings.

1.9. Brittle materials

A brittle material is one which exhibits relatively small extensions to fracture so that the partially plastic region of the tensile test graph is much reduced (Fig. 1.8). Whilst Fig. 1.3 referred to a low carbon steel, Fig. 1.8 could well refer to a much higher strength steel with a higher carbon content. There is little or no necking at fracture for brittle materials.

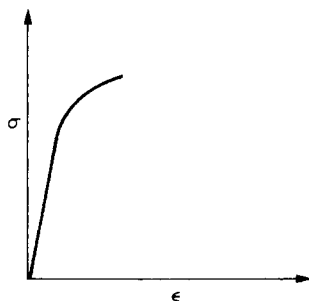


Fig. 1.8. Typical tensile test curve for a brittle material.

Typical variations of mechanical properties of steel with carbon content are shown in Fig. 1.9.

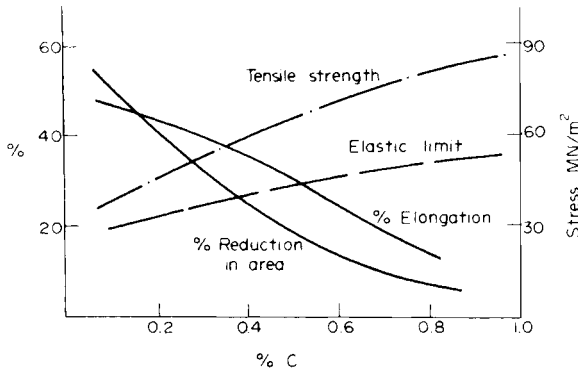


Fig. 1.9. Variation of mechanical properties of steel with carbon content.

1.10. Poisson's ratio

Consider the rectangular bar of Fig. 1.10 subjected to a tensile load. Under the action of this load the bar will increase in length by an amount δL giving a longitudinal strain in the bar of

$$\varepsilon_L = \frac{\delta L}{L}$$

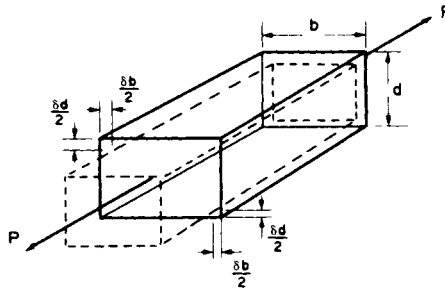


Fig. 1.10.

The bar will also exhibit, however, a *reduction* in dimensions laterally, i.e. its breadth and depth will both reduce. The associated lateral strains will both be equal, will be of opposite sense to the longitudinal strain, and will be given by

$$\varepsilon_{\text{lat}} = -\frac{\delta b}{b} = -\frac{\delta d}{d}$$

Provided the load on the material is retained within the elastic range the ratio of the lateral and longitudinal strains will always be constant. This ratio is termed *Poisson's ratio*.

i.e.
$$\text{Poisson's ratio } (\nu) = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{(-\delta d/d)}{\delta L/L} \quad (1.4)$$

The negative sign of the lateral strain is normally ignored to leave Poisson's ratio simply as

a ratio of strain magnitudes. It must be remembered, however, that the longitudinal strain induces a lateral strain of opposite sign, e.g. tensile longitudinal strain induces compressive lateral strain.

For most engineering materials the value of ν lies between 0.25 and 0.33. Since

$$\text{longitudinal strain} = \frac{\text{longitudinal stress}}{\text{Young's modulus}} = \frac{\sigma}{E} \quad (1.4a)$$

Hence

$$\text{lateral strain} = \nu \frac{\sigma}{E} \quad (1.4b)$$

1.11. Application of Poisson's ratio to a two-dimensional stress system

A two-dimensional stress system is one in which all the stresses lie within one plane such as the X - Y plane. From the work of §1.10 it will be seen that if a material is subjected to a tensile stress σ on one axis producing a strain σ/E and hence an extension on that axis, it will be subjected simultaneously to a lateral strain of ν times σ/E on any axis at right angles. This lateral strain will be compressive and will result in a compression or reduction of length on this axis.

Consider, therefore, an element of material subjected to two stresses at right angles to each other and let both stresses, σ_x and σ_y , be considered tensile, see Fig. 1.11.

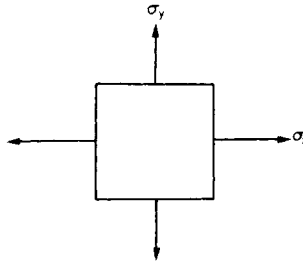


Fig. 1.11. Simple two-dimensional system of direct stresses.

The following strains will be produced:

- (a) in the X direction resulting from $\sigma_x = \sigma_x/E$,
- (b) in the Y direction resulting from $\sigma_y = \sigma_y/E$.
- (c) in the X direction resulting from $\sigma_y = -\nu(\sigma_y/E)$,
- (d) in the Y direction resulting from $\sigma_x = -\nu(\sigma_x/E)$.

strains (c) and (d) being the so-called *Poisson's ratio strain*, opposite in sign to the applied strains, i.e. compressive.

The total strain in the X direction will therefore be given by:

$$\epsilon_x = \frac{\sigma_x}{E} - \nu \frac{\sigma_y}{E} = \frac{1}{E}(\sigma_x - \nu\sigma_y) \quad (1.5)$$

and the total strain in the Y direction will be:

$$\epsilon_y = \frac{\sigma_y}{E} - \nu \frac{\sigma_x}{E} = \frac{1}{E} (\sigma_y - \nu \sigma_x) \quad (1.6)$$

If any stress is, in fact, compressive its value must be substituted in the above equations together with a negative sign following the normal sign convention.

1.12. Shear stress

Consider a block or portion of material as shown in Fig. 1.12a subjected to a set of equal and opposite forces Q . (Such a system could be realised in a bicycle brake block when contacted with the wheel.) There is then a tendency for one layer of the material to slide over another to produce the form of failure shown in Fig. 1.12b. If this failure is restricted, then a *shear stress* τ is set up, defined as follows:

$$\text{shear stress } (\tau) = \frac{\text{shear load}}{\text{area resisting shear}} = \frac{Q}{A}$$

This shear stress will always be *tangential* to the area on which it acts; direct stresses, however, are always *normal* to the area on which they act.

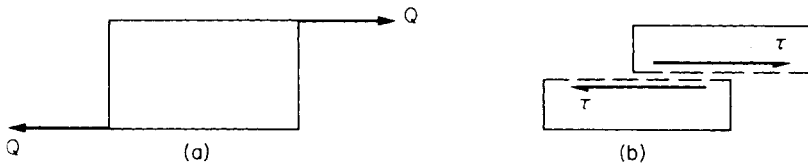


Fig. 1.12. Shear force and resulting shear stress system showing typical form of failure by relative sliding of planes.

1.13. Shear strain

If one again considers the block of Fig. 1.12a to be a bicycle brake block it is clear that the rectangular shape of the block will not be retained as the brake is applied and the shear forces introduced. The block will in fact change shape or “strain” into the form shown in Fig. 1.13. The angle of deformation γ is then termed the *shear strain*.

Shear strain is measured in radians and hence is non-dimensional, i.e. it has no units.

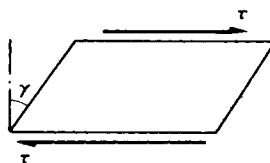


Fig. 1.13. Deformation (shear strain) produced by shear stresses.

1.14. Modulus of rigidity

For materials within the elastic range the shear strain is proportional to the shear stress producing it,
i.e.

$$\frac{\text{shear stress}}{\text{shear strain}} = \frac{\tau}{\gamma} = \text{constant} = G \quad (1.7)$$

The constant G is termed the *modulus of rigidity* or *shear modulus* and is directly comparable to the modulus of elasticity used in the direct stress application. The term *modulus* thus implies a ratio of stress to strain in each case.

1.15. Double shear

Consider the simple riveted lap joint shown in Fig. 1.14a. When load is applied to the plates the rivet is subjected to shear forces tending to shear it on one plane as indicated. In the butt joint with two cover plates of Fig. 1.14b, however, each rivet is subjected to possible shearing on two faces, i.e. *double shear*. In such cases twice the area of metal is resisting the applied forces so that the shear stress set up is given by

$$\text{shear stress } \tau \text{ (in double shear)} = \frac{P}{2A} \quad (1.8)$$

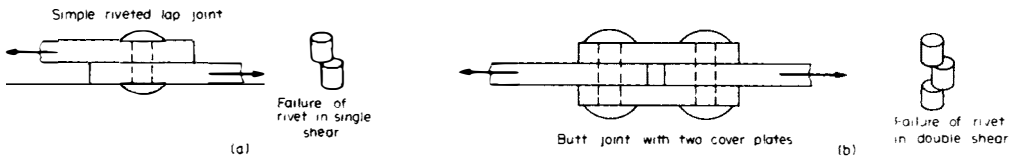
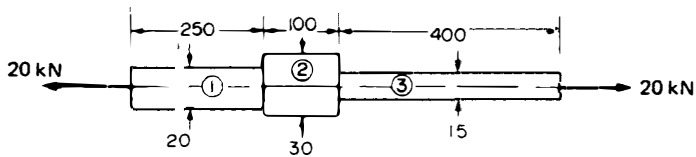


Fig. 1.14. (a) Single shear. (b) Double shear.

Examples

Example 1.1

Determine the stress in each section of the bar shown in Fig. 1.19 when subjected to an axial tensile load of 20 kN. The central section is 30 mm square cross-section; the other portions are of circular section, their diameters being indicated. What will be the total extension of the bar? For the bar material $E = 210 \text{ GN/m}^2$.



Not to scale
all dimensions mm

Fig. 1.19.

Solution

$$\text{Stress} = \frac{\text{force}}{\text{area}} = \frac{P}{A}$$

$$\text{Stress in section (1)} = \frac{20 \times 10^3}{\frac{\pi(20 \times 10^{-3})^2}{4}} = \frac{80 \times 10^3}{\pi \times 400 \times 10^{-6}} = 63.66 \text{ MN/m}^2$$

$$\text{Stress in section (2)} = \frac{20 \times 10^3}{30 \times 30 \times 10^{-6}} = 22.2 \text{ MN/m}^2$$

$$\text{Stress in section (3)} = \frac{20 \times 10^3}{\frac{\pi(15 \times 10^{-3})^2}{4}} = \frac{80 \times 10^3}{\pi \times 225 \times 10^{-6}} = 113.2 \text{ MN/m}^2$$

Now the extension of a bar can always be written in terms of the stress in the bar since

$$E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\delta/L}$$

$$\text{i.e.} \quad \delta = \frac{\sigma L}{E}$$

$$\therefore \text{extension of section (1)} = 63.66 \times 10^6 \times \frac{250 \times 10^{-3}}{210 \times 10^9} = 75.8 \times 10^{-6} \text{ m}$$

$$\text{extension of section (2)} = 22.2 \times 10^6 \times \frac{100 \times 10^{-3}}{210 \times 10^9} = 10.6 \times 10^{-6} \text{ m}$$

$$\text{extension of section (3)} = 113.2 \times 10^6 \times \frac{400 \times 10^{-3}}{210 \times 10^9} = 215.6 \times 10^{-6} \text{ m}$$

$$\begin{aligned} \therefore \text{total extension} &= (75.8 + 10.6 + 215.6) \times 10^{-6} \\ &= 302 \times 10^{-6} \text{ m} \\ &= \mathbf{0.302 \text{ mm}} \end{aligned}$$

Example 1.2

(a) A 25 mm diameter bar is subjected to an axial tensile load of 100 kN. Under the action of this load a 200 mm gauge length is found to extend 0.19×10^{-3} mm. Determine the modulus of elasticity for the bar material.

(b) If, in order to reduce weight whilst keeping the external diameter constant, the bar is bored axially to produce a cylinder of uniform thickness, what is the maximum diameter of bore possible given that the maximum allowable stress is 240 MN/m^2 ? The load can be assumed to remain constant at 100 kN.

(c) What will be the change in the outside diameter of the bar under the limiting stress quoted in (b)? ($E = 210 \text{ GN/m}^2$ and $\nu = 0.3$).

Solution

(a) From eqn. (1.2),

$$\begin{aligned}\text{Young's modulus } E &= \frac{PL}{A \delta L} \\ &= \frac{100 \times 10^3 \times 200 \times 10^{-3}}{\frac{\pi(25 \times 10^{-3})^2}{4} \times 0.19 \times 10^{-3}} \\ &= 214 \text{ GN/m}^2\end{aligned}$$

(b) Let the required bore diameter be d mm; the cross-sectional area of the bar will then be reduced to

$$A = \left[\frac{\pi \times 25^2}{4} - \frac{\pi d^2}{4} \right] 10^{-6} = \frac{\pi}{4} (25^2 - d^2) 10^{-6} \text{ m}^2$$

$$\therefore \text{ stress in bar } = \frac{P}{A} = \frac{4 \times 100 \times 10^3}{\pi(25^2 - d^2) 10^{-6}}$$

But this stress is restricted to a maximum allowable value of 240 MN/m^2 .

$$\therefore 240 \times 10^6 = \frac{4 \times 100 \times 10^3}{\pi(25^2 - d^2) 10^{-6}}$$

$$\therefore 25^2 - d^2 = \frac{4 \times 100 \times 10^3}{240 \times 10^6 \times \pi \times 10^{-6}} = 530.5$$

$$\therefore d^2 = 94.48 \quad \text{and} \quad d = 9.72 \text{ mm}$$

The maximum bore possible is thus **9.72 mm**.

(c) The change in the outside diameter of the bar will be obtained from the lateral strain,

$$\text{i.e. lateral strain} = \frac{\delta d}{d}$$

$$\text{But Poisson's ratio } \nu = \frac{\text{lateral strain}}{\text{longitudinal strain}}$$

$$\text{and longitudinal strain} = \frac{\sigma}{E} = \frac{240 \times 10^6}{210 \times 10^9}$$

$$\therefore \frac{\delta d}{d} = -\nu \frac{\sigma}{E} = -\frac{0.3 \times 240 \times 10^6}{210 \times 10^9}$$

$$\begin{aligned}\therefore \text{ change in outside diameter} &= -\frac{0.3 \times 240 \times 10^6}{210 \times 10^9} \times 25 \times 10^{-3} \\ &= -8.57 \times 10^{-6} \text{ m (a reduction)}\end{aligned}$$

Example 1.3

The coupling shown in Fig. 1.20 is constructed from steel of rectangular cross-section and is designed to transmit a tensile force of 50 kN. If the bolt is of 15 mm diameter calculate:

- the shear stress in the bolt;
- the direct stress in the plate;
- the direct stress in the forked end of the coupling.

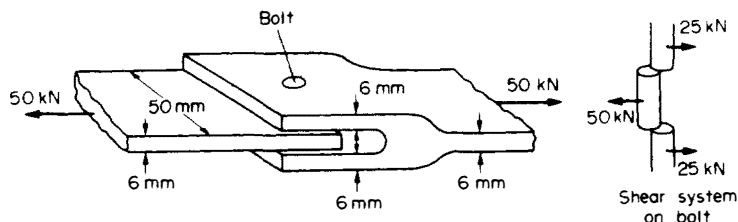


Fig. 1.20.

Solution

(a) The bolt is subjected to double shear, tending to shear it as shown in Fig. 1.14b. There is thus twice the area of the bolt resisting the shear and from eqn. (1.8)

$$\begin{aligned}\text{shear stress in bolt} &= \frac{P}{2A} = \frac{50 \times 10^3 \times 4}{2 \times \pi (15 \times 10^{-3})^2} \\ &= \frac{100 \times 10^3}{\pi (15 \times 10^{-3})^2} = \mathbf{141.5 \text{ MN/m}^2}\end{aligned}$$

(b) The plate will be subjected to a direct tensile stress given by

$$\sigma = \frac{P}{A} = \frac{50 \times 10^3}{50 \times 6 \times 10^{-6}} = \mathbf{166.7 \text{ MN/m}^2}$$

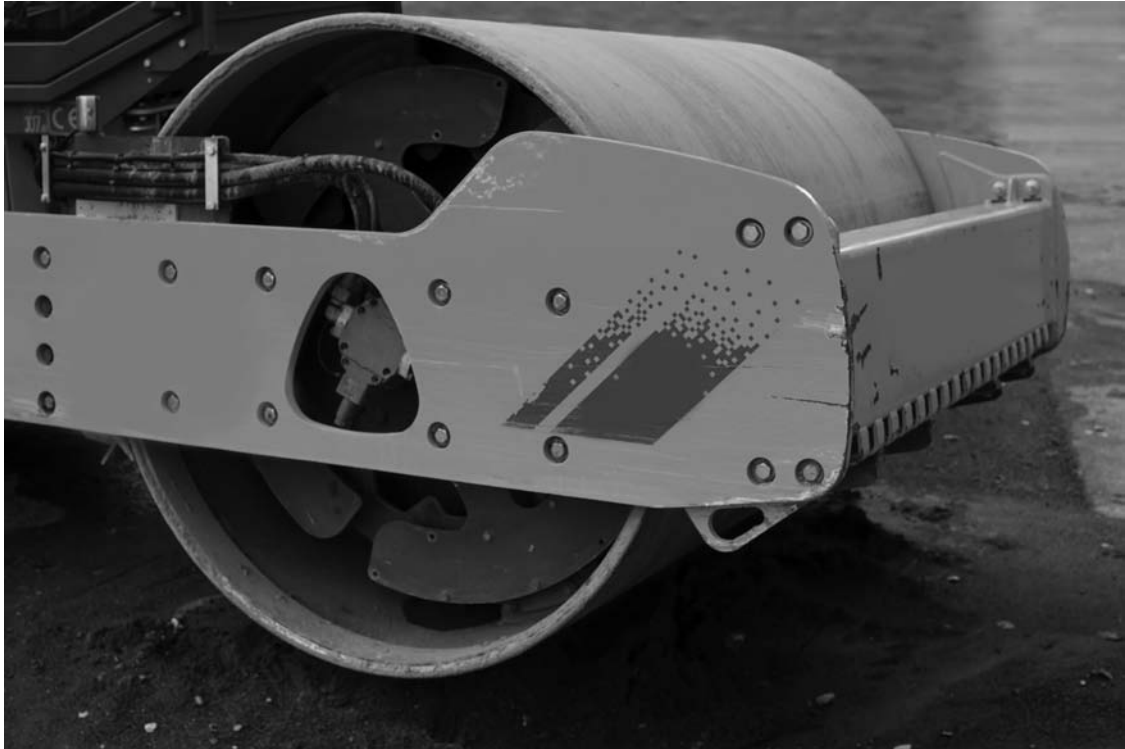
(c) The force in the coupling is shared by the forked end pieces, each being subjected to a direct stress

$$\sigma = \frac{P}{A} = \frac{25 \times 10^3}{50 \times 6 \times 10^{-6}} = \mathbf{83.3 \text{ MN/m}^2}$$

MODULE 2

16

Planar Kinematics of Rigid Bodies



Rolling without slipping is a commonly encountered problem of kinematics. This chapter contains several problems that analyze rolling, such as Sample Problem 16.4, Prob.16.20 and 16.23. (© Spe/Dreamstime.com)

16.1 *Introduction*

A body is said to be *rigid* if the distance between any two points in the body remains constant. In other words, a rigid body does not deform. Of course, the rigid-body concept is an idealization, because all bodies deform to some extent when subjected to forces. But if the deformation is sufficiently small (“small” usually means negligible when compared to the dimensions of the body), the assumption of rigidity is justified.

This chapter is concerned only with the kinematics of *plane motion* of rigid bodies. A body undergoes plane motion if all points in the body remain a constant distance from a fixed reference plane, called the *plane of motion* of the body.

In other words, all points in the body move in planes that are parallel to the plane of motion. There are three categories of plane motion:

- *Translation* is the special case in which the body moves without rotation; that is, any line in the body remains parallel to its initial position, as shown in Fig. 16.1(a).^{*} Because all points in the body have the same displacement, the motion of one point determines the motion of the entire body.

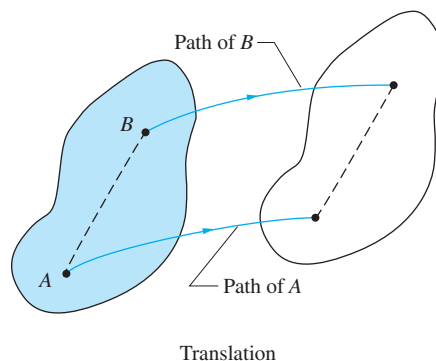


Fig. 16.1(a)

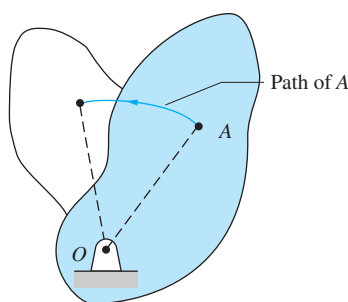


Fig. 16.1(b)

- *Rotation about a fixed axis* is the special case in which a line in the body, called the *axis of rotation*, is fixed in space. Consequently, each point not on the axis of rotation moves in a circle about the axis, as illustrated in Fig. 16.1(b) (the axis of rotation at O is perpendicular to the plane of the paper).
- *General plane motion* is the superposition of translation and rotation. The rolling disk in Fig. 16.1(c) is an example of such motion: the disk is translating and rotating simultaneously.

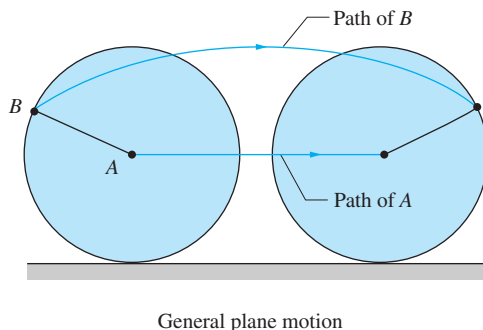


Fig. 16.1(c)

^{*}When illustrating plane motion, it is often convenient to show a representative cross section of the body that is parallel to the plane of motion.

16.2 Plane Angular Motion

In this article, we introduce the angular displacement, angular velocity, and angular acceleration of lines and rigid bodies that move in a plane.

Angular Position Consider the rigid body $\mathcal{B}$ in Fig. 16.2 that is moving in the xy -plane. Let AB be a line that is embedded in $\mathcal{B}$ and lies in the plane of motion.* The angle $\theta(t)$ between AB and a fixed reference line, such as the x -axis, is known as the *angular position coordinate of line AB* .

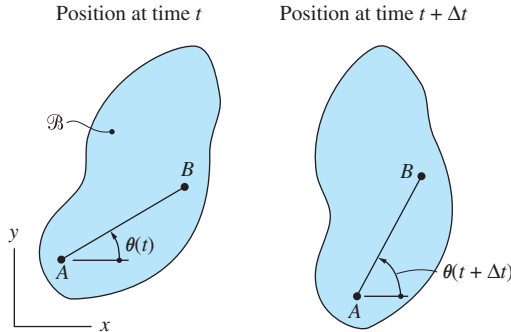


Fig. 16.2

Angular Displacement During the time interval Δt , the angular position coordinate of AB changes from $\theta(t)$ to $\theta(t + \Delta t)$, as shown in Fig. 16.2. The angular displacement of line AB during this time interval is defined as

$$\Delta\theta = \theta(t + \Delta t) - \theta(t) \quad (16.1)$$

We now show that all lines in $\mathcal{B}$ that lie in the plane of motion have the same angular displacement. Figure 16.3 shows two such lines AB and CD , with the

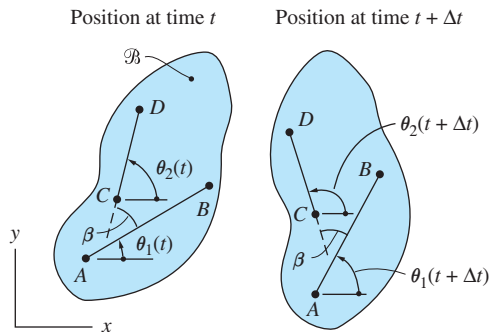


Fig. 16.3

*Here, and in the discussion that follows, the term “plane of motion” also applies to any plane that is parallel to the plane of motion.

angular position coordinates $\theta_1(t)$ and $\theta_2(t)$, respectively. Because the body is rigid, the angle β between the lines does not change between the two positions shown in the figure; that is, $\beta = \theta_2(t) - \theta_1(t) = \theta_2(t + \Delta t) - \theta_1(t + \Delta t)$. It follows that $\theta_1(t + \Delta t) - \theta_1(t) = \theta_2(t + \Delta t) - \theta_2(t)$, or

$$\Delta\theta_1 = \Delta\theta_2$$

Because AB and CD can be chosen arbitrarily in the plane of the motion, we conclude that all lines in the plane of the motion have the same angular displacement. Therefore, $\Delta\theta$ is also called the *angular displacement of body $\mathcal{B}$* .

Angular Velocity The *angular velocity ω of line AB* is defined as the time derivative of its angular position coordinate θ :

$$\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} = \dot{\theta} \quad (16.2)$$

Because all lines in the plane of motion have the same angular displacement $\Delta\theta$, they also have the same angular velocity. Therefore, ω is also called the *angular velocity of body $\mathcal{B}$* . Common units for angular velocity are rad/s and rev/min.

Angular Acceleration The *angular acceleration α of line AB* is defined to be the time derivative of its angular velocity:

$$\alpha = \frac{d\omega}{dt} = \dot{\omega} \quad \text{or} \quad \alpha = \frac{d^2\theta}{dt^2} = \ddot{\theta} \quad (16.3)$$

Because the angular velocity ω is the same for all lines in the plane of motion, α is also called the *angular acceleration of body $\mathcal{B}$* . The units of angular acceleration are usually taken to be rad/s².

The time t in Eq. (16.3) can be eliminated as an explicit variable by using the chain rule for differentiation:

$$\alpha = \frac{d\omega}{dt} = \frac{d\omega}{d\theta} \frac{d\theta}{dt} = \omega \frac{d\omega}{d\theta} \quad (16.4)$$

- Observe that positive directions of angular velocity and angular acceleration are the same as the assumed positive direction of θ .
- Equations (16.2)–(16.4) are analogous to Eqs. (12.9) and (12.10) for rectilinear motion of a particle: $v = \dot{x}$, $a = \dot{v} = \ddot{x}$, and $\alpha = v(dv/dx)$. Therefore, the mathematical methods presented in Art. 12.4 can also be used to analyze plane angular rotation.

Vector Representation of Angular Motion Because $\Delta\theta$, ω , and α possess magnitude and direction, it is sometimes convenient to represent them as vectors using

the right-hand rule. Assuming that we have a right-handed coordinate system (the positive z -axis in Fig. 16.2 would point out of the paper), we can write

$$\Delta\theta = \Delta\theta\mathbf{k} \quad \omega = \dot{\omega}\mathbf{k} = \dot{\theta}\mathbf{k} \quad \alpha = \dot{\alpha}\mathbf{k} = \dot{\omega}\mathbf{k} = \ddot{\theta}\mathbf{k} \quad (16.5)$$

Comment on Three-Dimensional Motion In planar motion, angular displacement is a vector quantity because it has magnitude ($\Delta\theta$), has direction (perpendicular to the plane of motion), and obeys the parallelogram law for addition (summing collinear vectors is a special case of parallelogram addition). It follows that angular velocity and angular acceleration are also vectors. However, if the motion is three-dimensional, angular displacements do not obey the parallelogram law for addition. Therefore, angular displacements, in general, are *not* vectors.

As an illustration of the nonvector character of angular displacements, consider a book that is placed in the initial position shown in Fig. 16.4(a) and given the angular rotations $\Delta\theta_x = 90^\circ$ (rotation about the x -axis) and $\Delta\theta_y = 90^\circ$ (rotation about the y -axis). Figure 16.4(b) shows the final position of the book if the rotations are performed in the order $\Delta\theta_x$ followed by $\Delta\theta_y$. Figure 16.4(c) shows the result if the rotations are performed in the order $\Delta\theta_y$ followed by $\Delta\theta_x$. From these figures, we see that the final orientation of the book depends on the order

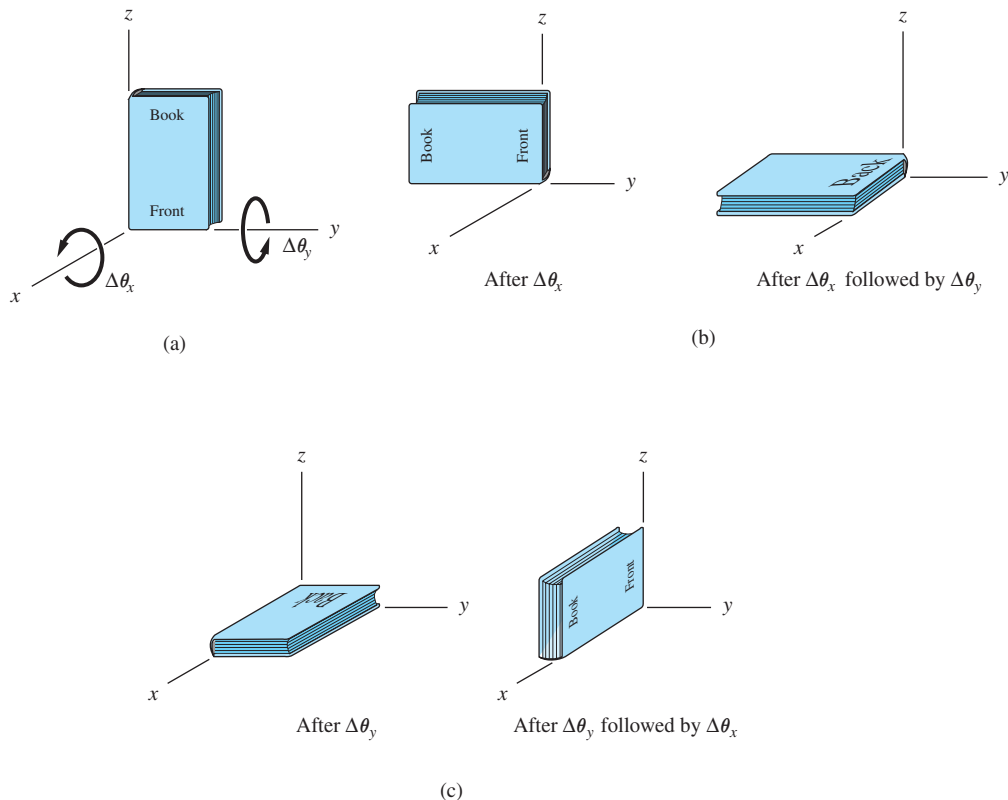


Fig. 16.4

in which the rotations are performed. We conclude that angular displacements generally are not vectors because they do not obey the commutative property of vector addition. However, it can be shown* that differential (infinitesimal) angular displacements are vectors, even in three-dimensional motion. For this reason, angular velocity and angular acceleration are *always* vector quantities.

16.3 Rotation about a Fixed Axis

Rotation about a fixed axis is the special case of plane motion in which one line in the body, called the *axis of rotation*, is fixed in space. Figure 16.5 shows a rigid body that is rotating about an axis. We let B be a point in the body that is a distance R from the axis. Because the body is rigid, the path of B is a circle of radius R that lies in a plane perpendicular to the axis of rotation. The center O of the circle lies on the axis, and the angular position coordinate of the radial line OB is denoted by θ . Since OB lies in the plane of motion, the angular velocity and angular acceleration of the body are, according to Eqs. (16.2) and (16.3),

$$\omega = \dot{\theta} \quad \alpha = \dot{\omega} = \ddot{\theta} \quad (16.6)$$

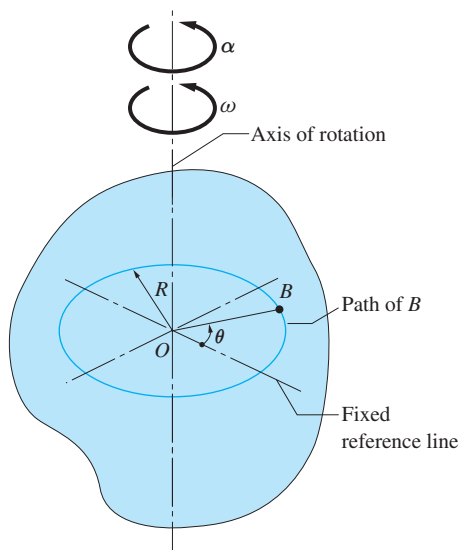


Fig. 16.5

Kinematics of a Point in the Body The motion of a particle along a circular path of radius R was previously discussed using path coordinates in Art. 13.2. With θ representing the angular velocity of the radial line to the particle,

*See *Principles of Dynamics*, 2e, Donald T. Greenwood, Prentice Hall, 1988, p. 350.

the velocity (v) and the normal (a_n) and tangential (a_t) components of acceleration of the particle were found to be

$$v = R\dot{\theta} \quad (13.10, \text{repeated})$$

$$a_n = R\dot{\theta}^2 \quad a_t = R\ddot{\theta} \quad (13.11, \text{repeated})$$

Substituting for $\dot{\theta}$ and $\ddot{\theta}$ from Eq. (16.6), the velocity and acceleration components of point B become

$$\begin{aligned} v &= R\omega \\ a_n &= R\omega^2 = \frac{v^2}{R} = v\omega \\ a_t &= R\alpha \end{aligned} \quad (16.7a)$$

The velocity and acceleration vectors of point B lie in the plane of motion (the plane of the circle), as shown in Fig. 16.6(a). The velocity v and the tangential component of acceleration a_t are tangent to the circle, pointing in the direction of increasing θ . The normal component of acceleration a_n is directed toward the center of the circular path.

Vector Representation of Kinematics It is sometimes convenient to compute the velocity and acceleration of a point using vector algebra.

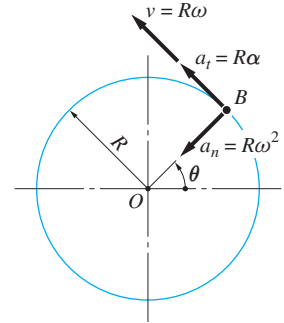
In Fig. 16.6(b), we let $\mathbf{r}$ be the position vector of B relative to an arbitrary reference point A that lies on the axis of rotation. The angle between $\mathbf{r}$ and the axis is denoted by β . We will show that the following vector equations give the magnitude and direction for the velocity and acceleration of point B :

$$\begin{aligned} \mathbf{v} &= \boldsymbol{\omega} \times \mathbf{r} \\ \mathbf{a}_n &= \boldsymbol{\omega} \times \mathbf{v} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}) \\ \mathbf{a}_t &= \boldsymbol{\alpha} \times \mathbf{r} \end{aligned} \quad (16.7b)$$

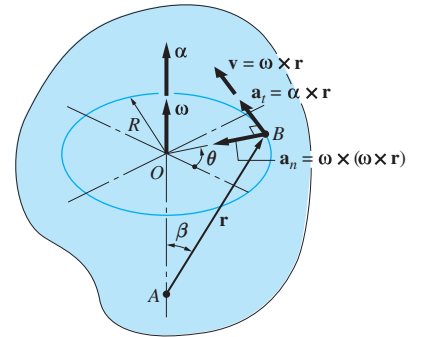
Using the properties of the cross product, the magnitudes of the vectors in Eqs. (16.7b) are

$$\begin{aligned} v &= |\boldsymbol{\omega} \times \mathbf{r}| = \omega r \sin \beta = R\omega \\ a_n &= |\boldsymbol{\omega} \times \mathbf{v}| = \omega v \sin 90^\circ = \omega v = R\omega^2 \\ a_t &= |\boldsymbol{\alpha} \times \mathbf{r}| = \alpha r \sin \beta = R\alpha \end{aligned}$$

These results agree with Eqs. (16.7a). Using the right-hand rule for the cross products, it is not difficult to see that the directions of the vectors in Eqs. (16.7b) are the same as those shown in Fig. 16.6(a).

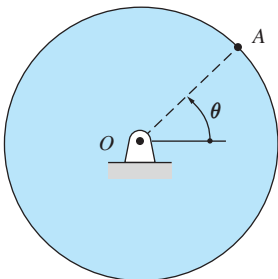


(a)



(b)

Fig. 16.6



Sample Problem 16.1

The disk rotates about a fixed axis at O . During the period $t = 0$ to $t = 4$ s, the angular position of the line OA in the disk varies as $\theta(t) = t^3 - 12t + 6$ rad, where t is in seconds. Determine (1) the angular velocity and the angular acceleration of the disk at the end of the period; (2) the angular displacement of the disk during the period; and (3) the total angle turned through by the disk during the period.

Solution

Part 1

The angular velocity and angular acceleration of the disk are

$$\omega = \dot{\theta} = 3t^2 - 12 \text{ rad/s} \quad \alpha = \dot{\omega} = 6t \text{ rad/s}^2$$

When $t = 4$ s, we have

$$\omega = 3(4)^2 - 12 = 36 \text{ rad/s (CCW)} \quad \text{Answer}$$

$$\alpha = 6(4) = 24 \text{ rad/s}^2 \text{ (CCW)} \quad \text{Answer}$$

Part 2

The angular positions of the line OA at the beginning and at the end of the period are

$$\theta|_{t=0} = 6 \text{ rad}$$

$$\theta|_{t=4 \text{ s}} = 4^3 - 12(4) + 6 = 22 \text{ rad}$$

Therefore, the angular displacement of the disk from $t = 0$ to $t = 4$ s is

$$\Delta\theta = \theta|_{t=4 \text{ s}} - \theta|_{t=0} = 22 - 6 = 16 \text{ rad (CCW)} \quad \text{Answer}$$

Part 3

Note that the direction of rotation of the disk changes when $\omega = 0$, that is, when

$$3t^2 - 12 = 0 \quad t = 2 \text{ s}$$

The angular position of OA at that time is

$$\theta|_{t=2 \text{ s}} = 2^3 - 12(2) + 6 = -10 \text{ rad}$$

We conclude that the disk rotates clockwise ($\omega < 0$) between $t = 0$ and $t = 2$ s, its angular displacement being

$$\Delta\theta_1 = \theta|_{t=2 \text{ s}} - \theta|_{t=0} = -10 - 6 = -16 \text{ rad}$$

Between $t = 2$ s and $t = 4$ s, the rotation is counterclockwise ($\omega > 0$); the corresponding angular displacement is

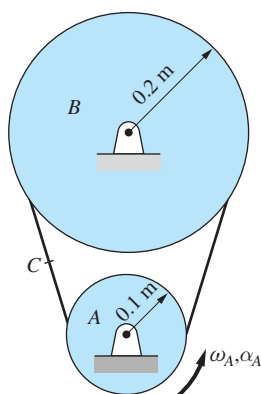
$$\Delta\theta_2 = \theta|_{t=4\text{ s}} - \theta|_{t=2\text{ s}} = 22 - (-10) = 32 \text{ rad}$$

Therefore, the total angle turned through by the disk from $t = 0$ to $t = 4$ s is

$$|\Delta\theta_1| + |\Delta\theta_2| = 16 + 32 = 48 \text{ rad} \quad \textbf{Answer}$$

Sample Problem 16.2

Pulley B is being driven by the motorized pulley A that is rotating at $\omega_A = 20$ rad/s. At time $t = 0$, the current in the motor is cut off, and friction in the bearings causes the pulleys to coast to a stop. The angular acceleration of A during the deceleration is $\alpha_A = -2.5t$ rad/s², where t is in seconds. Assuming that the drive belt does not slip on the pulleys, determine (1) the angular velocity of B as a function of time; (2) the angular displacement of B during the period of coasting; and (3) the acceleration of point C on the straight portion of the belt as a function of time.



Solution

Part 1

Because the belt does not slip, every point on the belt that is in contact with a pulley has the same velocity as the adjacent point on the pulley. Therefore, the speed of any point on the belt is

$$v = R_A\omega_A = R_B\omega_B \quad (\text{a})$$

so that

$$\omega_B = \frac{R_A}{R_B}\omega_A = \frac{0.10}{0.20}\omega_A = 0.5\omega_A$$

Differentiating with respect to time, we obtain for the angular acceleration of pulley B

$$\alpha_B = 0.5\alpha_A = 0.5(-2.5t) = -1.25t \text{ rad/s}^2$$

Because $\alpha_B = d\omega_B/dt$, we have $d\omega_B = \alpha_B dt$, or

$$\omega_B = \int \alpha_B dt = \int -1.25t dt = -0.625t^2 + C_1$$

The initial condition, $\omega_B = 20$ rad/s when $t = 0$, yields $C_1 = 20$ rad/s. Hence, the angular velocity of pulley B is

$$\omega_B = -0.625t^2 + 20 \text{ rad/s} \quad \text{Answer}$$

Part 2

We let θ_B be the angular position of a line in B measured from a fixed reference line. Recalling that $\omega_B = d\theta_B/dt$, we integrate $d\theta_B = \omega_B dt$ to obtain

$$\theta_B = \int \omega_B dt = \int (-0.625t^2 + 20)dt = -0.2083t^3 + 20t + C_2$$

Letting $\theta_B = 0$ when $t = 0$, we have $C_2 = 0$, which gives

$$\theta_B = -0.2083t^3 + 20t \text{ rad}$$

The pulley comes to rest when $\omega_B = -0.625t^2 + 20 = 0$, which yields $t = 5.657$ s. The corresponding angular position of the line in B is

$$\theta_B|_{t=5.657 \text{ s}} = -0.2083(5.657)^3 + 20(5.657) = 112.0 \text{ rad}$$

Therefore, the angular displacement of pulley B as it coasts to a stop is

$$\Delta\theta_B = \theta_B|_{t=5.657 \text{ s}} - \theta_B|_{t=0} = 112.0 - 0 = 112.0 \text{ rad} \quad \text{Answer}$$

Because the direction of rotation does not change, the total angle turned through by pulley B during the deceleration is also 112.0 rad.

Part 3

Substituting $R_B = 0.2$ m and $\omega_B = -0.625t^2 + 20$ rad/s into Eq. (a), the speed of point C (which is the same for all points on the belt) is

$$v_C = 0.2(-0.625t^2 + 20) = -0.125t^2 + 4 \text{ m/s} \quad \text{(b)} \quad \text{Answer}$$

Because the path of point C on the belt is a straight line, the acceleration of C is

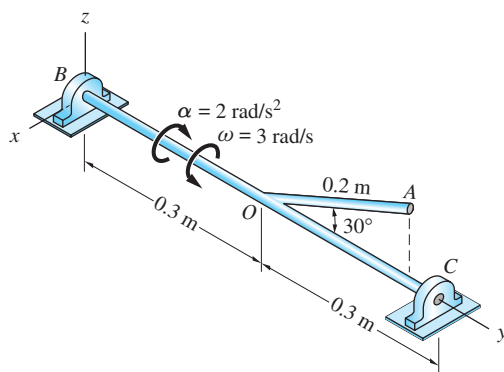
$$a_C = \dot{v}_C = -0.25t \text{ m/s}^2 \quad \text{Answer}$$

We could obtain the same result by observing that a_C is equal to the *tangential component* of acceleration of a point on the rim of pulley B (pulley A could also be used). Thus $a_C = R_B\alpha_B = 0.2(-1.25t) = -0.25t \text{ m/s}^2$.

Observe that the expression for v_C in Eq. (b) is valid for the entire deceleration period ($0 \leq t \leq 5.657$ s), whereas the answer for a_C applies only for those times when C is not in contact with either pulley. Points on the belt follow circular paths when they are in contact with a pulley.

Sample Problem 16.3

The rigid body is rotating about the y -axis. In the position shown in Fig. (a), the angular velocity and angular acceleration of the body are as specified in the figure. Determine the velocity and acceleration vectors of point A in this position using (1) vector equations; and (2) scalar equations.



(a)

Solution

Part 1

Using the right-hand rule, the angular velocity and acceleration vectors of the body are

$$\boldsymbol{\omega} = 3\mathbf{j} \text{ rad/s} \quad \boldsymbol{\alpha} = -2\mathbf{j} \text{ rad/s}^2$$

The position vector of A relative to point O on the axis of rotation is

$$\mathbf{r} = 0.2 \cos 30^\circ \mathbf{j} + 0.2 \sin 30^\circ \mathbf{k} = 0.1732\mathbf{j} + 0.1\mathbf{k} \text{ m}$$

The velocity and acceleration vectors of point A can now be computed from Eqs. (16.7b):

$$\mathbf{v}_A = \boldsymbol{\omega} \times \mathbf{r} = 3\mathbf{j} \times (0.1732\mathbf{j} + 0.1\mathbf{k}) = 0.3\mathbf{i} \text{ m/s} \quad \text{Answer}$$

$$\begin{aligned} \mathbf{a}_A &= \boldsymbol{\omega} \times \mathbf{v} + \boldsymbol{\alpha} \times \mathbf{r} = 3\mathbf{j} \times 0.3\mathbf{i} + (-2\mathbf{j}) \times (0.1732\mathbf{j} + 0.1\mathbf{k}) \\ &= -0.9\mathbf{k} - 0.2\mathbf{i} \text{ m/s}^2 \quad \text{Answer} \end{aligned}$$

Part 2

Figure (b) shows the plane of motion of point A (looking from C toward B). The path of A is a circle of radius $R = 0.2 \sin 38^\circ = 0.1$ m. The magnitudes of the velocity and acceleration components of A are, using Eqs. (16.7a),

$$v_A = R\omega = 0.1(3) = 0.3 \text{ m/s}$$

$$(a_A)_n = R\omega^2 = 0.1(3)^2 = 0.9 \text{ m/s}^2$$

$$(a_A)_t = R\alpha = 0.1(2) = 0.2 \text{ m/s}^2$$

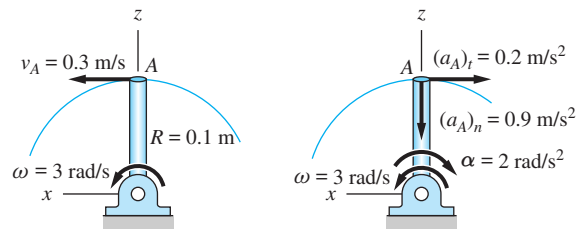
These components are shown in Fig. (b). Their directions were determined as follows:

- $\mathbf{v}_A$ is tangent to the path; its sense is consistent with the counterclockwise direction of ω .
- $(\mathbf{a}_A)_n$ is directed toward the center of the path.
- $(\mathbf{a}_A)_t$ is tangent to the path; its sense is consistent with the clockwise direction of α .

Therefore, the velocity and acceleration vectors of point A are

$$\mathbf{v}_A = 0.3\mathbf{i} \text{ m/s} \quad \text{Answer}$$

$$\mathbf{a}_A = -0.9\mathbf{k} - 0.2\mathbf{i} \text{ m/s}^2 \quad \text{Answer}$$



(b)

16.4 *Relative Motion of Two Points
in a Rigid Body*

The kinematics of relative motion of particles (points) was discussed in Art. 15.2. The definition of relative motion between points A and B yielded the following relationships

$$\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A} \quad (15.3, \text{ repeated})$$

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A} \quad (15.4, \text{ repeated})$$

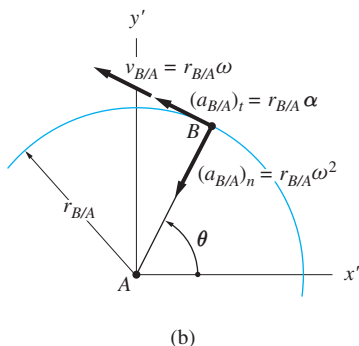
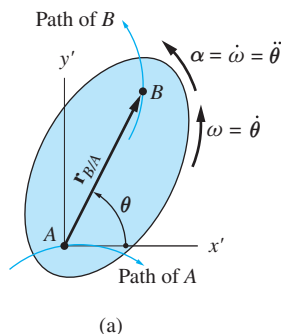


Fig. 16.7

where the subscript B/A denotes the motion of B relative to A . It is useful to recall that $\mathbf{v}_{B/A}$ and $\mathbf{a}_{B/A}$ can be viewed as the velocity and acceleration of B as seen by a *nonrotating* observer attached to point A .

If points A and B belong to the same *translating* (i.e., nonrotating) rigid body, their relative position vector $\mathbf{r}_{B/A}$ is constant, and there is no relative motion ($\mathbf{v}_{B/A} = \mathbf{a}_{B/A} = \mathbf{0}$). Therefore, all points in the translating body have the same velocities and accelerations.

Now consider the body in Fig. 16.7(a) that is undergoing *general plane motion* (simultaneous translation and rotation). The *nonrotating* $x'y'$ -axes are attached to point A . The angular velocity of the body is $\omega = \dot{\theta}$, and its angular acceleration is $\alpha = \dot{\omega} = \ddot{\theta}$, where θ is the angle between $\mathbf{r}_{B/A}$ and the x' -axis. As the body moves, the direction of $\mathbf{r}_{B/A}$ changes, but its magnitude $r_{B/A}$ (the distance between A and B) is constant. It follows that the path of B in the *translating* $x'y'$ -coordinate system is a circle of radius $r_{B/A}$ that lies in the plane of motion, as shown in Fig. 16.7(b). Thus, the motion of B relative to A is due only to the rotation of the body about an axis that passes through point A . Therefore, the relative velocity and relative acceleration between two points on a rigid body can be computed using the equations developed in the previous article for rotation about a fixed axis. With the appropriate changes in notation, Eqs. (16.7a) become

$$\begin{aligned} v_{B/A} &= r_{B/A}\omega \\ (a_{B/A})_n &= r_{B/A}\omega^2 = \frac{v_{B/A}^2}{R} = v_{B/A}\omega \\ (a_{B/A})_t &= r_{B/A}\alpha \end{aligned} \quad (16.8a)$$

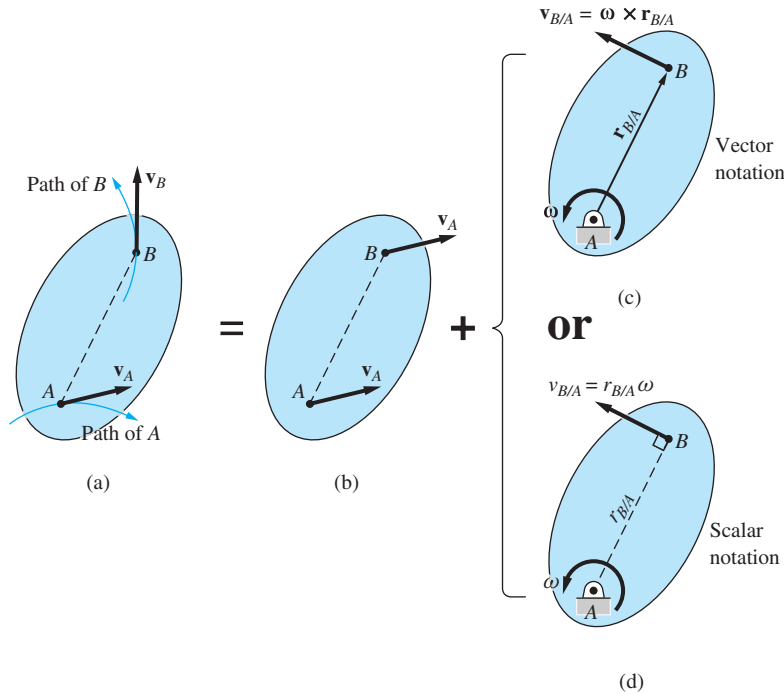
Figure 16.7(b) displays each of the terms in Eqs. (16.8a). This figure is identical to Fig. 16.6(a) for rotation about a fixed axis, except for the notational changes. It is important to understand that the *absolute* angular velocity ($\omega = \dot{\theta}$) and *absolute* angular acceleration ($\alpha = \dot{\omega} = \ddot{\theta}$) are used in the computations. Because the angle θ is measured from the fixed x' -direction, all angular measurements are absolute quantities.

The vector forms of Eqs. (16.8a) are

$$\begin{aligned} \mathbf{v}_{B/A} &= \boldsymbol{\omega} \times \mathbf{r}_{B/A} \\ (\mathbf{a}_{B/A})_n &= \boldsymbol{\omega} \times \mathbf{v}_{B/A} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}_{B/A}) \\ (\mathbf{a}_{B/A})_t &= \boldsymbol{\alpha} \times \mathbf{r}_{B/A} \end{aligned} \quad (16.8b)$$

16.5 Method of Relative Velocity

There are two methods for analyzing velocities associated with rigid bodies undergoing plane motion. This article discusses the *method of relative velocity*, which utilizes the equation $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$ for two points on the same rigid body. The other method, based on *instant centers for velocity*, is described in the next article.



Plane motion = Translation + Rotation about A

Fig. 16.8

Figure 16.8(a) shows a rigid body that is undergoing general plane motion. If A and B are points in the body, then according to Eqs. (16.8), the velocity of B relative to A is

$$\mathbf{v}_{B/A} = \boldsymbol{\omega} \times \mathbf{r}_{B/A} \quad (v_{B/A} = r_{B/A}\omega) \quad (16.9)$$

where $\boldsymbol{\omega}$ is the angular velocity of the body. Substituting Eq. (16.9) into $\mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$, we get

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A} \quad (16.10)$$

The physical interpretation of Eq. (16.10) is illustrated in Fig. 16.8. The figure shows that general plane motion is equivalent to the superposition of two simpler motions*.

1. A rigid-body translation, where the velocity of each point is equal to the velocity of the reference point A ($\mathbf{v}_B = \mathbf{v}_A$), as shown in Fig. 16.8(b).
2. A rigid-body rotation about a fixed axis at A ($\mathbf{v}_{B/A} = \boldsymbol{\omega} \times \mathbf{r}_{B/A}$), illustrated in Figs. 16.8(c) or (d). Showing a pin support at A reinforces the notion that A is considered to be fixed at the instant when the contribution of the rotation is computed.

*This representation of general plane motion is a two-dimensional form of Chasle's theorem: The most general rigid body displacement is equivalent to a translation of a point in the body plus a rotation about an axis through that point. For a proof of this theorem, see *Principles of Dynamics*, 2e, Donald T. Greenwood, Prentice Hall, 1988, p. 339.

The contribution of the rotation of the body to the velocity of B may be computed using either the vector notation in Fig. 16.8(c) or the scalar notation in Fig. 16.8(d). The choice of notation is a matter of personal preference. When using scalar notation, keep in mind that the direction of the velocity is perpendicular to AB , and its sense is determined by the direction of the angular velocity ω of the body (recall that ω is a property of the body and is independent of the choice of the reference point).

The method of relative velocity consists of writing Eq. (16.10), $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$, for two points in the *same rigid body*, and then solving for the unknowns. For plane motion, Eq. (16.10) is equivalent to two scalar equations (e.g., the equations that result from equating the horizontal and vertical components of both sides of the vector equation). The number of variables appearing in Eq. (16.10) is five (assuming that $\mathbf{r}_{B/A}$ is known):

- $\mathbf{v}_B$: two variables (magnitude and direction, or horizontal and vertical components)
- $\mathbf{v}_A$: two variables (magnitude and direction, or horizontal and vertical components)
- ω : one variable (magnitude ω of the angular velocity). Note that the direction of vector $\boldsymbol{\omega}$ is known because it is always perpendicular to the plane of motion.

Clearly, Eq. (16.10) cannot be solved unless three of the preceding five variables are known beforehand. Therefore, the choice of A and/or B is restricted to points whose velocities contain less than two unknowns. We refer to these points as a *kinematically important points for velocity*.^{*} A common example of a kinematically important point is a fixed point; that is, a point that is pinned to a support. Because the velocity of a fixed point is known to be zero, it contains no unknowns. A second example is a point that travels along a given path. Since the direction of the velocity is known to be tangent to the path, the velocity of such a point contains at most one unknown, namely its magnitude.

The steps in the application of the relative velocity method are as follows:

- Step 1: Identify two kinematically important points, say, A and B , on the same rigid body.
- Step 2: Write $\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}_{B/A}$, identifying the unknown variables (either vector or scalar notation can be used).
- Step 3: If the number of unknowns is two, solve the equation.

If the number of unknowns is greater than two, it may still be possible to solve the problem by considering the motion of other kinematically important points.

^{*}Since we are discussing velocities, we will refer to these points for the time being as simply “kinematically important points”.

Rolling Without Slipping Figure 16.9(a) shows a circular disk of radius R that is rolling on a horizontal surface with the angular velocity ω and angular acceleration α , both clockwise. Observe that the path of the center O is a straight line parallel to the surface. Rolling without slipping occurs if the contact point C on the disk has no velocity, that is, if the disk does not slide along the surface. This case deserves special attention because it occurs in many engineering applications.

Relating the velocities of points O and C , we have

$$\mathbf{v}_O = \mathbf{v}_C + \boldsymbol{\omega} \times \mathbf{r}_{O/C}$$

Substituting $\mathbf{v}_C = \mathbf{0}$, $\boldsymbol{\omega} = -\omega\mathbf{k}$ and $\mathbf{r}_{O/C} = R\mathbf{j}$, we obtain

$$\mathbf{v}_O = -\omega\mathbf{k} \times R\mathbf{j} = R\omega\mathbf{i} \quad (16.11a)$$

As expected, this result shows that the velocity of the center O is parallel to the surface on which the disk rolls, its magnitude being

$$v_O = R\omega \quad (16.11b)$$

as shown Fig. 16.9(b).

It is convenient here to also derive the acceleration of O , although this information will not be used until Art. 16.7. The acceleration of O can be obtained by differentiation of Eq. (16.11a). Noting that R and $\mathbf{i}$ are constants, we get

$$\mathbf{a}_O = \dot{\mathbf{v}}_O = R\alpha\mathbf{i} \quad (16.12a)$$

where $\alpha = \dot{\omega}$ is the angular acceleration of the disk. Thus the acceleration of O is parallel to the horizontal surface, and its magnitude is

$$a_O = R\alpha \quad (16.12b)$$

as shown in Fig. 16.9(c).

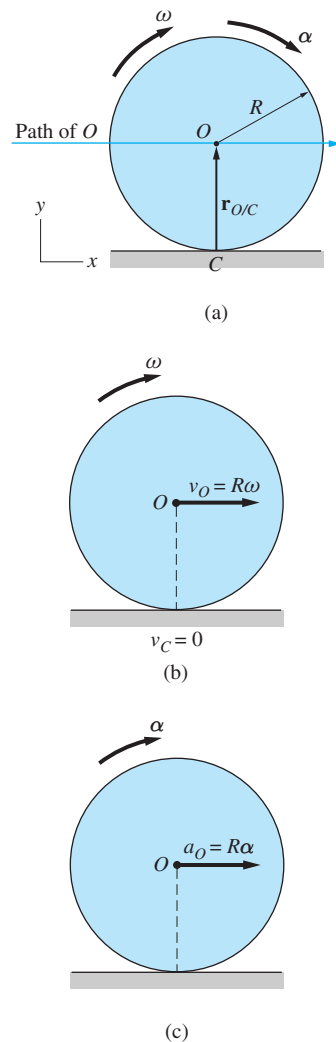


Fig. 16.9

Sample Problem 16.4

Figure (a) shows a wheel of radius R that is rolling without slipping with the clockwise angular velocity ω . For the position shown, determine the velocity vectors of (1) point A ; and (2) point B .

Solution

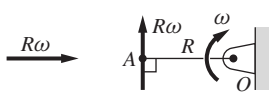
Introductory Comments

This problem will be solved using both scalar and vector notations. (The results are, of course, identical regardless of which notation is used.) We choose point O (the center of the wheel) as the reference point, because its velocity is known from Eq. (16.11) to be $\mathbf{v}_O = R\omega\mathbf{i}$, as shown in Fig. (b). The reader may find it instructive to repeat the solution using the point of contact C ($\mathbf{v}_C = \mathbf{0}$) as the reference point.

Solution I (using scalar notation)

Part 1

When scalar notation is used to relate the velocities of A and O , $\mathbf{v}_{A/O}$ is computed by assuming that point O is fixed. Therefore, the relative velocity equation becomes

$$\mathbf{v}_A = \mathbf{v}_O + \mathbf{v}_{A/O}$$


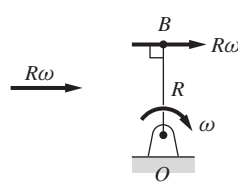
from which the velocity of A is found to be

$$\mathbf{v}_A = \sqrt{2}R\omega \quad \angle 45^\circ \quad \text{Answer}$$

as shown in Fig. (b).

Part 2

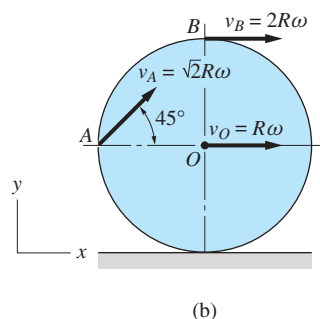
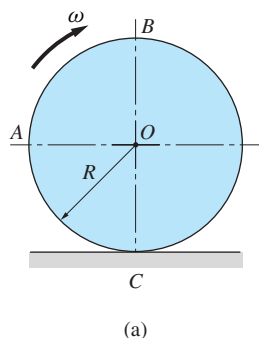
Using scalar notation, the velocities of points B and O are related by

$$\mathbf{v}_B = \mathbf{v}_O + \mathbf{v}_{B/O}$$


which yields

$$\mathbf{v}_B = 2R\omega \rightarrow \quad \text{Answer}$$

The velocity of B is also shown in Fig. (b).



Solution II (using vector notation)

Part 1

In vector notation, the relationship between the velocities of A and O is

$$\mathbf{v}_A = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_{A/O}$$

Substituting $\mathbf{v}_O = R\omega\mathbf{i}$, $\boldsymbol{\omega} = -\omega\mathbf{k}$, and $\mathbf{r}_{A/O} = -R\mathbf{i}$ [refer to Fig. (c)], we get

$$\mathbf{v}_A = R\omega\mathbf{i} + (-\omega\mathbf{k}) \times (-R\mathbf{i}) = R\omega\mathbf{i} + R\omega\mathbf{j} \quad \text{Answer}$$

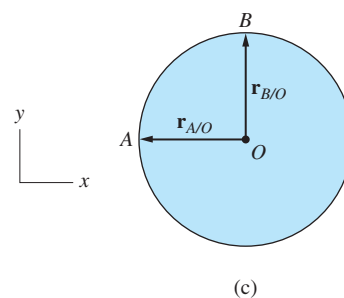
Part 2

The velocities of B and O are related by

$$\mathbf{v}_B = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_{B/O}$$

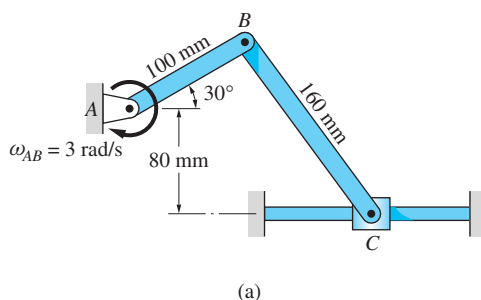
Substituting $\mathbf{v}_O = R\omega\mathbf{i}$, $\boldsymbol{\omega} = -\omega\mathbf{k}$, and $\mathbf{r}_{B/O} = R\mathbf{j}$ [refer to Fig. (c)] we obtain

$$\mathbf{v}_B = R\omega\mathbf{i} + (-\omega\mathbf{k}) \times (R\mathbf{j}) = 2R\omega\mathbf{i} \quad \text{Answer}$$



Sample Problem 16.5

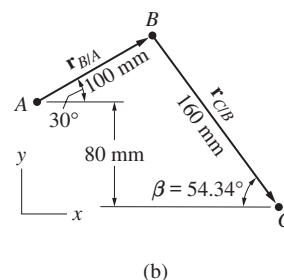
The angular velocity of bar AB in Fig. (a) is 3 rad/s clockwise in the position shown. Determine the angular velocity of bar BC and the velocity of the slider C in this position.



Solution

Introductory Comments

We note from the geometry in Fig. (b) that $100 \sin 30^\circ + 80 = 160 \sin \beta$, from which we find that $\beta = 54.34^\circ$ for the position shown.



The following are the kinematically important points:

A: It is a fixed point.

C: Its path is a horizontal straight line.

B: Its path is a circle with center at *A*; it also connects bars *AB* and *BC*.

Because *B* and *C* are points in the same rigid bar *BC*, it seems reasonable to investigate the equation $\mathbf{v}_C = \mathbf{v}_B + \mathbf{v}_{C/B}$ (the equivalent equation $\mathbf{v}_B = \mathbf{v}_C + \mathbf{v}_{B/C}$ could also be used).

Two solutions are presented below—one using scalar notation, the other employing vector notation. The relative position vectors used in the solutions are shown in Fig. (b).

Solution I (using scalar notation)

Using scalar notation, the velocities of *B* and *C* are related by

$$\mathbf{v}_C = \mathbf{v}_B + \mathbf{v}_{C/B} \quad (a)$$

Comments on Eq. (a):

- $\mathbf{v}_C$ is assumed to be directed to the right.
- $\mathbf{v}_B$ is found by recognizing that the path of *B* is a circle centered at *A*. Its magnitude is thus $v_B = r_{B/A}\omega_{AB} = 100(3) = 300$ mm/s, directed at right angles to *AB*. The sense of $\mathbf{v}_B$ is determined by the clockwise direction of ω_{AB} .
- $\mathbf{v}_{C/B}$ is obtained by considering *B* to be a fixed point at this instant, which gives $v_{C/B} = r_{C/B}\omega_{BC} = 160\omega_{BC}$. The direction of the vector is perpendicular to *BC*, with its sense determined by the (assumed) counterclockwise direction of ω_{BC} .

Because there are a total of two unknowns (v_C and ω_{BC}), Eq. (a) can be solved. Equating *x*- and *y*-components, we obtain

$$+\rightarrow v_C = 300 \sin 30^\circ + 160\omega_{BC} \sin 54.34^\circ \quad (b)$$

$$+\uparrow 0 = -300 \cos 30^\circ + 160\omega_{BC} \cos 54.34^\circ \quad (c)$$

The solution is

$$v_C = 512 \text{ mm/s} \quad \text{and} \quad \omega_{BC} = 2.785 \text{ rad/s} \quad \text{Answer}$$

The positive signs mean that the assumed directions of v_C and ω_{BC} are correct.

Solution II (using vector notation)

The vector equation relating the velocities of points B and C is

$$\mathbf{v}_C = \mathbf{v}_B + \mathbf{v}_{C/B} = \boldsymbol{\omega}_{AB} \times \mathbf{r}_{B/A} + \boldsymbol{\omega}_{BC} \times \mathbf{r}_{C/B} \quad (d)$$

From the given information and inspection of Fig. (b), we obtain

$$\mathbf{v}_C = v_C \mathbf{i} \quad (\text{assuming the velocity of } C \text{ to be directed to the right})$$

$$\boldsymbol{\omega}_{AB} = -3\mathbf{k} \text{ rad/s}$$

$$\mathbf{r}_{B/A} = 100 \cos 30^\circ \mathbf{i} + 100 \sin 30^\circ \mathbf{j} = 86.6\mathbf{i} + 50\mathbf{j} \text{ mm}$$

$$\boldsymbol{\omega}_{BC} = \omega_{BC} \mathbf{k} \quad (\text{assuming the direction of } \omega_{BC} \text{ to be counterclockwise})$$

$$\mathbf{r}_{C/B} = 160 \cos 54.34^\circ \mathbf{i} - 160 \sin 54.34^\circ \mathbf{j} = 93.28\mathbf{i} - 130\mathbf{j} \text{ mm}$$

Inspection of the above expressions reveals that there are only two unknowns: v_C and ω_{BC} . Therefore, Eq. (d) can be solved, because it is equivalent to two scalar equations.

Substituting the above expressions into Eq. (d) yields

$$\begin{aligned} v_C \mathbf{i} &= (-3\mathbf{k}) \times (86.6\mathbf{i} + 50\mathbf{j}) + (\omega_{BC} \mathbf{k}) \times (93.28\mathbf{i} - 130\mathbf{j}) \\ &= -259.8\mathbf{j} + 150\mathbf{i} + 93.28\omega_{BC}\mathbf{j} + 130\omega_{BC}\mathbf{i} \end{aligned}$$

Equating the coefficients of $\mathbf{i}$ and $\mathbf{j}$ yields

$$v_C = 150 + 130\omega_{BC}$$

$$0 = -259.8 + 93.28\omega_{BC}$$

Solving the two equations, we find that

$$v_C = 512 \text{ mm/s} \quad \text{and} \quad \omega_{BC} = 2.785 \text{ rad/s} \quad \textbf{Answer}$$

The positive signs indicate that the assumed directions of $\mathbf{v}_C$ and ω_{BC} are correct.

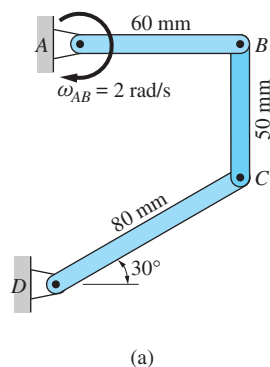
Sample Problem 16.6

In the position shown in Fig. (a), the angular velocity of bar AB is 2 rad/s clockwise. Calculate the angular velocities of bars BC and CD for this position.

Solution

Introductory Comments

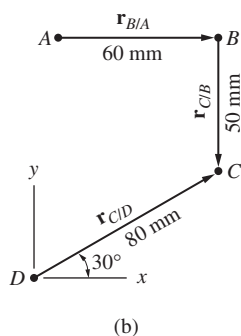
A mechanism of the type shown in Fig. (a) is called a *four-bar linkage*. (The ground joining the supports at A and D is considered to be the fourth bar.)



From Fig. (a) we observe that the following are the kinematically important points:

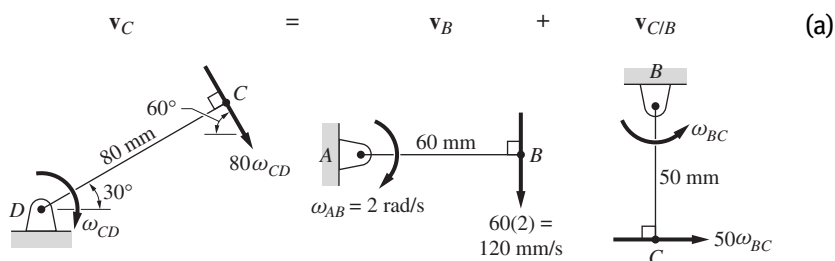
- A: It is a fixed point.
- B: Its path is a circle with center at A; it also connects bars AB and BC .
- D: It is a fixed point.
- C: Its path is a circle centered at D; it also connects bars BC and CD .

Because B and C belong to the same rigid body, we are led to consider the equation $\mathbf{v}_C = \mathbf{v}_B + \mathbf{v}_{C/B}$ (the equivalent equation $\mathbf{v}_B = \mathbf{v}_C + \mathbf{v}_{B/C}$ could also be used). Two solutions are presented—one using scalar notation, the other using vector notation. The relative position vectors used in the solution are shown in Fig. (b).



Solution I (using scalar notation)

The equation relating the velocities of B and C is



Comments on Eq. (a):

1. $\mathbf{v}_C$ is found by recognizing that the path of C is a circle centered at D . Therefore, its magnitude is $v_C = r_{C/D}\omega_{CD} = 80\omega_{CD}$, its direction being perpendicular to CD . The sense of $\mathbf{v}_C$ is found from the assumed (clockwise) direction of ω_{CD} .
2. $\mathbf{v}_B$ is found by noting that the path of B is a circle centered at A . Its magnitude is $v_B = r_{B/A}\omega_{AB} = 60(2) = 120$ mm/s. The direction of $\mathbf{v}_B$ is perpendicular to AB , and its sense is found from the given clockwise direction of ω_{AB} .
3. $\mathbf{v}_{C/B}$ is constructed by considering B to be a fixed point at this instant, which gives $v_{C/B} = r_{C/B}\omega_{BC} = 50\omega_{BC}$. The direction of $\mathbf{v}_{C/B}$ is perpendicular to BC , and its sense is found from the assumed (counterclockwise) direction of ω_{BC} .

Equation (a) contains a total of two unknowns, ω_{CD} and ω_{BC} , which can be found by solving the two equivalent scalar equations. Equating x - and y -components of Eq. (a), we obtain

$$\begin{array}{l} \rightarrow \\ + \end{array} \quad 80\omega_{CD} \cos 60^\circ = 0 + 50\omega_{BC} \quad (b)$$

$$\begin{array}{l} \uparrow \\ + \end{array} \quad -80\omega_{CD} \sin 60^\circ = -120 + 0 \quad (c)$$

Solving these equations, we get

$$\omega_{CD} = 1.732 \text{ rad/s} \quad \text{and} \quad \omega_{BC} = 1.386 \text{ rad/s} \quad \textbf{Answer}$$

The positive signs indicate that the assumed directions of ω_{CD} and ω_{BC} are correct.

Solution II (using vector notation)

The velocities of points B and C are related by

$$\begin{aligned} \mathbf{v}_C &= \mathbf{v}_B + \mathbf{v}_{C/B} \\ \omega_{CD} \times \mathbf{r}_{C/D} &= \omega_{AB} \times \mathbf{r}_{B/A} + \omega_{BC} \times \mathbf{r}_{C/B} \end{aligned} \quad \text{(d)}$$

Using the given information and the vectors shown in Fig. (b), we obtain

$$\begin{aligned} \omega_{CD} &= -\omega_{CD} \mathbf{k} \quad (\text{the direction of } \omega_{CD} \text{ is assumed to be clockwise}) \\ \mathbf{r}_{C/D} &= 80 \cos 30^\circ \mathbf{i} + 80 \sin 30^\circ \mathbf{j} = 69.28 \mathbf{i} + 40.00 \mathbf{j} \text{ mm} \\ \omega_{AB} &= -2 \mathbf{k} \text{ rad/s} \\ \mathbf{r}_{B/A} &= 60 \mathbf{i} \text{ mm} \\ \omega_{BC} &= \omega_{BC} \mathbf{k} \quad (\text{the direction of } \omega_{BC} \text{ is assumed to be counterclockwise}) \\ \mathbf{r}_{C/B} &= -50 \mathbf{j} \text{ mm} \end{aligned}$$

We see that the above expressions contain only two unknowns: ω_{CD} and ω_{BC} . Therefore, Eq. (d), being equivalent to two scalar equations, can be solved for these unknowns.

Substituting the above expressions into Eq. (d) yields

$$(-\omega_{CD} \mathbf{k}) \times (69.28 \mathbf{i} + 40.00 \mathbf{j}) = (-2 \mathbf{k}) \times (60 \mathbf{i}) + (\omega_{BC} \mathbf{k}) \times (-50 \mathbf{j})$$

which becomes

$$-69.28 \omega_{CD} \mathbf{j} + 40.00 \omega_{CD} \mathbf{i} = -120 \mathbf{j} + 50 \omega_{BC} \mathbf{i}$$

Equating the coefficients of $\mathbf{i}$ and $\mathbf{j}$ yields

$$\begin{aligned} 40.00 \omega_{CD} &= 50 \omega_{BC} \\ -69.28 \omega_{CD} &= -120 \end{aligned}$$

The solution of these two equations is

$$\omega_{CD} = 1.732 \text{ rad/s} \quad \text{and} \quad \omega_{BC} = 1.386 \text{ rad/s} \quad \textbf{Answer}$$

The positive answers indicate that the assumed directions of ω_{CD} and ω_{BC} are correct.

16.6 *Instant Center for Velocities*

The instant center for velocities of a body undergoing plane motion is defined to be the point that has zero velocity at the instant under consideration.* This point may be either in a body or outside the body (in the “body extended”). It is often convenient to use the instant center of the body in computing the velocities of points in the body.

*Three “centers” are sometimes used in the kinematic analysis of plane motion: the instant center of rotation for virtual motion (see Art. 10.6), the instant center for velocities, and the instant center for accelerations. Each of these points is called simply the instant center when it is clear from the context which center is being used. The discussion of instant center for velocities presented here parallels the discussion of instant center of rotation for virtual motion in Art. 10.6.

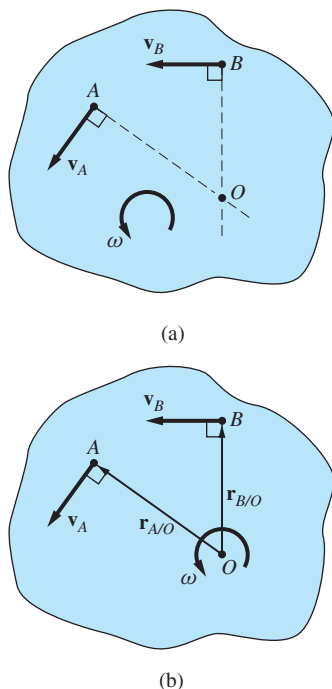


Fig. 16.10

Figure 16.10(a) shows a rigid body that is undergoing plane motion. It is assumed that the velocity vectors for points A and B are not parallel, and that the angular velocity ω of the body is counterclockwise. To locate the instant center for velocities, we construct a line through A that is perpendicular to $\mathbf{v}_A$, and a line through B that is perpendicular to $\mathbf{v}_B$. These two lines will intersect at a point labeled O in the figure. If point O does not lie in the body, we simply imagine that the body is enlarged to include it, the expanded body being called the *body extended*.

We now show that point O is the instant center—that is, that the velocity of O is zero. Because A , B , and O are points in the same rigid body (or body extended), we can write the following relative velocity equations:

$$\mathbf{v}_O = \mathbf{v}_A - \boldsymbol{\omega} \times \mathbf{r}_{A/O} \quad (\text{a})$$

$$\mathbf{v}_O = \mathbf{v}_B - \boldsymbol{\omega} \times \mathbf{r}_{B/O} \quad (\text{b})$$

where $\mathbf{r}_{A/O}$ and $\mathbf{r}_{B/O}$ are the relative position vectors shown in Fig. 16.10(b). Because both ω and $\mathbf{r}_{A/O}$ are perpendicular to $\mathbf{v}_A$, it follows that $\boldsymbol{\omega} \times \mathbf{r}_{A/O}$ in Eq. (a) is parallel to $\mathbf{v}_A$. This means that $\mathbf{v}_O$ is parallel to $\mathbf{v}_A$. Similarly, it can be shown from Eq. (b) that $\mathbf{v}_O$ is parallel to $\mathbf{v}_B$. Because a nonzero vector cannot be parallel to two different directions simultaneously, we conclude that $\mathbf{v}_O = \mathbf{0}$. Therefore, point O is indeed the instant center for velocities.

In general, the instant center for velocities is not a fixed point. Therefore, the acceleration of the instant center for velocities is not necessarily zero. However, if a point of a body is fixed, that point is obviously the instant center for both velocities and accelerations.

If O is the instant center, Eq. (a) reduces to

$$\mathbf{v}_A = \boldsymbol{\omega} \times \mathbf{r}_{A/O} \quad (v_A = r_{A/O}\omega) \quad (16.13)$$

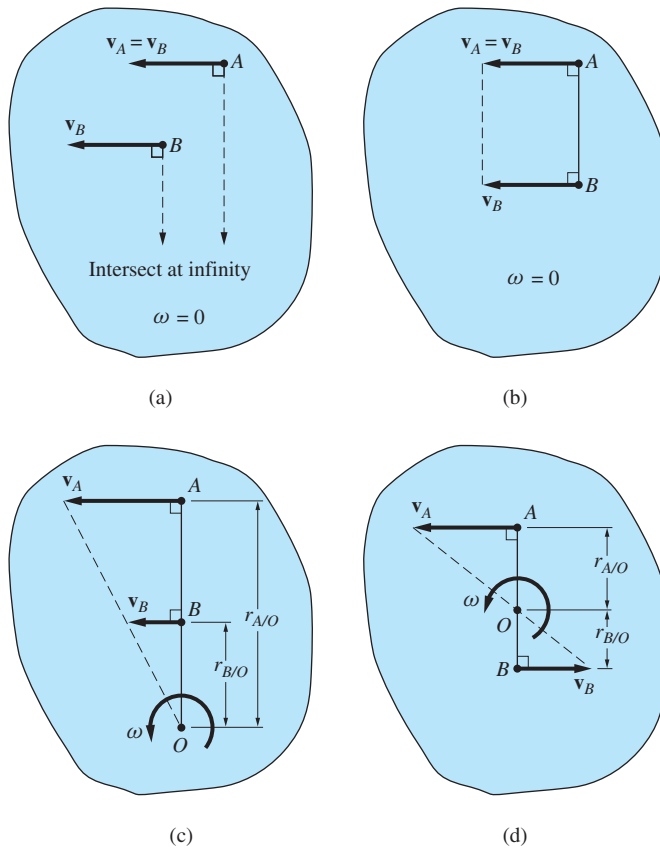
This equation has the same form as the first of Eqs. (16.8) for rotation about a fixed axis through O . We must reiterate, however, that for general plane motion the instant center is not a fixed point. Therefore, the analogy with rotation about a fixed axis is valid only for velocities at a given *instant* of time (hence the term “instant center”). The analogy does not apply to accelerations.

The following rules, which follow directly from Eq. (16.13), apply when the instant center is used.

1. The velocity of any point in the body is perpendicular to the line drawn from the point to the instant center.
2. The magnitude of the velocity of any point of the body is proportional to the distance of the point from the instant center.
3. The sense of the velocity vector of any point must be consistent with the sense of the angular velocity of the body.

The construction shown in Fig. 16.10(a) for locating the instant center is obviously valid only if $\mathbf{v}_A$ and $\mathbf{v}_B$ are not parallel. Figure 16.11 illustrates the methods for locating the instant center when $\mathbf{v}_A$ and $\mathbf{v}_B$ are parallel.

In Fig. 16.11(a), the perpendicular lines to the velocity vectors are parallel, which means that the instant center is an infinite distance from A and B . This leads to the conclusion that $\omega = 0$; that is, the body is translating with $\mathbf{v}_A = \mathbf{v}_B$.

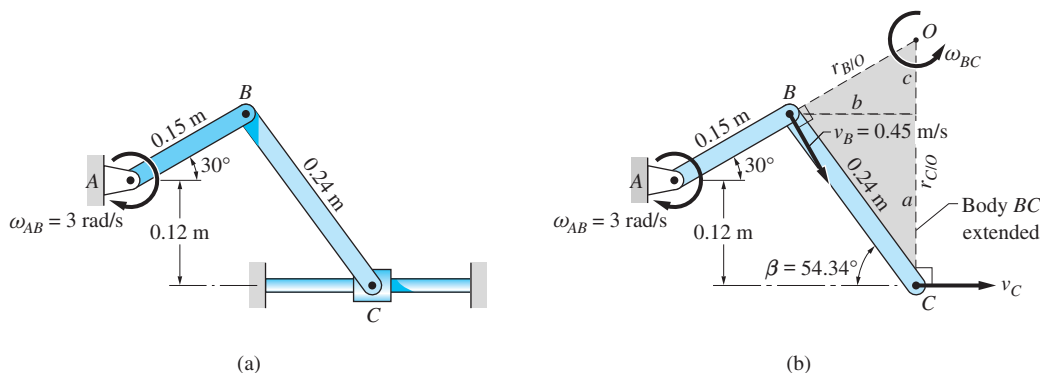
**Fig. 16.11**

In Fig. 16.11 parts (b) through (d), the lines joining A and B are perpendicular to both velocity vectors, and the usual construction for locating the instant center does not work. In the case where $\mathbf{v}_A = \mathbf{v}_B$, as in Fig. 16.11(b), the instant center is at infinity, so that $\omega = 0$ (i.e., the body is translating). If the magnitudes of the velocities are not equal, the instant center is located using the constructions in Fig. 16.11 parts (c) and (d), which are simply applications of rules 1 through 3 just stated.

The question arises whether the simplification of velocity analysis based on the instant center is worth the labor it takes to find the instant center in the first place. A general rule of thumb is to begin by marking the instant center on a sketch of the body. If the distances between the kinematically important points on the body (or body extended) and the instant center are fairly easy to calculate, the analysis is well suited for using the instant center. However, if a prohibitive amount of trigonometry is required to compute the location of the instant center, the relative velocity method of the preceding article would be the preferred method of solution. Of course, there are many problems that require the same amount of labor using either method, in which case personal preference would dictate which method to use.

Sample Problem 16.7

When the mechanism in Fig. (a) is in the position shown, the angular velocity of bar AB is $\omega_{AB} = 3 \text{ rad/s}$ clockwise. Using instant centers for velocities, calculate the angular velocity of bar BC and the velocity of slider C for this position. (This problem was solved previously as Sample Problem 16.5 by using the method of relative velocity.)



Solution

We must first locate the instant centers of the two rigid bodies AB and BC . Because A is a fixed point, it is obviously the instant center of bar AB . The instant center of BC , labeled O in Fig. (b), is located at the point of intersection of the lines that are perpendicular to the velocity vectors of B and C . Note that $\mathbf{v}_B$ is perpendicular to AB , and $\mathbf{v}_C$ is horizontal. Therefore, O is located at the intersection of line AB and the vertical line that passes through C .

From the geometry in Fig. (b), we note that $.15 \sin 30^\circ + 0.12 = 0.24 \sin \beta$, from which we find that $\beta = 54.34^\circ$. Therefore, the distances a , b , and c are

$$\begin{aligned} a &= 0.24 \sin 54.34^\circ = 0.195 \text{ m} \\ b &= 0.24 \cos 54.34^\circ = 0.140 \text{ m} \\ c &= b \tan 30^\circ = 0.140 \tan 30^\circ = 0.080 \text{ m} \end{aligned}$$

The distances to B and C from O are

$$\begin{aligned} r_{B/O} &= c / \sin 30^\circ = \frac{0.080}{\sin 30^\circ} = 0.16 \text{ m} \\ r_{C/O} &= a + c = 0.195 + 0.16 = 0.355 \text{ m} \end{aligned}$$

Now that the instant center for each bar has been found, we can compute the required velocities.

Considering the motion of AB (its instant center is at A), we find that $v_B = r_{B/A} \omega_{AB} = 0.15(3) = 0.45 \text{ m/s}$, directed as shown in Fig. (b). Analyzing the motion of BC (its instant center is at O) yields

$$\begin{aligned} \omega_{BC} &= \frac{v_B}{r_{B/O}} = \frac{0.45}{0.16} = 2.81 \text{ rad/s} \\ \omega_{BC} &= 2.81 \text{ rad/s} \odot \end{aligned}$$

Answer

and

$$v_C = r_{C/O} \omega_{BC} = 0.355(2.81) = 0.99 \text{ m/s}$$

$$\mathbf{v}_C = 0.99 \text{ m/s} \rightarrow$$

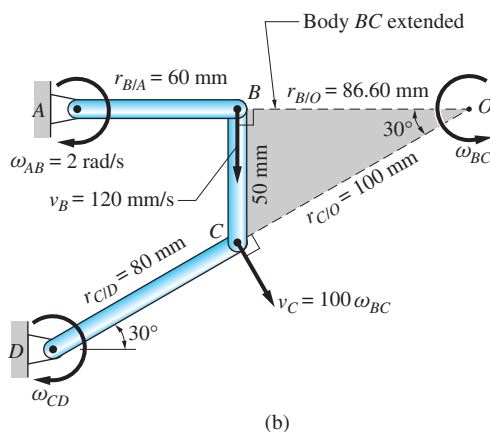
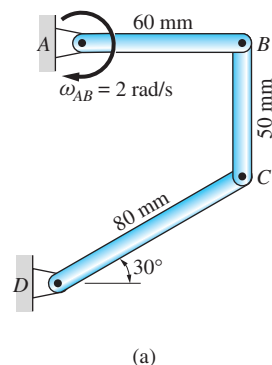
Answer

Sample Problem 16.8

When the linkage in Fig. (a) is in the position shown, the angular velocity of bar AB is $\omega_{AB} = 2 \text{ rad/s}$ clockwise. For this position, determine the angular velocities of bars BC and CD and the velocity of C using the instant centers for velocities. (This problem was solved previously as Sample Problem 16.6 using the method of relative velocity.)

Solution

Because A and D are fixed points, they are the instant centers for bars AB and CD , respectively. The instant center for bar BC , labeled O in Fig. (b), is located at the point of intersection of the lines that are perpendicular to the velocity vectors of B and C . Because $\mathbf{v}_B$ and $\mathbf{v}_C$ are perpendicular to AB and CD , respectively, the instant center is at the intersection of these two lines. Note that body BC must be “extended” to include point O .



The distances to B and C from O , found from the triangle OBC , are

$$r_{B/O} = 50 / \tan 30^\circ = 86.60 \text{ mm}$$

$$r_{C/O} = 50 / \sin 30^\circ = 100 \text{ mm}$$

The instant centers A , O , and D can now be used to compute the required angular velocities directly from Fig. (b).

Considering the motion of AB (its instant center is at A), we find that $v_B = r_{B/A}\omega_{AB} = 60(2) = 120$ mm/s, directed as shown in Fig. (b). Analyzing the motion of BC (its instant center is at O) yields

$$\omega_{BC} = \frac{v_B}{r_{B/O}} = \frac{120}{86.60} = 1.386 \text{ rad/s}$$

$$\omega_{BC} = 1.386 \text{ rad/s } \odot$$

Answer

and

$$v_C = r_{C/O}\omega_{BC} = 100(1.386) = 138.6 \text{ mm/s}$$

$$v_C = 138.6 \text{ mm/s } \nearrow 30^\circ$$

Answer

Because C is also a point on bar CD (its instant center is at D), the angular velocity of bar CD is

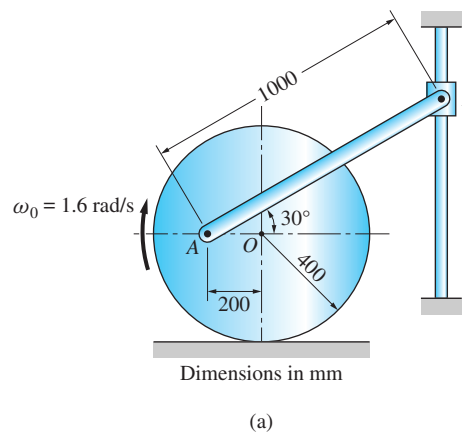
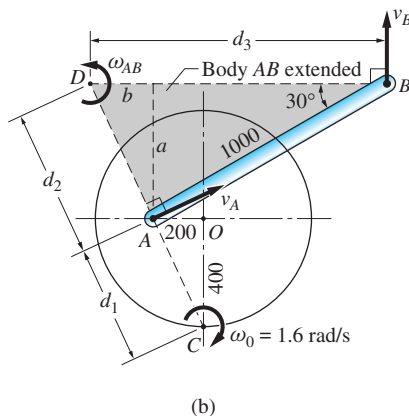
$$\omega_{CD} = \frac{v_C}{r_{C/D}} = \frac{138.6}{80} = 1.733 \text{ rad/s}$$

$$\omega_{CD} = 1.733 \text{ rad/s } \odot$$

Answer

Sample Problem 16.9

The wheel in Fig. (a) rolls without slipping with the constant clockwise angular velocity $\omega_0 = 1.6$ rad/s. Calculate the angular velocity of bar AB and the velocity of the slider B when the mechanism is in the position shown. Use the instant centers for velocities.



Solution

The velocity vectors, distances, and points required to solve this problem are shown in Fig. (b).

Because the wheel rolls without slipping, its instant center is at the point of contact C . Therefore, $\mathbf{v}_A$ is perpendicular to the line AC . Note that the slope of AC equals $\overline{OC}/\overline{AO} = 400/200 = 2$.

The instant center of bar AB , labeled D in Fig. (b), is located at the intersection of the lines that are perpendicular to $\mathbf{v}_A$ and $\mathbf{v}_B$. Because $\mathbf{v}_A$ is perpendicular to AC , D also lies on AC . The line BD , which is drawn perpendicular to $\mathbf{v}_B$, is horizontal because $\mathbf{v}_B$ is vertical (it is not necessary to know here that the sense of $\mathbf{v}_B$ is upward).

Referring to Fig. (b), the distances of interest are computed as follows.

$$d_1 = \sqrt{\overline{AO}^2 + \overline{OC}^2} = \sqrt{200^2 + 400^2} = 447.2 \text{ mm}$$

$$a = 1000 \sin 30^\circ = 500 \text{ mm}$$

$$b = a/2 = 250 \text{ mm} \quad (\text{the slope of the line } AC \text{ is } 2)$$

$$d_2 = \sqrt{a^2 + b^2} = \sqrt{500^2 + 250^2} = 559.0 \text{ mm}$$

$$d_3 = 1000 \cos 30^\circ + b = 1000 \cos 30^\circ + 250 = 1116 \text{ mm}$$

Using the distances d_1 , d_2 , and d_3 and $\omega_0 = 1.6 \text{ rad/s}$, we find that

$$v_A = d_1 \omega_0 = 447.2(1.6) = 715.5 \text{ mm/s}$$

$$\mathbf{v}_A = 716 \text{ mm/s} \quad \begin{array}{c} \nearrow \\ 1 \\ \hline 2 \end{array}$$

Answer

$$\omega_{AB} = \frac{v_A}{d_2} = \frac{715.5}{559.0} = 1.280 \text{ rad/s}$$

$$\omega_{AB} = 1.280 \text{ rad/s } \curvearrowright$$

Answer

$$v_B = d_3 \omega_{AB} = 1116(1.280) = 1428 \text{ mm/s}$$

$$\mathbf{v}_B = 1428 \text{ mm/s } \uparrow$$

Answer

The sense of $\mathbf{v}_A$ was found by considering that the angular velocity of the wheel is clockwise and its instant center is at C . The counterclockwise direction of ω_{AB} was deduced by inspection of the sense of $\mathbf{v}_A$ and the location of the instant center D . The direction of ω_{AB} and the location of B relative to D determine that the sense of $\mathbf{v}_B$ is upward.

It is frequently convenient to show the instant centers of more than one body on the same sketch, as is done in Fig. (b). However, one must be careful to use the proper instant center when discussing the velocity of a particular point. For example, a common error when referring to Fig. (b) would be to write $v_C = (d_1 + d_2)\omega_{AB}$, which is incorrect, because D is the instant center of bar AB , which does not include point C .

16.4 PLANE MOTION OF A RIGID BODY. D'ALEMBERT'S PRINCIPLE

Consider a rigid slab of mass m moving under the action of several external forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$, contained in the plane of the slab (Fig. 16.5). Substituting for $\mathbf{H}_G$ from Eq. (16.5) into Eq. (16.2) and writing the fundamental equations of motion (16.1) and (16.2) in scalar form, we have

$$\Sigma F_x = m\bar{a}_x \quad \Sigma F_y = m\bar{a}_y \quad \Sigma M_G = \bar{I}\alpha \quad (16.6)$$

Equations (16.6) show that the acceleration of the mass center G of the slab and its angular acceleration α are easily obtained once the resultant of the external forces acting on the slab and their moment resultant about G have been determined. Given appropriate initial conditions, the coordinates $\bar{x}$ and $\bar{y}$ of the mass center and the angular coordinate θ of the slab can then be obtained by integration at any instant t . Thus *the motion of the slab is completely defined by the resultant and moment resultant about G of the external forces acting on it.*

This property, which will be extended in Chap. 18 to the case of the three-dimensional motion of a rigid body, is characteristic of the motion of a rigid body. Indeed, as we saw in Chap. 14, the motion of a system of particles which are not rigidly connected will in general depend upon the specific external forces acting on the various particles, as well as upon the internal forces.

Since the motion of a rigid body depends only upon the resultant and moment resultant of the external forces acting on it, it follows that *two systems of forces which are equipollent*, i.e., which have the same resultant and the same moment resultant, *are also equivalent*; that is, they have exactly the same effect on a given rigid body.†

Consider in particular the system of the external forces acting on a rigid body (Fig. 16.6a) and the system of the effective forces associated with the particles forming the rigid body (Fig. 16.6b). It was shown in Sec. 14.2 that the two systems thus defined are equipollent. But since the particles considered now form a rigid body, it follows from the discussion above that the two systems are also equivalent. We can thus state that *the external forces acting on a rigid body are equivalent to the effective forces of the various particles forming the body.* This statement is referred to as *d'Alembert's principle* after the French mathematician Jean le Rond d'Alembert (1717–1783), even though d'Alembert's original statement was written in a somewhat different form.

The fact that the system of external forces is *equivalent* to the system of the effective forces has been emphasized by the use of a red equals sign in Fig. 16.6 and also in Fig. 16.7, where using results obtained earlier in this section, we have replaced the effective forces by a vector $m\bar{\mathbf{a}}$ attached at the mass center G of the slab and a couple of moment $\bar{I}\alpha$.

†This result has already been derived in Sec. 3.19 from the principle of transmissibility (Sec. 3.3). The present derivation is independent of that principle, however, and will make possible its elimination from the axioms of mechanics (Sec. 16.5).

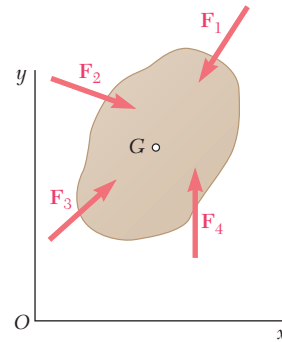


Fig. 16.5

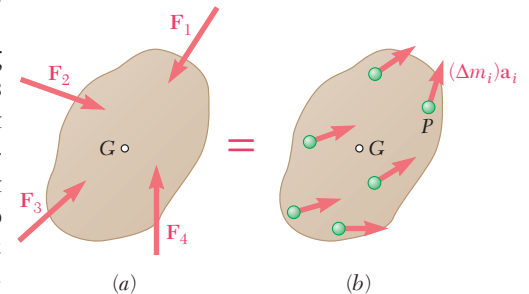


Fig. 16.6

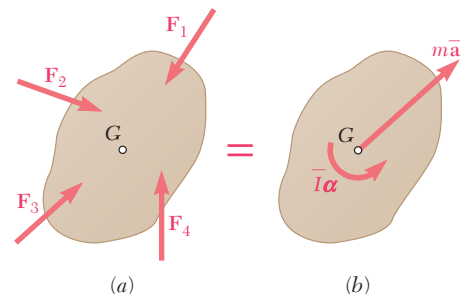


Fig. 16.7

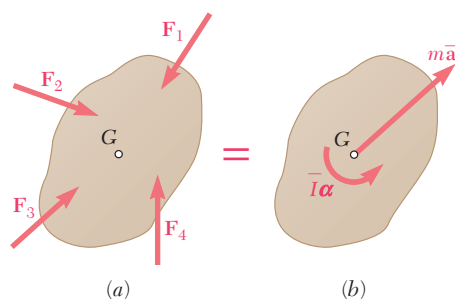


Fig. 16.7 (repeated)

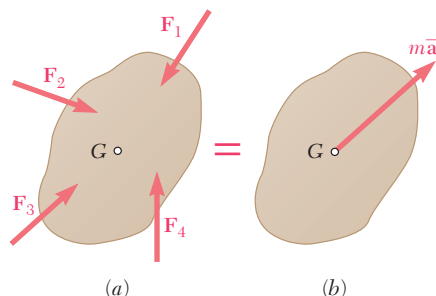


Fig. 16.8 Translation.

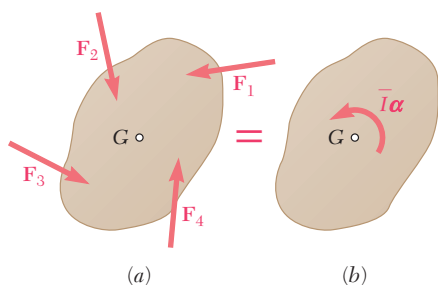


Fig. 16.9 Centroidal rotation.

Translation. In the case of a body in translation, the angular acceleration of the body is identically equal to zero and its effective forces reduce to the vector $m\bar{\mathbf{a}}$ attached at G (Fig. 16.8). Thus, the resultant of the external forces acting on a rigid body in translation passes through the mass center of the body and is equal to $m\bar{\mathbf{a}}$.

Centroidal Rotation. When a slab, or, more generally, a body symmetrical with respect to the reference plane, rotates about a fixed axis perpendicular to the reference plane and passing through its mass center G , we say that the body is in *centroidal rotation*. Since the acceleration $\bar{\mathbf{a}}$ is identically equal to zero, the effective forces of the body reduce to the couple $\bar{I}\alpha$ (Fig. 16.9). Thus, the external forces acting on a body in centroidal rotation are equivalent to a couple of moment $\bar{I}\alpha$.

General Plane Motion. Comparing Fig. 16.7 with Figs. 16.8 and 16.9, we observe that from the point of view of *kinetics*, the most general plane motion of a rigid body symmetrical with respect to the reference plane can be replaced by the sum of a translation and a centroidal rotation. We should note that this statement is more restrictive than the similar statement made earlier from the point of view of *kinematics* (Sec. 15.5), since we now require that the mass center of the body be selected as the reference point.

Referring to Eqs. (16.6), we observe that the first two equations are identical with the equations of motion of a particle of mass m acted upon by the given forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$. We thus check that *the mass center G of a rigid body in plane motion moves as if the entire mass of the body were concentrated at that point, and as if all the external forces acted on it.* We recall that this result has already been obtained in Sec. 14.4 in the general case of a system of particles, the particles being not necessarily rigidly connected. We also note, as we did in Sec. 14.4, that the system of the external forces does not, in general, reduce to a single vector $m\bar{\mathbf{a}}$ attached at G . Therefore, in the general case of the plane motion of a rigid body, *the resultant of the external forces acting on the body does not pass through the mass center of the body.*

Finally, it should be observed that the last of Eqs. (16.6) would still be valid if the rigid body, while subjected to the same applied forces, were constrained to rotate about a fixed axis through G . Thus, *a rigid body in plane motion rotates about its mass center as if this point were fixed.*

*16.5 A REMARK ON THE AXIOMS OF THE MECHANICS OF RIGID BODIES

The fact that two equipollent systems of external forces acting on a rigid body are also equivalent, i.e., have the same effect on that rigid body, has already been established in Sec. 3.19. But there it was derived from the *principle of transmissibility*, one of the axioms used in our study of the statics of rigid bodies. It should be noted that this axiom has not been used in the present chapter because Newton's second and third laws of motion make its use unnecessary in the study of the dynamics of rigid bodies.

In fact, the principle of transmissibility may now be *derived* from the other axioms used in the study of mechanics. This principle

stated, without proof (Sec. 3.3), that the conditions of equilibrium or motion of a rigid body remain unchanged if a force $\mathbf{F}$ acting at a given point of the rigid body is replaced by a force $\mathbf{F}'$ of the same magnitude and same direction, but acting at a different point, provided that the two forces have the same line of action. But since $\mathbf{F}$ and $\mathbf{F}'$ have the same moment about any given point, it is clear that they form two equipollent systems of external forces. Thus, we may now *prove*, as a result of what we established in the preceding section, that $\mathbf{F}$ and $\mathbf{F}'$ have the same effect on the rigid body (Fig. 3.3).

The principle of transmissibility can therefore be removed from the list of axioms required for the study of the mechanics of rigid bodies. These axioms are reduced to the parallelogram law of addition of vectors and to Newton's laws of motion.

16.6 SOLUTION OF PROBLEMS INVOLVING THE MOTION OF A RIGID BODY

We saw in Sec. 16.4 that when a rigid body is in plane motion, there exists a fundamental relation between the forces $\mathbf{F}_1, \mathbf{F}_2, \mathbf{F}_3, \dots$, acting on the body, the acceleration $\bar{\mathbf{a}}$ of its mass center, and the angular acceleration α of the body. This relation, which is represented in Fig. 16.7 in the form of a *free-body-diagram equation*, can be used to determine the acceleration $\bar{\mathbf{a}}$ and the angular acceleration α produced by a given system of forces acting on a rigid body or, conversely, to determine the forces which produce a given motion of the rigid body.

The three algebraic equations (16.6) can be used to solve problems of plane motion.[†] However, our experience in statics suggests that the solution of many problems involving rigid bodies could be simplified by an appropriate choice of the point about which the moments of the forces are computed. It is therefore preferable to remember the relation existing between the forces and the accelerations in the pictorial form shown in Fig. 16.7 and to derive from this fundamental relation the component or moment equations which fit best the solution of the problem under consideration.

The fundamental relation shown in Fig. 16.7 can be presented in an alternative form if we add to the external forces an inertia vector $-m\bar{\mathbf{a}}$ of sense opposite to that of $\bar{\mathbf{a}}$, attached at G , and an inertia couple $-\bar{I}\alpha$ of moment equal in magnitude to $\bar{I}\alpha$ and of sense opposite to that of α (Fig. 16.10). The system obtained is equivalent to zero, and the rigid body is said to be in *dynamic equilibrium*.

Whether the principle of equivalence of external and effective forces is directly applied, as in Fig. 16.7, or whether the concept of dynamic equilibrium is introduced, as in Fig. 16.10, the use of free-body-diagram equations showing vectorially the relationship existing between the forces applied on the rigid body and the resulting linear and angular accelerations presents considerable advantages over the blind application of formulas (16.6). These advantages can be summarized as follows:

1. The use of a pictorial representation provides a much clearer understanding of the effect of the forces on the motion of the body.

[†]We recall that the last of Eqs. (16.6) is valid only in the case of the plane motion of a rigid body symmetrical with respect to the reference plane. In all other cases, the methods of Chap. 18 should be used.

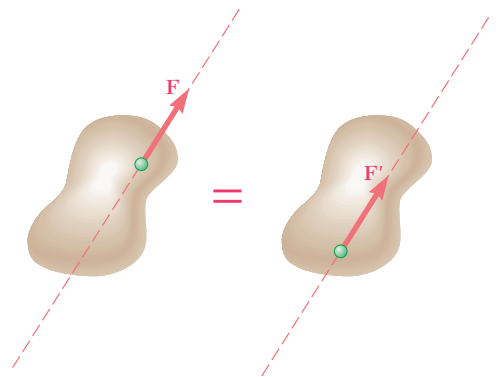


Fig. 3.3 (repeated)

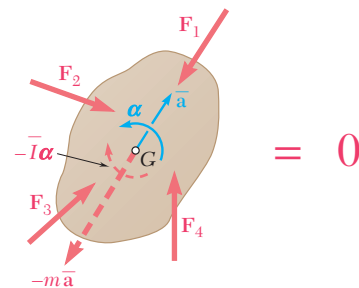


Fig. 16.10

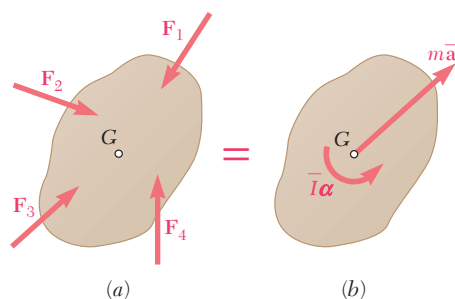


Fig. 16.7 (repeated)

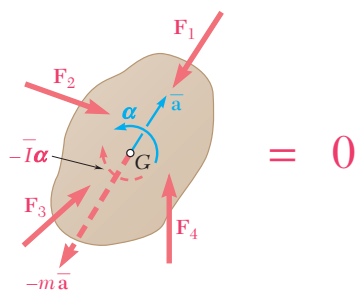


Fig. 16.10 (repeated)

2. This approach makes it possible to divide the solution of a dynamics problem into two parts: In the first part, the analysis of the kinematic and kinetic characteristics of the problem leads to the free-body diagrams of Fig. 16.7 or 16.10; in the second part, the diagram obtained is used to analyze the various forces and vectors involved by the methods of Chap. 3.
3. A unified approach is provided for the analysis of the plane motion of a rigid body, regardless of the particular type of motion involved. While the kinematics of the various motions considered may vary from one case to the other, the approach to the kinetics of the motion is consistently the same. In every case a diagram will be drawn showing the external forces, the vector $m\bar{a}$ associated with the motion of G , and the couple $\bar{I}\alpha$ associated with the rotation of the body about G .
4. The resolution of the plane motion of a rigid body into a translation and a centroidal rotation, which is used here, is a basic concept which can be applied effectively throughout the study of mechanics. It will be used again in Chap. 17 with the method of work and energy and the method of impulse and momentum.
5. As you will see in Chap. 18, this approach can be extended to the study of the general three-dimensional motion of a rigid body. The motion of the body will again be resolved into a translation and a rotation about the mass center, and free-body-diagram equations will be used to indicate the relationship existing between the external forces and the rates of change of the linear and angular momentum of the body.

16.7 SYSTEMS OF RIGID BODIES

The method described in the preceding section can also be used in problems involving the plane motion of several connected rigid bodies. For each part of the system, a diagram similar to Fig. 16.7 or Fig. 16.10 can be drawn. The equations of motion obtained from these diagrams are solved simultaneously.

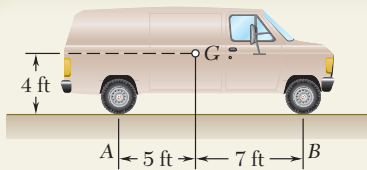
In some cases, as in Sample Prob. 16.3, a single diagram can be drawn for the entire system. This diagram should include all the external forces, as well as the vectors $m\bar{a}$ and the couples $\bar{I}\alpha$ associated with the various parts of the system. However, internal forces such as the forces exerted by connecting cables, can be omitted since they occur in pairs of equal and opposite forces and are thus equipollent to zero. The equations obtained by expressing that the system of the external forces is equipollent to the system of the effective forces can be solved for the remaining unknowns.†

It is not possible to use this second approach in problems involving more than three unknowns, since only three equations of motion are available when a single diagram is used. We need not elaborate upon this point, since the discussion involved would be completely similar to that given in Sec. 6.11 in the case of the equilibrium of a system of rigid bodies.



Photo 16.3 The forklift and moving load can be analyzed as a system of two connected rigid bodies in plane motion.

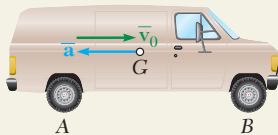
†Note that we cannot speak of *equivalent* systems since we are not dealing with a single rigid body.



SAMPLE PROBLEM 16.1

When the forward speed of the truck shown was 30 ft/s, the brakes were suddenly applied, causing all four wheels to stop rotating. It was observed that the truck skidded to rest in 20 ft. Determine the magnitude of the normal reaction and of the friction force at each wheel as the truck skidded to rest.

SOLUTION



Kinematics of Motion. Choosing the positive sense to the right and using the equations of uniformly accelerated motion, we write

$$\begin{aligned} \bar{v}_0 &= +30 \text{ ft/s} & \bar{v}^2 &= \bar{v}_0^2 + 2\bar{a}\bar{x} & 0 &= (30)^2 + 2\bar{a}(20) \\ \bar{a} &= -22.5 \text{ ft/s}^2 & \bar{a} &= 22.5 \text{ ft/s}^2 \leftarrow \end{aligned}$$

Equations of Motion. The external forces consist of the weight $\mathbf{W}$ of the truck and of the normal reactions and friction forces at the wheels. (The vectors $\mathbf{N}_A$ and $\mathbf{F}_A$ represent the sum of the reactions at the rear wheels, while $\mathbf{N}_B$ and $\mathbf{F}_B$ represent the sum of the reactions at the front wheels.) Since the truck is in translation, the effective forces reduce to the vector $m\bar{\mathbf{a}}$ attached at G . Three equations of motion are obtained by expressing that the system of the external forces is equivalent to the system of the effective forces.

$$+\uparrow \Sigma F_y = \Sigma (F_y)_{\text{eff}}: \quad N_A + N_B - W = 0$$

Since $F_A = \mu_k N_A$ and $F_B = \mu_k N_B$, where μ_k is the coefficient of kinetic friction, we find that

$$F_A + F_B = \mu_k(N_A + N_B) = \mu_k W$$

$$\begin{aligned} +\rightarrow \Sigma F_x &= \Sigma (F_x)_{\text{eff}}: \quad -(F_A + F_B) = -m\bar{a} \\ -\mu_k W &= -\frac{W}{32.2 \text{ ft/s}^2} (22.5 \text{ ft/s}^2) \\ \mu_k &= 0.699 \end{aligned}$$

$$\begin{aligned} +\uparrow \Sigma M_A &= \Sigma (M_A)_{\text{eff}}: \quad -W(5 \text{ ft}) + N_B(12 \text{ ft}) = m\bar{a}(4 \text{ ft}) \\ -W(5 \text{ ft}) + N_B(12 \text{ ft}) &= \frac{W}{32.2 \text{ ft/s}^2} (22.5 \text{ ft/s}^2)(4 \text{ ft}) \\ N_B &= 0.650W \end{aligned}$$

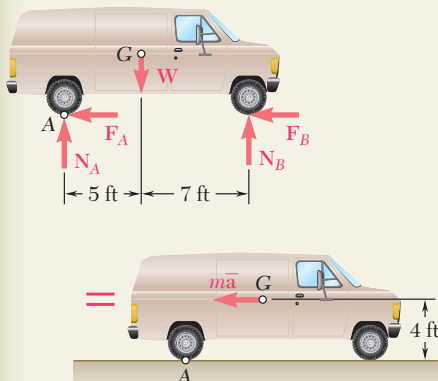
$$F_B = \mu_k N_B = (0.699)(0.650W) \quad F_B = 0.454W$$

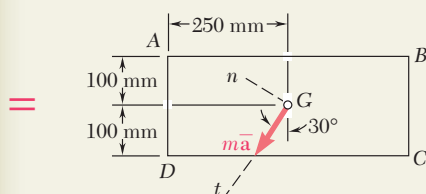
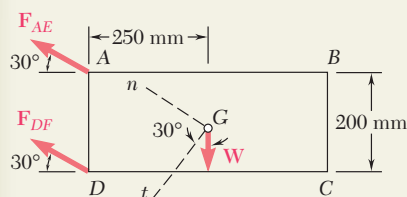
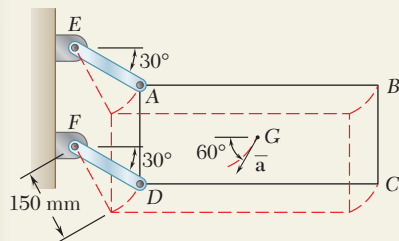
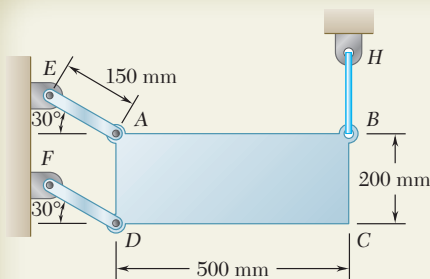
$$\begin{aligned} +\uparrow \Sigma F_y &= \Sigma (F_y)_{\text{eff}}: \quad N_A + N_B - W = 0 \\ N_A + 0.650W - W &= 0 \\ N_A &= 0.350W \end{aligned}$$

$$F_A = \mu_k N_A = (0.699)(0.350W) \quad F_A = 0.245W$$

Reactions at Each Wheel. Recalling that the values computed above represent the sum of the reactions at the two front wheels or the two rear wheels, we obtain the magnitude of the reactions at each wheel by writing

$$\begin{aligned} N_{\text{front}} &= \frac{1}{2}N_B = 0.325W & N_{\text{rear}} &= \frac{1}{2}N_A = 0.175W \quad \blacktriangleleft \\ F_{\text{front}} &= \frac{1}{2}F_B = 0.227W & F_{\text{rear}} &= \frac{1}{2}F_A = 0.122W \quad \blacktriangleleft \end{aligned}$$





SAMPLE PROBLEM 16.2

The thin plate $ABCD$ of mass 8 kg is held in the position shown by the wire BH and two links AE and DF . Neglecting the mass of the links, determine immediately after wire BH has been cut (a) the acceleration of the plate, (b) the force in each link.

SOLUTION

Kinematics of Motion. After wire BH has been cut, we observe that corners A and D move along parallel circles of radius 150 mm centered, respectively, at E and F . The motion of the plate is thus a curvilinear translation; the particles forming the plate move along parallel circles of radius 150 mm .

At the instant wire BH is cut, the velocity of the plate is zero. Thus the acceleration $\bar{a}$ of the mass center G of the plate is tangent to the circular path which will be described by G .

Equations of Motion. The external forces consist of the weight $\mathbf{W}$ and the forces $\mathbf{F}_{AE}$ and $\mathbf{F}_{DF}$ exerted by the links. Since the plate is in translation, the effective forces reduce to the vector $m\bar{a}$ attached at G and directed along the t axis. A free-body-diagram equation is drawn to show that the system of the external forces is equivalent to the system of the effective forces.

a. Acceleration of the Plate.

$$+\nearrow \Sigma F_t = \Sigma (F_t)_{\text{eff}}:$$

$$W \cos 30^\circ = m\bar{a}$$

$$mg \cos 30^\circ = m\bar{a}$$

$$\bar{a} = g \cos 30^\circ = (9.81 \text{ m/s}^2) \cos 30^\circ \quad (1)$$

$$\bar{a} = 8.50 \text{ m/s}^2 \nearrow 60^\circ \quad \blacktriangleleft$$

b. Forces in Links AE and DF .

$$+\nearrow \Sigma F_n = \Sigma (F_n)_{\text{eff}}: \quad F_{AE} + F_{DF} - W \sin 30^\circ = 0 \quad (2)$$

$$+\downarrow \Sigma M_G = \Sigma (M_G)_{\text{eff}}:$$

$$(F_{AE} \sin 30^\circ)(250 \text{ mm}) - (F_{AE} \cos 30^\circ)(100 \text{ mm}) + (F_{DF} \sin 30^\circ)(250 \text{ mm}) + (F_{DF} \cos 30^\circ)(100 \text{ mm}) = 0$$

$$38.4F_{AE} + 211.6F_{DF} = 0$$

$$F_{DF} = -0.1815F_{AE} \quad (3)$$

Substituting for F_{DF} from (3) into (2), we write

$$F_{AE} - 0.1815F_{AE} - W \sin 30^\circ = 0$$

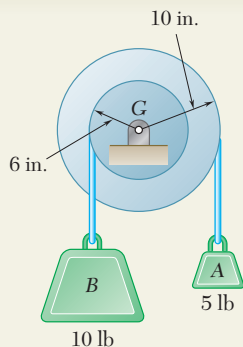
$$F_{AE} = 0.6109W$$

$$F_{DF} = -0.1815(0.6109W) = -0.1109W$$

Noting that $W = mg = (8 \text{ kg})(9.81 \text{ m/s}^2) = 78.48 \text{ N}$, we have

$$F_{AE} = 0.6109(78.48 \text{ N}) \quad F_{AE} = 47.9 \text{ N } T \quad \blacktriangleleft$$

$$F_{DF} = -0.1109(78.48 \text{ N}) \quad F_{DF} = 8.70 \text{ N } C \quad \blacktriangleleft$$



SAMPLE PROBLEM 16.3

A pulley weighing 12 lb and having a radius of gyration of 8 in. is connected to two blocks as shown. Assuming no axle friction, determine the angular acceleration of the pulley and the acceleration of each block.

SOLUTION

Sense of Motion. Although an arbitrary sense of motion can be assumed (since no friction forces are involved) and later checked by the sign of the answer, we may prefer to determine the actual sense of rotation of the pulley first. The weight of block B required to maintain the equilibrium of the pulley when it is acted upon by the 5-lb block A is first determined. We write

$$+\circlearrowleft \Sigma M_G = 0: \quad W_B(6 \text{ in.}) - (5 \text{ lb})(10 \text{ in.}) = 0 \quad W_B = 8.33 \text{ lb}$$

Since block B actually weighs 10 lb, the pulley will rotate counterclockwise.

Kinematics of Motion. Assuming α counterclockwise and noting that $a_A = r_A \alpha$ and $a_B = r_B \alpha$, we obtain

$$\mathbf{a}_A = \left(\frac{10}{12} \text{ ft}\right) \alpha \uparrow \quad \mathbf{a}_B = \left(\frac{6}{12} \text{ ft}\right) \alpha \downarrow$$

Equations of Motion. A single system consisting of the pulley and the two blocks is considered. Forces external to this system consist of the weights of the pulley and the two blocks and of the reaction at G. (The forces exerted by the cables on the pulley and on the blocks are internal to the system considered and cancel out.) Since the motion of the pulley is a centroidal rotation and the motion of each block is a translation, the effective forces reduce to the couple $\bar{I}\alpha$ and the two vectors $m\mathbf{a}_A$ and $m\mathbf{a}_B$. The centroidal moment of inertia of the pulley is

$$\bar{I} = m\bar{k}^2 = \frac{W}{g}\bar{k}^2 = \frac{12 \text{ lb}}{32.2 \text{ ft/s}^2} \left(\frac{8}{12} \text{ ft}\right)^2 = 0.1656 \text{ lb} \cdot \text{ft} \cdot \text{s}^2$$

Since the system of the external forces is equipollent to the system of the effective forces, we write

$$+\circlearrowleft \Sigma M_G = \Sigma (M_G)_{\text{eff}}:$$

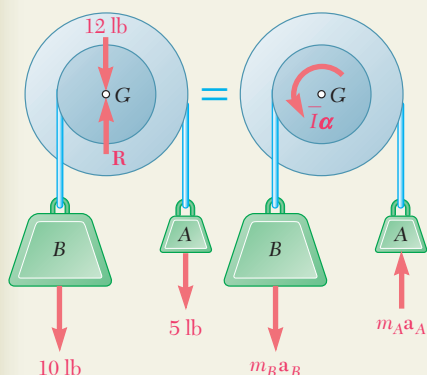
$$(10 \text{ lb})\left(\frac{6}{12} \text{ ft}\right) - (5 \text{ lb})\left(\frac{10}{12} \text{ ft}\right) = +\bar{I}\alpha + m_B a_B \left(\frac{6}{12} \text{ ft}\right) + m_A a_A \left(\frac{10}{12} \text{ ft}\right)$$

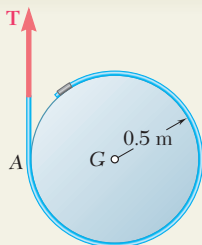
$$(10)\left(\frac{6}{12}\right) - (5)\left(\frac{10}{12}\right) = 0.1656\alpha + \frac{10}{32.2}\left(\frac{6}{12}\alpha\right) + \frac{5}{32.2}\left(\frac{10}{12}\alpha\right)$$

$$\alpha = +2.374 \text{ rad/s}^2 \quad \alpha = 2.37 \text{ rad/s}^2 \uparrow$$

$$a_A = r_A \alpha = \left(\frac{10}{12} \text{ ft}\right)(2.374 \text{ rad/s}^2) \quad a_A = 1.978 \text{ ft/s}^2 \uparrow$$

$$a_B = r_B \alpha = \left(\frac{6}{12} \text{ ft}\right)(2.374 \text{ rad/s}^2) \quad a_B = 1.187 \text{ ft/s}^2 \downarrow$$

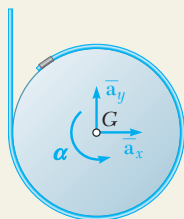




SAMPLE PROBLEM 16.4

A cord is wrapped around a homogeneous disk of radius $r = 0.5$ m and mass $m = 15$ kg. If the cord is pulled upward with a force $\mathbf{T}$ of magnitude 180 N, determine (a) the acceleration of the center of the disk, (b) the angular acceleration of the disk, (c) the acceleration of the cord.

SOLUTION

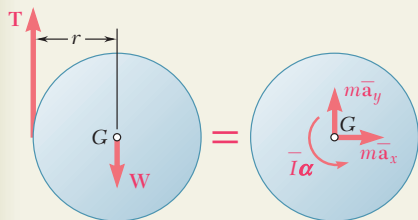


Equations of Motion. We assume that the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of the center are directed, respectively, to the right and upward and that the angular acceleration of the disk is counterclockwise. The external forces acting on the disk consist of the weight $\mathbf{W}$ and the force $\mathbf{T}$ exerted by the cord. This system is equivalent to the system of the effective forces, which consists of a vector of components $m\bar{a}_x$ and $m\bar{a}_y$ attached at G and a couple $\bar{I}\alpha$. We write

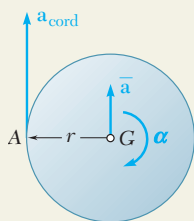
$$\begin{aligned} \rightarrow \Sigma F_x &= \Sigma (F_x)_{\text{eff}}: & 0 &= m\bar{a}_x & \bar{a}_x &= 0 \quad \blacktriangleleft \\ + \uparrow \Sigma F_y &= \Sigma (F_y)_{\text{eff}}: & T - W &= m\bar{a}_y \\ & & \bar{a}_y &= \frac{T - W}{m} \end{aligned}$$

Since $T = 180$ N, $m = 15$ kg, and $W = (15 \text{ kg})(9.81 \text{ m/s}^2) = 147.1$ N, we have

$$\bar{a}_y = \frac{180 \text{ N} - 147.1 \text{ N}}{15 \text{ kg}} = +2.19 \text{ m/s}^2 \quad \bar{a}_y = 2.19 \text{ m/s}^2 \uparrow \quad \blacktriangleleft$$

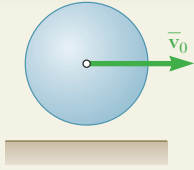


$$\begin{aligned} + \curvearrowright \Sigma M_G &= \Sigma (M_G)_{\text{eff}}: & -Tr &= \bar{I}\alpha \\ & & -Tr &= (\tfrac{1}{2}mr^2)\alpha \\ \alpha &= -\frac{2T}{mr} = -\frac{2(180 \text{ N})}{(15 \text{ kg})(0.5 \text{ m})} = -48.0 \text{ rad/s}^2 \\ & & \alpha &= 48.0 \text{ rad/s}^2 \downarrow \quad \blacktriangleleft \end{aligned}$$



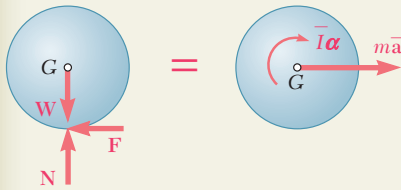
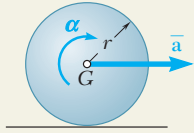
Acceleration of Cord. Since the acceleration of the cord is equal to the tangential component of the acceleration of point A on the disk, we write

$$\begin{aligned} \mathbf{a}_{\text{cord}} &= (\mathbf{a}_A)_t = \bar{\mathbf{a}} + (\mathbf{a}_{A/G})_t \\ &= [2.19 \text{ m/s}^2 \uparrow] + [(0.5 \text{ m})(48 \text{ rad/s}^2) \uparrow] \\ \mathbf{a}_{\text{cord}} &= 26.2 \text{ m/s}^2 \uparrow \quad \blacktriangleleft \end{aligned}$$



SAMPLE PROBLEM 16.5

A uniform sphere of mass m and radius r is projected along a rough horizontal surface with a linear velocity $\bar{v}_0$ and no angular velocity. Denoting by μ_k the coefficient of kinetic friction between the sphere and the floor, determine (a) the time t_1 at which the sphere will start rolling without sliding, (b) the linear velocity and angular velocity of the sphere at time t_1 .



SOLUTION

Equations of Motion. The positive sense is chosen to the right for $\bar{a}$ and clockwise for α . The external forces acting on the sphere consist of the weight $\mathbf{W}$, the normal reaction $\mathbf{N}$, and the friction force $\mathbf{F}$. Since the point of the sphere in contact with the surface is sliding to the right, the friction force $\mathbf{F}$ is directed to the left. While the sphere is sliding, the magnitude of the friction force is $F = \mu_k N$. The effective forces consist of the vector $m\bar{a}$ attached at G and the couple $\bar{I}\alpha$. Expressing that the system of the external forces is equivalent to the system of the effective forces, we write

$$\begin{aligned} +\uparrow \Sigma F_y &= \Sigma (F_y)_{\text{eff}}: & N - W &= 0 \\ & N = W = mg & F &= \mu_k N = \mu_k mg \\ +\rightarrow \Sigma F_x &= \Sigma (F_x)_{\text{eff}}: & -F &= m\bar{a} & -\mu_k mg &= m\bar{a} & \bar{a} &= -\mu_k g \\ +\curvearrowright \Sigma M_G &= \Sigma (M_G)_{\text{eff}}: & Fr &= \bar{I}\alpha \end{aligned}$$

Noting that $\bar{I} = \frac{2}{5}mr^2$ and substituting the value obtained for F , we write

$$(\mu_k mg)r = \frac{2}{5}mr^2\alpha \quad \alpha = \frac{5}{2} \frac{\mu_k g}{r}$$

Kinematics of Motion. As long as the sphere both rotates and slides, its linear and angular motions are uniformly accelerated.

$$t = 0, \bar{v} = \bar{v}_0 \quad \bar{v} = \bar{v}_0 + \bar{a}t = \bar{v}_0 - \mu_k gt \quad (1)$$

$$t = 0, \omega = 0 \quad \omega = \omega_0 + \alpha t = 0 + \left(\frac{5}{2} \frac{\mu_k g}{r} \right) t \quad (2)$$

The sphere will start rolling without sliding when the velocity $\mathbf{v}_C$ of the point of contact C is zero. At that time, $t = t_1$, point C becomes the instantaneous center of rotation, and we have

$$\bar{v}_1 = r\omega_1 \quad (3)$$

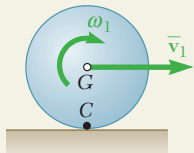
Substituting in (3) the values obtained for $\bar{v}_1$ and ω_1 by making $t = t_1$ in (1) and (2), respectively, we write

$$\bar{v}_0 - \mu_k gt_1 = r \left(\frac{5}{2} \frac{\mu_k g}{r} t_1 \right) \quad t_1 = \frac{2}{7} \frac{\bar{v}_0}{\mu_k g} \quad \blacktriangleleft$$

Substituting for t_1 into (2), we have

$$\omega_1 = \frac{5}{2} \frac{\mu_k g}{r} t_1 = \frac{5}{2} \frac{\mu_k g}{r} \left(\frac{2}{7} \frac{\bar{v}_0}{\mu_k g} \right) \quad \omega_1 = \frac{5}{7} \frac{\bar{v}_0}{r} \quad \omega_1 = \frac{5}{7} \frac{\bar{v}_0}{r} \quad \blacktriangleleft$$

$$\bar{v}_1 = r\omega_1 = r \left(\frac{5}{7} \frac{\bar{v}_0}{r} \right) \quad \bar{v}_1 = \frac{5}{7} \bar{v}_0 \quad \mathbf{v}_1 = \frac{5}{7} \bar{v}_0 \rightarrow \quad \blacktriangleleft$$



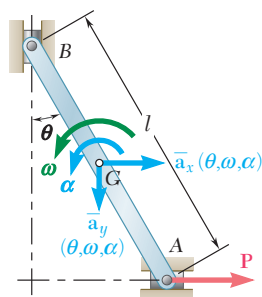


Fig. 16.11

16.8 CONSTRAINED PLANE MOTION

Most engineering applications deal with rigid bodies which are moving under given constraints. For example, cranks must rotate about a fixed axis, wheels must roll without sliding, and connecting rods must describe certain prescribed motions. In all such cases, definite relations exist between the components of the acceleration $\bar{\mathbf{a}}$ of the mass center G of the body considered and its angular acceleration α ; the corresponding motion is said to be a *constrained motion*.

The solution of a problem involving a constrained plane motion calls first for a *kinematic analysis* of the problem. Consider, for example, a slender rod AB of length l and mass m whose extremities are connected to blocks of negligible mass which slide along horizontal and vertical frictionless tracks. The rod is pulled by a force $\mathbf{P}$ applied at A (Fig. 16.11). We know from Sec. 15.8 that the acceleration $\bar{\mathbf{a}}$ of the mass center G of the rod can be determined at any given instant from the position of the rod, its angular velocity, and its angular acceleration at that instant. Suppose, for example, that the values of θ , ω , and α are known at a given instant and that we wish to determine the corresponding value of the force $\mathbf{P}$, as well as the reactions at A and B . We should first *determine the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of the mass center G* by the method of Sec. 15.8. We next apply d'Alembert's principle (Fig. 16.12), using the expressions obtained for $\bar{a}_x$ and $\bar{a}_y$. The unknown forces $\mathbf{P}$, $\mathbf{N}_A$, and $\mathbf{N}_B$ can then be determined by writing and solving the appropriate equations.

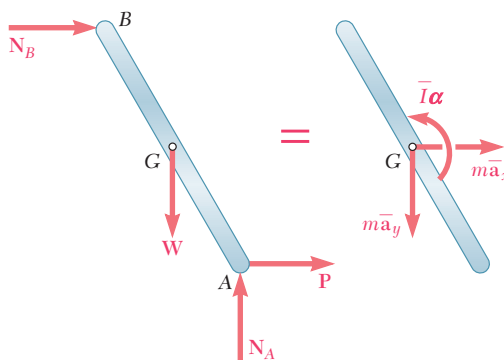


Fig. 16.12

Suppose now that the applied force $\mathbf{P}$, the angle θ , and the angular velocity ω of the rod are known at a given instant and that we wish to find the angular acceleration α of the rod and the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of its mass center at that instant, as well as the reactions at A and B . The preliminary kinematic study of the problem will have for its object to *express the components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of G in terms of the angular acceleration α of the rod*. This will be done by first expressing the acceleration of a suitable reference point such as A in terms of the angular acceleration α . The components $\bar{a}_x$ and $\bar{a}_y$ of the acceleration of G can then be determined in terms of α , and the expressions obtained carried into Fig. 16.12. Three equations can then be derived in terms of α , N_A , and N_B and solved for the three unknowns (see Sample

Prob. 16.10). Note that the method of dynamic equilibrium can also be used to carry out the solution of the two types of problems we have considered (Fig. 16.13).

When a mechanism consists of *several moving parts*, the approach just described can be used with each part of the mechanism. The procedure required to determine the various unknowns is then similar to the procedure followed in the case of the equilibrium of a system of connected rigid bodies (Sec. 6.11).

Earlier, we analyzed two particular cases of constrained plane motion: the translation of a rigid body, in which the angular acceleration of the body is constrained to be zero, and the centroidal rotation, in which the acceleration $\bar{\mathbf{a}}$ of the mass center of the body is constrained to be zero. Two other particular cases of constrained plane motion are of special interest: the *noncentroidal rotation* of a rigid body and the *rolling motion* of a disk or wheel. These two cases can be analyzed by one of the general methods described above. However, in view of the range of their applications, they deserve a few special comments.

Noncentroidal Rotation. The motion of a rigid body constrained to rotate about a fixed axis which does not pass through its mass center is called *noncentroidal rotation*. The mass center G of the body moves along a circle of radius $\bar{r}$ centered at the point O , where the axis of rotation intersects the plane of reference (Fig. 16.14). Denoting, respectively, by $\boldsymbol{\omega}$ and $\boldsymbol{\alpha}$ the angular velocity and the angular acceleration of the line OG , we obtain the following expressions for the tangential and normal components of the acceleration of G :

$$\bar{a}_t = \bar{r}\alpha \quad \bar{a}_n = \bar{r}\omega^2 \quad (16.7)$$

Since line OG belongs to the body, its angular velocity $\boldsymbol{\omega}$ and its angular acceleration $\boldsymbol{\alpha}$ also represent the angular velocity and the angular acceleration of the body in its motion relative to G . Equations (16.7) define, therefore, the kinematic relation existing between the motion of the mass center G and the motion of the body about G . They should be used to eliminate $\bar{a}_t$ and $\bar{a}_n$ from the equations obtained by applying d'Alembert's principle (Fig. 16.15) or the method of dynamic equilibrium (Fig. 16.16).

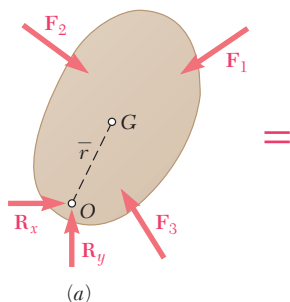


Fig. 16.15

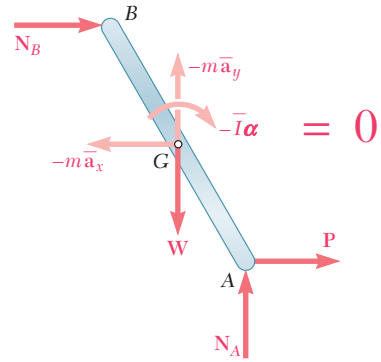
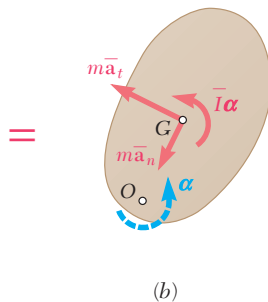


Fig. 16.13

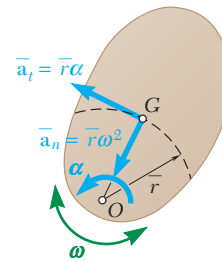


Fig. 16.14

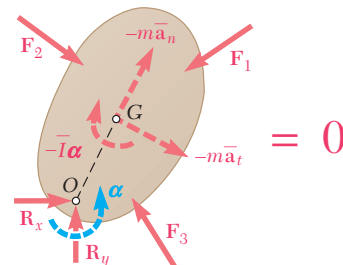


Fig. 16.16

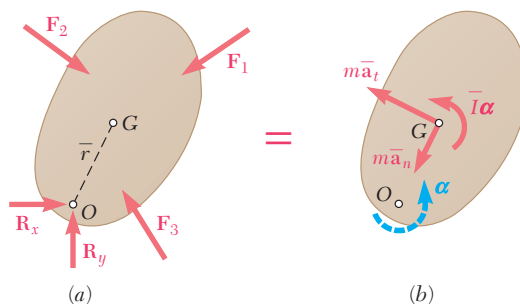


Fig. 16.15 (repeated)

An interesting relation is obtained by equating the moments about the fixed point O of the forces and vectors shown, respectively, in parts a and b of Fig. 16.15. We write

$$+\sum M_O = \bar{I}\alpha + (m\bar{r}\alpha)\bar{r} = (\bar{I} + m\bar{r}^2)\alpha$$

But according to the parallel-axis theorem, we have $\bar{I} + m\bar{r}^2 = I_O$, where I_O denotes the moment of inertia of the rigid body about the fixed axis. We therefore write

$$\Sigma M_O = I_O \alpha \quad (16.8)$$

Although formula (16.8) expresses an important relation between the sum of the moments of the external forces about the fixed point O and the product $I_O\alpha$, it should be clearly understood that this formula does not mean that the system of the external forces is equivalent to a couple of moment $I_O\alpha$. The system of the effective forces, and thus the system of the external forces, reduces to a couple only when O coincides with G —that is, *only when the rotation is centroidal* (Sec. 16.4). In the more general case of noncentroidal rotation, the system of the external forces does not reduce to a couple.

A particular case of noncentroidal rotation is of special interest—the case of *uniform rotation*, in which the angular velocity ω is constant. Since α is zero, the inertia couple in Fig. 16.16 vanishes and the inertia vector reduces to its normal component. This component (also called *centrifugal force*) represents the tendency of the rigid body to break away from the axis of rotation.

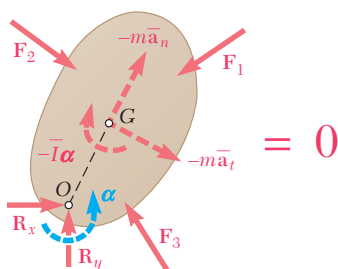


Fig. 16.16 (repeated)

Rolling Motion. Another important case of plane motion is the motion of a disk or wheel rolling on a plane surface. If the disk is constrained to roll without sliding, the acceleration $\bar{a}$ of its mass center G and its angular acceleration α are not independent. Assuming that the disk is balanced, so that its mass center and its geometric center coincide, we first write that the distance $\bar{x}$ traveled by G during a rotation θ of the disk is $\bar{x} = r\theta$, where r is the radius of the disk. Differentiating this relation twice, we write

$$\bar{a} = r\alpha \quad (16.9)$$

Recalling that the system of the effective forces in plane motion reduces to a vector $m\bar{\mathbf{a}}$ and a couple $\bar{I}\alpha$, we find that in the particular case of the rolling motion of a balanced disk, the effective forces reduce to a vector of magnitude $mra\alpha$ attached at G and to a couple of magnitude $\bar{I}\alpha$. We may thus express that the external forces are equivalent to the vector and couple shown in Fig. 16.17.

When a disk *rolls without sliding*, there is no relative motion between the point of the disk in contact with the ground and the ground itself. Thus as far as the computation of the friction force $\mathbf{F}$ is concerned, a rolling disk can be compared with a block at rest on a surface. The magnitude F of the friction force can have any value, as long as this value does not exceed the maximum value $F_m = \mu_s N$, where μ_s is the coefficient of static friction and N is the magnitude of the normal force. In the case of a rolling disk, the magnitude F of the friction force should therefore be determined independently of N by solving the equation obtained from Fig. 16.17.

When *sliding is impending*, the friction force reaches its maximum value $F_m = \mu_s N$ and can be obtained from N .

When the disk *rotates and slides* at the same time, a relative motion exists between the point of the disk which is in contact with the ground and the ground itself, and the force of friction has the magnitude $F_k = \mu_k N$, where μ_k is the coefficient of kinetic friction. In this case, however, the motion of the mass center G of the disk and the rotation of the disk about G are independent, and $\bar{a}$ is not equal to $r\alpha$.

These three different cases can be summarized as follows:

Rolling, no sliding:	$F \leq \mu_s N$	$\bar{a} = r\alpha$
Rolling, sliding impending:	$F = \mu_s N$	$\bar{a} = r\alpha$
Rotating and sliding:	$F = \mu_k N$	$\bar{a}$ and α independent

When it is not known whether or not a disk slides, it should first be assumed that the disk rolls without sliding. If F is found smaller than or equal to $\mu_s N$, the assumption is proved correct. If F is found larger than $\mu_s N$, the assumption is incorrect and the problem should be started again, assuming rotating and sliding.

When a disk is *unbalanced*, i.e., when its mass center G does not coincide with its geometric center O , the relation (16.9) does not hold between $\bar{a}$ and α . However, a similar relation holds between the magnitude a_O of the acceleration of the geometric center and the angular acceleration α of an unbalanced disk which rolls without sliding. We have

$$a_O = r\alpha \quad (16.10)$$

To determine $\bar{a}$ in terms of the angular acceleration α and the angular velocity ω of the disk, we can use the relative-acceleration formula

$$\begin{aligned} \bar{\mathbf{a}} &= \bar{\mathbf{a}}_G = \mathbf{a}_O + \mathbf{a}_{G/O} \\ &= \mathbf{a}_O + (\mathbf{a}_{G/O})_t + (\mathbf{a}_{G/O})_n \end{aligned} \quad (16.11)$$

where the three component accelerations obtained have the directions indicated in Fig. 16.18 and the magnitudes $a_O = r\alpha$, $(a_{G/O})_t = (OG)\alpha$, and $(a_{G/O})_n = (OG)\omega^2$.

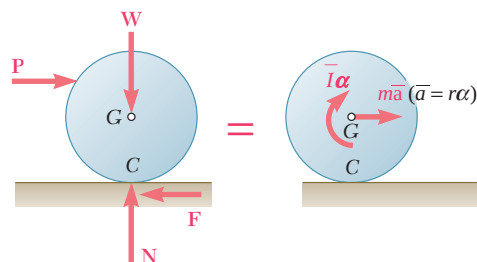


Fig. 16.17



Photo 16.4 As the ball hits the bowling alley, it first spins and slides, then rolls without sliding.

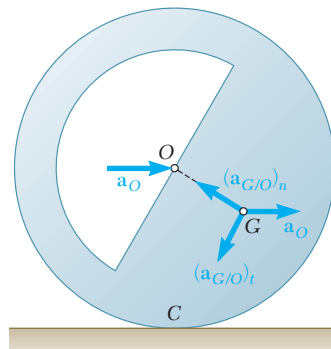
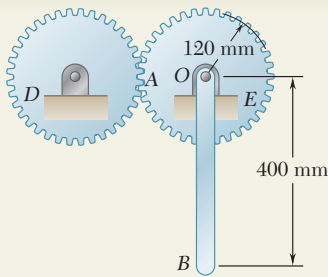


Fig. 16.18



SAMPLE PROBLEM 16.6

The portion AOB of a mechanism consists of a 400-mm steel rod OB welded to a gear E of radius 120 mm which can rotate about a horizontal shaft O. It is actuated by a gear D and, at the instant shown, has a clockwise angular velocity of 8 rad/s and a counterclockwise angular acceleration of 40 rad/s². Knowing that rod OB has a mass of 3 kg and gear E a mass of 4 kg and a radius of gyration of 85 mm, determine (a) the tangential force exerted by gear D on gear E, (b) the components of the reaction at shaft O.

SOLUTION

In determining the effective forces of the rigid body AOB, gear E and rod OB will be considered separately. Therefore, the components of the acceleration of the mass center G_{OB} of the rod will be determined first:

$$(\bar{a}_{OB})_t = \bar{r}\alpha = (0.200 \text{ m})(40 \text{ rad/s}^2) = 8 \text{ m/s}^2$$

$$(\bar{a}_{OB})_n = \bar{r}\omega^2 = (0.200 \text{ m})(8 \text{ rad/s})^2 = 12.8 \text{ m/s}^2$$

Equations of Motion. Two sketches of the rigid body AOB have been drawn. The first shows the external forces consisting of the weight W_E of gear E, the weight W_{OB} of the rod OB, the force F exerted by gear D, and the components R_x and R_y of the reaction at O. The magnitudes of the weights are, respectively,

$$W_E = m_E g = (4 \text{ kg})(9.81 \text{ m/s}^2) = 39.2 \text{ N}$$

$$W_{OB} = m_{OB} g = (3 \text{ kg})(9.81 \text{ m/s}^2) = 29.4 \text{ N}$$

The second sketch shows the effective forces, which consist of a couple $\bar{I}_E \alpha$ (since gear E is in centroidal rotation) and of a couple and two vector components at the mass center of OB. Since the accelerations are known, we compute the magnitudes of these components and couples:

$$\bar{I}_E \alpha = m_E \bar{k}_E^2 \alpha = (4 \text{ kg})(0.085 \text{ m})^2 (40 \text{ rad/s}^2) = 1.156 \text{ N} \cdot \text{m}$$

$$m_{OB}(\bar{a}_{OB})_t = (3 \text{ kg})(8 \text{ m/s}^2) = 24.0 \text{ N}$$

$$m_{OB}(\bar{a}_{OB})_n = (3 \text{ kg})(12.8 \text{ m/s}^2) = 38.4 \text{ N}$$

$$\bar{I}_{OB} \alpha = (\frac{1}{12} m_{OB} L^2) \alpha = \frac{1}{12} (3 \text{ kg})(0.400 \text{ m})^2 (40 \text{ rad/s}^2) = 1.600 \text{ N} \cdot \text{m}$$

Expressing that the system of the external forces is equivalent to the system of the effective forces, we write the following equations:

$$+\uparrow \Sigma M_O = \Sigma (M_O)_{\text{eff}}:$$

$$F(0.120 \text{ m}) = \bar{I}_E \alpha + m_{OB}(\bar{a}_{OB})_t(0.200 \text{ m}) + \bar{I}_{OB} \alpha$$

$$F(0.120 \text{ m}) = 1.156 \text{ N} \cdot \text{m} + (24.0 \text{ N})(0.200 \text{ m}) + 1.600 \text{ N} \cdot \text{m}$$

$$F = 63.0 \text{ N}$$

$$\mathbf{F} = 63.0 \text{ N} \downarrow \quad \blacktriangleleft$$

$$+\rightarrow \Sigma F_x = \Sigma (F_x)_{\text{eff}}:$$

$$R_x = m_{OB}(\bar{a}_{OB})_t$$

$$R_x = 24.0 \text{ N}$$

$$\mathbf{R}_x = 24.0 \text{ N} \rightarrow \quad \blacktriangleleft$$

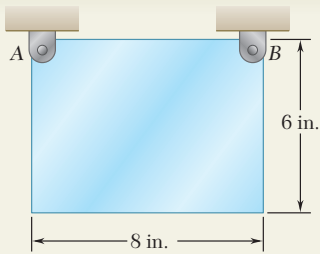
$$+\uparrow \Sigma F_y = \Sigma (F_y)_{\text{eff}}:$$

$$R_y - F - W_E - W_{OB} = m_{OB}(\bar{a}_{OB})_n$$

$$R_y - 63.0 \text{ N} - 39.2 \text{ N} - 29.4 \text{ N} = 38.4 \text{ N}$$

$$R_y = 170.0 \text{ N}$$

$$\mathbf{R}_y = 170.0 \text{ N} \uparrow \quad \blacktriangleleft$$



SAMPLE PROBLEM 16.7

A 6×8 in. rectangular plate weighing 60 lb is suspended from two pins A and B. If pin B is suddenly removed, determine (a) the angular acceleration of the plate, (b) the components of the reaction at pin A, immediately after pin B has been removed.

SOLUTION

a. Angular Acceleration. We observe that as the plate rotates about point A, its mass center G describes a circle of radius $\bar{r}$ with center at A.

Since the plate is released from rest ($\omega = 0$), the normal component of the acceleration of G is zero. The magnitude of the acceleration $\bar{a}$ of the mass center G is thus $\bar{a} = \bar{r}\alpha$. We draw the diagram shown to express that the external forces are equivalent to the effective forces:

$$+\downarrow \Sigma M_A = \Sigma (M_A)_{\text{eff}}: \quad W\bar{x} = (m\bar{a})\bar{r} + \bar{I}\alpha$$

Since $\bar{a} = \bar{r}\alpha$, we have

$$W\bar{x} = m(\bar{r}\alpha)\bar{r} + \bar{I}\alpha \quad \alpha = \frac{W\bar{x}}{\frac{W}{g}\bar{r}^2 + \bar{I}} \quad (1)$$

The centroidal moment of inertia of the plate is

$$\begin{aligned} \bar{I} &= \frac{m}{12}(a^2 + b^2) = \frac{60 \text{ lb}}{12(32.2 \text{ ft/s}^2)}[(\frac{8}{12} \text{ ft})^2 + (\frac{6}{12} \text{ ft})^2] \\ &= 0.1078 \text{ lb} \cdot \text{ft} \cdot \text{s}^2 \end{aligned}$$

Substituting this value of $\bar{I}$ together with $W = 60 \text{ lb}$, $\bar{r} = \frac{5}{12} \text{ ft}$, and $\bar{x} = \frac{4}{12} \text{ ft}$ into Eq. (1), we obtain

$$\alpha = +46.4 \text{ rad/s}^2 \quad \alpha = 46.4 \text{ rad/s}^2 \downarrow \quad \blacktriangleleft$$

b. Reaction at A. Using the computed value of α , we determine the magnitude of the vector $m\bar{a}$ attached at G .

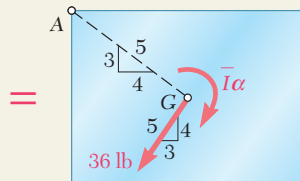
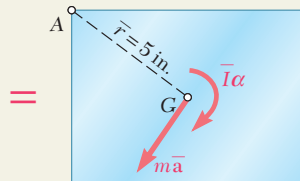
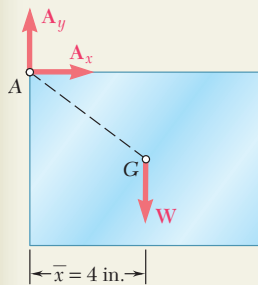
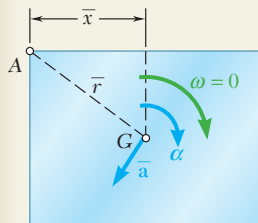
$$m\bar{a} = m\bar{r}\alpha = \frac{60 \text{ lb}}{32.2 \text{ ft/s}^2}(\frac{5}{12} \text{ ft})(46.4 \text{ rad/s}^2) = 36.0 \text{ lb}$$

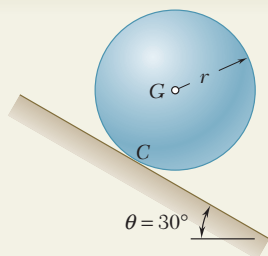
Showing this result on the diagram, we write the equations of motion

$$+\rightarrow \Sigma F_x = \Sigma (F_x)_{\text{eff}}: \quad A_x = -\frac{3}{5}(36 \text{ lb}) = -21.6 \text{ lb} \quad \mathbf{A}_x = 21.6 \text{ lb} \leftarrow \quad \blacktriangleleft$$

$$+\uparrow \Sigma F_y = \Sigma (F_y)_{\text{eff}}: \quad A_y - 60 \text{ lb} = -\frac{4}{5}(36 \text{ lb}) \quad \mathbf{A}_y = 31.2 \text{ lb} \uparrow \quad \blacktriangleleft$$

The couple $\bar{I}\alpha$ is not involved in the last two equations; nevertheless, it should be indicated on the diagram.

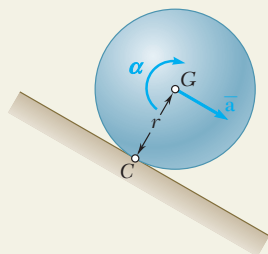




SAMPLE PROBLEM 16.8

A sphere of radius r and weight W is released with no initial velocity on the incline and rolls without slipping. Determine (a) the minimum value of the coefficient of static friction compatible with the rolling motion, (b) the velocity of the center G of the sphere after the sphere has rolled 10 ft, (c) the velocity of G if the sphere were to move 10 ft down a frictionless 30° incline.

SOLUTION



a. Minimum μ_s for Rolling Motion. The external forces $\mathbf{W}$, $\mathbf{N}$, and $\mathbf{F}$ form a system equivalent to the system of effective forces represented by the vector $m\bar{\mathbf{a}}$ and the couple $\bar{I}\alpha$. Since the sphere rolls without sliding, we have $\bar{\mathbf{a}} = r\alpha$.

$$+\downarrow \Sigma M_C = \Sigma (M_C)_{\text{eff}}: \quad (W \sin \theta)r = (m\bar{\mathbf{a}})r + \bar{I}\alpha$$

$$(W \sin \theta)r = (mr\alpha)r + \bar{I}\alpha$$

Noting that $m = W/g$ and $\bar{I} = \frac{2}{5}mr^2$, we write

$$(W \sin \theta)r = \left(\frac{W}{g}r\alpha\right)r + \frac{2}{5}\frac{W}{g}r^2\alpha \quad \alpha = +\frac{5g \sin \theta}{7r}$$

$$\bar{\mathbf{a}} = r\alpha = \frac{5g \sin \theta}{7} = \frac{5(32.2 \text{ ft/s}^2) \sin 30^\circ}{7} = 11.50 \text{ ft/s}^2$$

$$+\searrow \Sigma F_x = \Sigma (F_x)_{\text{eff}}: \quad W \sin \theta - F = m\bar{\mathbf{a}}$$

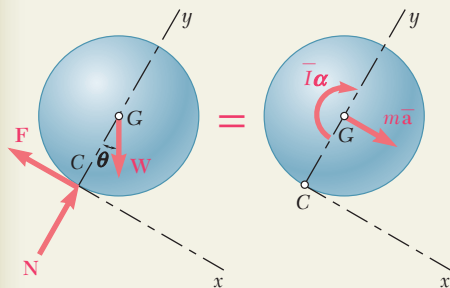
$$W \sin \theta - F = \frac{W}{g} \frac{5g \sin \theta}{7}$$

$$F = +\frac{2}{7}W \sin \theta = \frac{2}{7}W \sin 30^\circ \quad \mathbf{F} = 0.143W \nearrow 30^\circ$$

$$+\nearrow \Sigma F_y = \Sigma (F_y)_{\text{eff}}: \quad N - W \cos \theta = 0$$

$$N = W \cos \theta = 0.866W \quad \mathbf{N} = 0.866W \nearrow 60^\circ$$

$$\mu_s = \frac{F}{N} = \frac{0.143W}{0.866W} \quad \mu_s = 0.165 \quad \blacktriangleleft$$



b. Velocity of Rolling Sphere. We have uniformly accelerated motion:

$$\bar{v}_0 = 0 \quad \bar{\mathbf{a}} = 11.50 \text{ ft/s}^2 \quad \bar{x} = 10 \text{ ft} \quad \bar{x}_0 = 0$$

$$\bar{v}^2 = \bar{v}_0^2 + 2\bar{\mathbf{a}}(\bar{x} - \bar{x}_0) \quad \bar{v}^2 = 0 + 2(11.50 \text{ ft/s}^2)(10 \text{ ft})$$

$$\bar{v} = 15.17 \text{ ft/s} \quad \bar{\mathbf{v}} = 15.17 \text{ ft/s} \searrow 30^\circ \quad \blacktriangleleft$$

c. Velocity of Sliding Sphere. Assuming now no friction, we have $F = 0$ and obtain

$$+\downarrow \Sigma M_G = \Sigma (M_G)_{\text{eff}}: \quad 0 = \bar{I}\alpha \quad \alpha = 0$$

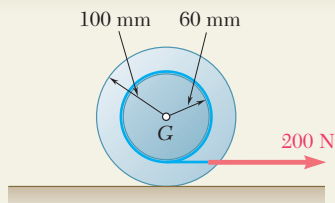
$$+\searrow \Sigma F_x = \Sigma (F_x)_{\text{eff}}: \quad W \sin 30^\circ = m\bar{\mathbf{a}} \quad 0.50W = \frac{W}{g}\bar{\mathbf{a}}$$

$$\bar{\mathbf{a}} = +16.1 \text{ ft/s}^2 \quad \bar{\mathbf{a}} = 16.1 \text{ ft/s}^2 \searrow 30^\circ$$

Substituting $\bar{\mathbf{a}} = 16.1 \text{ ft/s}^2$ into the equations for uniformly accelerated motion, we obtain

$$\bar{v}^2 = \bar{v}_0^2 + 2\bar{\mathbf{a}}(\bar{x} - \bar{x}_0) \quad \bar{v}^2 = 0 + 2(16.1 \text{ ft/s}^2)(10 \text{ ft})$$

$$\bar{v} = 17.94 \text{ ft/s} \quad \bar{\mathbf{v}} = 17.94 \text{ ft/s} \searrow 30^\circ \quad \blacktriangleleft$$



SAMPLE PROBLEM 16.9

A cord is wrapped around the inner drum of a wheel and pulled horizontally with a force of 200 N. The wheel has a mass of 50 kg and a radius of gyration of 70 mm. Knowing that $\mu_s = 0.20$ and $\mu_k = 0.15$, determine the acceleration of G and the angular acceleration of the wheel.

SOLUTION

a. Assume Rolling without Sliding. In this case, we have

$$\bar{a} = r\alpha = (0.100 \text{ m})\alpha$$

We can determine whether this assumption is justified by comparing the friction force obtained with the maximum available friction force. The moment of inertia of the wheel is

$$\bar{I} = m\bar{k}^2 = (50 \text{ kg})(0.070 \text{ m})^2 = 0.245 \text{ kg} \cdot \text{m}^2$$

Equations of Motion

$$\begin{aligned} +\downarrow \Sigma M_C &= \Sigma (M_C)_{\text{eff}}: & (200 \text{ N})(0.040 \text{ m}) &= m\bar{a}(0.100 \text{ m}) + \bar{I}\alpha \\ 8.00 \text{ N} \cdot \text{m} &= (50 \text{ kg})(0.100 \text{ m})\alpha(0.100 \text{ m}) + (0.245 \text{ kg} \cdot \text{m}^2)\alpha \\ \alpha &= +10.74 \text{ rad/s}^2 \\ \bar{a} &= r\alpha = (0.100 \text{ m})(10.74 \text{ rad/s}^2) = 1.074 \text{ m/s}^2 \end{aligned}$$

$$\begin{aligned} \rightarrow \Sigma F_x &= \Sigma (F_x)_{\text{eff}}: & F + 200 \text{ N} &= m\bar{a} \\ F + 200 \text{ N} &= (50 \text{ kg})(1.074 \text{ m/s}^2) \\ F &= -146.3 \text{ N} & \mathbf{F} &= 146.3 \text{ N} \leftarrow \end{aligned}$$

$$\begin{aligned} +\uparrow \Sigma F_y &= \Sigma (F_y)_{\text{eff}}: & N - W &= 0 \\ N - W &= 0 & N - W &= mg = (50 \text{ kg})(9.81 \text{ m/s}^2) = 490.5 \text{ N} \\ N &= 490.5 \text{ N} \uparrow \end{aligned}$$

Maximum Available Friction Force

$$F_{\text{max}} = \mu_s N = 0.20(490.5 \text{ N}) = 98.1 \text{ N}$$

Since $F > F_{\text{max}}$, the assumed motion is impossible.

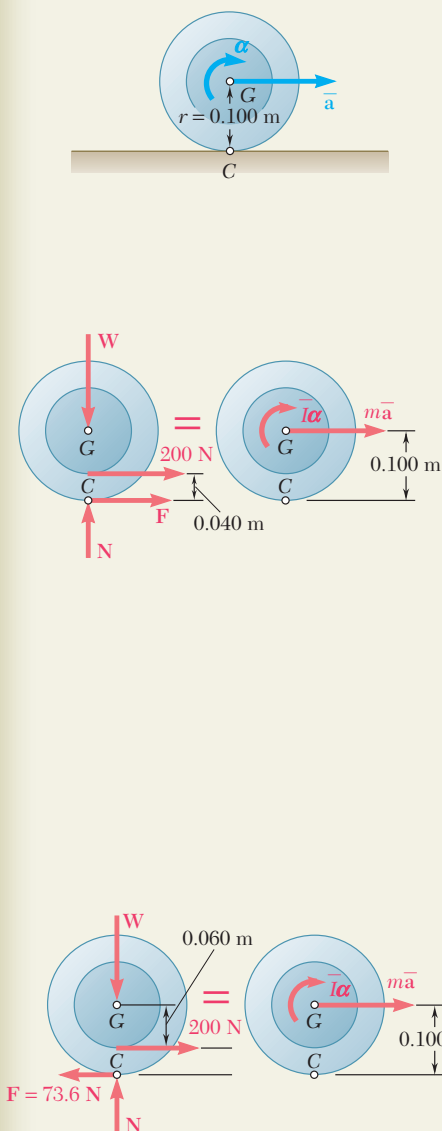
b. Rotating and Sliding. Since the wheel must rotate and slide at the same time, we draw a new diagram, where $\bar{a}$ and α are independent and where

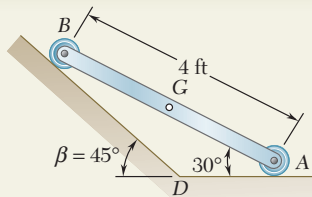
$$F = F_k = \mu_k N = 0.15(490.5 \text{ N}) = 73.6 \text{ N}$$

From the computation of part *a*, it appears that $\mathbf{F}$ should be directed to the left. We write the following equations of motion:

$$\begin{aligned} \rightarrow \Sigma F_x &= \Sigma (F_x)_{\text{eff}}: & 200 \text{ N} - 73.6 \text{ N} &= (50 \text{ kg})\bar{a} \\ \bar{a} &= +2.53 \text{ m/s}^2 & \bar{a} &= 2.53 \text{ m/s}^2 \rightarrow \end{aligned}$$

$$\begin{aligned} +\downarrow \Sigma M_G &= \Sigma (M_G)_{\text{eff}}: & (73.6 \text{ N})(0.100 \text{ m}) - (200 \text{ N})(0.060 \text{ m}) &= (0.245 \text{ kg} \cdot \text{m}^2)\alpha \\ \alpha &= -18.94 \text{ rad/s}^2 & \alpha &= 18.94 \text{ rad/s}^2 \curvearrowright \end{aligned}$$





SAMPLE PROBLEM 16.10

The extremities of a 4-ft rod weighing 50 lb can move freely and with no friction along two straight tracks as shown. If the rod is released with no velocity from the position shown, determine (a) the angular acceleration of the rod, (b) the reactions at A and B.

SOLUTION

Kinematics of Motion. Since the motion is constrained, the acceleration of G must be related to the angular acceleration α . To obtain this relation, we first determine the magnitude of the acceleration $\mathbf{a}_A$ of point A in terms of α . Assuming that α is directed counterclockwise and noting that $a_{B/A} = 4\alpha$, we write

$$\mathbf{a}_B = \mathbf{a}_A + \mathbf{a}_{B/A}$$

$$[a_B \searrow 45^\circ] = [a_A \rightarrow] + [4\alpha \nearrow 60^\circ]$$

Noting that $\phi = 75^\circ$ and using the law of sines, we obtain

$$a_A = 5.46\alpha \quad a_B = 4.90\alpha$$

The acceleration of G is now obtained by writing

$$\bar{\mathbf{a}} = \mathbf{a}_G = \mathbf{a}_A + \mathbf{a}_{G/A}$$

$$\bar{\mathbf{a}} = [5.46\alpha \rightarrow] + [2\alpha \nearrow 60^\circ]$$

Resolving $\bar{\mathbf{a}}$ into x and y components, we obtain

$$\begin{aligned} \bar{a}_x &= 5.46\alpha - 2\alpha \cos 60^\circ = 4.46\alpha & \bar{\mathbf{a}}_x &= 4.46\alpha \rightarrow \\ \bar{a}_y &= -2\alpha \sin 60^\circ = -1.732\alpha & \bar{\mathbf{a}}_y &= 1.732\alpha \downarrow \end{aligned}$$

Kinetics of Motion. We draw a free-body-diagram equation expressing that the system of the external forces is equivalent to the system of the effective forces represented by the vector of components $m\bar{\mathbf{a}}_x$ and $m\bar{\mathbf{a}}_y$ attached at G and the couple $\bar{I}\alpha$. We compute the following magnitudes:

$$\bar{I} = \frac{1}{12}ml^2 = \frac{1}{12} \frac{50 \text{ lb}}{32.2 \text{ ft/s}^2} (4 \text{ ft})^2 = 2.07 \text{ lb} \cdot \text{ft} \cdot \text{s}^2 \quad \bar{I}\alpha = 2.07\alpha$$

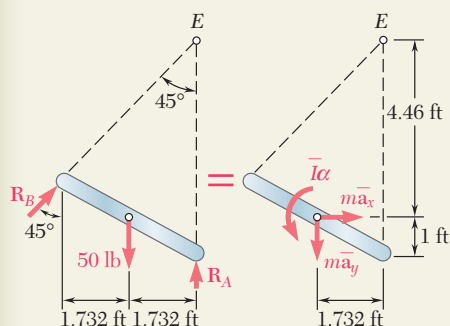
$$m\bar{a}_x = \frac{50}{32.2} (4.46\alpha) = 6.93\alpha \quad m\bar{a}_y = -\frac{50}{32.2} (1.732\alpha) = -2.69\alpha$$

Equations of Motion

$$\begin{aligned} +\curvearrowright \Sigma M_E &= \Sigma (M_E)_{\text{eff}}: \\ (50)(1.732) &= (6.93\alpha)(4.46) + (2.69\alpha)(1.732) + 2.07\alpha \\ \alpha &= +2.30 \text{ rad/s}^2 \quad \alpha = 2.30 \text{ rad/s}^2 \nearrow \end{aligned}$$

$$\begin{aligned} \pm \Sigma F_x &= \Sigma (F_x)_{\text{eff}}: & R_B \sin 45^\circ &= (6.93)(2.30) = 15.94 \\ & & R_B &= 22.5 \text{ lb} \quad \mathbf{R}_B = 22.5 \text{ lb} \nearrow 45^\circ \end{aligned}$$

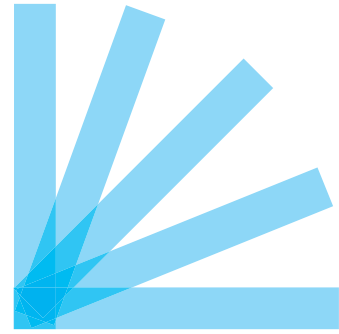
$$\begin{aligned} +\uparrow \Sigma F_y &= \Sigma (F_y)_{\text{eff}}: & R_A + R_B \cos 45^\circ - 50 &= -(2.69)(2.30) \\ R_A &= -6.19 - 15.94 + 50 = 27.9 \text{ lb} \quad \mathbf{R}_A = 27.9 \text{ lb} \uparrow \end{aligned}$$



MODULE 3

7

Dry Friction



7.1 Introduction

In most of the equilibrium problems that we have analyzed up to this point, the surfaces of contact have been frictionless. The reactive forces were, therefore, normal to the contact surfaces. The concept of a frictionless surface is, of course, an idealization. All real surfaces also provide a force component that is tangent to the surface, called the *friction force*, that resists sliding.

In many situations, friction forces are helpful. For example, friction enables you to walk without slipping, it holds nails and screws in place, and it allows us

The force required to turn the handle of a screw press, such as the antique wine press shown here, depends on the coefficient of friction between the screw threads. This effect is illustrated Sample Problem 7.12.
Andrey Kudinov/Shutterstock

to transmit power by means of clutches and belts. On the other hand, friction can also be detrimental: It causes wear in machinery and reduces efficiency in the transmission of power by converting mechanical energy into heat.

Dry friction refers to the friction force that exists between two unlubricated solid surfaces. *Fluid friction* acts between moving surfaces that are separated by a layer of fluid. The friction in a lubricated journal bearing is classified as fluid friction, because the two halves of the bearing are not in direct contact but are separated by a thin layer of liquid lubricant. In this chapter, we consider only dry friction.* A study of fluid friction involves hydrodynamics, which is beyond the scope of this text.

7.2 Coulomb's Theory of Dry Friction

Dry friction is a complex phenomenon that is not yet completely understood. This article introduces a highly simplified theory, known as Coulomb's theory of dry friction, that has been found to give satisfactory results in many practical problems.

Coulomb's theory is best explained by considering two bodies that are in contact with each other, as shown in Fig. 7.1 (a). Although a single point of contact is indicated in this figure, the following discussion also applies for a finite contact area. The *plane of contact* shown in Fig. 7.1(a) is tangent to both bodies at the point of contact. Figure 7.1(b) displays the free-body diagrams of the bodies, where N is the normal contact force and F is the friction force. The force N is perpendicular to the plane of contact, whereas F lies in the plane of contact. Coulomb's theory consists of several postulates that are explained in the following.

a. Static case

Coulomb proposed the following law: If there is no relative motion between two surfaces that are in contact, the normal force N and the friction force F satisfy the following relationship.

$$F \leq F_{\max} = \mu_s N \quad (7.1)$$

where $F_{\max}$ is the *maximum* static friction force that *can* exist between the contacting surfaces and μ_s is known as the *coefficient of static friction*. The coefficient of static friction is an experimental constant that depends on the composition and roughness of the contacting surfaces. Typical values of μ_s are listed in Table 7.1. Observe that Eq. (7.1) states simply that the friction force F that exists under static conditions (no relative motion) has an upper limit that is proportional to the normal force.

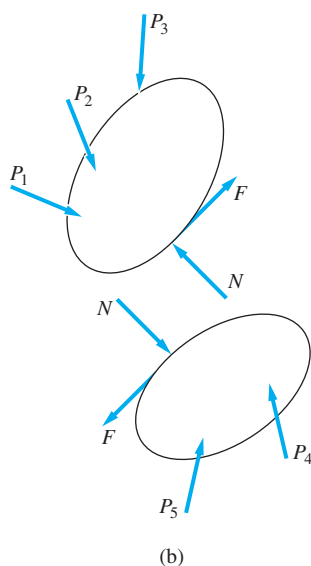
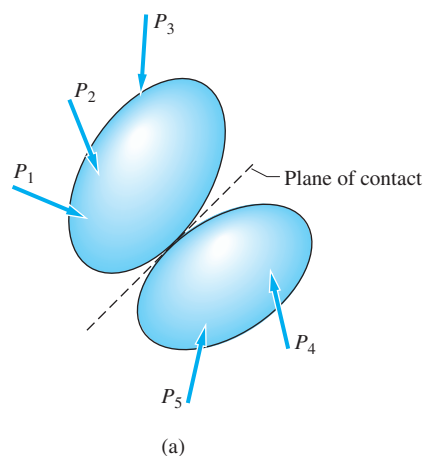


Fig. 7.1

*Dry friction is also known as Coulomb friction, after C.-A. de Coulomb (1736–1806), the first investigator to completely state the laws of dry friction.

Text not available due to copyright restrictions

b. Impending sliding

Consider the static case in which the friction force equals its limiting value; that is,

$$F = F_{\max} = \mu_s N \quad (7.2)$$

For this condition, the surfaces are on the verge of sliding, a condition known as *impending sliding*. When sliding impends, the surfaces are at rest relative to each other. However, any change that would require an increase in the friction force would cause sliding. The direction for $F_{\max}$ can be determined from the observation that

$F_{\max}$ always opposes impending sliding

c. Dynamic case

If the two contact surfaces are sliding relative to each other, the friction force F is postulated to be

$$F = F_k = \mu_k N \quad (7.3)$$

where N is the contact normal force; μ_k is an experimental constant called the *coefficient of kinetic friction*; and F_k is referred to as the *kinetic*, or *dynamic friction force*. As indicated in Table 7.1, the coefficient of kinetic friction is usually smaller than its static counterpart. As in the static case,

F_k always opposes sliding

d. Further discussion of Coulomb friction

When applying Coulomb's theory, the difference between $F_{\max}$ and F_k must be clearly understood: $F_{\max}$ is the *maximum* friction force that *can* exist under static conditions; F_k is the friction force that *does* exist during sliding.

To illustrate Coulomb's laws of friction, consider the situation depicted in Fig. 7.2 (a). The block of weight W is assumed to be at rest on a horizontal surface when it is subjected to the horizontal force P . (We limit our attention here to sliding motion; the possibility that the block may tip about its corner is considered

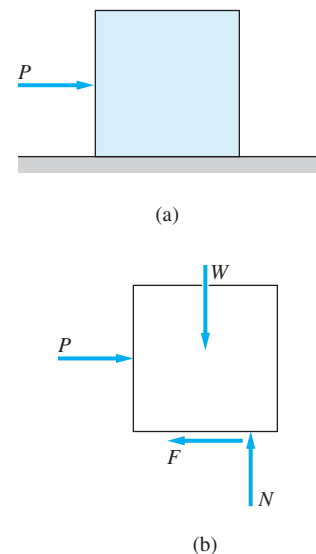


Fig. 7.2

later.) The free-body diagram of the block is shown in Fig. 7.2(b). Because the friction force F resists the tendency of the block to slide, F is directed opposite to P . We now examine the variation of F with P as the latter increases slowly from zero.

If P is relatively small, the block will remain at rest, and the force equations of equilibrium, $\Sigma F_x = 0$ and $\Sigma F_y = 0$, yield $F = P$ and $N = W$.^{*} Therefore, as long as the block remains at rest, the friction force F equals the applied force P . Figure 7.3 shows the plot of F versus P . In the static region, $0 \leq F \leq F_{\max}$; the variation is a straight line with unit slope. When $F = F_{\max}$, the block is still in static equilibrium, but sliding impends. However, the slightest increase in P would result in sliding. In Fig. 7.3, the point referring to impending sliding marks the end of the static region. When P exceeds $F_{\max}$, the block starts to slide, and the friction force F drops to its kinetic value F_k . If P is further increased, F remains constant at F_k . Consequently, the plot of F versus P is a horizontal line in the dynamic range.

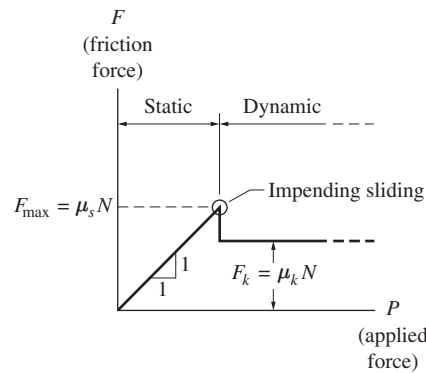


Fig. 7.3

e. Limitations

Because there is no theoretical explanation that accurately describes friction phenomena, engineers must rely on empirical constants, such as the coefficient of friction. Handbook values for the coefficients of friction should be treated as approximate values. Experimental results indicate that the coefficients may vary widely with environmental conditions, such as humidity, the cleanliness of the surfaces, and so on.

The theory of dry friction is applicable only to surfaces that are dry or that contain only a small amount of lubricant. If there is relative motion between the surfaces of contact, the theory is valid for low speeds only. If the surfaces are well lubricated and are moving with high relative speeds, the frictional characteristics are best described by the theories of fluid friction, which are beyond the scope of this text.

^{*}The moment equation of equilibrium would determine the line of action of the normal force N , an important consideration in the analysis of tipping. However, because we are delaying a discussion of tipping, this equation is not of interest at the present time.

It is interesting to note that Coulomb's theory of dry friction does not depend on the area of contact. There are, of course, many situations where this is not the case. For example, the traction (friction force) between an automobile tire and the pavement can be increased under certain conditions by letting a small amount of air out of the tire, thus increasing the contact area. Obviously, Coulomb's theory of dry friction is not applicable in this situation. The maximum traction in this case is also influenced by factors, such as surface adhesion, that depend on the area of contact.

7.3 Problem Classification and Analysis

The analysis of equilibrium problems that involve friction can be somewhat complicated because Coulomb's law, Eq. (7.1), is an inequality. It does not tell us the friction force; it tells us only the *largest possible* friction force. The equality $F = F_{\max} = \mu_s N$ can be used only if slipping is known to impend. Because F is not necessarily equal to $F_{\max}$ at a friction surface, it is not possible to develop a single method of analysis that is valid for all friction problems. However, friction problems can be classified into three types, and a separate method of solution can be outlined for each type.

Type I *The problem statement does not specify impending motion.* In problems of this type, we do not know whether or not the body is in equilibrium. Therefore, the analysis must begin with an assumption about equilibrium.

Method of Analysis

1. **Assume equilibrium** You are strongly advised to write down this assumption as a reminder that the solution will not be complete unless the assumption has been checked. The sense of each friction force can be assumed because the solution of the equilibrium equations will determine the correct sense.
2. **Solve the equilibrium equations** for the friction forces required for equilibrium.*
3. **Check the assumption** If the friction forces required for equilibrium do not exceed their limits (i.e., if $F \leq \mu_s N$ at each friction surface), then the assumption is correct, and the remaining unknowns can be computed using equilibrium analysis. (Note that if $F = \mu_s N$ at a surface, which would imply impending sliding, then the assumption is still correct.) If equilibrium requires that $F > \mu_s N$ at any friction surface (which is physically impossible), the assumption of equilibrium is incorrect. Therefore, we have a dynamics problem in which the friction forces at the sliding surfaces are $F = F_k = \mu_k N$.

See Sample Problems 7.1 and 7.4 for examples of Type I problems.

*This analysis presupposes that the friction forces are statically determinate. Statically indeterminate friction forces are omitted from the present discussion. Problems of this type are best solved using the principle of virtual work (see Chapter 10).

Type II The problem statement implies impending sliding, and the surfaces where sliding impends are known. Friction problems of this type have the most straightforward analyses, because no assumptions and, therefore, no checks are required. It is not necessary to assume equilibrium—a body known to be in a state of impending sliding is in equilibrium by definition.

Method of Analysis

1. Set $F = F_{\max} = \mu_s N$ at the surfaces where sliding impends. Make sure that the sense of each $F_{\max}$ is correctly shown on the FBD (opposing impending sliding), because the solution of the equilibrium equations may depend on the assumed directions of the friction forces.
2. Solve for the unknowns using the equilibrium equations.

See Sample Problems 7.2, 7.5, and 7.6 for examples of Type II problems.

Type III The problem statement implies impending sliding, but the surfaces at which sliding impends are not known. Problems of this type are the most tedious to analyze, because the surfaces at which sliding impends must be identified by trial and error. Once an assumption has been made, the analysis is similar to that for Type II problems. Two methods of analysis can be used here, both of which are described in the following.

Method of Analysis 1

1. Determine all possible ways in which sliding can impend.
2. For each case, set $F = F_{\max}$ at the surfaces where sliding impends and solve the equilibrium equations. Again, the sense of each $F_{\max}$ should be correct on the FBD. In general, a different solution is obtained for each mode of impending sliding.
3. Choose the correct answer by inspection of the solutions.

Method of Analysis 2

1. Determine all possible ways in which sliding can impend.
2. For one of the cases, set $F = F_{\max}$ at the surfaces where sliding impends and solve the equilibrium equations.
3. Check the solution by comparing the friction force at each of the other surfaces with its limiting value. If all these forces are less than or equal to their maximum permissible values, then the solution is correct. If a friction force exceeds its limiting value $\mu_s N$, the solution is invalid and another mode of impending sliding must be analyzed. This procedure must be continued until the correct solution is found.

See Sample Problems 7.3 and 7.7 for examples of Type III problems.

Caution Remember that the equation $F = \mu_s N$ is valid only in the special case of impending sliding. Many difficulties encountered by students can be traced to the incorrect assumption that the equation $F = \mu_s N$ is always true.

Sample Problem 7.1

The 50-kg block in Fig. (a) is initially at rest on a horizontal plane. Determine the friction force between the block and the surface after P was gradually increased from 0 to 150 N.

Solution

From the problem statement we conclude that this is a Type I problem (impending motion is not specified). Furthermore, we do not know if the block will even remain at rest in static equilibrium when $P = 150$ N.

Assume Equilibrium

Once we have assumed that the body remains at rest, the equilibrium equations for the free-body diagram in Fig. (b) can be used to calculate the two unknowns (N and F), as follows.

$$\Sigma F_y = 0 \quad +\uparrow \quad N - 490.5 = 0$$

$$N = 490.5 \text{ N}$$

$$\Sigma F_x = 0 \quad +\rightarrow \quad P - F = 0$$

$$F = P = 150 \text{ N}$$

Before we can accept this solution, the assumption of equilibrium must be checked.

Check

The maximum static friction force is

$$F_{\max} = \mu_s N = 0.5(490.5) = 245.25 \text{ N}$$

Because $F < F_{\max}$, we conclude that the block is in static equilibrium, and the correct value of the friction force is

$$F = 150 \text{ N}$$

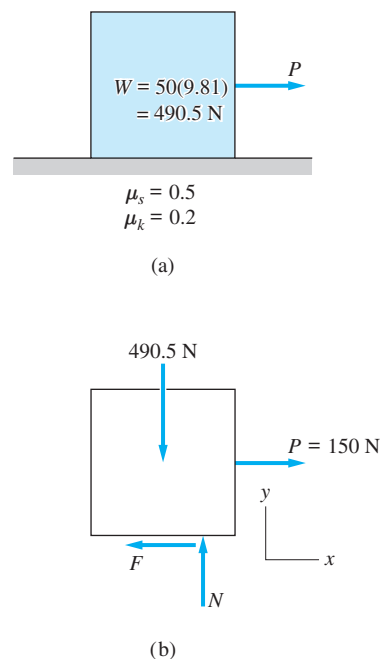
Answer

Comment

If the coefficient of static friction had been 0.25, instead of 0.5, the block would not be in equilibrium. The 150-N friction force required for equilibrium would be greater than $F_{\max} = 0.25(490.5) = 122.625$ N; therefore, the friction force would be

$$F = F_k = \mu_k N = 0.2(490.5) = 98.1 \text{ N}$$

and the block would be sliding to the right.



Sample Problem 7.2

The 50-kg block in Fig. (a) is at rest on a rough horizontal plane before the force P is applied. Determine the magnitude of P that would cause impending sliding to the right.

Solution

The problem statement clearly specifies that sliding impends. Because we know where it impends (there is only one friction surface), we conclude that this is a Type II problem.

The free-body diagram of the block is shown in Fig. (b), where the friction force is shown acting to the left, opposite the direction of impending sliding. There are three unknowns in this FBD: P , N , and F . There are also three independent equations: two equilibrium equations and Coulomb's law for impending sliding.

From the FBD we see that the equilibrium equations give $N = 490.5$ N and $P = F$. Coulomb's law then yields

$$P = F = F_{\max} = \mu_s N = 0.5(490.5) = 245.25 \text{ N} \quad \text{Answer}$$

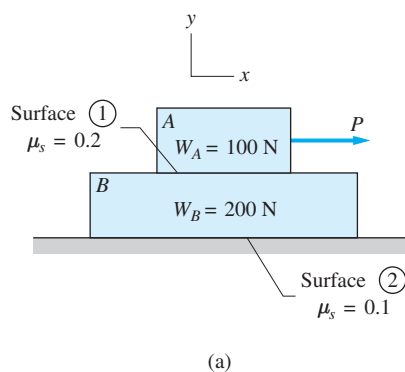
This completes the solution. Because there were no assumptions, no checks are necessary.

Comment

Note that Fig. (a) in both Sample Problems 7.1 and 7.2 is identical. The differences are revealed only in the problem statements. This shows that you must read each problem statement very carefully, because it determines the problem type.

A problem statement can *imply* impending sliding. For example, the following are equivalent to the original statement of this problem: They both imply that P is to be calculated for impending sliding.

1. Determine the largest force P that can be applied without causing the block to slide to the right.
2. Determine the smallest force P that will cause the block to slide to the right.



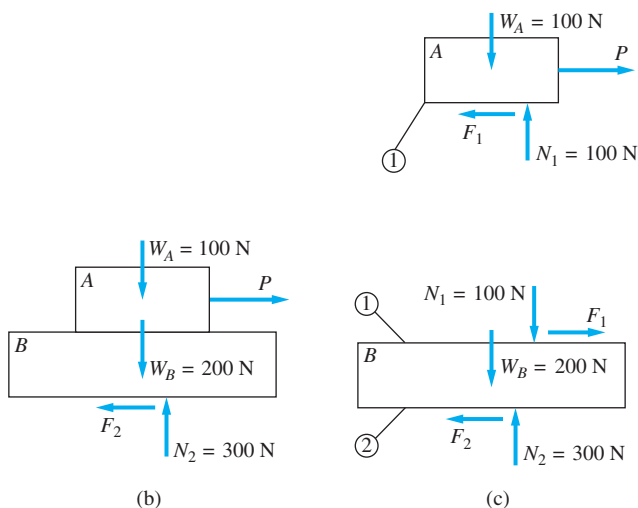
Sample Problem 7.3

Determine the maximum force P that can be applied to block A in Fig. (a) without causing either block to move.

Solution

The problem statement indicates that we are to find P that would cause impending motion. However, there are two possible ways in which motion can impend: impending sliding at surface 1, or impending sliding at surface 2. Because impending sliding is specified but not its location, this is a Type III problem.

The free-body diagrams of the entire system and each block are shown in Figs. (b) and (c), respectively. Note that the equilibrium of each block yields



$N_1 = 100 \text{ N}$ and $N_2 = 300 \text{ N}$, as shown on the FBDs. Attention should be paid to the friction forces. The friction force F_2 on the bottom of block B is directed to the left, opposite the direction in which sliding would impend. At surface 1, block A would tend to slide to the right, across the top of block B . Therefore, F_1 is directed to the left on block A , and to the right on block B . The tendency of F_1 to slide B to the right is resisted by the friction force F_2 . Note that F_1 and N_1 do not appear in the FBD in Fig. (b), because they are internal to the system of both blocks.

Two solutions are presented here to illustrate both methods of analysis described in Art. 7.3.

Method of Analysis 1

First, assume impending sliding at surface 1. Under this assumption we have

$$F_1 = (F_1)_{\max} = (\mu_s)_1 N_1 = 0.2(100) = 20 \text{ N}$$

The FBD of block A then gives

$$\begin{aligned} \Sigma F_x = 0 \quad \rightarrow \quad P - F_1 &= 0 \\ P &= F_1 = 20 \text{ N} \end{aligned}$$

Next, assume impending sliding at surface 2, which gives

$$F_2 = (F_2)_{\max} = (\mu_s)_2 N_2 = 0.1(300) = 30 \text{ N}$$

From the FBD of the entire system, Fig. (b), we then obtain

$$\begin{aligned} \Sigma F_x = 0 \quad \rightarrow \quad P - F_2 &= 0 \\ P &= F_2 = 30 \text{ N} \end{aligned}$$

So far, we have determined that $P = 20 \text{ N}$ will cause motion to impend at surface 1 and that $P = 30 \text{ N}$ will cause motion to impend at surface 2. Therefore, the largest force that can be applied without causing either block to move is

$$P = 20 \text{ N} \qquad \text{Answer}$$

with sliding impending at surface 1.

Be sure you understand that the largest force that can be applied is the smaller of the two values determined in the preceding calculations. If sliding impends when $P = 20 \text{ N}$, then the system would not be at rest when $P = 30 \text{ N}$.

Method of Analysis 2

Assume impending motion at surface 1. We would then obtain $P = (F_1)_{\max} = 20 \text{ N}$, as determined in Method of Analysis 1. Next, we check the assumption.

Check

The assumption of impending motion at surface 1 is checked by comparing the friction force F_2 with $(F_2)_{\max}$, its maximum possible value. Using the FBD of block B , we obtain

$$\begin{aligned} \Sigma F_x = 0 \quad \rightarrow \quad F_1 - F_2 &= 0 \\ F_1 = F_2 &= 20 \text{ N} \end{aligned}$$

Because $(F_2)_{\max} = (\mu_s)_2 N_2 = 0.1(300) = 30 \text{ N}$, we have $F_2 < (F_2)_{\max}$. Consequently, we conclude that impending motion at surface 1 is the correct assumption, so that the answer is $P = 20 \text{ N}$.

Had F_2 turned out to be greater than $(F_2)_{\max}$, we would know that sliding would first impend at surface 2, and the problem would have to be solved again making use of this fact.

Comment

There are five unknowns in this problem: P , N_1 , F_1 , N_2 , and F_2 . There are four independent equilibrium equations: two for each block. The assumption of impending motion at one surface provides the fifth equation, $F = \mu_s N$, making the problem statically determinate.

In our solution, we have considered two possible modes of impending motion—impending sliding at surface 1 and impending sliding at surface 2. Impending sliding at both surfaces at the same time is obviously a third possibility, but it need not be examined independently. Both of the foregoing analyses would determine if simultaneous impending sliding is indeed the case. In Method of Analysis 1 the two computed values of P would be equal. In Method of Analysis 2 the check would reveal that $F = F_{\max}$ at both surfaces.

Caution A mistake that is often made in the analysis of Type III problems is to assume that motion impends at the surface with the smallest coefficient of static friction. The solution to this problem illustrates that this need not be the case.

Sample Problem 7.4

Can the system in Fig. (a) be in static equilibrium in the position shown? The uniform bar AB 500 kg and the mass of block C is 300 kg. Friction at A is negligible, and the coefficient of static friction is 0.4 at the other two contact surfaces.

Solution

Because it is not known whether motion impends, we identify this as a Type I problem. Note that the FBDs of the bar and the block, Figs. (b) and (c), contain five unknowns: N_A , N_B , F_B , N_C , and F_C .

Assume Equilibrium

Under this assumption, there are five equilibrium equations: three for the bar AB and two for the block C . The unknowns may be computed by the following procedure.

FBD of AB [Fig. (b)]

$$\begin{aligned}\Sigma M_B = 0 \quad \curvearrowright \quad & N_A \sin 40^\circ (5 \cos 30^\circ) + N_A \cos 40^\circ (5 \sin 30^\circ) \\ & - 4905 (2.5 \cos 30^\circ) = 0 \\ & N_A = 2260.24 \text{ N}\end{aligned}$$

$$\begin{aligned}\Sigma F_x = 0 \quad \rightarrow \quad & F_B - N_A \cos 40^\circ = 0 \\ & F_B = 2260.24 \cos 40^\circ = 1731.44 \text{ N}\end{aligned}$$

$$\begin{aligned}\Sigma F_y = 0 \quad \uparrow \quad & N_B + N_A \sin 40^\circ - 4905 = 0 \\ & N_B = -2260.24 \sin 40^\circ + 4905 = 3452.15 \text{ N}\end{aligned}$$

FBD of Block C [Fig. (c)]

$$\begin{aligned}\Sigma F_y = 0 \quad \uparrow \quad & N_C - N_B - 2943 = 0 \\ & N_C = 3452.15 + 2943 = 6395.15 \text{ N}\end{aligned}$$

$$\begin{aligned}\Sigma F_x = 0 \quad \rightarrow \quad & F_C - F_B = 0 \\ & F_C = F_B = 1731.44 \text{ N}\end{aligned}$$

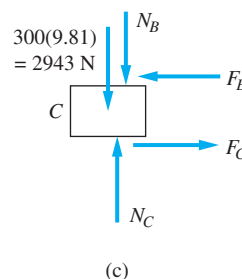
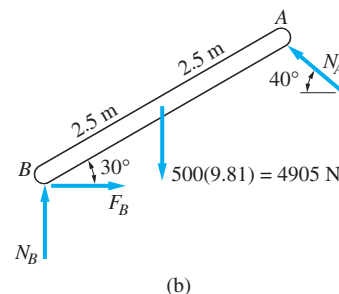
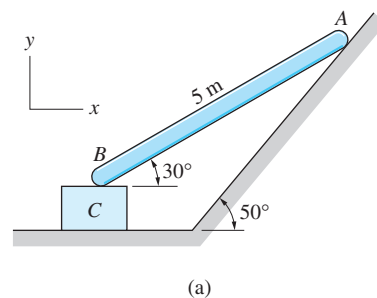
Check

To check the assumption of equilibrium, we must compare each of the friction forces against its maximum static value.

$$\begin{aligned}(F_B)_{\max} &= 0.4N_B = 0.4(3452.15) = 1380.86 \text{ N} < F_B = 1731.44 \text{ N} \\ (F_C)_{\max} &= 0.4N_C = 0.4(6395.15) = 2558.06 \text{ N} > F_C = 1731.44 \text{ N}\end{aligned}$$

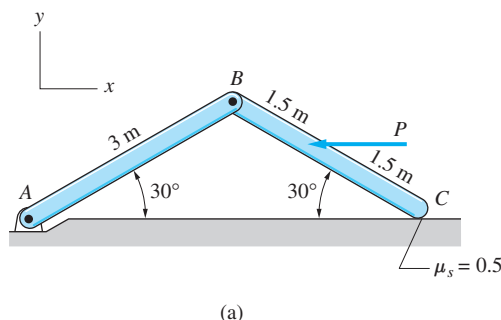
Answer

We conclude that the system cannot be in equilibrium. Although there is sufficient friction beneath B , the friction force under C exceeds its limiting value.



Sample Problem 7.5

Determine the largest and smallest values of the force P for which the system in Fig. (a) will be in static equilibrium. The homogeneous bars AB and BC are identical, each having a mass of 100 kg. The coefficient of static friction between the bar at C and the horizontal plane is 0.5.

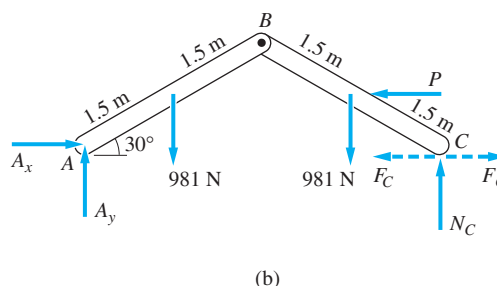


Solution

This is a Type II problem because impending sliding at C is implied. However, finding the largest and smallest values of P are two separate problems.

Note that the weights of the bars have a tendency to slide C to the right. Therefore, impending sliding of C to the right corresponds to the *smallest* P . The *largest* P occurs when sliding of C impends to the left; in this case, P must overcome both the friction and the tendency of the weights to slide C to the right. Consequently, the only difference between the two problems is the direction of the friction force at C .

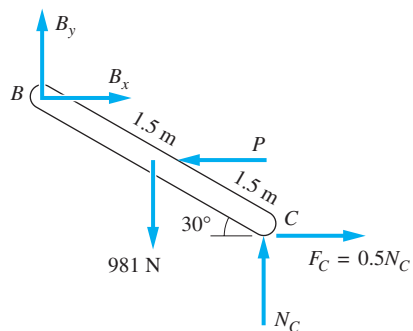
The FBD of the system consisting of both bars is shown in Fig. (b); the two directions of F_C are indicated by dashed lines. The weight of each bar, $W = mg = 100(9.81) = 981$ N, is also shown on the diagram.



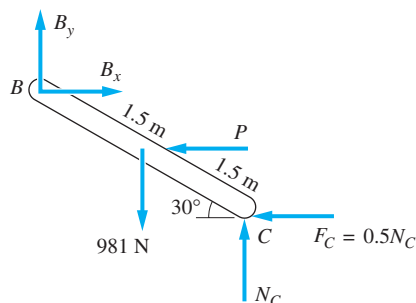
An equation involving only N_C and P is obtained by summing moments about A in Fig. (b):

$$\begin{aligned} \Sigma M_A = 0 \quad (+) \quad & N_C(6 \cos 30^\circ) + P(1.5 \sin 30^\circ) \\ & - 981(1.5 \cos 30^\circ) - 981(4.5 \cos 30^\circ) = 0 \end{aligned} \quad (a)$$

The FBDs of bar BC corresponding to the largest and smallest values of P are shown in Figs. (c) and (d), respectively. In both cases, F_C is set equal to $(F_C)_{\max}$



(c) Largest P



(d) Smallest P

because sliding impends. Summing moments about B yields another equation containing N_C and P .

$$\begin{aligned} \Sigma M_B = 0 \quad \curvearrowright \quad & N_C(3 \cos 30^\circ) - 981(1.5 \cos 30^\circ) \\ & - P(1.5 \sin 30^\circ) \pm 0.5N_C(3 \sin 30^\circ) = 0 \end{aligned} \quad (b)$$

where the positive (negative) sign on the last term corresponds to the largest (smallest) value of P .

Solving Eqs. (a) and (b) gives

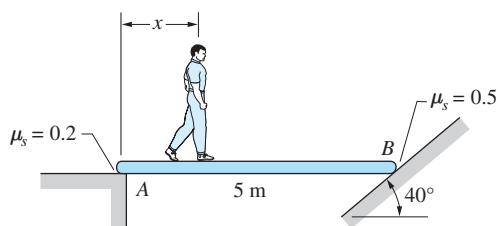
$$\begin{array}{ll} \text{largest } P = 1630 \text{ N} & \text{Answer} \\ \text{smallest } P = 530 \text{ N} & \text{Answer} \end{array}$$

Therefore, the system is in static equilibrium for values of P in the range $530 \text{ N} \leq P \leq 1630 \text{ N}$.

The solution of this sample problem clearly illustrates that the directions of the friction forces must be shown correctly on the free-body diagrams when sliding impends.

Sample Problem 7.6

The uniform 50-kg plank in Fig. (a) is resting on friction surfaces at A and B . The coefficients of static friction are shown in the figure. If a 80-kg man starts walking from A toward B , determine the distance x when the plank will start to slide.



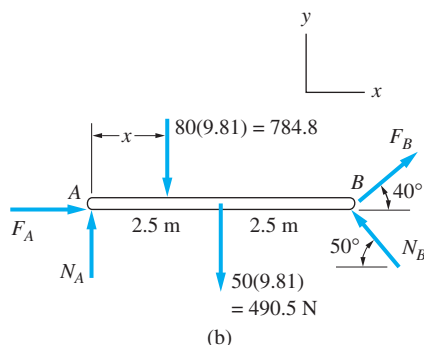
(a)

Solution

This is a Type II problem. When the plank is on the verge of moving, sliding must impend at both A and B . Impending sliding at A only, or at B only, would be physically impossible. Because the plank is a rigid body, any movement of end A must be accompanied by a movement of end B .

The FBD of the plank is shown in Fig. (b). Observe that the friction forces are shown acting in their correct directions. When the plank is ready to move, the direction of impending sliding of end B is down the inclined plane. Consequently, end A would tend to slide to the left. The directions of F_A and F_B must oppose these motions. Showing either of the friction forces in the opposite direction would lead to incorrect results.

Inspection of the FBD in Fig. (b) reveals that there are five unknowns: N_A , F_A , N_B , F_B , and x . There are also five equations: three equilibrium equations and two friction equations. Therefore, all the unknowns can be computed from the FBD as follows.



$$\begin{aligned} \Sigma M_A = 0 \quad \curvearrowright \quad & N_B \sin 50^\circ (5) + F_B \sin 40^\circ (5) \\ & - 784.8x - 490.5(2.5) = 0 \end{aligned} \quad (a)$$

$$\Sigma F_x = 0 \quad \rightarrow \quad F_A - N_B \cos 50^\circ + F_B \cos 40^\circ = 0 \quad (b)$$

$$\Sigma F_y = 0 \quad \uparrow \quad N_A - 784.8 - 490.5 + N_B \sin 50^\circ + F_B \sin 40^\circ = 0 \quad (c)$$

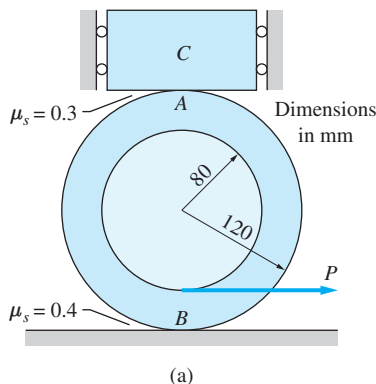
Substituting the friction equations, $F_A = 0.2N_A$ and $F_B = 0.5N_B$, and solving Eqs. (a)–(c) give $N_A = 694.14$ N, $N_B = 534.43$ N, and

$$x = 2.55 \text{ m}$$

Answer

Sample Problem 7.7

The spool in Fig. (a) weighs 25 N, and its center of gravity is located at the geometric center. The weight of block C is 50 N. The coefficients of static friction at the two points of contact are as shown. Determine the largest horizontal force P that can be applied without disturbing the equilibrium of the system.



Solution

The key to the analysis of this problem is understanding that there are two possibilities for impending motion (both could occur simultaneously): (1) impending sliding at A with impending rolling (without sliding) at B , and (2) impending sliding at B with impending rolling (without sliding) at A . Because it is initially not known which of these possibilities represents the actual mode of impending motion, this is a Type III problem.

The free-body diagrams for the block and the spool are shown in Fig. (b). Observe that both friction forces have been shown in their correct directions. The force P tends to slide points A and B on the spool to the right. Therefore, both friction forces are shown acting to the left on the FBD of the spool.

Inspecting the FBDs in Fig. (b), we conclude from $\Sigma F_y = 0$ that $N_A = 50$ N and $N_B = 75$ N. At this stage three unknowns remain in the FBD of the spool: F_A , F_B , and P . Because only two equilibrium equations are left ($\Sigma F_y = 0$ has already been used), the remainder of the solution depends on the assumption regarding impending motion.

Assume Impending Sliding at A

This assumption gives us the additional equation $F_A = 0.3N_A = 0.3(50) = 15$ N. The FBD of the spool then yields

$$\Sigma M_B = 0 \quad \curvearrowright \quad F_A(240) - P(40) = 0$$

which gives

$$P = 6F_A = 6(15) = 90.0 \text{ N}$$

Assume Impending Sliding at B

This assumption gives $F_B = 0.4N_B = 0.4(75) = 30$ N. From the FBD of the spool, we now obtain

$$\Sigma M_A = 0 \quad \curvearrowright \quad -F_B(240) + P(200) = 0$$

which gives

$$P = 1.2F_B = 1.2(30) = 36.0 \text{ N}$$

Choose the Correct Answer

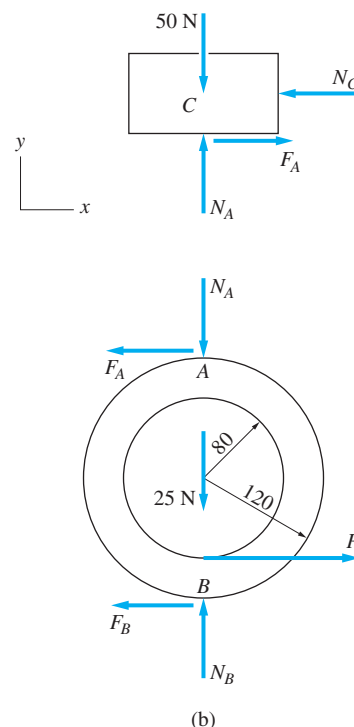
Up to this point, the analysis has determined that sliding impends at A if $P = 90.0$ N and at B if $P = 36.0$ N. Consequently, the largest force P that can be applied without disturbing the static equilibrium of the spool is

$$P = 36.0 \text{ N}$$

Answer

with sliding impending at B .

An alternate method for solving this problem is to assume impending sliding at one surface and then to compare the friction force at the other surface with its limiting static value.



7.4 *Impending Tipping*

In the preceding article, we restricted our attention to sliding; the possibility of tipping was neglected. We now discuss problems that include both sliding and tipping as possible motions.

Consider again a homogeneous block on a friction surface being pushed by a force P , as shown in Fig. 7.4 (a). We assume that the weight W of the block, and the dimensions b , h , and d are known. We wish to determine the magnitude of P that will cause impending motion of the block, either impending sliding or impending tipping.

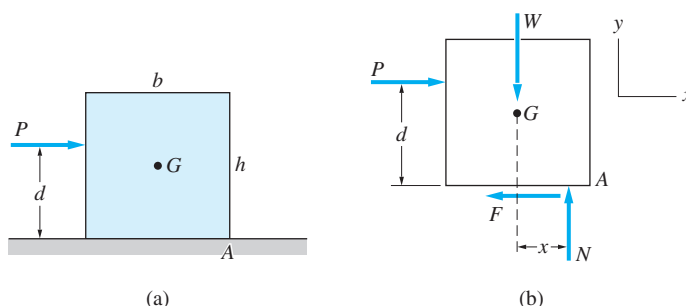


Fig. 7.4

We can gain insight into the solution by comparing the number of unknowns with the number of available equilibrium equations. From the free-body diagram of the block, Fig. 7.4(b), we see that there are four unknowns: the applied force P , the resultant normal force N , the friction force F , and the distance x that locates the line of action of N . Because there are only three independent equilibrium equations, an additional equation must be found before all unknowns can be calculated. If we assume impending sliding, the additional equation is $F = F_{\max} = \mu_s N$. On the other hand, if impending tipping about corner A is assumed, the additional equation is $x = b/2$, because N acts at the corner of the block when tipping impends.

In the preceding article, three classes of friction problems were introduced for impending sliding. This classification can be easily reworded to include the possibility of impending tipping.

Type I The problem statement does not specify impending motion (sliding or tipping).

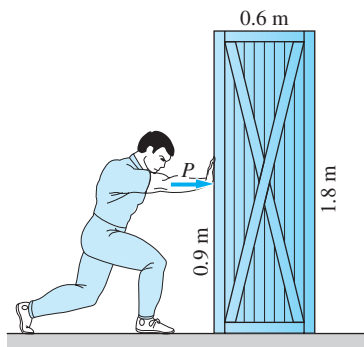
Type II The problem statement implies impending motion, and the type of motion (sliding at known surfaces, or tipping) is known.

Type III The problem statement implies impending motion, but the type of motion (sliding or tipping) and/or the surfaces where sliding impends are not known.

Examples of the three types of problems are given in the sample problems that follow.

Sample Problem 7.8

The man in Fig. (a) is trying to move a packing crate across the floor by applying a horizontal force P . The center of gravity of the 250-N crate is located at its geometric center. Does the crate move if $P = 60$ N? The coefficient of static friction between the crate and the floor is 0.3.

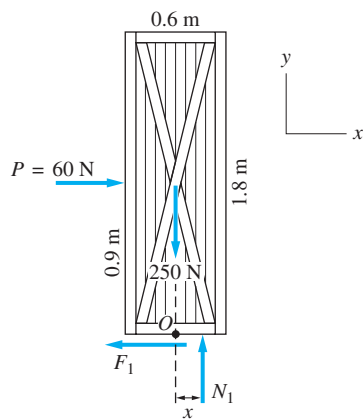


(a)

Solution

This is a Type I problem because the problem statement does not specify impending motion. To determine if the crate moves for the conditions stated, we first assume equilibrium and then check the assumption. However, the check must answer two questions—(1) does the crate slide and (2) does the crate tip?

The free-body diagram of the crate is shown in Fig. (b). If the block is assumed to remain in equilibrium, the three equilibrium equations can be used to calculate the three unknowns: the normal force N_1 , the friction force F_1 , and the distance x locating the line of action of N_1 , as shown in the following.



(b)

Assume Equilibrium

$$\begin{aligned}\Sigma F_x = 0 \quad \rightarrow \quad P - F_1 &= 0 \\ F_1 &= P = 60 \text{ N}\end{aligned}$$

$$\begin{aligned}\Sigma F_y = 0 \quad \uparrow \quad N_1 - 250 &= 0 \\ N_1 &= 250 \text{ N}\end{aligned}$$

$$\Sigma M_O = 0 \quad \curvearrowright \quad N_1 x - P(0.9) = 0$$

which gives

$$x = P(0.9)/N_1 = 60(0.9)/250 = 0.216 \text{ m}$$

Check

The largest possible value for x is 0.3 m (half the width of the crate). Because $x = 0.216$ m, as obtained from equilibrium analysis, is smaller than that, we conclude that the block *will not tip*.

The limiting static friction force is $(F_1)_{\max} = \mu_s N_1 = 0.3(250) = 75.0$ N, which is larger than the force $F_1 = 60$ N that is required for equilibrium. We therefore conclude that the crate *will not slide*.

Crate will not move when $P = 60$ N

Answer

Sample Problem 7.9

Calculate the force P required to cause tipping of the packing crate in Sample Problem 7.8. Also determine the minimum coefficient of static friction that permits tipping.

Solution

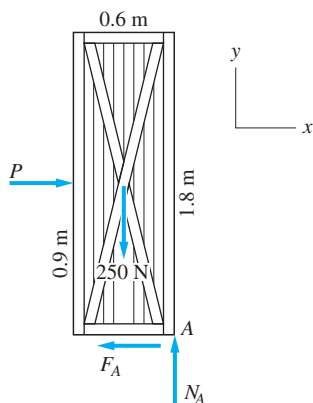
This is a Type II problem because impending tipping is specified. The free-body diagram for the crate is shown in the figure. Note that when the crate is on the verge of tipping, the normal force N_A acts at corner A. There are three equilibrium equations that can be solved for the unknowns P , N_A , and F_A :

$$\begin{aligned}\Sigma M_A = 0 \quad \curvearrowright \quad 250(0.3) - P(0.9) &= 0 \\ P &= 83.3 \text{ N}\end{aligned}$$

Answer

$$\begin{aligned}\Sigma F_x = 0 \quad \rightarrow \quad P - F_A &= 0 \\ F_A &= P = 83.3 \text{ N}\end{aligned}$$

$$\begin{aligned}\Sigma F_y = 0 \quad \uparrow \quad N_A - 250 &= 0 \\ N_A &= 250 \text{ N}\end{aligned}$$



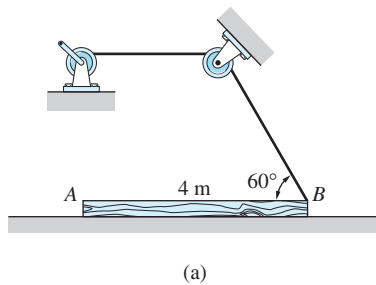
The minimum coefficient of static friction that permits tipping is

$$\mu_s = F_A/N_A = 83.3/250 = 0.333 \quad \text{Answer}$$

Note that if the coefficient of static friction were exactly 0.333, then the force $P = 83.3$ N would result in simultaneous impending sliding and impending tipping.

Sample Problem 7.10

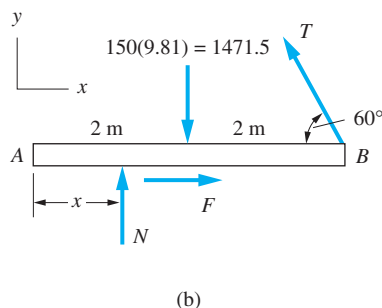
The winch in Fig. (a) is used to move the 150-kg uniform log AB . Compute the largest tension in the cable for which the log remains at rest. The coefficient of static friction between the log and the plane is 0.4.



Solution

Although we are asked to find the cable tension that would cause impending motion, we do not know whether sliding or tipping impends. Therefore, this is a Type III problem.

The free-body diagram of the log in Fig. (b) contains four unknowns: tension T , resultant normal force N , friction force F , and x (the distance from A to the line of action of N). Because there are only three independent equilibrium equations, all unknowns cannot be calculated unless an assumption is made concerning the type of impending motion.



Assume Impending Sliding

Under this assumption, we have $F = F_{\max} = 0.4 \text{ N}$, and the force equilibrium equations for the FBD in Fig. (b) are as follows.

$$\Sigma F_x = 0 \quad \rightarrow \quad 0.4 \text{ N} - T \cos 60^\circ = 0 \quad (\text{a})$$

$$\Sigma F_y = 0 \quad \uparrow \quad N - 1471.5 + T \sin 60^\circ = 0 \quad (\text{b})$$

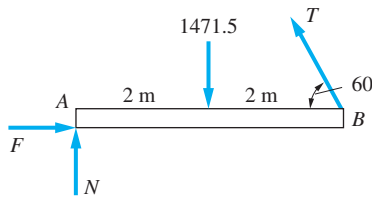
Solving Eqs. (a) and (b) simultaneously, we obtain $T = 695.4 \text{ N}$ and $N = 869.3 \text{ N}$.

Assume Impending Tipping

Under this assumption, N will act at A , as shown in the FBD in Fig. (c). The cable tension T can be computed from the moment equation

$$\Sigma M_A = 0 \quad \curvearrowright \quad T \sin 60^\circ (4) - 1471.5(2) = 0$$

$$T = 849.6 \text{ N}$$



(c)

Choose the Correct Answer

Because $T = 695.4 \text{ N}$ for impending sliding and $T = 849.6 \text{ N}$ for impending tipping, the maximum tension that can be applied without moving the log is

$$T = 695.4 \text{ N}$$

Answer

Alternate Solutions

As with most equilibrium problems, there are several equivalent methods of analysis that could be used. Two such methods are

1. Assume impending sliding, and solve for T . Continue the equilibrium analysis to find x . Then check to see if this value of x is physically possible.
2. Assume impending tipping, and solve for T . Continue the equilibrium analysis to find F . Then check to see if $F \leq F_{\max}$.

7.5 *Angle of Friction; Wedges and Screws*

a. *Angle of friction*

Figure 7.5 shows a block on a friction surface subjected to the horizontal force P . As seen in the free-body diagram, we let ϕ be the angle between the contact force R and the normal n to the contact surface. The angle ϕ is given by $\tan \phi = F/N$, where N and F are the normal and friction forces, respectively. The upper limit of ϕ , denoted by ϕ_s , is reached at impending sliding when $F = F_{\max} = \mu_s N$. Therefore, we have

$$\tan \phi_s = \mu_s \quad (7.4)$$

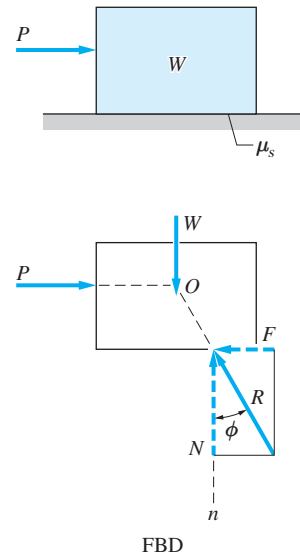


Fig. 7.5

The angle ϕ_s is called the *angle of static friction*. Note that $\phi \leq \phi_s$ signifies equilibrium and that $\phi = \phi_s$ indicates impending sliding. Therefore, the direction of the contact force R is known at all surfaces where sliding impends. This knowledge can be frequently utilized to gain insight into problems involving two- and three-force bodies.

In Fig. 7.5, the friction force F opposes the tendency of P to slide the block to the right. If the direction of P is reversed, the direction of F would also be reversed. This leads to the conclusion that the block can be in equilibrium only if the line of action of R stays within the sector AOB (bounded by $\pm\phi_s$), as shown in Fig. 7.6. For more general loadings, the line of action of R must lie within the cone, called the *cone of static friction*, that is formed by rotating sector AOB about the normal n . Observe that the vertex angle of the cone of static friction is $2\phi_s$.

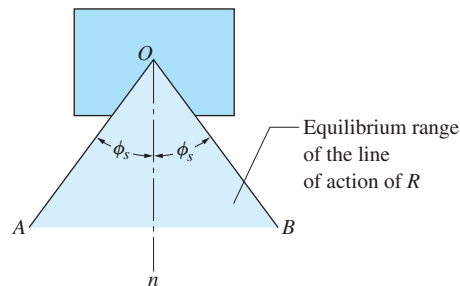


Fig. 7.6

When sliding occurs, the friction force is $F = \mu_k N$, and the value of ϕ that specifies the direction of R is given by

$$\tan \phi_k = \mu_k \quad (7.5)$$

The angle ϕ_k is called the *angle of kinetic friction*. For this case, the cone of static friction is replaced by the smaller *cone of kinetic friction*, for which the vertex angle is $2\phi_k$.

b. Wedges

A wedge is a simple device that is used for the same purpose as a lever—to create a mechanical advantage. Consider, for example, the wedge shown in Fig. 7.7(a) that is being forced into a crack by the applied force P . The angle formed by the tip of the wedge is 2β , where β is called the *wedge angle*. Neglecting the weight, the free-body diagram of the wedge at impending sliding is shown in Fig. 7.7(b). As before, we let ϕ be the angle between the contact force R and the normal n to the contact surface. Because sliding impends, $\phi = \phi_s$, where $\phi_s = \tan^{-1} \mu_s$ is the angle of friction. From the force diagram in Fig. 7.7(c) we see that $R = P/[2 \sin(\phi_s + \beta)]$, which is substantially larger than P if the wedge angle β is small and the sides of the wedge are lubricated (giving a small value for ϕ_s).

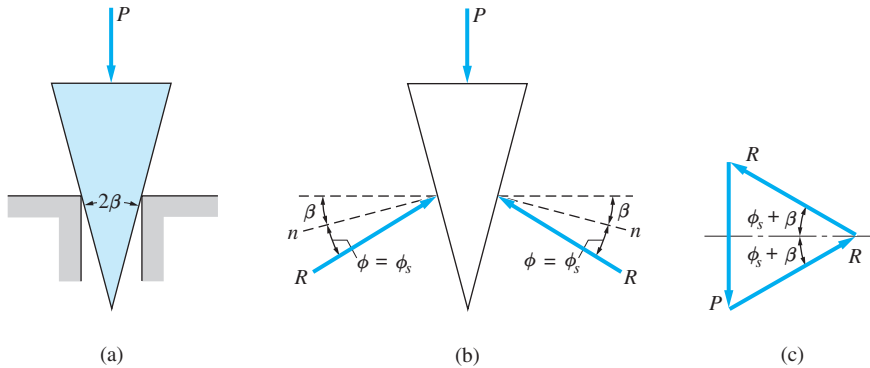


Fig. 7.7

Ideally, a wedge should be slippery enough to be easily driven into the crack, but have enough friction so that it stays in place when the driving force is removed. In the absence of P , the wedge becomes a two-force body. Therefore, the contact forces R must be collinear, as indicated in the free-body diagram in Fig. 7.8, where now $\phi = \beta$. Recalling that equilibrium can exist only if $\phi \leq \phi_s$, we conclude that the wedge will stay in place provided that $\beta \leq \phi_s$.

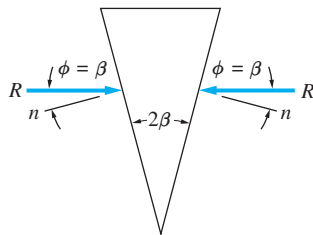


Fig. 7.8

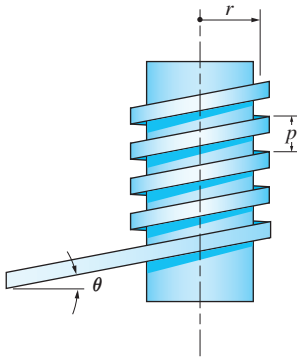


Fig. 7.9

c. Square-threaded screws

Screws with square threads are employed in jacks, vises, and other devices that produce a large axial force by applying a relatively small couple about the axis of the screw. A square-threaded screw can be viewed as a bar of rectangular cross section wrapped around a cylinder in a helical fashion, as shown in Fig. 7.9. The helix angle θ is called the *lead angle*, the distance p between the threads is known as the *pitch*, and the mean radius of the threads is denoted by r . It can be seen that these parameters are related by

$$p = 2\pi r \tan \theta \quad (7.6)$$

Figure 7.10 (a) shows a screw being employed as a jack. Assuming that the couple C_0 , called the *torque*, is large enough, it will cause the screw to advance, thereby elevating the weight W . The analysis of this problem is simplified if we recall that in Coulomb's theory the friction force is independent of the contact area. Therefore, we can assume the contact area to be very small, as illustrated in Fig. 7.10(b). Note that the entire weight W is carried by the contact area and that the horizontal force $Q = C_0/r$ is supplied by the applied torque C_0 . We can now see that this problem is identical to the one shown in Fig. 7.11(a)—namely, a block of weight W being pushed up an incline by the horizontal force Q .

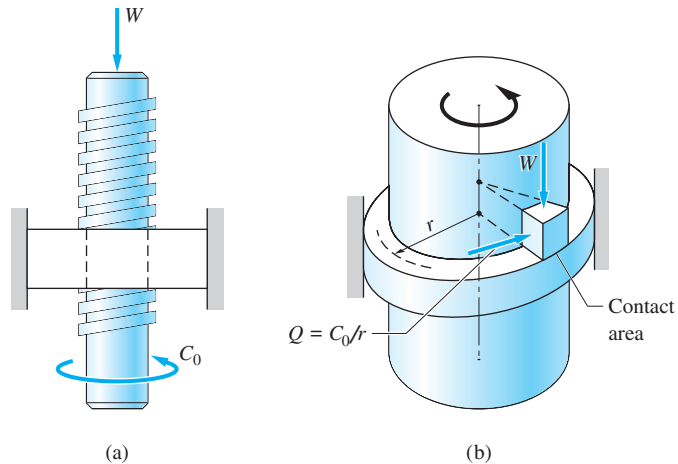


Fig. 7.10

The smallest torque required to start the weight W moving *upward* can now be obtained from the FBD in Fig. 7.11(b). Note that at impending sliding the angle between R and the normal n to the contact surface is $\phi = \phi_s$, and that the direction of ϕ_s relative to the normal n indicates that the impending motion is directed up the incline. For equilibrium of the block, we have

$$\Sigma F_x = 0 \quad \rightarrow \quad \frac{C_0}{r} - R \sin(\phi_s + \theta) = 0 \quad (a)$$

$$\Sigma F_y = 0 \quad \uparrow \quad R \cos(\phi_s + \theta) - W = 0 \quad (b)$$

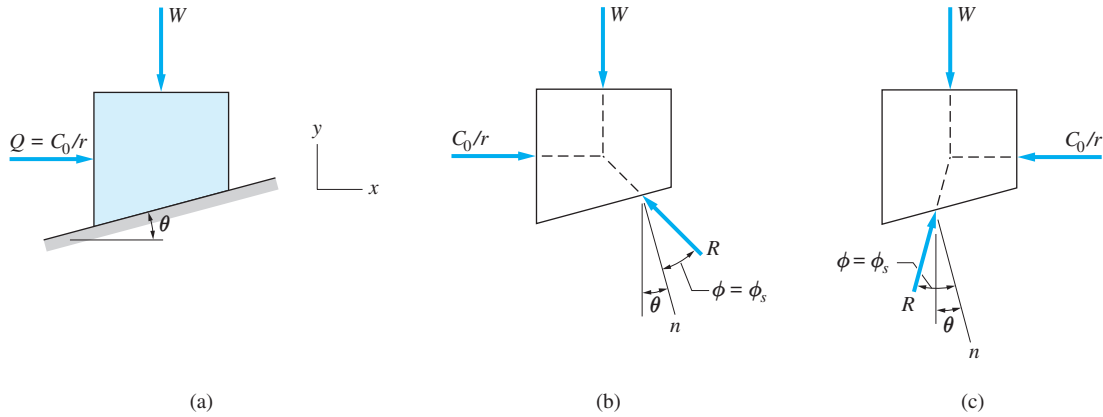


Fig. 7.11

Solving Eqs. (a) and (b), we find that the smallest torque that will cause the weight W to move *upward* is

$$(C_0)_{\text{up}} = Wr \tan(\phi_s + \theta) \quad (7.7a)$$

If we reverse the direction of C_0 and assume impending motion down the incline, the FBD in Fig. 7.11(c) must be used. It is seen from the equilibrium equations that the torque required to cause the weight W to move *downward* is

$$(C_0)_{\text{down}} = Wr \tan(\phi_s - \theta) \quad (7.7b)$$

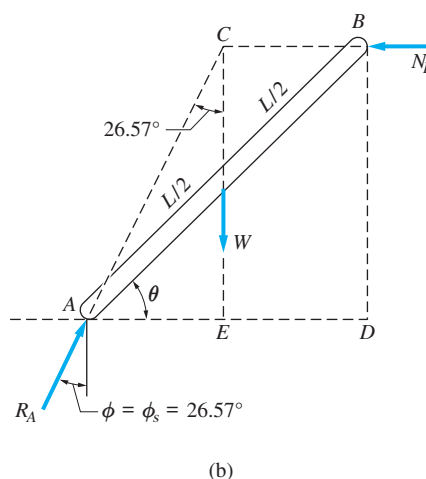
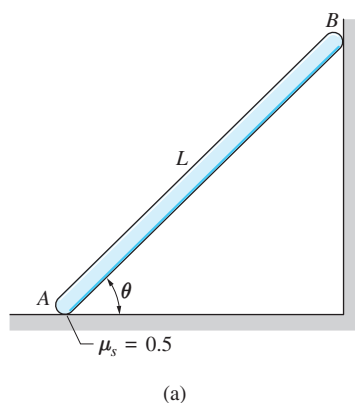
If $\phi_s \geq \theta$, the torque C_0 in Eq. (7.7b) is positive, which means that the weight W remains at rest if C_0 is removed. In this case, the screw is said to be *self-locking*. On the other hand, if $\phi_s < \theta$, the torque C_0 in Eq. (7.7b) is negative, indicating that the weight W would come down by itself in the absence of C_0 .

Sample Problem 7.11

Using the angle of friction, determine the smallest angle θ for which the slender bar shown in Fig. (a) can remain at rest. The bar is homogeneous of weight W and length L . Neglect friction between the bar and wall at B .

Solution

As shown in the FBD, Fig. (b), the bar is acted on by three forces: the weight W acting at the midpoint of the bar, the horizontal normal force N_B , and the reaction R_A at the horizontal surface. Because impending motion is specified, the angle ϕ between R_A and the normal to the contact surface is equal to its limiting value: $\phi = \phi_s = \tan^{-1} \mu_s = \tan^{-1} 0.5 = 26.57^\circ$. Because the bar is a three-force member, the forces intersect at point C .



From triangle ABD we see that

$$\tan \theta = \frac{\overline{BD}}{\overline{AD}} \quad (a)$$

From triangle ACE we obtain $\overline{AE} = \overline{CE} \tan 26.57^\circ$. Because $\overline{AD} = 2\overline{AE}$, this becomes

$$\overline{AD} = 2\overline{CE} \tan 26.57^\circ \quad (b)$$

Substituting Eq. (b) into Eq. (a) together with $\overline{BD} = \overline{CE}$, we get

$$\tan \theta = \frac{\overline{CE}}{2\overline{CE} \tan 26.57^\circ} = \frac{1}{2 \tan 26.57^\circ}$$

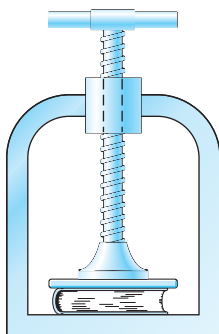
which yields

$$\theta = 45.0^\circ \quad \text{Answer}$$

By using the angle of friction and by recognizing the bar to be a three-force member, we were able to find θ from geometry, without having to write the equilibrium equations.

Sample Problem 7.12

The screw press shown is used in bookbinding. The screw has a mean radius of 10 mm and its pitch is 5 mm. The static coefficient of friction between the threads is 0.18. If a clamping force of 1000 N is applied to the book, determine (1) the torque that was applied to the handle of the press; and (2) the torque required to loosen the press.



Solution

The lead angle of the screw is computed from Eq. (7.6).

$$\theta = \tan^{-1} \frac{p}{2\pi r} = \tan^{-1} \frac{5}{2\pi(10)} = 4.550^\circ$$

The friction angle is

$$\phi_s = \tan^{-1} \mu_s = \tan^{-1} 0.18 = 10.204^\circ$$

Part 1

The torque required to apply the force $W = 1000$ N can be calculated from Eq. (7.7a).

$$C_0 = Wr \tan(\phi_s + \theta) = 1000(0.01) \tan(10.204^\circ + 4.550^\circ)$$

$$C_0 = 2.63 \text{ N} \cdot \text{m} \quad \text{Answer}$$

Part 2

The torque needed to loosen the press is obtained from Eq. (7.7b).

$$C_0 = Wr \tan(\phi_s - \theta) = 1000(0.01) \tan(10.204^\circ - 4.550^\circ)$$

$$C_0 = 0.990 \text{ N} \cdot \text{m} \quad \text{Answer}$$

***7.6** *Ropes and Flat Belts*

The theory of Coulomb friction can also be used to analyze the forces acting between a flexible body, such as a rope or belt, and a friction surface.

Figure 7.12 shows a weight W that is held in static equilibrium by a rope that passes over a peg. If the peg is frictionless, then $P = W$; that is, the peg simply reverses the direction of the rope without changing its tension. If the contact surface between the peg and the rope has friction, the friction force will help to keep the weight from falling. In this case, it is possible to have $P < W$ and still

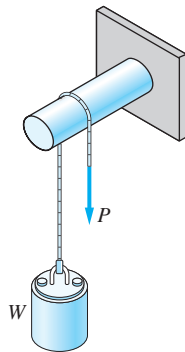


Fig. 7.12

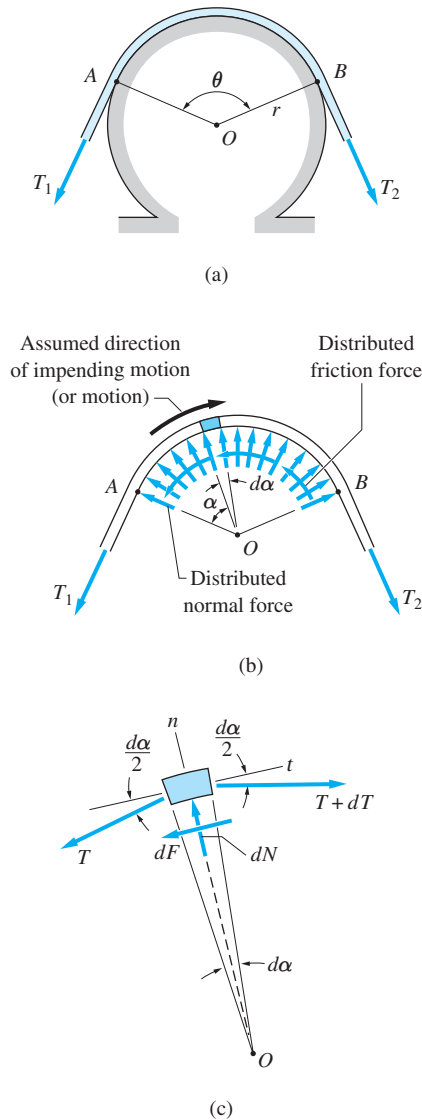


Fig. 7.13

maintain equilibrium. A good example of this principle in action is the capstan—a device for fastening a ship to the dock. Other applications are belt drives and band brakes. In a belt drive, the friction between the belt and the pulleys enables power to be transmitted between rotating shafts. Band brakes use friction between a band (belt) and a cylindrical drum to reduce the speed of rotating machinery.

Figure 7.13 (a) shows a thin, flat belt that passes over a cylinder of radius r . Assuming that there is friction between the belt and the cylinder, the tensions T_1 and T_2 are not equal. According to the theory of dry friction, the belt is in one of the following states depending on the values of T_1 and T_2 : equilibrium without impending slipping, equilibrium with impending slipping, or slipping. The analysis that follows determines the relationship between T_1 and T_2 for the last two cases.

The forces acting on the belt are shown in Fig. 7.13(b), where the weight of the belt is assumed to be negligible. The cylinder exerts normal and friction forces on the belt, both of which are distributed along the contact area between A and B. Because the direction of impending motion (or motion) of the belt is assumed to be clockwise, equilibrium implies that $T_2 > T_1$.^{*} Because the contact forces are distributed, it is necessary to perform the equilibrium analysis on an infinitesimal (differential) length of the belt that subtends the differential angle $d\alpha$.

The free-body diagram of the differential element is shown in Fig. 7.13(c), where n and t refer to the directions that are normal and tangent to the cylindrical surface at the center of the element. The belt tension on the left side of the element is denoted by T , and the tension on the right side by $(T + dT)$, where the differential change in the tension is $dT = (dT/d\alpha) d\alpha$. The angle between each tension and the t -direction is $d\alpha/2$. The element is also subjected to the normal force dN and to the friction force dF , acting in the n - and t -directions, respectively.

Equilibrium of forces in the tangential direction yields

$$\Sigma F_t = 0 \quad \nearrow \quad (T + dT) \cos \frac{d\alpha}{2} - T \cos \frac{d\alpha}{2} - dF = 0 \quad (a)$$

Because the cosine of an infinitesimal angle equals 1, Eq. (a) reduces to

$$dF = dT \quad (b)$$

The balance of forces in the normal direction gives

$$\Sigma F_n = 0 \quad \nwarrow \quad dN - (T + dT) \sin \frac{d\alpha}{2} - T \sin \frac{d\alpha}{2} = 0 \quad (c)$$

Assuming that α is measured in radians, $\sin(d\alpha/2)$ can be replaced by $d\alpha/2$, an identity that is valid for infinitesimal angles. Making this substitution gives

$$dN - (T + dT) \frac{d\alpha}{2} - T \frac{d\alpha}{2} = 0 \quad (d)$$

^{*}The equilibrium equations are applicable even if the belt is moving. Because the weight of the belt is assumed to be negligible, inertial effects can be omitted except for very high speeds.

Neglecting the product of differentials ($dT d\alpha$) compared to $T d\alpha$ (this is not an approximation), we have

$$dN = T d\alpha \quad (e)$$

If the belt is slipping or if motion impends, we have the additional equation $dF = \mu dN$, where $\mu = \mu_k$ (slipping), or $\mu = \mu_s$ (impending slipping). Substituting this for dF in Eq. (b) and eliminating dN between Eqs. (b) and (e), we obtain

$$\frac{dT}{T} = \mu d\alpha \quad (f)$$

Integrating both sides of Eq. (f) over the contact angle θ shown in Fig. 7.13(a), we get

$$\int_0^\theta \frac{dT}{T} = \mu \int_0^\theta d\alpha$$

Noting that $T = T_1$ when $\alpha = 0$, and $T = T_2$ when $\alpha = \theta$, integration yields

$$\ln(T_2/T_1) = \mu\theta$$

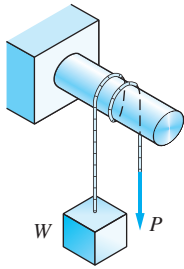
which can be written as

$$T_2 = T_1 e^{\mu\theta} \quad (7.8)$$

where $e = 2.718 \dots$ is the base of natural (Napierian) logarithms. If the tension in one side of the belt is known, Eq. (7.8) can be used to calculate the belt tension in the other side.

The following points should be kept in mind when using Eq. (7.8).

- T_2 is the belt tension that is directed opposite the belt friction. Thus, T_2 must always refer to the larger of the two tensions.
- For impending motion, use $\mu = \mu_s$. If there is relative motion between the belt and cylinder, use $\mu = \mu_k$.
- The angle of contact θ must be expressed in radians.
- Because Eq. (7.8) is independent of r , its use is not restricted to circular contact surfaces; it may also be used for a surface of arbitrary shape.



Sample Problem 7.13

The block of weight W is supported by a rope that is wrapped one-and-one-half times around the circular peg. Determine the range of values of P for which the block remains at rest. The coefficient of static friction between the rope and the peg is 0.2.

Solution

The tension in the portion of the rope that is attached to the block is obviously equal to W . Because motion impends, Eq. (7.8) can be used to relate this tension to P . Because the angle of contact is $\theta = 1.5(2\pi) = 3\pi$ rad, Eq. (7.8) becomes

$$T_2 = T_1 e^{\mu_s \theta} = T_1 e^{0.2(3\pi)} = 6.59 T_1$$

Recall that in this equation, T_2 refers to the larger of the two tensions.

The largest value of P for equilibrium occurs when the block is on the verge of moving upward. For this case we must substitute $T_1 = W$ and $T_2 = P$ into the preceding equation. The result is $P = 6.59W$.

The smallest value of P corresponds to impending motion of the block downward, when W will be larger than P . Substituting $T_1 = P$ and $T_2 = W$, we have $W = 6.59P$, or $P = W/6.59 = 0.152W$.

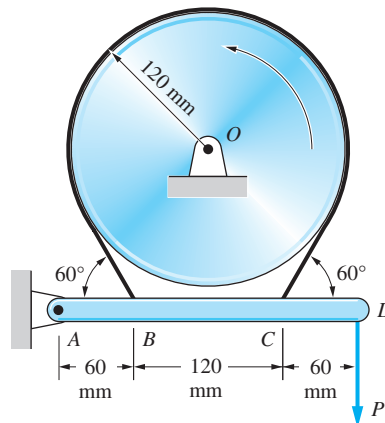
Therefore, the block is at rest if P is in the range

$$0.152W \leq P \leq 6.59W$$

Answer

Sample Problem 7.14

As shown in Fig. (a), a flexible belt placed around a rotating drum of 120-mm radius acts as a brake when the arm $ABCD$ is pulled down by the force P . The coefficient of kinetic friction between the belt and the drum is 0.2. Determine the force P that would result in a braking torque of $12000\text{-N} \cdot \text{mm}$, assuming that the drum is rotating counterclockwise. Neglect the weight of the brake arm.



(a)

Solution

The free-body diagram of the *belt* is shown in Fig. (b). The distributed contact forces exerted by the drum have been replaced by the equivalent force-couple system at O . The resultant force R is not of interest to us, but the couple C^R represents the braking torque; that is, $C^R = 12000 \text{ N} \cdot \text{mm}$. Note that C^R has the same sense as the rotation of the drum—namely, counterclockwise. The moment equation of equilibrium, with O as the moment center, is

$$\Sigma M_O = 0 \quad \curvearrowleft \quad (T_C - T_B) 120 - 12000 = 0 \quad (\text{a})$$

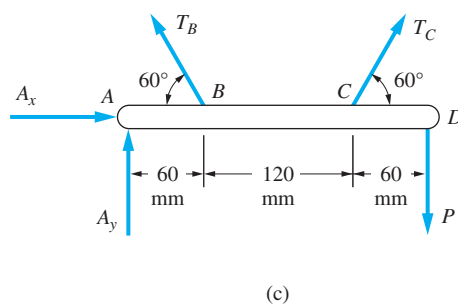
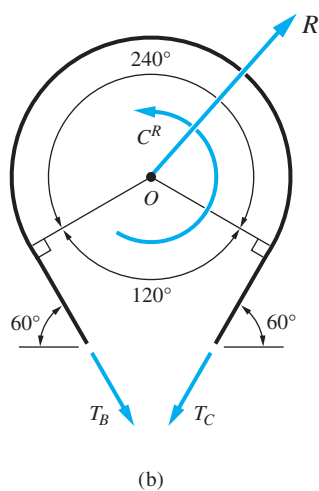
Equation (7.8) provides us with another relationship between the belt tensions. Substituting $\mu = \mu_k = 0.2$, $T_1 = T_B$, $T_2 = T_C$ (note that $T_C > T_B$), and $\theta = 240(\pi/180) = 1.333\pi$ rad, Eq. (7.8) becomes

$$T_C = T_B e^{0.2(1.333\pi)} = 2.311 T_B \quad (\text{b})$$

The solution of Eqs. (a) and (b) is $T_B = 76.3 \text{ N}$ and $T_C = 176.3 \text{ N}$.

The force P can now be found by the balance of moments about A on the FBD of the brake arm, shown in Fig. (c):

$$\Sigma M_A = 0 \quad \curvearrowleft \quad T_B \sin 60^\circ (60) + T_C \sin 60^\circ (180) - P(240) = 0$$



Substituting the values for T_B and T_C , and solving for P , gives

$$P = 131.0 \text{ N} \quad \text{Answer}$$

MODULE 4

Boilers

11.1 INTRODUCTION

Steam is extensively used for various applications such as power production, industrial processes, work interaction, heating etc. With the increasing use of steam in different engineering systems the steam generation technology has also undergone various developments starting from 100 B.C. when Hero of Alexandria invented a combined reaction turbine and boiler.

Boiler, also called steam generator is the engineering device which generates steam at constant pressure. It is a closed vessel, generally made of steel in which vaporization of water takes place. Heat required for vaporization may be provided by the combustion of fuel in furnace, electricity, nuclear reactor, hot exhaust gases, solar radiations etc.

Earlier boilers were closed vessels made from sheets of wrought iron which were lapped, riveted and formed into shapes of simple sphere type or complex sections such as the one shown in Fig. 11.1. It is the 'Wagon boiler' of Watt developed in 1788.

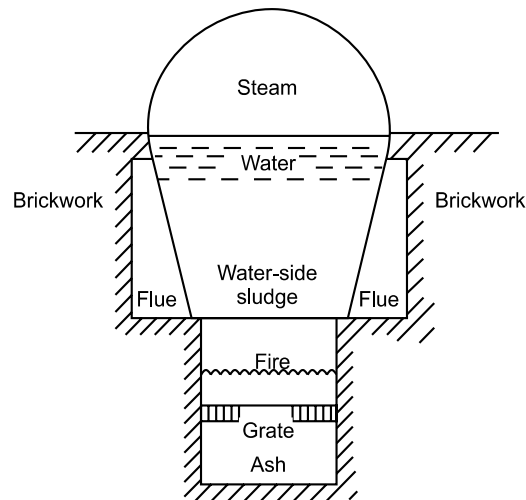


Fig. 11.1 Wagon boiler of Watt, (1788)

According to A.S.M.E. (American Society of Mechanical Engineers, U.S.A.) code a boiler is defined as a combination of apparatus for producing, furnishing or recovering heat together with the apparatus for transferring the heat so made available to water which could be heated and vaporised to steam form.

Boiler technology got revolutionized during second world war, when the need arose for the boilers to supply steam to field installations. Field requirements were critical as the boiler installation and commissioning should take place in minimum time. Therefore the 'Package boilers' which were complete with all auxiliaries as one unit came up and gradually transformed into modern boiler having lot of accessories and mountings. Thus in a boiler other than heat supplying unit, shell and tubes, a number of other devices are used for its control, safe and efficient operation. Devices which are mounted on boiler for its control and safe operation are called "mountings" while devices which are mounted on boiler for improving its performance are called "accessories". Thus boiler mountings are necessary while boiler accessories are optional.

11.2 TYPES OF BOILERS

Boilers are of many types. Depending upon their features they can be classified as given under:

- (a) Based upon the orientation/axis of the shell: According to the axis of shell boiler can be classified as vertical boiler and horizontal boiler.
 - (i) *Vertical boiler* has its shell vertical.
 - (ii) *Horizontal boiler* has its shell horizontal.
 - (iii) *Inclined boiler* has its shell inclined.
- (b) Based upon utility of boiler: Boilers can be classified as
 - (i) *Stationery boiler*, such boilers are stationery and are extensively used in power plants, industrial processes, heating etc.
 - (ii) *Portable boiler*, such boilers are portable and are of small size. These can be of the following types,
 - Locomotive boiler, which are exclusively used in locomotives.
 - Marine boiler, which are used for marine applications.
- (c) Based on type of firing employed: According to the nature of heat addition process boilers can be classified as,
 - (i) Externally fired boilers, in which heat addition is done externally i.e. furnace is outside the boiler unit. Such as Lanchashire boiler, Locomotive boiler etc.
 - (ii) Internally fired boilers, in which heat addition is done internally i.e. furnace is within the boiler unit. Such as Cochran boiler, Babcock Wilcox boiler etc.
- (d) Based upon the tube content: Based on the fluid inside the tubes, boilers can be,
 - (i) *Fire tube boilers*, such boilers have the hot gases inside the tube and water is outside surrounding them. Examples for these boilers are, Cornish boiler, Cochran boiler, Lancashire boiler, Locomotive boiler etc.
 - (ii) *Water tube boilers*, such boilers have water flowing inside the tubes and hot gases surround them. Examples for such boilers are Babcock-Wilcox boiler, Stirling boiler, La-Mont boiler, Benson boiler etc.
- (e) Based on type of fuel used: According to the type of fuel used the boilers can be,
 - (i) *Solid fuel fired boilers*, such as coal fired boilers etc.
 - (ii) *Liquid fuel fired boilers*, such as oil fired boilers etc.
 - (iii) *Gas fired boilers*, such as natural gas fired boilers etc.
- (f) Based on circulation: According to the flow of water and steam within the boiler circuit the boilers may be of following types,
 - (i) *Natural circulation boilers*, in which the circulation of water/steam is caused by the density difference which is due to the temperature variation.

- (ii) *Forced circulation boilers*, in which the circulation of water/steam is caused by a pump i.e. externally assisted circulation.
- (g) Based on extent of firing: According to the extent of firing the boilers may be,
 - (i) *Fired boilers*, in which heat is provided by fuel firing.
 - (ii) *Unfired boilers*, in which heat is provided by some other source except fuel firing such as hot flue gases etc.
 - (iii) *Supplementary fired boilers*, in which a portion of heat is provided by fuel firing and remaining by some other source.

11.3 REQUIREMENTS OF A GOOD BOILER

Different requirements of a good boiler are given below. In general boiler is supposed to generate large quantity of steam at desired pressure and temperature quickly and efficiently.

- (a) It should be capable of generating steam at desired rate at desired pressure and temperature with minimum fuel consumption and cost.
- (b) It should have sufficient steam and water storage capacity to meet fluctuation in demand and to prevent fluctuation in steam pressure or water level.
- (c) Boiler should have a constant and thorough circulation of water.
- (d) It should be equipped with all necessary mountings.
- (e) Boiler should have capability to get started quickly from cold.
- (f) Its construction should be simple and have good workmanship for the ease of inspection and repairs i.e. easily accessible parts.
- (g) Boiler should have its heating surface nearly at right angle to the current of hot gases for good heat transfer.
- (h) There should be minimum frictional power loss during flow of hot gases and water/steam i.e. pressure drop throughout the system should be minimum.
- (i) Tubes should be so designed so as to have minimum soot deposition and good strength against wear. Boiler should have a mud drum to receive all impurities.
- (j) Boiler should have strength to withstand excessive thermal stresses.
- (k) Boiler should occupy less floor area and space.

Boilers may be selected for a particular applications considering above general requirements and constraints, if any. For deciding the boiler for any application, generally following criterion are made;

- (i) Steam pressure requirement
- (ii) Steam temperature requirement
- (iii) Steam generation rate
- (iv) Initial cost and constraints
- (v) Running and maintenance costs
- (vi) Availability of fuel and water
- (vii) Inspection and maintenance requirements.

11.4 FIRE TUBE AND WATER TUBE BOILERS

Fire tube boilers are those boilers in which hot gases (combustion products) flow inside the tubes and water surrounds them. Water extracts heat for its phase transformation from the hot gases flowing inside the tubes, thus heat is indirectly transferred from hot gas to water through a metal interface.

Such boilers came up in eighteenth century and were extensively used for steam generation in variety of applications. With the passage of time and coming up of another types of boilers the fire tube

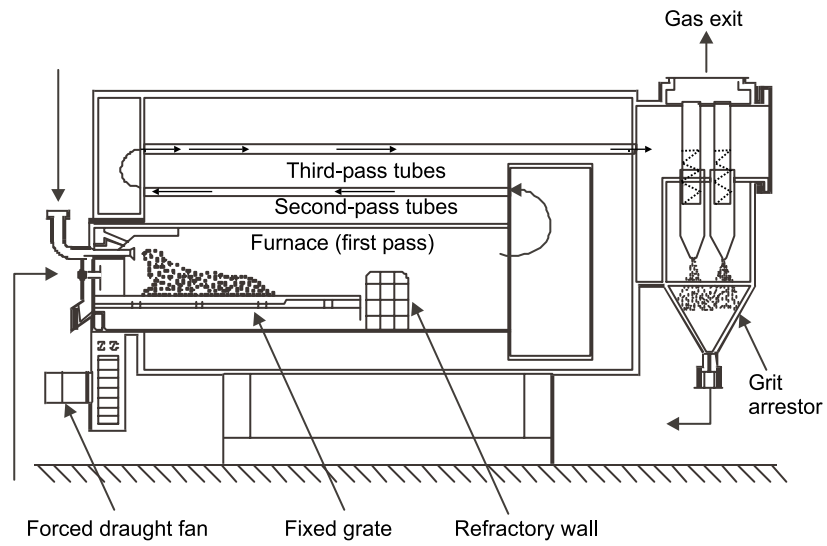


Fig. 11.2 *Fire tube boiler*

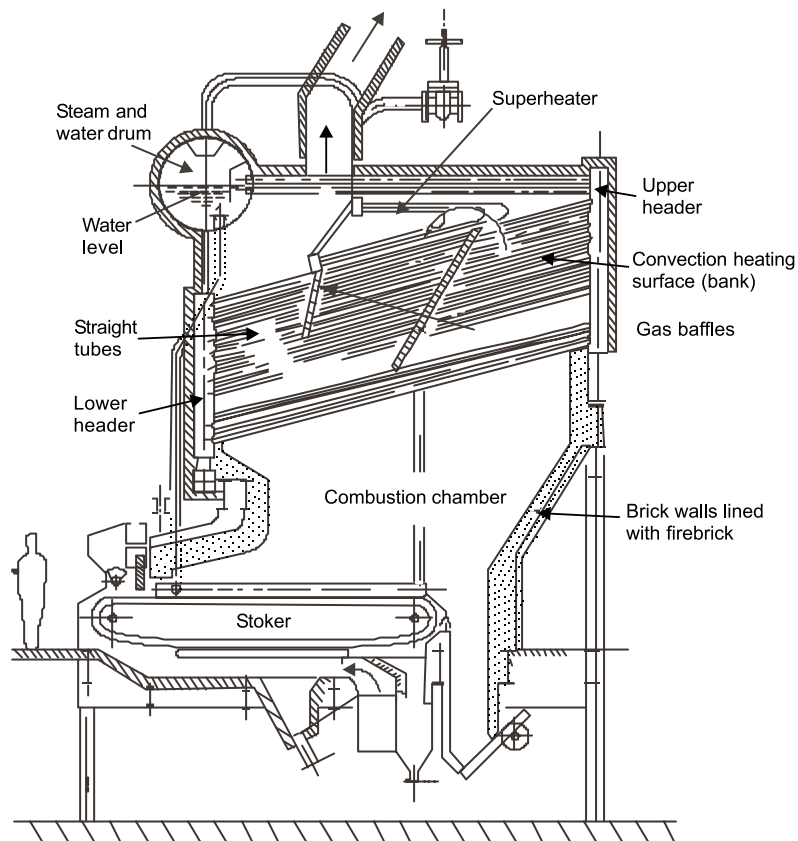


Fig. 11.3 *Water tube boiler*

boilers have lost their charm to some extent due to limitations in terms of steam pressure. Fire tube boilers are used for applications having small steam requirement. Different types of fire tube boilers have been discussed ahead.

Water tube boilers are those boilers in which water flows inside the tubes and hot gases surround them. This type of boilers came up as a solution to the problem of explosion faced in fire tube boilers when the pressure and steam generation capacity were increased. In such boilers the shell behaved as heated pressure vessel subjected to internal pressure which set up tensile stresses (hoop stress) in walls. Mathematically, this stress can be given as,

$$\text{Hoop stress} = \frac{P \times D}{2t} \text{ where } P \text{ is internal working pressure, } D \text{ is diameter of shell and } t \text{ is}$$

thickness of shell wall.

Above expression shows that if ' P ' (pressure) increases then either ' D ' (diameter) should be decreased or ' t ' (thickness) be increased to keep stress within acceptable limits. While increasing thickness the mass of boiler and cost of manufacturing both increase therefore the reduction of ' D ' (diameter) is an attractive option. This became the basis for water tube boilers in which small diameter of tube facilitated quite high pressure steam generation.

Such boilers came up in late eighteenth and nineteenth century. George Babcock and Stephen Wilcox gave straight-tube boiler of water tube boiler type in 1867 which was subsequently modified and developed as present 'Babcock and Wilcox boiler'.

Water tube boilers may be further classified based on type of tubes employed. These can be *Straight water tube boilers* and *Bent water tube boilers*. Straight water tube boilers are those in which tubes carrying water are straight from one end to the other end. At the two ends headers are provided. In general water comes down from drum into down header and after passing through tubes get heated and evaporated to steam which is carried back to drum through upcomer header or riser. Circulation of water is caused by the density difference as density of feed water is more than density of hot water/wet/dry steam due to lower temperature of feed water.

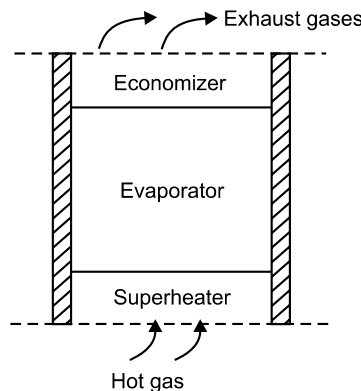


Fig. 11.4 Relative position of boiler sections

Bent water tube boilers are those in which bent tubes are employed for carrying water. Bent water tubes are advantageous over straight water tubes in many respects. Bent tubes offer better access into boiler and ease of inspection and maintenance. Also tube arrangement can be modified so as to maximize heating surface and exposure of tubes to hot gases.

Circulation is better in case of bent tube boilers as compared to straight tube, since the orientation of tubes in case of former is generally at inclination from vertical while for later it is horizontal. Stirling

boiler is one such boiler. In water tube boilers the heat distribution generally occurs amongst economiser tubes, evaporator tubes, superheater tubes. Hottest gases are designed to come in contact with superheater tubes. The evaporator tubes are in between superheater and economizer tubes. Relative position of three sections shall be as shown here.

A comparative study between fire tube and water tube boiler is presented below to understand relative merits of one over the other.

Advantages of fire tube boilers

- (a) Fire tube boilers are more flexible and can meet sudden steam demand without much pressure fluctuations. It is because of the large volume of water contained by these boilers and heat energy stored in hot water. It may be noted that energy stored in a definite volume of water at given pressure and temperature shall be more than that stored in same volume of steam at same thermodynamic states.
- (b) Fire tube boilers are less sensitive to the failure of feed water supply as they have large capacity of water stored. Such feed water supply failure is very damaging in water tube boilers due to small storage capacity.
- (c) Fire tube boilers are rigid and simple in construction, therefore have great reliability and less initial cost. Number of parts in fire tube boilers is less than those in water tube boilers so maintenance cost is also small. Since thickness of boiler shell is large enough so the problems of pitting and erosion are less. Also the large drum of boiler provides ample water space and desired conditions for dry steam generation.

Advantages of water tube boilers

- (a) Steam generation rate is large in water tube boilers as compared to fire tube boilers due to small quantity of water contained, large heating surface, better circulation of water etc. Water tube boilers are made in bigger sizes with very high limit to maximum output due to smaller drum, circulation etc.
- (b) Maximum pressure of steam generation is quite high in water tube boilers (125 bar and above) compared to fire tube boilers (up to 20 bar) due to fluid flowing through tubes of small diameter and diameter of drum being relatively small.
- (c) In case of explosion the steam generation may not stop in water tube boilers as the place of explosion in tubes can be plugged easily. While in fire tube boilers the explosion is very dangerous due to large quantity of water flashing into steam.
- (d) Water tube boilers are easy to fabricate and transport due to the small size of drum. The shell of fire tube boiler shall be nearly twice or thrice of the shell of water tube boiler for same power.
- (e) Water tube boilers are generally externally fired and various parts of boiler are more readily accessible for cleaning, inspection and maintenance, compared to fire tube boilers.

Characteristics of fire tube and water tube boilers are tabulated as under

Table 11.1

<i>Characteristics</i>	<i>Fire tube boiler</i>	<i>Water tube boiler</i>
(a) Steam Pressure	It is limited to 20–30 bar. In case of waste heat boilers, it can be more.	It is virtually unlimited within metallurgical and design limits.
(b) Unit output	Limited to about 20 MW. within design limits.	It is virtually unlimited
(c) Fuel	All commercial fuels and treated waste can be used.	Any fuel can be used. Also the furnace size is large.

Characteristics	Fire tube boiler	Water tube boiler
(d) Erection	It is packaged ready for work site.	It is to be shop assembled or erected at site.
(e) Efficiency	Normally 80–85%, gross calorific value, but can be further increased using accessories.	Normally 85–90%, gross calorific but can be further increased using accessories.
(f) Application	Generally for heat supply.	Generally for power and heat supply together.
(g) Inspection requirement.	Frequent inspection requirement. It is more than in water tube	Inspection requirement is less than in fire tube boiler. boilers.

Composite boilers: These are the boilers developed as combination of fire tube and water tube boilers so as to derive advantages of both designs. In these boilers there is no restriction in pressure and output capabilities. Such boilers are usually transported to site in sections/modules and assembled at site. Generic arrangement in composite boiler is shown in Fig. 11.5.

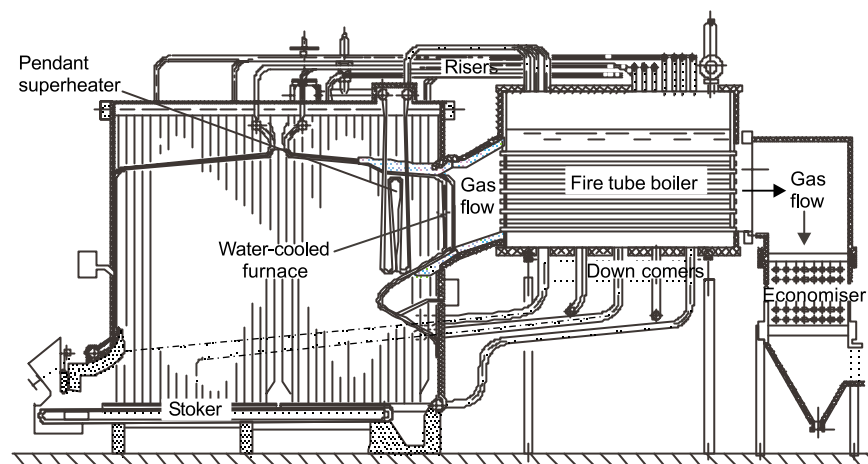


Fig. 11.5 Composite boiler

11.5 SIMPLE VERTICAL BOILER

Simple vertical boiler shown in Fig. 11.6 has a vertical boiler shell of cylindrical shape. It has fire box of cylindrical type inside the shell. Vertical passage of tubular type called uptake is provided over fire box for exhaust of flue gases. Cross tubes are provided for improving water circulation and increasing heating surface. At the bottom of fire box a fire grate is provided for burning fuel. Total heating surface area is about 7–10 times grate area. Man hole and hand holes are provided in the shell for access to inside of shell. Hot gases raising from fire grate go upwardly and heat the water contained in shell and tubes. Steam generated in shell can be tapped through a steam stop valve placed on the crown of shell. Such boilers have steam generation capacity up to 1000 kg per hour and maximum steam pressure up to 10 bar. Size of the boiler ranges from 0.6 m diameter to 2 m diameter and height from 1.2 m to 4 m high. Boiler efficiency is nearly 50%.

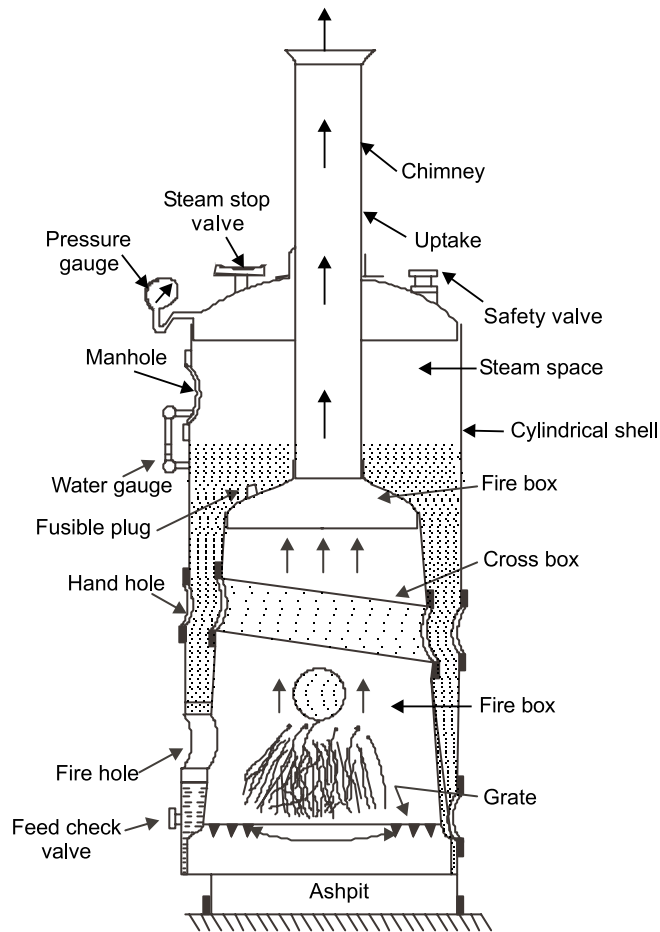


Fig. 11.6 Simple vertical boiler

11.6 COCHRAN BOILER

This is a fire tube boiler of vertical type and came up as a modification over the simple vertical boiler in order to maximize heating surface. Total heating surface area is 10–25 times the grate area. It has cylindrical shell with hemispherical crown. Hemispherical geometry offers maximum volume space for given mass of material and is also very good for strength and maximization of radiant heat absorption. Figure. 11.7 shows the schematic of Cochran boiler with various mountings upon it. Fire box is also of hemispherical form. Flue gases flow from fire box to refractory material lined combustion chamber through a flue pipe. Incomplete combustion if any can get completed in combustion chamber and hot gases subsequently enter into tubes. After coming out of fire tubes hot gases enter into smoke box having chimney upon it. As the fire box is separately located so any type of fuel such as wood, paddy husk, oil fuel etc. can be easily burnt.

These boilers are capable of generating steam up to pressure of 20 bar and steam generating capacity from 20 kg/hr to 3000 kg/hr. Boilers have dimensions ranging from 1 m diameter and 2 m height to 3 m diameter and 6 m height. Efficiency of such boilers ranges between 70 and 75%.

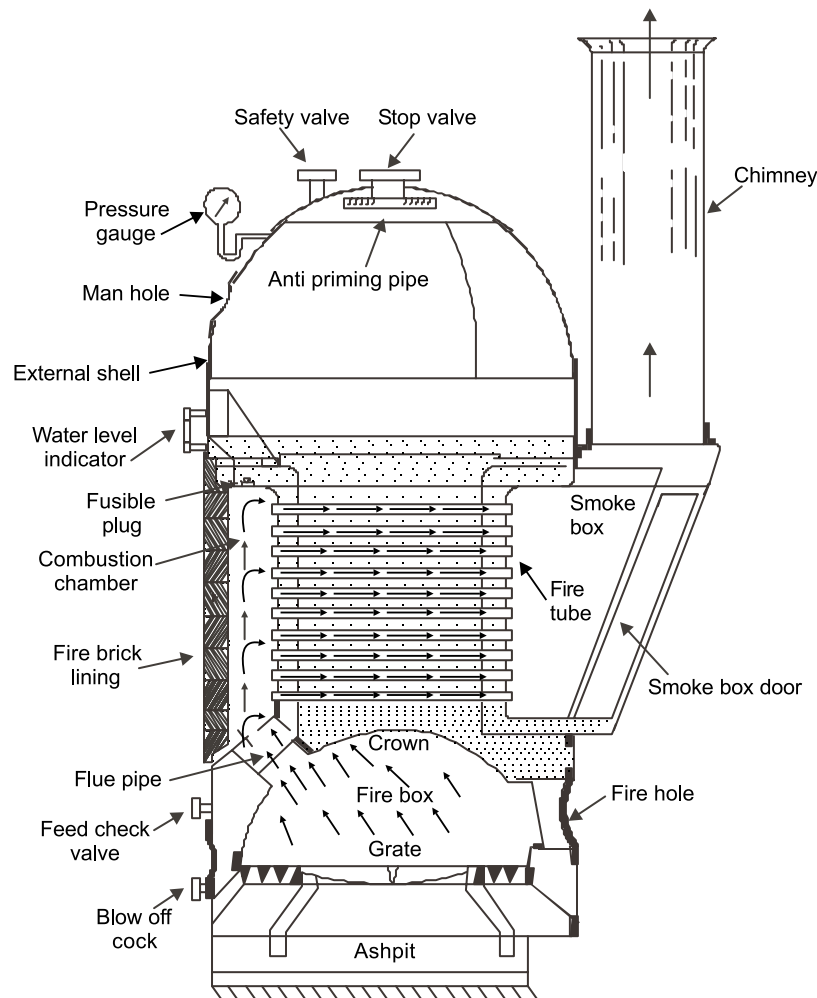


Fig. 11.7 Cochran boiler

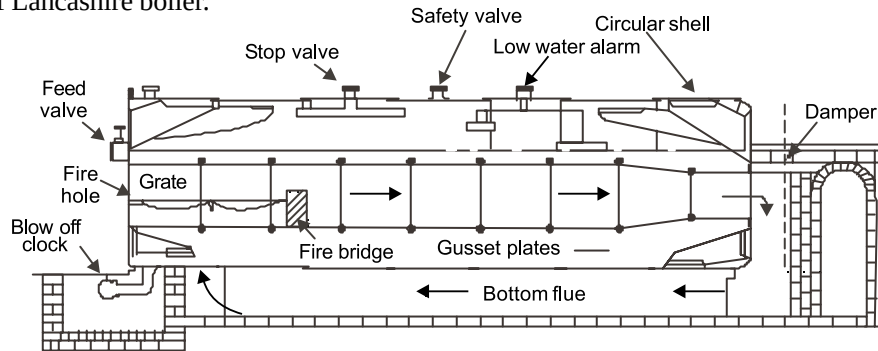
11.7 LANCASHIRE BOILER

It is a horizontal fire tube boiler. General arrangement in the boiler is shown in Fig. 11.8.

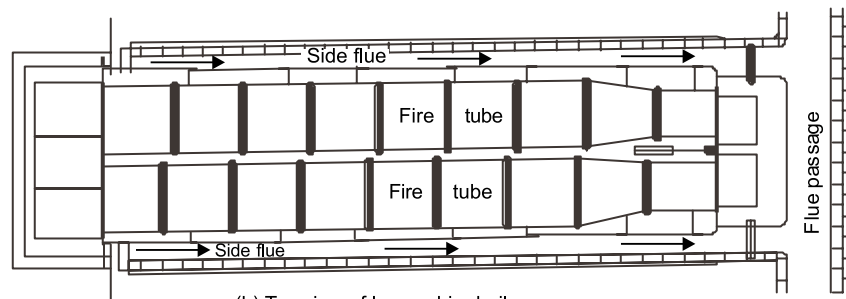
Boiler is mounted on a *brickwork setting* with front end of shell sloping about 1 : 250 for emptying the shell. It has a circular shell connected to *end plates* supported by gusset plates. Two *fire tubes* run throughout the length of the boiler. Fire tubes are of diameter less than half the diameter of shell and diameter of fire tubes is reduced as shown to have access to lower side of boiler.

Fire bridge is provided to prevent fuel from falling over the end of furnace. Fire bridge also helps in producing a better mixture of air and gases for perfect combustion by partly enveloping the combustion space. Hot gases start from grate area, enter into fire tubes and come out at back of boiler from where these gases flow towards the front of boiler through *bottom flue*. Upon reaching the front these hot gases flow through the *side flues* and enter the *main outlet*. Outlet passage may also be used commonly by more than one boilers. About 85% of actual heat transferred is transferred through surface of fire tubes while 15% is transferred through bottom and side flues.

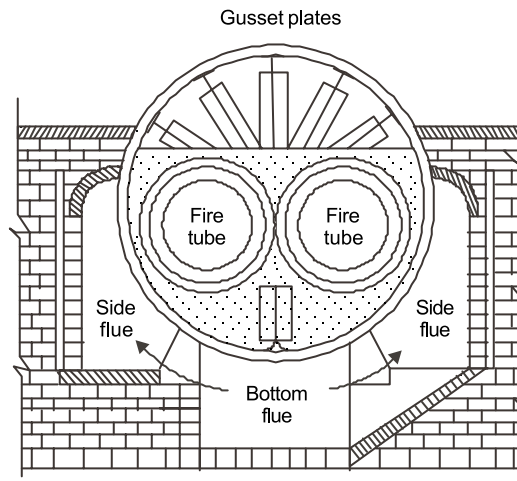
Plan, elevation and side views of Lancashire boiler shown in figure explain the furnace, different firetubes, bottom flues, side flues etc. Dampers are provided at the end of side flues for regulating the pressure difference (draught) for exit of burnt gases. Other mountings and accessories are shown in the elevation of Lancashire boiler.



(a) Front view of Lancashire boiler



(b) Top view of Lancashire boiler



(c) Side view of Lancashire boiler

Fig. 11.8 Lancashire boiler

Working pressure in these boilers are in the range of 0.7 MPa to 2 MPa and efficiency of the boiler is about 65%–70%. Size of these boiler depends upon size of shell which may be 2 m to 3 m in diameter and 6m to 10m in length.

11.11 BABCOCK AND WILCOX BOILER

It is a water tube boiler suitable for meeting demand of increased pressure and large evaporation capacity or large sized boiler units. Figure 11.12 shows the Babcock and Wilcox boiler. It has three main parts:

- (i) Steam and water drum
- (ii) Water tubes
- (iii) Furnace.

Steam and water drum is a long drum fabricated using small shells riveted together. End *cover plates* can be opened as and when required. Mountings are mounted on drum as shown. Drum is followed by water tubes which are arranged below drum and connected to one another and drum through headers. Header in which water flows from drum to tubes is called *down take header* while headers in which flow is from tubes to drum is called *uptake header*.

Soot deposition takes place in *mud box* which is connected to *downtake header*. “Blow off cock” for blowing out the sediments settled in *mud box* is shown in figure. Superheater tubes are also shown in the arrangement, which are U-shape tubes placed horizontally between drum and water tubes. Superheating of steam is realized in superheater tubes.

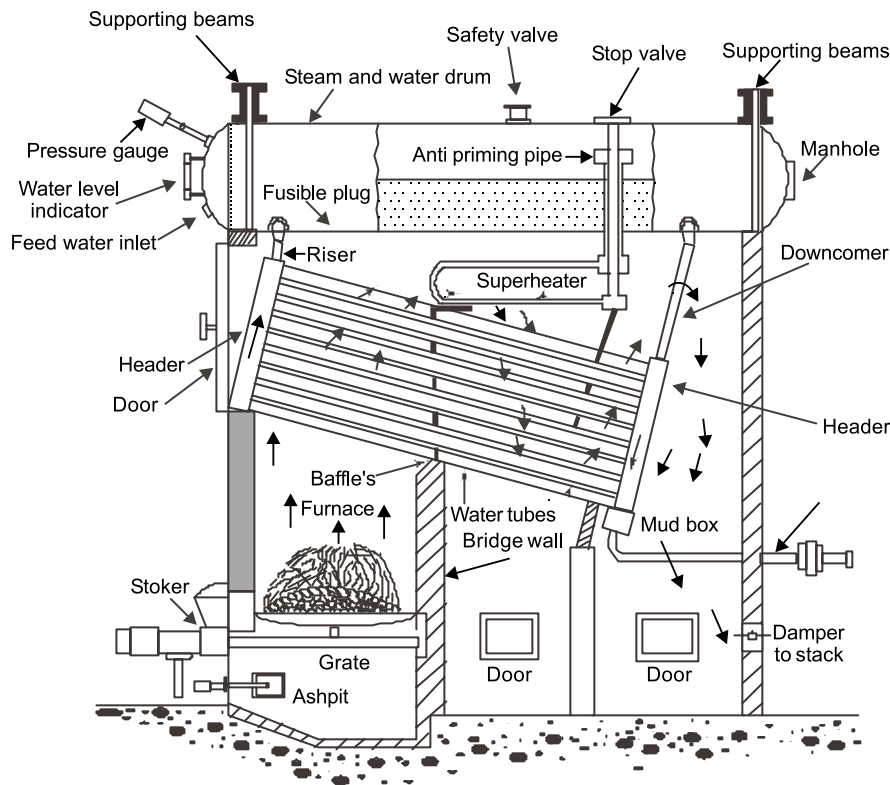


Fig. 11.12 Babcock and Wilcox boiler

Below the superheater and water tubes is the *furnace*, at the front of which *fuel feed hopper* is attached. *Mechanical stoker* is arranged below the hopper for feeding fuel. Bridge wall and baffles made of fire resistant bricks are constructed so as to facilitate hot gases moving upward from the *grate* area, then downwards and again upwards before escaping to the chimney. A *smoke box* is put at the back of furnace through which smoke goes out via *chimney*, put at top of smoke box. A *damper* is used for regulating pressure difference (draught) causing expulsion of hot gases.

The complete boiler unit with all mountings and accessories is suspended by steel slings from girders resting on steel columns. It is done so as to permit free expansion and contraction of boiler parts with temperature.

11.20 BOILER MOUNTINGS AND ACCESSORIES

Boiler mountings and accessories have been defined earlier and shown on the different boilers. Different mountings are

- (i) Water level indicator
- (ii) Safety valves
- (iii) High steam and low water safety valves
- (iv) Fusible plug
- (v) Pressure gauge
- (vi) Stop valve
- (vii) Feed check valve
- (viii) Blow off cock
- (ix) Manhole and mud box

Various boiler accessories are:

- (i) Superheater
- (ii) Economiser
- (iii) Air preheater
- (iv) Feed pump

Water level indicator: It is used for knowing the level of water in boiler as water level inside boiler should not go below a certain limit. General arrangement is shown in Fig. 11.24 with the different parts in it.

It has two tubes one is front glass tube while other is metal tube. Water level is seen through glass tube which is made strong enough to withstand high steam pressure and temperature. Two control cocks

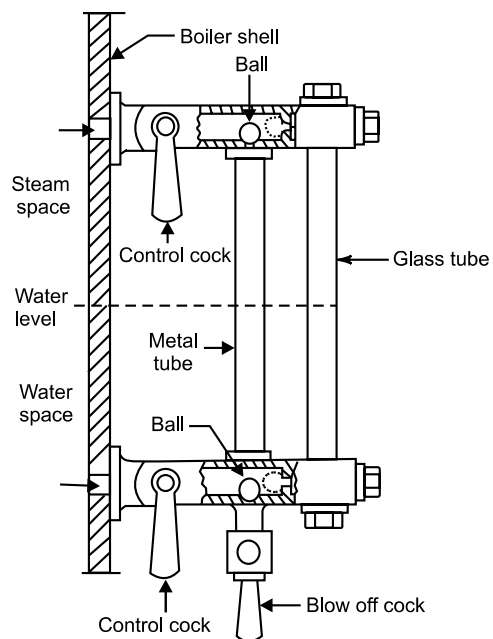


Fig. 11.24 Water level indicator

are provided for regulating steam and water passage from boiler to glass tube. For blow off purpose a blowing cock is also provided as shown. In case of breakage of glass tube the possibility of accident is prevented by providing two balls. As glass tube breaks the rush of water and steam carries the two balls with it and closes the openings for glass tube, thus water and steam flowing out can be prevented.

Number of other types of water level indicators are also available.

Safety valve: Its function is to prevent the steam pressure from exceeding a limiting maximum pressure value. Safety valve should operate automatically by releasing excess steam and bring pressure down within safe limits. These are of different types such as 'dead weight safety valve', 'lever safety valve' 'spring loaded safety valve' etc. Figure 11.25 gives the general description of 'dead weight safety valve'.

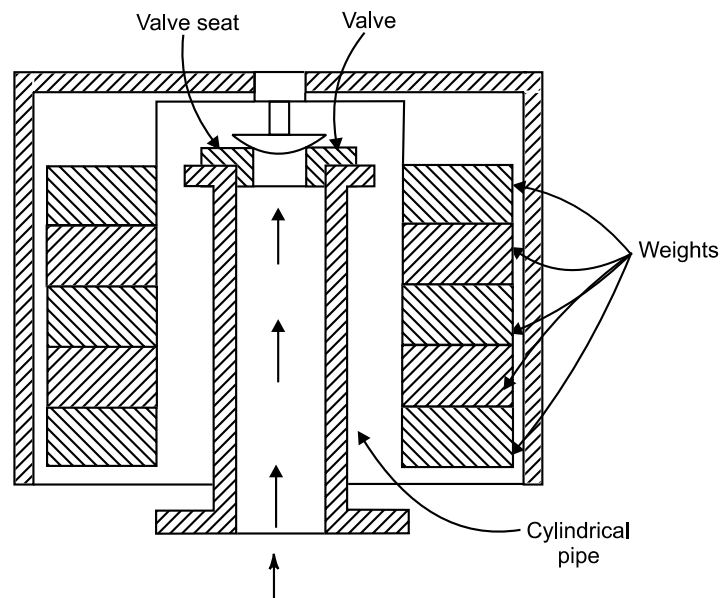


Fig. 11.25 *Dead weight safety valve*

It has a large vertical pipe on the top of which a valve seat is fixed. Valve rests upon this valve seat. A weight carrier is hung on the top of valve upon which cast iron rings enclosed in cast iron cover are placed in weight carrier as dead weight.

When the pressure of steam exceeds the total weight of valve, it is lifted and falls back as steam pressure gets reduced.

High steam and low water safety valve: This is a combined form of safety valve and low water level indicator. Figure. 11.26 shows the high steam and low water safety valve. It prevents from excessive pressure as it has a simple lever safety valve loaded by two weights as shown. Low water safety arrangement is activated through float put in boiler shell and prevents from overheating due to low water. It has two distinct valves as shown in Fig. 11.26. When the pressure inside goes beyond limiting value then high steam valve gets lifted up and excess pressure of steam is released. When the water level goes below critical level then low water valve gets raised up as it is fulcrumed and is linked to float on one end and balance weight on other end. With raising of low water valve the hemispherical valve shifts up from valve seat and steam escapes out with hissing sound. This hissing sound is loud enough to warn

attendant about the low water level in boiler. When the water level is correct then high steam valve simply acts as the dead weight safety valve. A drain pipe is also provided so that the steam getting condensed can be drained out. Condensation of steam is possible due to throttle of steam during its release from valves.

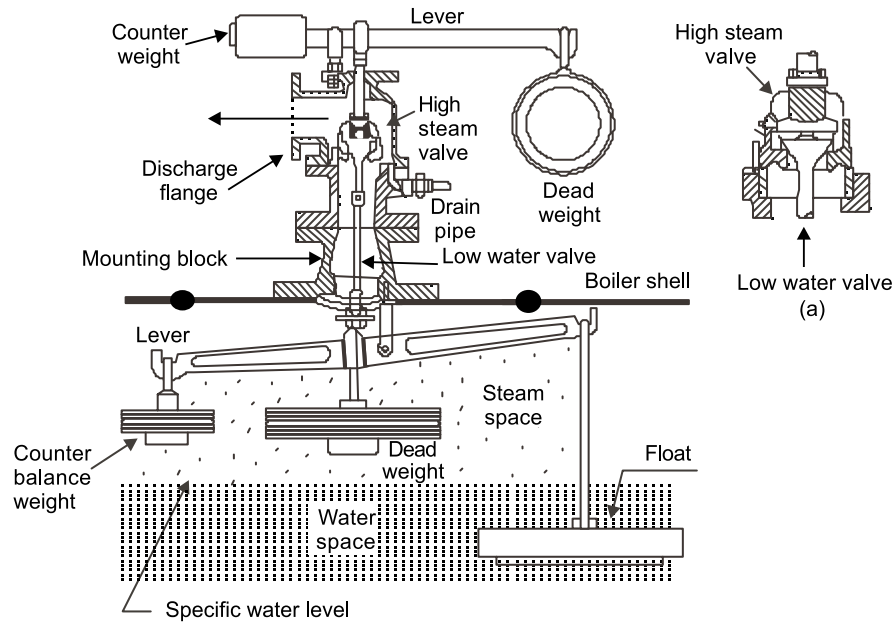


Fig. 11.26 High steam and low water safety valve

Fusible plug: It is a safety device used for preventing the level of water from going down below a critical point and thus avoid overheating. Fusible plug is mounted at crown plate of combustion chamber.

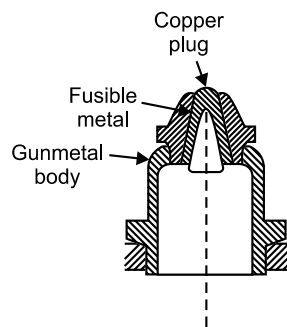


Fig. 11.27 Fusible plug

Fusible plug has gun metal body and a copper plug put with fusible metal at interface of copper plug and gun metal body. As water level goes down the heat available from furnace could not be completely utilized for steam formation and so the overheating may cause melting of fusible metal. Fusible metal is a low melting point metal. Thus upon melting of lining the copper plug falls down and

water falls from this opening onto furnace and thus quenches fire.

Pressure gauge: It is mounted at front top. Generally Bourdon type pressure gauge are being used for pressure measurement. Pressure is continuously monitored so as to avoid occurrence of over shooting of boiler pressure. Although safety devices to protect boiler against pressure rising beyond a limit are provided but pressure gauges are also used for monitoring pressure.

Stop valve: It regulates the flow of steam from the boiler as shown in Fig 11.28. This is generally mounted on highest part of boiler shell and performs function of regulating the flow of steam from boiler. Stop valve generally has main body of cast steel, valve, valve seat and nut etc. are of brass. Stop valve can be easily operated by rotating the hand wheel which causes lifting or lowering of spindle, thus causing opening or closing of valve.

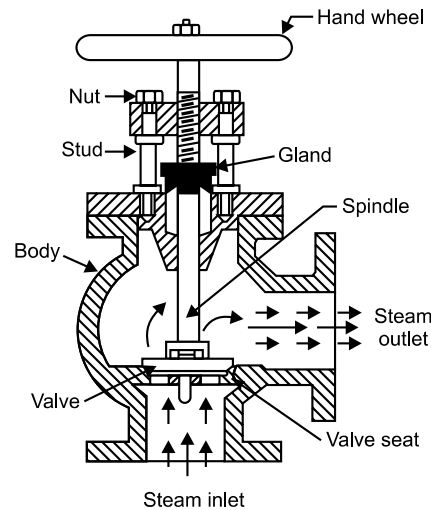


Fig. 11.28 Stop valve

Feed check valve: It is a non return valve at the end of delivery pipe from feed water pump and is placed on boiler shell slightly below normal water level. Figure 11.29 shows the arrangement in a feed check valve. It has a check valve whose opening and closing are regulated by the position of spindle. By

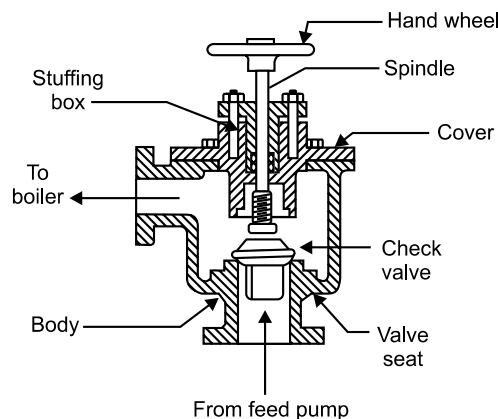


Fig. 11.29 Feed check valve

hand wheel rotation the position of spindle can be altered suitably. Feed check valve permits unidirec-

tional flow of water from feed pump to be boiler shell. Under normal running the pressure of feed water coming from pump is more than pressure inside the boiler and so the feed water continues to enter the shell. While during the non working of feed pump the pressure in boiler shell is more and so the check valve gets closed.

Blow off cock: It is used for periodical cleaning by discharging the water and sediments from bottom of boiler. Figure 11.29 shows the blow off cock. Blow off cock is fitted to the bottom of boiler shell. Blow off cock has a plug of conical type put into the mating casing. Plug position is altered for opening and closing the flow. Plug has rectangular opening which when comes in line with inlet and outlet passage then blow off cock is open and when opening is not in line then cock is closed. Plug is rotated by spindle.

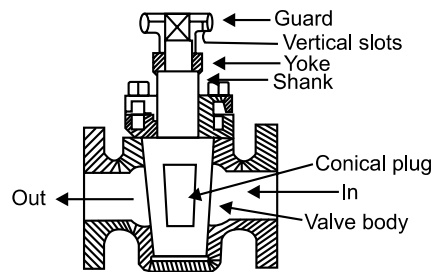
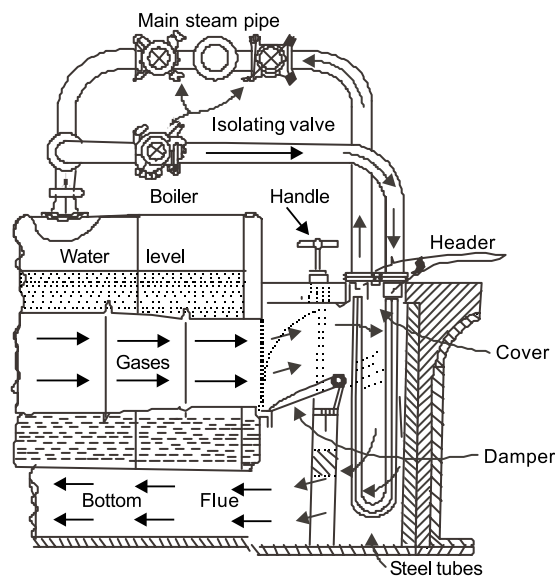


Fig. 11.30 Blow off cock

Blow off cock also helps in regulating the salt concentration as frequent draining helps in throwing out the salt deposited over period of time. Opening blow off cock removes deposited sediments in boiler.

Manhole and mud box: Manhole provides opening for cleaning, inspection and maintenance purpose. Mud box is a collection chamber (as shown in Babcock and Wilcox boiler) for collecting the mud.

Superheater: Its purpose is to super heat steam and is a type of heat exchanger in which steam flows inside tubes and hot gases surround it. Figure 11.31 shows the smooth tube hairpin type superheater (Sudgen's superheater) and convective and radiant superheater.



(a) Smooth tube hairpin type superheater

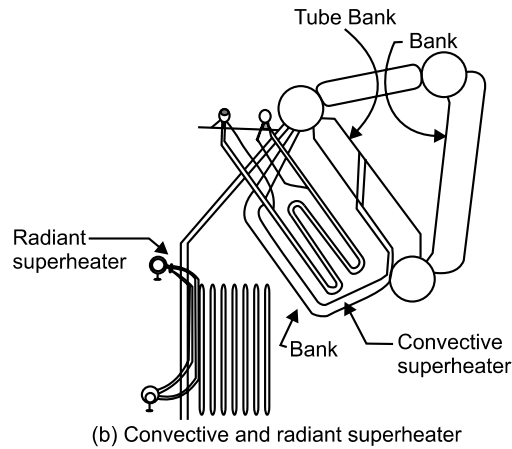


Fig. 11.31 Superheater

In hair pin superheater the steam generated is passed through isolating valve to U-shaped steel tubes. Superheated steam leaves superheater through tube connected to steam stop valve. Hot gases from fire tube are diverted over superheater tubes by damper as shown. These hot gases upon passing over steel tubes leave boiler through bottom flue. The convective and radiant superheater as shown has two set of tubes picking up heat through convection and radiation.

Economizer: It is also a heat recovery device in which feed water is heated from heat available with exhaust gases. Thus hot feed water available from economizer lowers the fuel requirement in

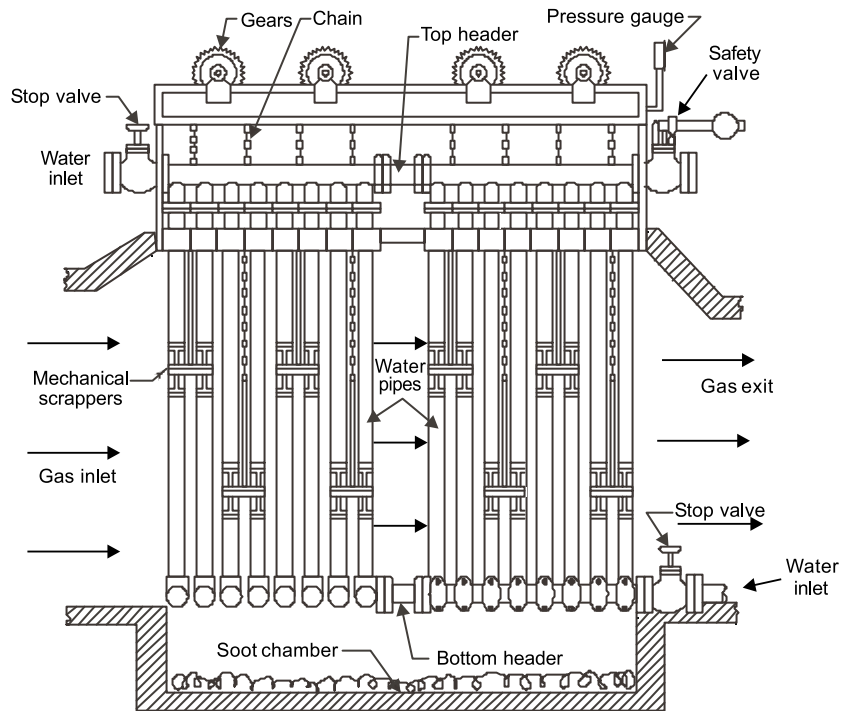


Fig. 11.32 Economizer

combustion. It is also a type of heat exchanger having exhaust gas and feed water as two fluids. General arrangement in economizer is shown in Fig. 11.32. Economizer also helps in removal of dissolved gases by preheating of water and thus minimizes tendency of corrosion and pitting. Hotter feed water also reduces thermal strain in boiler parts.

Economizer is located in the boiler structure so as to expose the economizer surface to hot gases. Its location varies with the boiler designs. Typical economizer called Green's economizer as shown in Fig. 11.32 has vertical pipes of cast iron fitted with two headers at bottom and top respectively. Feed water passes through bottom header, economizer tubes and top header to boiler. Thus economizer is simply a heat exchanger where heat is transferred from hot flue gases to water inside the tubes through metal interface. Top header is also provided with a safety valve so as to avoid explosion due to excessive pressure of water developing inside economizer tubes. Bottom header is also provided with a blow off valve so as to throw out the sediments deposited in feed water. Economizer is also provided with scrapers fitted to clean pipes from the deposition of soot carried by the flue gases. Continuous scrapping is always desired so as to maximize heat transfer rate. Economizer also has a by pass provided so that flue gases can be diverted when economizer is out of full or part operation due to failure or cleaning purpose or feed water temperature control.

Air preheater: It is used for recovering the heat going along with exhaust gases by the air before being sent to furnace. Heat is recovered by passing exhaust gases through an air to air heat exchanger as shown in Fig. 11.33. Air preheaters are generally placed after economizer and before chimney. Air when preheated before supply to furnace/combustion chamber helps in achieving 'faster rate of combustion', 'possibility of burning inferior quality coal/fuel' and 'increased rate of evaporation from boiler' etc.

Air preheaters are of tubular type, plate type and regenerative type. This classification of air preheaters bases upon the kind of arrangement used for heat exchange between two fluids. Generally, tubular type air preheater are generally used in small boilers. Tubular air preheater has hot flue gases passing inside tubes and air blown over these tubes.

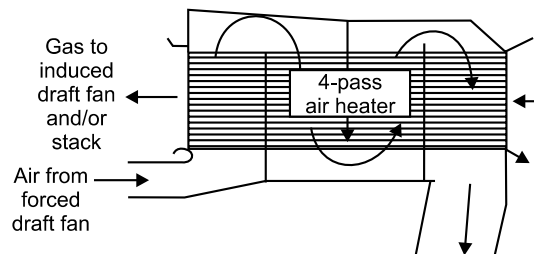


Fig. 11.33 Tubular air preheater

In case of plate type air preheater there are number of plates having air and flue gases flowing through alternative spacings. In regenerative type air preheater there is a wire mesh rotor which is alternatively heated and cooled by the hot flue gases and air to be used for combustion.

Feed pump: Feed pump is used for sending water into boiler at the pressure at which steam generation takes place. It is generally of three types i.e. centrifugal pump, reciprocating pump and injectors.

A reciprocating type feed pump is shown in Fig. 11.34. In boilers the pumps raise feed water pressure to the value more than the highest operating pressure of boiler. Pumps also have capability to deliver feed water in excess to the maximum evaporation rate of boiler. This excess capacity of feed pump is generally 15–20% of maximum continuous rating and is required to meet one or more of following situations.

- (i) Sometimes excessive steam demand may occur.

- (ii) Since boilers are to be blown out frequently to remove depositions and salts, therefore excess capacity is required.
- (iii) Malfunctioning of boiler may cause carrying away of water with steam, thereby causing water shortage in boiler.
- (iv) Over a period of time pump capacity decreases and so excess pump capacity is desired.

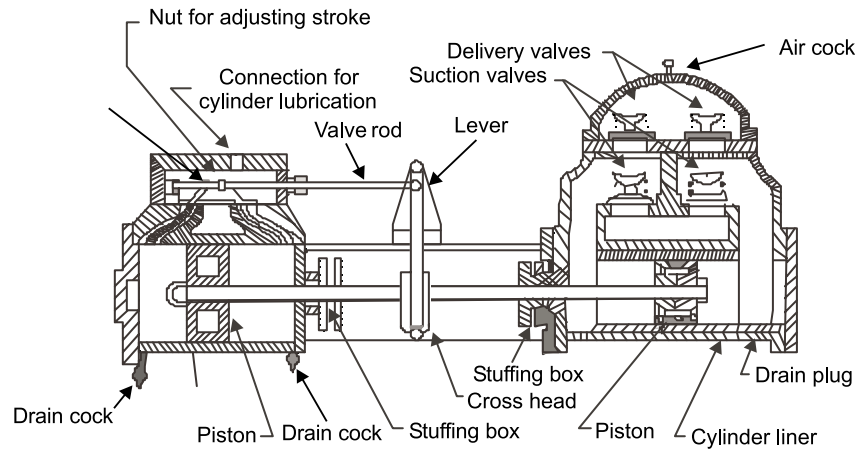


Fig. 11.34 *Reciprocating type pump, Duplex feed pump*

Now a days multistage centrifugal type pumps are used, which have favourable pressure/volume characteristics.

Steam trap: Steam traps perform function of catching steam getting condensed in the form of condensate (water) due to partial condensation of steam in pipes, steam jackets etc. Figure 11.35 shows a bucket type steam trap. Water available due to partial condensation enters steam trap at inlet A. Steam

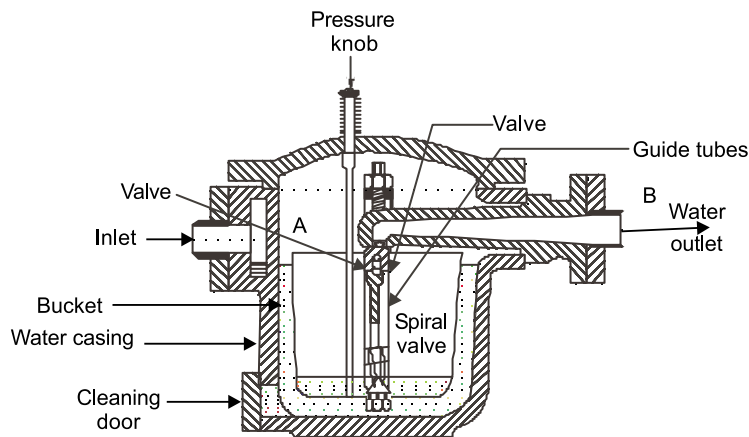


Fig. 11.35 *Bucket type steam trap*

trap casing already has water in it and bucket keeps on floating. As the water level in steam trap casing rises to the extent that water overflows from the bucket, due to excess weight of water in bucket the bucket sinks down and discharge valve opens causing water to leave through outlet passage B. After sufficient water is drained out the weight of water in bucket reduces and the bucket starts floating again. This rise of bucket closes the discharge valve again.

11.25 BOILER EFFICIENCY

Boiler efficiency quantifies, how effectively the heat is being used in boiler. Thus it could be given by the ratio of heat actually used for steam generation and total heat available due to combustion of fuel in boiler.

$$\begin{aligned}\text{Boiler efficiency} &= \frac{\text{Heat used in steam generation}}{\text{Total heat available due to fuel burning}} \\ &= \frac{m(h - h_w)}{m_f \times \text{C.V.}}\end{aligned}$$

here m_f is the mass of fuel burnt per hour, C.V. is calorific value of fuel used (kcal/kg), m is mass of steam generated per hour and enthalpies h and h_w are that of final steam and feed water, kcal/kg.

Generally high heating value of fuel is used as calorific value of fuel.

Introduction to Internal Combustion Engines

17.1 INTRODUCTION

History of internal combustion engines dates back to year 1680 when Christian Huygens developed an engine using gun powder for explosion inside a cylindrical tube. Subsequently, in year 1860 a non compression engine utilising coal gas was developed by 'Lenoir' in France. Engine was called Lenoir engine and it was operationally similar to steam engine. In 1866 'Otto-Langen free piston engine' was developed in Germany and it had thermal efficiency more than 'Lenoir engine.' In 1876 four stroke engine based on Otto cycle was developed by Nikolous Otto in Germany which revolutionised the developments of internal combustion engines and are even used till date. The compression ignition engine technology and engines based on it evolved in year 1892. Such engines are credited to Rudolf Diesel another German engineer and are so named as Diesel engine. Two stroke engines came up in 1881 in Scotland and were first developed by Dugald Clerke. 'Brayton engine' based on Brayton cycle and 'Atkinson engine' based on Atkinson cycle were developed in year 1873 and 1885 respectively. Apart from conventional two stroke and four stroke engines described earlier the 'Wankel engine' which was rotary engine and the 'Stirling engine' which was external combustion engine came up in years 1957 and 1938 respectively. In 1923 an engine called as free piston engine was developed by Pateras Pescara in France. Free piston engine is actually a combination of reciprocating piston-cylinder engine and turbine. "Engine refers to a device which transforms one form of energy into the other form". "Heat engine is a modified form of engine used for transforming chemical energy of fuel into thermal energy and subsequently for producing work". Based on the mechanism used for adding thermal energy they can be classified into the following:

- (a) External combustion engine
- (b) Internal combustion engine.

External combustion engine have combustion occurring outside engine and adding heat to the working fluid used in the engine. Thus, in external combustion engines heat released during combustion is indirectly utilized by the working fluid in external combustion engine. Internal combustion engines have combustion occurring in engine itself and heat released during combustion is directly utilized for getting shaft work. Internal combustion engines have numerous advantages over external combustion engines such as lower weight to power output ratio, simplicity, smaller initial cost, and higher efficiency etc. Due to indirect heat transfer the external combustion engines permit for the use of any cheaper fuel such as coal, wood, oil etc. for combustion.

Internal combustion engines are exhaustively used in automobiles, gas turbine etc. and external combustion engines are used in steam turbine, steam engine, nuclear power plant etc. Internal combustion

engines are efficient than external combustion engines and also are smaller than external combustion engine of similar capacity.

17.2 CLASSIFICATION OF IC ENGINES

Internal combustion engines can be classified on the following basis.

- (a) *Based on number of strokes* : Number of strokes involved in a cycle of IC engine can be two strokes or four strokes. Such engine can be;
 - (i) Two stroke engines
 - (ii) Four stroke engines
- (b) *Based on thermodynamic cycle* : Depending upon thermodynamic cycle used in the internal combustion engines these can be classified as:
 - (i) Engines based on Otto cycle ('Spark-Ignition engine')
 - (ii) Engines based on Diesel or Dual cycle ('Compression-Ignition engine')
- (c) *Based on mechanism of ignition*: Internal combustion engines have combustion as the basic process. Combustion process may be initiated using externally assisted ignition (spark ignition) or it may get initiated on its' own due to excessive compression (compression ignition). Such engines are called:
 - (i) Spark ignition engines
 - (ii) Compression ignition engines.The spark ignition engines may have "magneto ignition system" or "battery ignition system" for creating necessary electric potential for producing spark.
- (d) *Based on type of fuel used*: IC engines may be classified depending upon the type of fuel being used. These can be:
 - (i) Petrol engines (petrol being used as fuel)
 - (ii) Gas engines (gaseous fuel being used)
 - (iii) Diesel engines (diesel being used as fuel)
 - (iv) Multi-fuel engines (more than one fuel being used)
- (e) *Based on fuel admission*: IC engines can be of different types depending upon arrangement used for fuel admission:
 - (i) Carburettor type engines (use carburettor fuel metering)
 - (ii) Injection type engines (use fuel injector and injection system)
- (f) *Based on type of cooling*: IC engines have inherent requirement of continuous cooling of engine.
Based on type of cooling these can be classified as:
 - (i) Air cooled engines (Generally used in small sized engines)
 - (ii) Water cooled engines (Generally used in large sized engines)
- (g) *Based on type of motion*: IC engines may have reciprocating motion of piston or it may also have rotary motion. Such engines can be:
 - (i) Reciprocating engines
 - (ii) Rotary enginesReciprocating engines may have different cylinder arrangements such as:
 - (i) Opposed cylinder engines
 - (ii) Inclined cylinder engines
 - (iii) V-shaped cylinder arrangement.Rotary engines may be further classified as single rotor engines or multirotor engines i.e.
 - (i) Single rotor engine
 - (ii) Multi rotor engine

17.3 IC ENGINE TERMINOLOGY

Some of the basic components and generally used terms in internal combustion engines are given as under.

1. Cylinder: It is a cylindrical block having cylindrical space inside for piston to make reciprocating motion. Upper portion of cylinder which covers it from the top is called cylinder head. This is manufactured by casting process and materials used are cast iron or alloy steel.

2. Piston and Piston rings: Piston is a cylindrical part which reciprocates inside the cylinder and is used for doing work and getting work. Piston has piston rings tightly fitted in groove around piston and provide a tight seal so as to prevent leakage across piston and cylinder wall during piston's reciprocating motion. Pistons are manufactured by casting or forging process. Pistons are made of cast iron, aluminum alloy. Piston rings are made of silicon, cast iron, steel alloy by casting process.

3. Combustion space: It is the space available between the cylinder head and top of piston when piston is at farthest position from crankshaft (TDC).

4. Intake manifold: It is the passage/duct connecting intake system to the inlet valve upon cylinder. Through intake manifold the air/air-fuel mixture goes into cylinder.

5. Exhaust manifold: It is the passage/duct connecting exhaust system to the exhaust valve upon cylinder. Through exhaust manifold burnt gases go out of cylinder.

6. Valves: Engine has both intake and exhaust type of valves which are operated by valve operating mechanism comprising of cam, camshaft, follower, valve rod, rocker arm, valve spring etc. Valves are generally of spring loaded type and made out of special alloy steels by forging process.

7. Spark plug: It is the external ignitor used for initiating combustion process. Spark plug is activated by electrical energy fed by electrical system with engine. It delivers spark with suitable energy to initiate combustion at appropriate time for suitable duration.

8. Bearing: Bearings are required to support crank shaft. Bearings are made of white metal leaded bronze.

9. Connecting rod: It is the member connecting piston and crankshaft. It has generally I section and is made of steel by forging process.

10. Crank: It is the rigid member connecting the crankshaft and connecting rod. Crank is mounted on crankshaft. Crank transfers motion from connecting rod to crankshaft as it is linked to connecting rod through crank pin.

11. Crankshaft: It is the shaft at which useful positive work is available from the piston-cylinder arrangement. Reciprocating motion of piston gets converted into rotary motion of crankshaft. Crankshaft are manufactured by forging process from alloy steel.

12. Crankcase: Crankcase actually acts like a sump housing crank, crankshaft, connecting rod and is attached to cylinder. These are made of aluminium alloy, steel, cast iron etc. by casting process.

13. Gudgeon pin: It is the pin joining small end of the connecting rod and piston. This is made of steel by forging process.

14. Cams and Camshafts: Cams are mounted upon camshaft for opening and closing the valves at right timings and for correct duration. Camshaft gets motion from crankshaft through timing gears.

15. Carburettor: Carburettor is device to prepare the air fuel mixture in right proportion and supply at right time.

16. Bore: It is nominal inner diameter of the cylinder.

17. Piston area: It is the area of a circle of diameter equal to bore.

18. Stroke: It is the nominal distance travelled by the piston between two extreme positions in the cylinder.

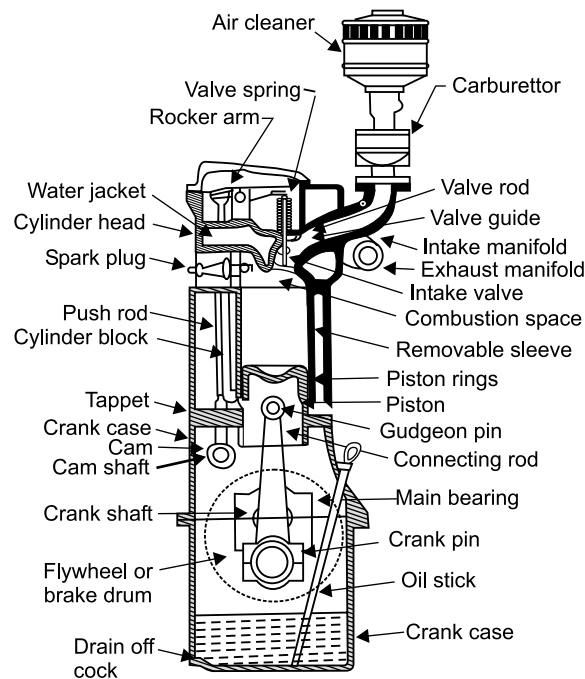


Fig. 17.1 Internal combustion engine

19. Dead centre: It refers to the extreme end positions inside the cylinder at which piston reverses its motion. Thus, there are two dead centres in cylinder, called as 'top dead centre' or 'inner dead centre' and 'bottom dead centre' or 'outer dead centre'.

Top dead centre (TDC) is the farthest position of piston from crankshaft. It is also called inner dead centre (IDC).

Bottom dead centre (BDC) refers to the closed position of piston from crankshaft. It is also called outer dead center (ODC).

20. Swept volume : It is the volume swept by piston while travelling from one dead centre to the other. It may also be called stroke volume or displacement volume.

Mathematically, Swept volume = Piston area $\times$ Stroke

21. Clearance volume: It is the volume space above the piston inside cylinder, when piston is at top dead centre. It is provided for cushioning considerations and depends, largely upon compression ratio.

22. Compression ratio: It is the ratio of the total cylinder volume when piston is at BDC to the clearance volume.

$$\text{Compression ratio} = \frac{\text{Swept volume} + \text{Clearance volume}}{\text{Clearance volume}}$$

17.4 4-STROKE SI ENGINE

Spark ignition (SI) engines employ external ignition system for initiating the combustion process. Spark plug is the most commonly used ignition method used in spark ignition engines. Let us first understand

working of 4-stroke SI engine. Figure 17.3 shows the different strokes and processes associated with them.

State	Valve position		Action	Work	Heat
	Inlet	Exhaust			
1	Just opened	Just closed	Air-fuel intake	Air to piston	None
1–2	Open	Closed			
2	Just closed	Closed	Compression	Piston to air	None
2–3	Closed	Closed			
3	Closed	Closed	Ignition Combustion Max. T , P Expansion	Gas to piston	Input to gas
3–4	Closed	Closed			
4	Closed	Closed			None
4–5	Closed	Closed			
5	Closed	Just opened	Exhaust blowdown	None	Gas to atmosphere
5–6	Closed	Open			
6	Closed	Open	Pressure drop to atmosphere Exhaust	None	Gas to atmosphere
6–1	Closed	Open			

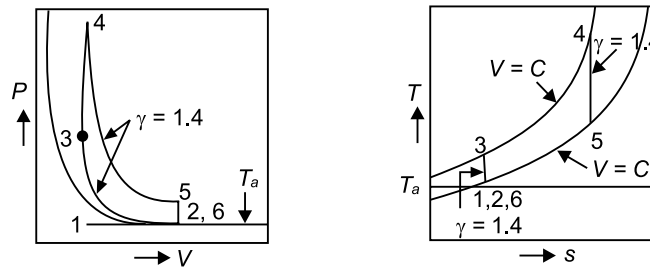
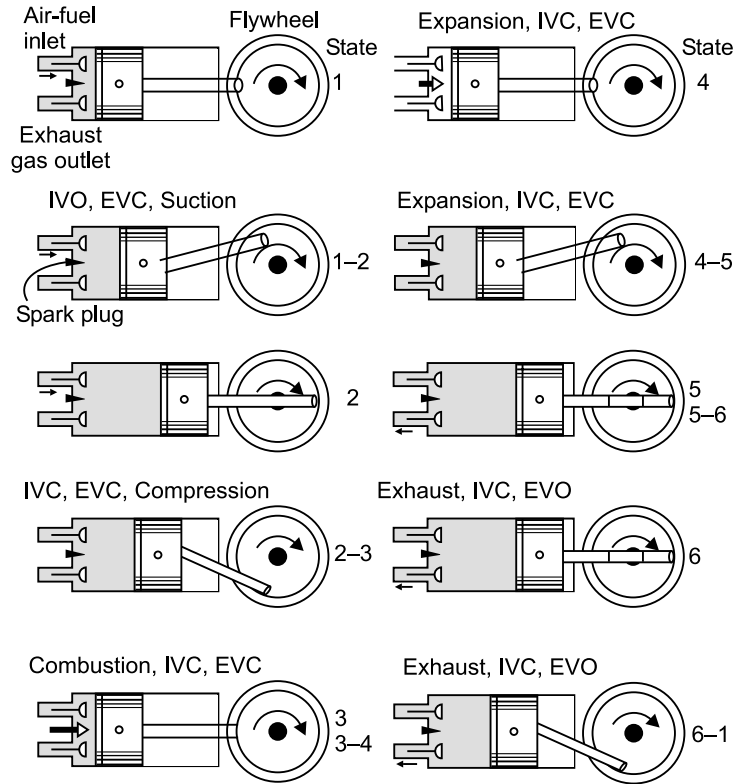


Fig. 17.2 P-V and T-s diagram for SI engine cycle (Otto cycle)

Let us start with piston at TDC, state 1. As piston moves for TDC to BDC, the inlet valve gets opened and fresh air-fuel mixture prepared in carburettor enters the cylinder. This supply of air-fuel mixture into cylinder is called suction process or suction stroke during which inlet valve is open while exhaust valve remains closed. After the piston reaches BDC, it reverses its motion and moves towards TDC. During this piston travel both inlet and exhaust valves remain closed. Thus, the air-fuel mixture inside cylinder gets compressed till piston reaches TDC. This is the second stroke and called compression stroke or compression process (2–3).



IVC : Inlet valve closed, IVO: Inlet valve open, EVC: Exhaust valve closed, EVO: Exhaust valve open.

Fig. 17.3 Working of 4-stroke S.I. engine

Now highly compressed air-fuel mixture is available inside the cylinder and ready for combustion. With piston at TDC, the spark plug is activated and it releases spark for igniting air-fuel mixture. Spark plug used in the SI engines gets activated by suitable mechanism in the engine and provides suitable amount of energy in the form of spark for initiating combustion process. This burning of mixture is accompanied by sudden increase in pressure and temperature while piston passes the TDC position. Here constant volume heat addition takes place and state 4 is attained. Due to this release of fuel energy, the combustion products try to expand and piston moves from 4–5 i.e. TDC to BDC. During this travel the inlet and exhaust valves remain closed. This stroke is called expansion stroke or power stroke or expansion process. This is the stroke accompanied by positive work available at shaft. Now while piston is at BDC the exhaust valve gets opened and combustion products are exhausted out during 5–6. Cylinder is further emptied and made ready for being recharged while piston travels from BDC to TDC, 6–1 pushing out burnt gases. This is called exhaust stroke.

Here we have seen that out of suction, compression, expansion and exhaust strokes only expansion stroke is accompanied by the production of positive work, rest three strokes are work absorbing strokes. Work requirement for the three strokes is met from the work available during expansion stroke. For

storing the excess energy and releasing it when required, a flywheel is mounted over the crankshaft. Cycle gets completed in two revolutions of crankshaft. Examples of four stroke engines are petrol engines used in cars etc.

17.5 2-STROKE SI ENGINE

2-stroke SI engine is a modified form of 4-stroke engine where all the four processes required for completion of one cycle of SI engine get completed in two strokes. Thus, obviously in each stroke two processes get completed. Figure 17.4 shows the line diagram of 2-stroke SI engine.

General arrangement shows that here there are no valves as in case of 4-stroke engines, instead it has exhaust and suction ports. Piston has a projection on its top, which acts like deflector. Mixture of air-fuel goes into crank case first and then gets transferred to top of piston at appropriate time. Let us start piston movement from TDC to BDC. When piston reverses its motion from BDC to TDC then the suction port gets uncovered and fresh mixture enters and goes into crank case. With piston moving from TDC to BDC and during covered position

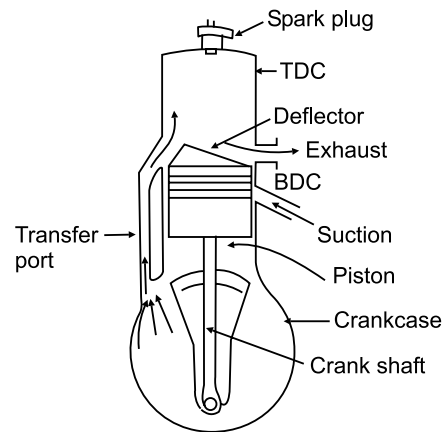


Fig. 17.4 2-stroke SI engine

of suction port the mixture gets transferred to the top of piston through transfer port. Upon reversal during piston travelling from BDC to TDC, the air fuel mixture on top of piston gets compressed and subsequently gets ignited by spark from spark plug. The combustion of fuel-air mixture results in release of excessive energy which forces piston to move from TDC to BDC. Simultaneously as piston uncovers exhaust port the burnt gases go out through exhaust port. Again when piston reaches BDC it reverses its motion and during travel from BDC to TDC the suction takes place as explained above at the bottom of piston while compression of fuel air mixture takes place on top of piston. Thus, suction and compression, both processes get completed during travel of piston from BDC to TDC. Expansion and exhaust processes occur during travel of piston from TDC to BDC along with transfer of fresh fuel air mixture from crankcase to top of piston.

Here all four processes occur during two strokes and one revolution of crank shaft. Thermodynamic cycle followed by 2-stroke SI engine is Otto cycle. Scooter engines are generally two stroke engines. 2-stroke SI engines are used for smaller applications.

17.6 4-STROKE CI ENGINE

Compression ignition (CI) engines operate generally on “Diesel”/“Dual” cycle. In these engines the combustion is realized due to excessive compression and are so called compression ignition engines. Here air alone is sucked inside the cylinder during suction stroke and compressed. Degree of compression is much more than that of spark ignition (SI) engines. After compression of air the fuel is injected into the high pressure and high temperature compressed air. Due to high temperature of air the combustion of fuel gets set on its own. Self ignition of fuel takes place due to temperature of air-fuel mixture being higher than self ignition temperature of fuel. Thus in CI engines, larger amount of compression causes high temperature, therefore unassisted combustion.

Schematic of 4-stroke CI engine is quite similar to that of 4-stroke SI engine with the only major difference that spark plug is replaced by fuel injector for injecting fuel at high pressure into compressed air. 4-Stroke CI engine works with following four processes getting completed in separate strokes. General arrangement in CI engine is similar to that of SI engine with spark plug replaced by fuel injector. *Stroke 1:* Piston travels from TDC to BDC and air is sucked.

Stroke 2: Piston travels from BDC to TDC, while air is compressed with inlet and exit passages closed.

Stroke 3: Piston reaches TDC and air gets compressed. Fuel injector injects fuel into compressed air for certain duration. Ignition of fuel also takes place simultaneously as air temperature is much higher than self ignition temperature of fuel. Burning of fuel results in release of fuel chemical energy, which forces piston to travel from TDC to BDC. Contrary to SI engine where heat addition gets completed near instantaneously, in CI engines fuel injection and thus heat addition is spread in certain stroke travel of piston i.e. heat addition takes place at constant pressure during which piston travels certain stroke length as decided by cut-off ratio. This is expansion process and piston comes down to BDC with both inlet and exit valves closed.

Stroke 4: After expansion piston reverses its motion upon reaching BDC and travels up to TDC with exit passage open. During this piston travel burnt gases are expelled out of cylinder i.e. exhaust stroke.

Completion of above four stroke requires two revolutions of crankshaft.

17.7 2-STROKE CI ENGINE

Fig. 17.5 shows the general arrangement in 2-stroke CI engine. Here the structure of CI engine is very much similar to that of SI engine with the major difference that spark plug is replaced by fuel injector, structure is made more sturdy to withstand high compression ratio.

Working of 2-stroke compression ignition engine is shown in (a), (b), (c), (d) explaining the suction, compression, expansion and exhaust processes. During piston travel from BDC to TDC air enters crankcase. When piston reaches TDC and reverses its motion to BDC air in crankcase gets partly compressed and is transferred from crankcase to top of piston through transfer port. Upon reversal of piston motion from BDC to TDC the compression of air occurs by the top side of piston while on the bottom side of piston air again enters into crankcase. Upon piston reaching TDC fuel is injected into compressed air which is at high temperature and pressure. As fuel is injected into compressed air the fuel ignition gets set on its own due to temperature being more than self ignition temperature of fuel, i.e. compression ignition. Fuel injection is continued for some duration along with its ignition which causes release of excessive fuel energy. This energy release causes piston to go back from TDC to BDC, i.e. the expansion process as shown in (c). As piston reaches BDC it simultaneously forces air in crank case to get transferred to cylinder space and forces burnt gases out of cylinder i.e. exhaust process.

Here also cycle gets completed in single revolution of crankshaft i.e two processes occurring simultaneously in each stroke.

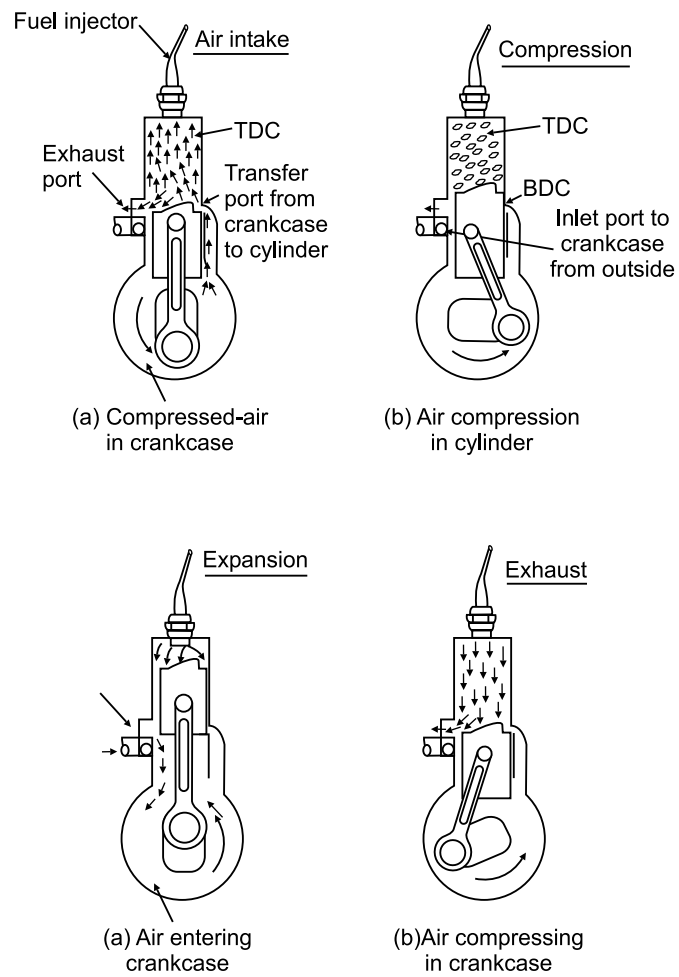


Fig. 17.5 2-stroke CI engine

MODULE 5

2.13 INTRODUCTION TO VARIOUS NON CONVENTIONAL (RENEWABLE) SOURCES OF ENERGY

Renewable energy development programme is gaining momentum in India. It has emerged as a viable option to achieve the goal of sustainable development. However, Indian renewable energy programme need more thrust at this stage. India has now the world's largest programme for deployment of renewable energy products and systems, the spread of various renewable energy technologies in the country has been supported by a variety of incentives and policy measures.

Power generation from non-conventional renewable sources has assumed significance in the context of environmental hazards posed by the excessive use of conventional fossil fuels. Renewable energy technologies have proved viable for power generation not so much as a substitute, but as supplement to conventional power generation. Currently renewables contribute over 3500 MW, which represents almost 3.5 percent of the total installed generating capacity of one lakh MW from all sources. Of this, wind power alone accounts for 1617 MW, while biomass power accounts for 450 MW and small hydros 1438 MW. An additional 4000 MW of power from renewable sources is to be added during the Tenth Five Year Plan period (2002–07) mainly through wind, biomass, small hydros, waste energy and solar energy system. Further, India has set a goal elevating the share of renewable energy sources in power generation up to 10 percent share of new capacity addition or 10,000 MW to come from renewables by 2012.

Today, India has the largest decentralised solar energy programme, the second largest biogas and improved stove programmes and the fifth largest wind energy programme in the world. A substantial manufacturing base, has been created in a variety of renewable energy technologies placing India in a position not only to export technologies; but also offer technical expertise to other countries.

Table 2.3

NON-CONVENTIONAL ENERGY POTENTIAL AND ACHIEVEMENTS		
SOURCE/SYSTEM	POTENTIAL	ACHIEVEMENTS (As on 31-03-2002)
Biogas Plants	120 lakh	33 lakh
Improved Chulhas	1,200 lakh	350 lakh
Wind	45,000 MW	1,617 MW
Small hydro	15,000 MW	1,438 MW
Biomass Power	19,500 MW	391 MW
Biomass Gasifieres		51 MW
Solar PV	20 MW/Sq Km	85 MW
Waste-to-energy	1,700 MW	22 MW
Solar Water Heating	1,400 lakh sqm collector area	6 lakh sqm collector area

2.15 WIND ENERGY

Wind energy is another potential source of energy. Winds are the motion of air caused by uneven heating of the earth's surface by the sun and rotation of the earth. It generates due to various global phenomena such as 'air-temperature difference' associated with different rates of solar heating. Since the earth's surface is made up of land, desert, water, and forest areas, the surface absorbs the sun's radiation differently. Locally, the strong winds are created by sharp temperature difference between the land and the sea. Wind resources in India are tremendous. They are mainly located near the sea coasts. Its potential in India is estimated to be of 25×10^3 mW. According to a news release from American Wind Energy Association the installed wind capacity in India in the year 2000 was 1167 mW and the wind energy production was 2.33×10^6 mWh. This is 0.6% of the total electricity production.

During the day, air above the land heats more quickly than air above water. The hot air over the land expands and rises, and the heavier, cooler air over a body of water rushes in to take its place, creating local winds. At night, the winds are reversed because air-cools more rapidly over land than over water. Similarly, the large atmospheric winds that circle the earth are created because land near the equator is heated more by the sun than land near the North and South Poles.

Today people can use wind energy to produce electricity. Wind is called a renewable energy source because we will never run out of it. Winds are natural phenomena in the atmosphere and have two different origins.

(1) Planetary winds are caused by daily rotation of earth around its polar axis and unequal temperature between Polar Regions and equatorial regions.

(2) Local Winds are caused by unequal heating and cooling of ground surface of ocean and lake surfaces during day and night.

2.15.1 WIND MACHINE FUNDAMENTALS

Throughout history people have harnessed the wind. Over 5,000 years ago, the ancient Egyptians used wind power to sail their ships on the Nile River. Later people built windmills to grind their grain. The earliest known windmills were in Persia (the area now occupied by Iran). The early windmills looked like large paddle wheels. Centuries later, the people in Holland improved the windmill. They gave it propeller type blades and made it so it could be turned to face the wind. They have been used for pumping water or grinding grain. Windmills helped Holland become one of the world's most industrialized countries by the 17th century. Today, the windmill's modern equivalent — a *wind turbine* — can use the wind's energy to generate electricity.

American colonists used windmills to grind wheat and corn, to pump water, and to cut wood at sawmills.

In this century, people used windmills to generate electricity in rural areas that did not have electric service. When power lines began to transport electricity to rural areas in the 1930s, the electric windmills were used less and less.

Then in the early 1970s, oil shortages created an environment eager for alternative energy sources, paving the way for the re-entry of the electric windmill on the American landscape.

Today's wind machine is very different from yesterday's windmill. Along with the change in name have come changes in the use and technology of the windmill. While yesterday's machines were used primarily to convert the wind's kinetic energy into mechanical power to grind grain or pump water, today's wind machines are used primarily to generate electricity. Like old-fashioned windmills, today's wind machines still use blades to collect the wind's kinetic energy. Windmills work because they slow down the speed of the wind. The wind flows over the airfoil shaped blades causing lift, like the effect on airplane wings, causing them to turn. The blades are connected to a drive shaft that turns an electric generator to produce electricity.

Modern wind machines are still wrestling with the problem of what to do when the wind isn't blowing. Large turbines are connected to the utility power network-some other type of generator picks up the load when there is no wind. Small turbines are often connected to diesel/electric generators or sometimes have a battery to store the extra energy they collect when the wind is blowing hard.

2.15.2 AEROFOIL DESIGN

A wind turbine changes the kinetic energy of the wind into rotary motion (or torque) that can do work. It could power a water pump, or turn a generator.

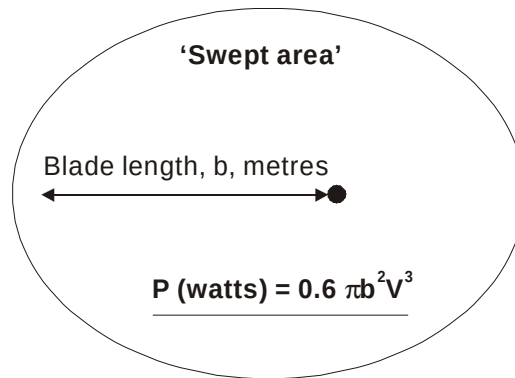


Fig. 2.5

Winds are the motion of air caused by uneven heating of the earth’s surface by the sun and rotation of the earth. There is a direct relationship between the swept area of the turbine blades and the turbine’s power output (see above). The estimated total power capacity of the winds passing over the land is about $1e^{15}$ W. But the total exploitable wind power is only $2e^{13}$ W.

The theoretical wind power can be estimated as:

$$\text{Power density} = 0.6 k \cdot \rho \cdot v^3 = 0.6 \pi b^2 v^3$$

where; k = Energy pattern factor (depends on type of wind)

ρ = Wind density

v = The average wind velocity.

Of the theoretical quantity energy that can be extracted from the wind, large commercial wind turbines are unlikely to get more than 25% of this. Small and less high-tech. designs might only get 15%. But the effect of this equation is that if the wind speed doubles, the power output increases eight-fold. So small increases in wind velocity can create large increases in power output.

The large amounts of energy that are produced at very high wind speeds means that most wind turbines have a pre-designed maximum power output to prevent the machinery ripping itself apart. Large wind turbines rated 150 kW and above are very complex machines. All wind turbines must ‘feather’ the blades turning then slightly out of the wind as wind speed increases in order to prevent the turbine running away. If this didn’t happen, the centripetal force could rip the blades off. But large turbines also have complex automatic gearboxes that keep the generator turning at the optimum speed for power generation. Rarely does the wind blow constantly. This means that the rated output of the turbine will never be achieved as a constant output. On average turbines produce about 30% of their rated capacity as continuous power. So, to compare wind turbines to continuous power sources, you have to multiply the continuous capacity by 3.33 to get the amount of wind turbine capacity required to produce the same power output. For example, a 1,000 mW coal-fired power station would require 3,333 mW of wind capacity to provide the same average power output. As yet there is no efficient form of large scale power storage that would allow the variations in wind turbine output to be evened out

over time. Reversible hydrogen fuel cells may be an option in the future — they can produce hydrogen to store energy, and then use the hydrogen to produce power at other times when the wind turbine output is low.

The larger the wind turbine, the higher the tower that supports the blades must be. This isn't just to keep the blades a safe distance off the ground. The higher you go from ground level, the faster and more uniform the wind so you get more power. So, as the power output of turbines increases, so does their size. The only restriction on the size of a wind turbine is the strength of the materials from which it is built. This is a serious engineering problem because whilst the turbine must be light enough to turn in a light breeze, the structure must be strong enough to withstand storm force winds.

Forces on the Blades. There are two types of forces operating on the blades of a propeller-type wind turbine. They are the circumferential forces in the direction of wheel rotation that provide the torque and the axial forces in the direction of the wind stream that provide an axial thrust that must be counteracted by proper mechanical design.

The circumferential force, or torque, T is obtained from

$$T = \frac{P}{\omega} = \frac{P}{\pi DN} \quad \dots(1)$$

where

T = torque, N or lb,

ω = angular velocity of turbine wheel, m/s

D = diameter of turbine wheel = $\sqrt{4A/\pi}$, m

N = wheel revolutions per unit time, s^{-1}

For a turbine operating at power P , the torque is given by

$$T = \eta \frac{1}{8g_c} \frac{\rho D V_1^3}{N} \quad \dots(2)$$

For a turbine operating at maximum efficiency $\eta_{\max} = 16/27$, the torque is given by $T_{\max}$,

$$T_{\max} = \frac{2}{27g_c} \frac{\rho D V_1^3}{N}$$

The axial force, or axial thrust, is

$$F_a = \frac{1}{2g_c} \rho A (V_1^2 - V_2^2) = \frac{\pi}{8g_c} \rho D^2 (V_1^2 - V_2^2)$$

The axial force on a turbine wheel operating at maximum efficiency where $V_e = 1/3 V_i$ is given by

$$F_{a, \max} = \frac{4}{9g_c} \rho A V_1^2 = \frac{\pi}{9g_c} \rho D^2 V_1^2$$

The axial forces are proportional to the square of the diameter of the turbine wheel which makes them difficult to cope with in extremely large-diameter machines. There is thus an upper limit of diameter that must be determined by design and economical considerations.

The performance of a wind mill rotor stated as coefficient of performance is expressed as:

$$C_p = A/P_{\max} \\ = A / (1/2 \rho V^3)$$

where ρ = Density of air
 A = Swept area
 V = Velocity of the wind

Further the tip speed ratio being the function of speed at the tip of the rotor to the wind speed, *i.e.* U/V and in most of the parts of India, the wind velocity being low (through the wind energy average around 3 kWh/m² day) The exploitation of wind mills in India is feasible. Depending upon the survey of velocity in a region the appropriate value of design parameter may be computed.

2.15.3 WIND POWER SYSTEMS

You usually stand in an open space to enjoy the wind. You know how wind originates. Moving air is wind. Since the wind has velocity it has kinetic energy. This is the energy of the wind. We shall see how the kinetic energy of the wind can be used to produce electricity. For that, we can use windmills. Windmills are devices, which work on wind. How the kinetic energy of the wind is made use of in windmills shall be looked into. We shall examine the working of a windmill. The important part of a windmill is a structure with large leaves, fixed at the top of a high tower.

What will happen when wind blows on these leaves? You may have seen paper fans available at festival places, rotating when the wind blows. In a similar manner the speed of leaves changes with the speed of the wind. What happens if the rotation of the windmill is given to the rotor of a generator? Rotor also rotates. Then electricity is obtained from the generator. What happens if the windmill is connected to a water pump? As the leaves of the windmill rotate pump works pumping out water.

Wind machines are just as efficient as coal plants. Wind plants convert 30 percent of the wind's kinetic energy into electricity. A coal-fired power plant converts about 30-35 percent of the heat energy in coal into electricity. It is the capacity factor of wind plants that puts them a step behind other power plants. Capacity factor refers to the capability of a plant to produce energy. A plant with a 100 percent capacity rating would run all day, every day at full power. There would be no down time for repairs or refueling, an impossible dream for any plant. Wind plants have about a 25 percent capacity rating because wind machines only run when the wind is blowing around nine mph or more. In comparison, coal plants typically have a 75 percent capacity rating since they can run day or night, during any season of the year.

One wind machine can produce 275-500 thousand kilowatt-hours (kWh) of electricity a year. That is enough electricity for about 50 homes per year.

In this country, wind machines produce about three billion kWh of energy a year. Wind energy provides 0.12% of the nation's electricity, a very small amount. Still, that is enough electricity to serve

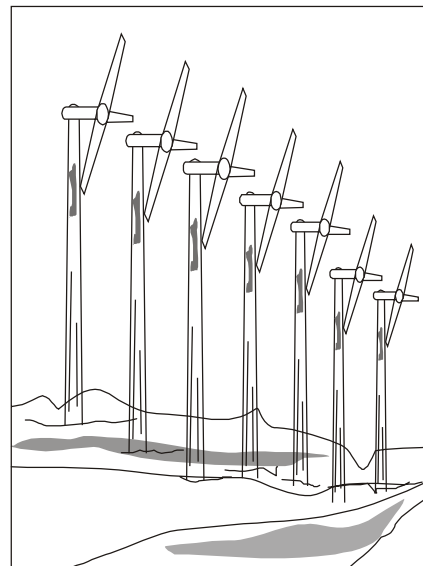


Fig. 2.6. Windmill farm.

more than 300,000 households, as many as in a city the size of San Francisco or Washington, D.C. California produces more electricity from the wind than any other state of USA. It produces 98 percent of the electricity generated from the wind in the United States. Some 16,000 wind machines produce more than one percent of California's electricity. (This is about half as much electricity as is produced by one nuclear power plant.) In the next 15 years, wind machines could produce five percent of California's electricity. The United States is the world's leading wind energy producer. The U.S. produces about half of the world's wind power. Other countries that have invested heavily in wind power research are Denmark, Japan, Germany, Sweden, The Netherlands, United Kingdom, and Italy. The American Wind Energy Association (AWEA) estimates wind energy could produce more than 10 percent of the nation's electricity within the next 30 years.

So, wind energy may be an important alternative energy source in the future, but it will not be the sole answer to our energy problems. We will still need other energy sources to meet our growing demand for electricity.

2.15.5 SELECTION OF WIND MILL

Wind power plants, or wind farms or wind mill as they are sometimes called, are clusters of wind machines used to produce electricity. A wind farm usually has hundreds of wind machines in all shapes and sizes.

Unlike coal or nuclear plants, public utility companies do not own most wind plants. Instead they are owned and operated by business people who sell the electricity produced on the wind farm to electric utilities. These private companies are known as Independent Power Producers.

Operating a wind power plant is not as simple as plunking down machines on a grassy field. Wind plant owners must carefully plan where to locate their machines. They must consider wind availability (how much the wind blows), local weather conditions, nearness to electrical transmission lines, and local zoning codes.

Wind plants also need a lot of land. One wind machine needs about two acres of land to call its own. A wind power plant takes up hundreds of acres. On the plus side, farmers can grow crops around the machines once they have been installed.

After a plant has been built, there are still maintenance costs. In some states, maintenance costs are offset by tax breaks given to power plants that use renewable energy sources. The Public Utility Regulatory Policies Act, or PURPA; also requires utility companies to purchase electricity from independent power producers at rates that are fair and nondiscriminatory.

2.15.6 RECENT DEVELOPMENTS

The present windmill technology is inadequate for the low wind speed regions in the plains. Special development projects in the following areas must be taken up so that wind energy can also be used in the low wind speed regions.

Artificial Winds. Generation of artificial winds to drive windmills by heating large surfaces with favorable thermodynamic properties is technically feasible. A project report has been prepared to heat a large surface in which case the resulting current (artificial wind) can drive turbines. The efforts needed to pursue the project in the form of money, manpower and time are huge.

Aeroelectric Plant. The low wind velocity in the plains can be augmented by the use of diffuser at intake to wind mills. Besides the propellers, Madaras and Darrieus, there has been a plethora of designs for wind machines. One intriguing power plant design, called the aeroelectric plant, uses the flow up a tower that looks like a cooling tower as shown in Fig. 2.10. Its walls are heated by solar radiation. Since the walls are circular, the sun's rays need not be tracked as it changes position in the sky during the day. The heated walls, in turn, heat the inside air and a flow up the tower is established. This air flow is made to drive a number of air turbines located near the top of the tower. The driving pressure causing air flow is given by the well-known chimney effect.

P. Carlson of California has proposed a slightly modified form in which, the interior air in a very tall tower would be cooled by pumping water to the top. The water evaporates in the low pressure air there, causing a downward flow of cooled air. The driving pressure can be calculated in a manner similar to that for wet cooling towers. A conceptual design of such a plant called for 2.4 km high, 300 m diameter tower located in a hot desert and 10 wind turbines surrounding the tower periphery at the bottom producing 2500 MW.

Low Wind Speed Turbines. The turbines available in India and abroad are suitable for a rated wind speed of 3.5 m/s or more whereas low wind speed turbines for rated values of 1.5 – 2 m/s are needed for plain areas. Special efforts are, therefore, needed to develop cheap and simple rotors, which can cut in at low wind speeds available in the plains. The Savonius rotor and American multi-blade type windmills have optimum power coefficients at a very low tip-speed and can therefore be used as starting point to develop windmills suitable for low wind speeds.

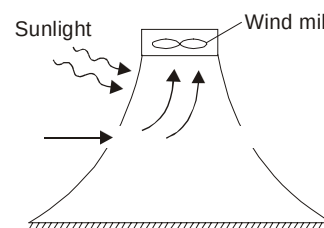


Fig. 2.10. Aeroelectric Plant

2.16 SOLAR ENERGY

The sun is the source of the vast majority of the energy we use on earth. Most of the energy we use has undergone various transformations before it is finally utilized, but it is also possible to tap this source of solar energy as it arrives on the earth's surface. There are many applications for the direct use of solar thermal energy, space heating and cooling, water heating, crop drying and solar cooking. It is a technology, which is well understood and widely used in many countries throughout the world. Most solar thermal technologies have been in existence in one form or another for centuries and have a well-established manufacturing base in most sun-rich developed countries.

The most common use for solar thermal technology is for domestic water heating. Hundreds of thousands of domestic hot water systems are in use throughout the world, especially in areas such as the Mediterranean and Australia where there is high solar insulation (the total energy per unit area received from the sun). As world oil prices vary, it is a technology, which is rapidly gaining acceptance as an energy saving measure in both domestic and commercial water heating applications. Presently, domestic water heaters are usually only found amongst wealthier sections of the community in developing countries. Other technologies exist which take advantage of the free energy provided by the sun. Water heating technologies are usually referred to as **active solar** technologies, whereas other technologies, such as space heating or cooling, which passively absorb the energy of the sun and have no moving components, are referred to as **passive solar** technologies. More sophisticated solar technologies exist for providing power for electricity generation. We will look at these briefly later in this fact sheet.

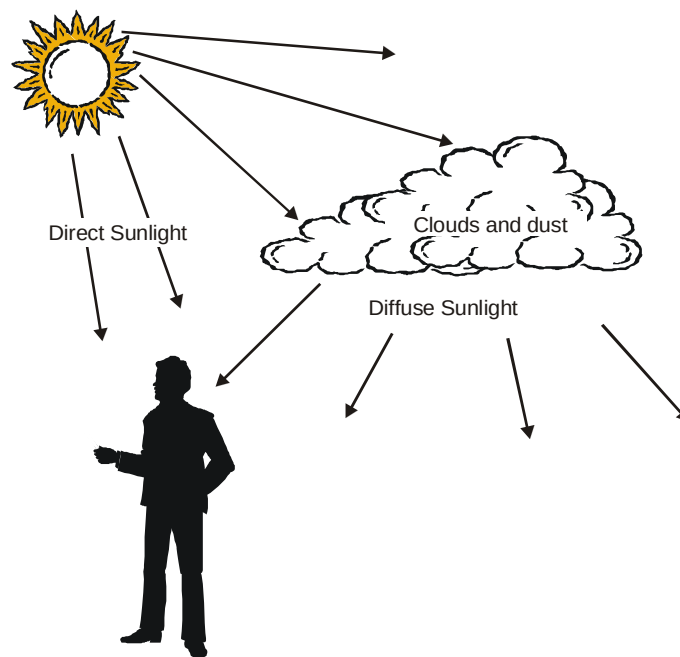


Fig. 2.11. Direct and Diffuse Solar Radiation.

Sun is the source of many forms of energy available to us. Do you know how energy is obtained from the sun? The most abundant element in sun is hydrogen. It is in a plasma state. This hydrogen at high temperature, high pressure and high density undergoes nuclear fusion and hence releases an enor-

mous amount of energy. This energy is emitted as radiations of different forms in the electromagnetic spectrum.

Out of these X-rays, gamma rays and most of ultraviolet rays do not pass through the earth's atmosphere. But heat energy and light energy are the main radiations that reach the earth. This energy is the basis for the existence of life on earth.

Sun is a sphere of intensely hot gaseous matter with a diameter of 1.39×10^9 m and 1.5×10^{11} m away from earth. Sun has an effective black body temperature of 5762 K and has a temperature of 8×10^6 K to 40×10^6 K. The sun is a continuous fusion reactor in which hydrogen (4 protons) combines to form helium (one He nucleus). The mass of the He nucleus is less than that of the four protons, mass having been lost in the reaction and converted to energy. The energy received from the sun on a unit area perpendicular to the direction of propagation of radiation outside atmosphere is called solar constant, and has a value 1353 Wm^{-2} . This radiation when received on the earth has a typical value of 1100 Wm^{-2} and is variable. The wavelength range is 0.29 to 2.5 micro meters. This energy is typically converted into usual energy form through natural and man-made processes. Natural processes include wind and biomass. Man-made processes include conversion into heat and electricity.

2.16.1 SOLAR RADIATIONS

Radiation from sun on entering the earth's atmosphere gets scattered by the atmospheric gas molecules and dust particles and received on earth from all directions and is called diffuse radiation. The portion of radiation received on earth from sun without change in original quality is called beam or direct radiation.

The earth revolves about the sun in an approximately circular path, with the sun located slightly off center of the circle. The earth's axis of rotation is tilted 23.5 degrees with respect to its plane of revolution about the sun, the position of the earth relative to the sun's rays at the time of winter solstice when the North Pole is inclined 23.5 degree away from the sun. All points on the earth's surface north of 66.5 N latitude are in total darkness while all regions within 23.5 degree of the South Pole receive continuous sunlight. At the time of the summer solstice, the situation is reversed. At the time of the two equinoxes, both poles are equidistant from the sun and all points on the earth's surface have 12 hours of daylight and 12 hours of darkness. The sun's ray passing through the center of the earth lies in the equatorial plane at the time of equinoxes. From vernal equinox to autumnal equinox, the rays lie north of the equatorial plane. From autumnal equinox to vernal equinox, the rays lie south of the equatorial plane. The average direction of the sun's rays for the entire year lies in the equatorial plane. Accordingly to intercept maximum amount of solar energy over the whole year, a solar collector in the northern hemisphere should be tilted and face due south.

The Nature and Availability of Solar Radiation. Solar radiation arrives on the surface of the earth at a maximum power density of approximately 1 kilowatt per metre squared (kWm^{-2}). The actual usable radiation component varies depending on geographical location, cloud cover, hours of sunlight each day, etc. In reality, the solar flux density (same as power density) varies between 250 and 2500 kilowatt hours per metre squared per year (kWhm^{-2} per year). As might be expected the total solar radiation is highest at the equator, especially in sunny, desert areas. Solar radiation arrives at the earth's outer atmosphere in the form of a direct beam. This light is then partially scattered by cloud, smog, dust or other atmospheric phenomenon. We therefore receive solar radiation either as direct radiation or scattered or diffuse radiation, the ratio depending on the atmospheric conditions. Both direct and diffuse components of radiation are useful, the only distinction between the two being that diffuse radiation cannot be concentrated for use.

Solar radiation arriving from the sun reaches the earth's surface as short wave radiation. All of the energy arriving from the sun is eventually re-radiated into deep space otherwise the temperature of the earth would be constantly increasing. This heat is radiated away from the earth as long-wave radiation. The art of extracting the power from the solar energy source is based around the principle of capturing the short wave radiation and preventing it from being reradiated directly to the atmosphere. Glass and other selective surfaces are used to achieve this. Glass has the ability to allow the passage of short wave radiation whilst preventing heat from being radiated in the form of long wave radiation. For storage of this trapped heat, a liquid or solid with a high thermal mass is employed. In a water heating system this will be the fluid that runs through the collector, whereas in a building the walls will act as the thermal mass. Pools or lakes are sometimes used for seasonal storage of heat.

2.16.2 SOLAR THERMAL POWER PLANT

In the solar power plant, solar energy is used to generate electricity. Sunrays are focused using concave reflectors on to copper tubes filled with water and painted black outside. The water in the tubes then boils and become steam. This steam is used to drive steam turbine, which in turn causes the generator to work. A plant using this principle is working on experimental basis in Gurgaon in Haryana. Its capacity is 500 kilowatt. Another plant of similar type is being constructed in Jodhpur in Rajasthan.

Many power plants today use fossil fuels as a heat source to boil water. The steam from the boiling water rotates a large turbine, which activates a generator that produces electricity. However, a new generation of power plants, with concentrating solar power systems, uses the sun as a heat source. There are three main types of concentrating solar power systems: parabolic-trough, dish/engine, and power tower.

Parabolic-trough systems concentrate the sun's energy through long rectangular, curved (U-shaped) mirrors. The mirrors are tilted toward the sun, focusing sunlight on a pipe that runs down the center of the trough. This heats the oil flowing through the pipe. The hot oil then is used to boil water in a conventional steam generator to produce electricity.

A dish/engine system uses a mirrored dish (similar to a very large satellite dish). The dish-shaped surface collects and concentrates the sun's heat onto a receiver, which absorbs the heat and transfers it to fluid within the engine. The heat causes the fluid to expand against a piston or turbine to produce mechanical power. The mechanical power is then used to run a generator or alternator to produce electricity.

A power tower system uses a large field of mirrors to concentrate sunlight onto the top of a tower, where a receiver sits. This heats molten salt flowing through the receiver. Then, the salt's heat is used to generate electricity through a conventional steam generator. Molten salt retains heat efficiently, so it can be stored for days before being converted into electricity. That means electricity can be produced on cloudy days or even several hours after sunset.

'Solar Power Tower' Power Plant The first is the 'Solar Power Tower' design which uses thousands of sun-tracking reflectors or heliostats to direct and concentrate solar radiation onto a boiler located atop a tower. The temperature in the boiler rises to 500 – 7000°C and the steam raised can be used to drive a turbine, which in turn drives an electricity producing turbine. There are also called central Receiver Solar Power Plants.

It can be divided into solar plant and conventional steam power plant. The flow diagram is given in Fig. 2.12.

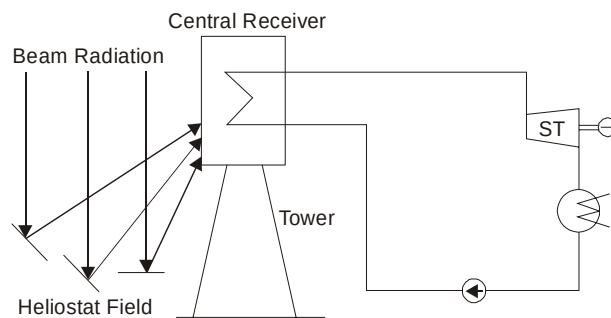


Fig. 2.12. Central Receiver Solar Power Plant.

A heliostat field consists of a large number of flat mirrors of 25 to 150 m² area which reflects the beam radiations onto a central receiver mounted on a tower. Each mirror is tracked on two axis. The absorber surface temperature may be 400 to 1000°C. The concentration ratio (total mirror area divided by receiver area) may be 1500. Steam, air or liquid metal may be used as working fluid. Steam is raised for the conventional steam power plant.

‘Distributed (Parabolic) Collector System’ Power Plant. The second type is the distributed collector system. It is also called solar farm power plant as a number of solar modules consisting of parabolic trough solar collectors are interconnected. This system uses a series of specially designed ‘Trough’ collectors which have an absorber tube running along their length. Large arrays of these collectors are coupled to provide high temperature water for driving a steam turbine. Such power stations can produce many megawatts (mW) of electricity, but are confined to areas where there is ample solar insulation.

Every module consists of a collector as shown in Figs. 2.13 and 2.14. It is rotated about one axis by a sun tracking mechanism. Thermo-oil is mostly used as heating fluid as it has very high boiling

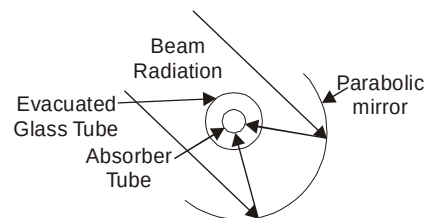


Fig. 2.13. Distributed (Parabolic) Solar Collector.

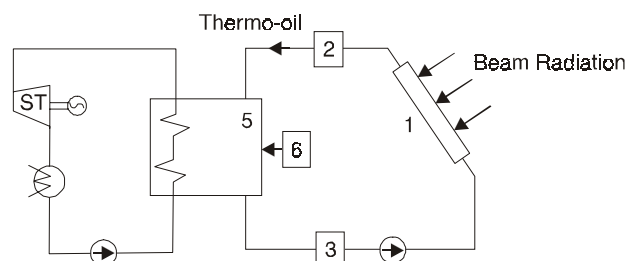


Fig. 2.14. Distributed (Parabolic) Trough Solar Power Plant.

point. Water/steam working fluid can also be used. The tubes have evacuated glass enclosure to reduce the losses. The concentration ratio is between 40 and 100. The maximum oil temperature is limited to 400°C as oil degrades above this temperature. Alternately steam at 550°C can be directly generated in the absorber tube.

These are commercially under operation. Fig. 2.14. shows a flow diagram of parabolic trough solar power plant. The working fluid is heated in collectors and collected in hot storage tank (2). The hot thermo-oil is used in boiler (5) to raise steam for the steam power plant. The boiler also is provided with a back-up unit (6) fired with natural gas. The cooled oil is stored in tank (3) and pumped (4) back to collector (1). Solar thermal power plants with a generating capacity of 80 MW are functioning in the USA.

Solar Chimney Power Plant. The air stream is heated by solar radiation absorbed by the ground and covered by a transparent cover. The hot air flow through or chimney which gives the air a certain velocity due to pressure drop caused by the chimney effect. The hot air flows through an air turbine to generate power.

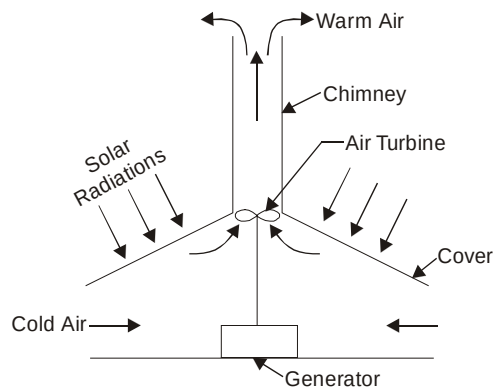


Fig. 2.15 Chimney Solar Power Plant.

2.16.3 SOLAR ENERGY STORAGE

It is well known that human beings have been using solar energy for different uses, from ancient days. Find examples of these uses and add to the list given below.

1. To get salt from sea water.
2. To dry wet clothes
3. To dry firewood
4. To dry cereals
5. To dry fish
6. To dry leather

We now use several appliances which work using solar energy. Appliances like solar cooker and solar heater absorb solar radiations and convert it into heat.

Then what about a solar cell? Solar energy is converted into electrical energy and it is directly used or stored in a battery.

There are eight possible pathways for conversion of solar radiation to useful energy. Solar thermal conversion method converts radiation to heat using solar flat collectors. Solar thermo chemical conversion method converts radiation to heat and produce steam then to kinetic energy using a pump or turbine. Solar thermal electric conversion method converts radiation to steam and to kinetic and electrical energy through a turbine and generator to electrical energy. The above route through a further electrolysis process gives chemical energy (H_2 fuel). A high temperature catalytic conversion process produces chemical energy (H_2 fuel) directly. Photovoltaic conversion of solar radiation gives direct electrical energy. Photosynthesis process produces chemical energy directly from radiation. Chemical energy (H_2 fuel) is directly produced from solar radiation using the electricity produced by the photovoltaic method. A few of these methods are dealt in detail further.

Commercial and industrial buildings may use the same solar technologies photovoltaic, passive heating, day lighting, and water heating that are used for residential buildings. These nonresidential buildings can also use solar energy technologies that would be impractical for a home. These technologies include ventilation air preheating, solar process heating and solar cooling.

Many large buildings need ventilated air to maintain indoor air quality. In cold climates, heating this air can use large amounts of energy. A solar ventilation system can preheat the air, saving both energy and money. This type of system typically uses a transpired collector, which consists of a thin, black metal panel mounted on a south-facing wall to absorb the sun's heat. Air passes through the many small holes in the panel. A space behind the perforated wall allows the air streams from the holes to mix together. The heated air is then sucked out from the top of the space into the ventilation system.

Solar process heating systems are designed to provide large quantities of hot water or space heating for nonresidential buildings. A typical system includes solar collectors that work along with a pump, a heat exchanger, and/or one or more large storage tanks. The two main types of solar collectors used an evacuated tube collector and a parabolic trough collector can operate at high temperatures with high efficiency. An evacuated-tube collector is a shallow box full of many glass, double-walled tubes and reflectors to heat the fluid inside the tubes. A vacuum between the two walls insulates the inner tube, holding in the heat. Parabolic troughs are long, rectangular, curved (U-shaped) mirrors tilted to focus sunlight on a tube, which runs down the center of the trough. This heats the fluid within the tube.

The heat from a solar collector can also be used to cool a building. It may seem impossible to use heat to cool a building, but it makes more sense if you just think of the solar heat as an energy source. Your familiar home air conditioner uses an energy source, electricity, to create cool air. Solar absorption coolers use a similar approach, combined with some very complex chemistry tricks, to create cool air from solar energy. Solar energy can also be used with evaporative coolers (also called "swamp coolers") to extend their usefulness to more humid climates, using another chemistry trick called desiccant cooling.

SPACE HEATING

In colder areas of the world (including high altitude areas within the tropics) space heating is often required during the winter months. Vast quantities of energy can be used to achieve this. If buildings are carefully designed to take full advantage of the solar insolation which they receive then much of the heating requirement can be met by solar gain alone. By incorporating certain simple design principles a new dwelling can be made to be fuel efficient and comfortable for habitation. The bulk of these technologies are architecture based and passive in nature. The use of building materials with a high thermal mass (which stores heat), good insulation and large glazed areas can increase a buildings capacity to capture and store heat from the sun. Many technologies exist to assist with diurnal heating needs but seasonal storage is more difficult and costly.

For passive solar design to be effective certain guidelines should be followed:

1. A building should have large areas of glazing facing the sun to maximise solar gain
2. Features should be included to regulate heat intake to prevent the building from overheating
3. A building should be of sufficient mass to allow heat storage for the required period
4. Contain features which promote the even distribution of heat throughout the building.

One example of a simple passive space heating technology is the Trombe wall. A massive black painted wall has a double glazed skin to prevent captured heat from escaping. The wall is vented to allow the warm air to enter the room at high level and cool air to enter the cavity between the wall and the glazing. Heat stored during the wall during the day is radiated into the room during the night. This type of technology is useful in areas where the nights are cold but the days are warm and sunny.

SPACE COOLING

The majority of the worlds developing countries, however, lie within the tropics and have little need of space heating. There is a demand, however, for space cooling. The majority of the worlds warm-climate cultures have again developed traditional, simple, elegant techniques for cooling their dwellings, often using effects promoted by passive solar phenomenon. There are many methods for minimising heat gain. These include siting a building in shade or near water, using vegetation or landscaping to direct wind into the building, good town planning to optimise the prevailing wind and available shade. Buildings can be designed for a given climate domed roofs and thermally massive structures in hot arid climates, shuttered and shaded windows to prevent heat gain, open structure bamboo housing in warm, humid areas. In some countries dwellings are constructed underground and take advantage of the relatively low and stable temperature of the surrounding ground. There are as many options as there are people.

2.16.4 RECENT DEVELOPMENTS IN SOLAR POWER PLANTS

Solar Thermal Applications. The applications include water heating for domestic, commercial and industrial use, space heating and drying, solar distillation, solar cooling through absorption & adsorption cycles, solar water pumping and solar power generation.

Solar Photovoltaics. Photovoltaic (PV) or solar cells refers to the creation of voltage from light. A solar cell is a converter; it changes the light energy into electrical energy. A cell does not store any energy, so when the source of light (typically the sun) is removed, there is no electrical current from the cell. If electricity is needed in the night, a battery must be included in the circuit. There are many materials that can be used to make solar cells, but the most common is the element silicon. A typical solar cell is 3-6 inches in diameter and are now available in various shapes like circular, square, etc. The conversion processes occurs instantly whenever there is light falling on the surface of a cell. And the output of the cell is proportional to the input light.

2.21 OCEAN ENERGY

The ocean can produce two types of energy: thermal energy from the sun's heat, and mechanical energy from the tides and waves. Oceans cover more than 70% of Earth's surface, making them the world's largest solar collectors. The sun's heat warms the surface water a lot more than the deep ocean water, and this temperature difference creates thermal energy. Just a small portion of the heat trapped in the ocean could power the world.



Fig. 2.43. Ocean Wave.

Ocean mechanical energy is quite different from ocean thermal energy. Even though the sun affects all ocean activity, tides are driven primarily by the gravitational pull of the moon, and waves are driven primarily by the winds. As a result, tides and waves are intermittent sources of energy, while ocean thermal energy is fairly constant. Also, unlike thermal energy, the electricity conversion of both tidal and wave energy usually involves mechanical devices.

A barrage (dam) is typically used to convert tidal energy into electricity by forcing the water through turbines, activating a generator. For wave energy conversion, there are three basic systems:

Channel systems that funnel the waves into reservoirs;

Float systems that drive hydraulic pumps; and

Oscillating water column systems that use the waves to compress air within a container.

The mechanical power created from these systems either directly activates a generator or transfers to a working fluid, water, or air, which then drives a turbine/generator.

The availability of ocean thermal energy on the earth can be calculated as;

Let,

E_t = Total terrestrial ocean thermal energy incidence

E_r = Total extraterrestrial solar energy received by the earth = 5.457×10^{18} MJ/year

C_r = Average clearness index = 0.5

F_A = fraction of the area of ocean = 0.7

$$\begin{aligned}
 E_t &= E_r \times C_r \times F_A \\
 &= 5.457 \times 1018 \times 0.5 \times 0.7 \\
 &= 1.9 \times 1018 \text{ MJ/year}
 \end{aligned}$$

This corresponds to an average terrestrial incidence on the waters of the solar constant
 $S = 1353 \text{ W/m}^2 \times 0.5 = 676 \text{ W/m}^2$.

This energy is not totally absorbed by the water because some of it is reflected back to the sky.
 A good estimate of the amount absorbed is obtained from the annual evaporation of water,

Annual evaporation (EV) = 1.20 m

Average water surface temperature (T) = 20°C

The latent heat of vaporization(C) = 2454 kJ/kg

Seawater density (Ω)= 1000 kg/m³.

$$\begin{aligned}
 \text{The annual energy absorbed } (E_{ab}) &= E_v \times \rho \times C \\
 &= 1.20 \times 1000 \times 2454 \\
 &= 3 \times 106 \text{ kJ/m}^2 = 95 \text{ W/m}^2.
 \end{aligned}$$

This amount is, of course, replenished by rainfall back on the water and by runoff from land.

Absorbed energy as % of incident energy = $E_{ab}/S = (95/676) \times 100 = 14\%$

2.21.1 POWER PLANTS BASED ON OCEAN ENERGY

Ocean thermal energy is used for many applications, including electricity generation. There are three types of electricity conversion systems: closed-cycle, open-cycle, and hybrid.

Closed-cycle systems use the ocean's warm surface water to vaporize a working fluid, which has a low-boiling point, such as ammonia. The vapor expands and turns a turbine. The turbine then activates a generator to produce electricity. Open-cycle systems actually boil the seawater by operating at low pressures. This produces steam that passes through a turbine/generator. Hybrid systems combine both closed-cycle and open-cycle systems.

Depending Upon these electricity conversion systems the Ocean power plant can be divided mainly in to two groups.

The Open or Claude OTEC Cycle Power Plant. The Frenchman Georges Claude constructed the first OTEC plant in 1929 on the Mantanzas Bay in Cuba.

The Claude plant used an open cycle in which seawater itself plays the multiple role of heat source, working fluid, coolant, and heat sink.

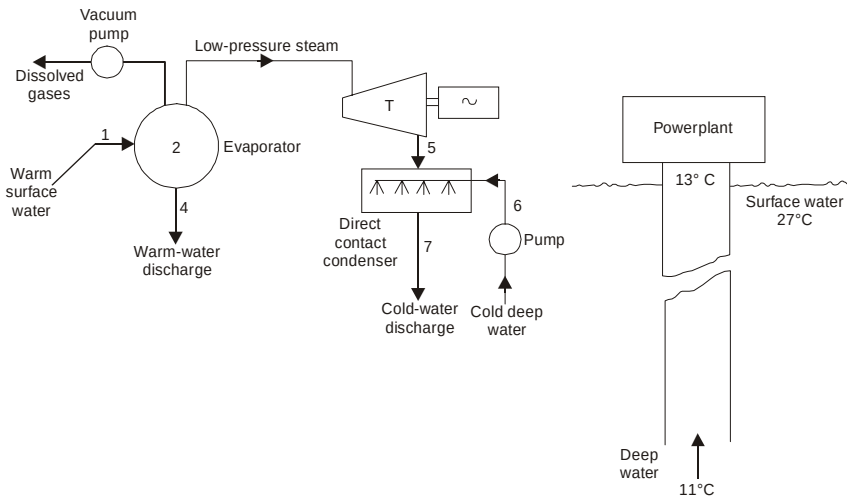


Fig. 2.44. Flow diagram and schematic of a Claude (open-cycle) OTEC power plant.

In the cycle warm surface water at 27°C is admitted into an evaporator in which the pressure is maintained at a value slightly below the saturation pressure corresponding to that water temperature. Water entering the evaporator, there four, finds itself “superheated” at the new pressure.

This temporarily superheated water undergoes volume boiling causing that water to partially flash to steam to an equilibrium two-phase condition at the new pressure and temperature. The low pressure in the evaporator is maintained by a vacuum pump that also removes the dissolved noncondensable gases from the evaporator.

The evaporator now contains a mixture of water and steam of very low quality at 2. The steam is separated from the water as saturated vapor at 3. The remaining water is saturated at 4 and is discharged as brine back to the ocean. The steam at 3 is, by conventional power plant standards, a very low-pressure, very high specific-volume working fluid (0.0317 bar, 43.40 m³/kg, compared to about 160 bar, 0.021 m³/kg for modern fossil power plants). It expands in a specially designed turbine that can handle such conditions to 5. Since the turbine exhaust system will be discharged back to the ocean in the open cycle, a direct-contact condenser is used, in which the exhaust at 5 is mixed with cold water from the deep cold-water pipe at 6, which results in a near-saturated water at 7. That water is now discharged to the ocean.

The cooling water reaching the condenser at 13°C is obtained from deep water at 11°C (51.8°F). This rise in temperature is caused by heat transfer between the progressively warmer outside water and the cooling water inside the pipe as it ascends the cold water pipe.

There are thus three temperature differences, all about 2°C: one between warm surface water and working steam, one between exhaust steam and cooling water, and one between cooling water reaching the condenser and deep water. These represent external irreversibility's that reduce the overall temperature difference between heat source and sink from 27 – 11 = 16°C (28.8°F) to 25 – 15 = 10°C (18°F) as the temperature difference available for cycle work. It is obvious that because of the very low temperature differences available to produce work, the external differences must be kept to absolute minimum to realize as high efficiency as possible. Such a necessary approach, unfortunately,

also results in very large warm and cold water flows and hence pumping power, as well as large heavy cold water pipes.

The Closed or Anderson, OTEC Cycle Power Plant. d'Arsonval's original concept in 1881 was that of a closed cycle that also utilizes the ocean's warm surface and cool deep waters as heat source and sink, respectively, but requires a separate working fluid that receives and rejects heat to the source and sink via heat exchangers (boiler and surface condenser). The working fluid may be ammonia, propane, or a Freon. The operating (saturation) pressures of such fluids at the boiler and condenser temperatures are much higher than those of water, being roughly 10 bar at the boiler, and their specific volumes are much lower, being comparable to those of steam in conventional power plants.

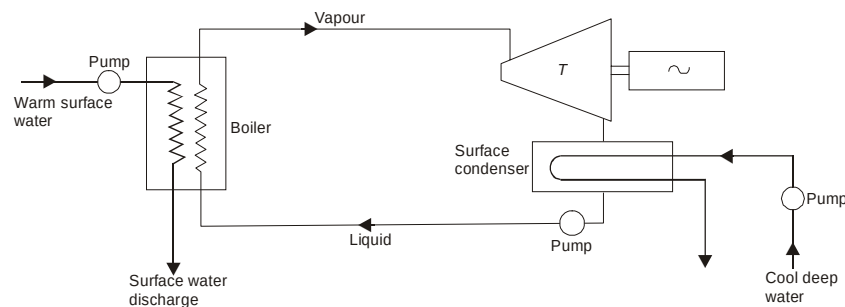


Fig. 2.45. Schematic of a closed-cycle OTEC power plant.

Such pressures and specific volumes result in turbines that are much smaller and hence less costly than those that use the low-pressure steam of the open cycle. The closed cycle also avoids the problems of the evaporator. It, however, requires the use of very large heat exchangers (boiler and condenser) because, for an efficiency of about 2 percent, the amounts of heat added and rejected are 50 times the output of the plant.

In addition, the temperature differences in the boiler and condenser must be kept as low as possible to allow for the maximum possible temperature difference across the turbine, which also contributes to the large surfaces of these units.

Barjot first proposed the closed-cycle approach in 1926, but the most recent design was by Anderson and Anderson in the 1960s. The closed cycle is sometimes referred to as the **Anderson cycle**. The Andersons chose propane as the working fluid with a 20°C temperature difference between warm surface and cool water, the latter some 600 m deep. Propane is vaporized in the boiler at 10 bars or more and exhausted in the condenser at about 5 bars.

In order to minimize the mass and the amount of material (and hence cost) used to manufacture the immensely large heat exchangers, the Anderson OTEC system employs thin plate-type heat exchangers instead of the usual heavier and more expensive shell-and-tube heat exchangers. To help reduce the thickness of the plates, the heat exchangers are placed at depths where the static pressure of the water in either ex-changer roughly equals the pressure of the working fluid. Thus if propane is the working fluid in the boiler at 26.7°C and 9.9 bar the boiler.